

POSTS AND TELECOMMUNICATIONS TRANSFORMATION STRATEGY 2025–2030





Special Assistant to the Chief Adviser

Ministry of Posts, Telecommunications and Information Technology

It is with a profound sense of responsibility and a clear vision for the future that I present the **Posts and Telecommunications Transformation Strategy (2025–2030)**.

My tenure as the Special Assistant to the Chief Adviser was defined by a singular mission: to dismantle a legacy of regulatory capture and administrative opacity that had long stifled innovation and prioritized political patronage over public service. This was no easy task and I am greatly thankful to my technical and support team for their hard work as they went above and beyond with minimal support. We prioritized digital rights, institutional autonomy, and technical sovereignty. We inherited an ecosystem characterized by a fragmented licensing regime with nearly 3,500 entities and vast arrays of idle assets. This strategy represents our collective response to those institutional failures, providing a roadmap to transition Bangladesh into a unified, resilient, and rights-based digital powerhouse.

The philosophy underpinning this strategy is anchored in the absolute protection of the citizen. We have enacted a foundational legal trinity—the Cyber Safety, Personal Data Protection, and National Data Governance Ordinances of 2025, which institutionalizes data governance as a matter of national law for the first time in our history as well as introducing long overdue protection mechanisms for personal data. Most significantly, through the Bangladesh Telecommunication (Amendment) Ordinance 2025, we have provided a statutory shield for our democracy by permanently prohibiting internet shutdowns. We have transitioned connectivity from a discretionary privilege into a protected public necessity. Furthermore, we have ensured that state infrastructure can never again be used for political coercion by introducing clear check and balance frameworks in telecommunication surveillance, to protect citizen's privacy from illegal and unwarranted surveillance.

We believe that institutional rebirth is the prerequisite for operational excellence. We have initiated a strategic turnaround for our state-owned enterprises, shifting them from fiscal liabilities to competitive infrastructure anchors. We restored the core licensing authority of the regulator, ensuring that spectrum management and licensing are driven by technical rigor rather than political directives. We have pivoted Teletalk toward a lean, rental-based network model and repealed the 127-year-old Post Office Act to redefine the national post as a modern logistics and financial engine. By rationalizing satellite pricing, we have finally made high-tech connectivity an affordable reality for maritime security and disaster response.

Our vision for technical sovereignty is embodied in the National Responsible Digital Architecture and Exchange hub. We have moved decisively to end the era of vendor lock-in by establishing the nation's first National Source Code Repository, ensuring that the state owns and controls the code essential for its services. By optimizing our National Fiber Bank and fiberizing our mobile towers, we are building the digital rails required for a truly inclusive economy. This strategy is ultimately about trust. By mandating public hearings every four months and maintaining a transparent Action Tracker, we have made the regulator directly accountable to the people it serves.

To our successor and the dedicated public servants of the division, we leave behind a framework designed to outlast political cycles. The responsibility now lies in the discipline of implementation. We must remain vigilant against the return of entrenched interests and stay focused on the master targets for 2030: zero shutdowns, universal broadband access, and a digitally sovereign Bangladesh. The journey from a captured past to a sovereign digital future has begun. Let us ensure that no citizen is left behind.

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Disclaimer

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Executive Summary

This document presents the Digital Transformation Strategy for the Posts and Telecommunications Division (PTD), the central government agency responsible for oversight, policy direction, and governance of Bangladesh's telecommunications infrastructure, postal service network, and associated state-owned enterprises. The strategy is the product of a comprehensive sectoral review conducted between June and December 2025, drawing on institutional visits, Task Force findings, project documentation, regulatory analysis, international benchmarking, media analysis, and over 650 citizen submissions.

The strategy spans ten entities under PTD's purview: the Bangladesh Telecommunication Regulatory Commission (BTRC), Bangladesh Telecommunications Company Limited (BTCL), Teletalk Bangladesh Limited, Bangladesh Submarine Cable Company Limited (BSCPLC), Bangladesh Satellite Company Limited (BSCL), Bangladesh Cable Shilpa Limited (BCSL), Telephone Shilpa Sangstha Limited (TSS), the Department of Telecommunications (DoT), the Mailing Operator and Courier Services Licensing Authority (MOCSLA), and the Bangladesh Post Office. Together, these entities manage infrastructure assets exceeding BDT 579 billion, employ over 55,000 personnel, regulate a telecommunications market serving 190 million subscribers, and operate a postal network of 9,800 offices reaching every corner of the country.

The diagnosis reveals a sector of immense strategic importance that has been systematically undermined by governance failures, regulatory capture, institutional fragmentation, and fiscal indiscipline accumulated over the period 2009 to 2024. The strategy proposes a coherent reform architecture organized across nine thematic chapters, each addressing a distinct dimension of the transformation agenda. This executive summary presents the core findings and recommendations of each chapter.

Strategic Vision and Context (Chapter 1)

The strategy is anchored in a vision of the Posts and Telecommunications Division as the institutional steward of Bangladesh's digital connectivity ecosystem, delivering affordable, reliable, secure, and inclusive telecommunications and postal services that underpin national economic competitiveness, citizen welfare, and sovereign digital capacity. The mission translates this vision into operational terms: to reform, modernize, and govern PTD entities through evidence-based policy, institutional professionalism, regulatory independence, and measurable performance accountability.

Nine guiding principles structure the entire strategy. Sovereignty requires that critical digital infrastructure, spectrum assets, and data governance remain under national control. Inclusion demands that connectivity and digital services reach every citizen regardless of geography, income, or ability. Transparency mandates that regulatory decisions, licensing processes,

financial management, and performance outcomes operate under public scrutiny. Sustainability insists that infrastructure investment and institutional design outlast political cycles and electoral transitions. Accountability ensures that every reform commitment is tied to measurable indicators, responsible entities, and time-bound milestones.

The strategic context reflects Bangladesh's position as a lower-middle-income country with 190 million mobile subscribers, broadband penetration reaching approximately 56 percent, an internet economy growing at over 20 percent annually, and a youthful population with median age below 28 years. However, this potential is constrained by a telecommunications sector where regulatory independence has been eroded, spectrum pricing is designed for revenue extraction rather than deployment efficiency, over twenty-three fragmented licence categories create intermediary rent extraction, and the Social Obligation Fund intended for digital inclusion has been subject to discretionary management. The strategy documents these failures in detail and proposes the most comprehensive restructuring of the sector since the corporatization of the Bangladesh Telegraph and Telephone Board.

Legal Landscape (Chapter 2)

The legal architecture underpinning the strategy rests on two landmark instruments enacted in 2025, supported by the broader national legal triad governing the digital economy.

The Bangladesh Telecommunication (Amendment) Ordinance 2025, approved in December 2025, represents the single most consequential legislative intervention in the sector in over a decade. Its provisions permanently prohibit government-directed internet shutdowns, a reform with no precedent in South Asia. The Ordinance restores BTRC's operational autonomy by separating the regulatory function from direct ministerial control, establishes a Parliamentary Accountability Committee to provide democratic oversight without executive interference, extends BTRC's regulatory scope to cover digital platforms and over-the-top services, and introduces a mandatory 72-hour data breach notification regime aligned with international standards such as the EU's General Data Protection Regulation.

The Telecommunications Network and Licensing Policy (TNL Policy) 2025, gazetted in September 2025, replaces the fragmented legacy licensing regime with four technology-neutral licence categories: Access Network Service Provider (ANSP), National Infrastructure and Connectivity Service Provider (NICSP), International Connectivity Service Provider (ICSP), and Non-Telecommunication Network Service Provider (NTNSP). This consolidation eliminates the intermediary layers, specifically the IGW Operators Forum, International Internet Gateway, and International Terrestrial Cable operators, that have extracted rents by standing between international bandwidth supply and domestic demand. The TNL Policy introduces performance-linked licence conditions including 50 percent tower fiberization within 18 months, coverage obligations for underserved areas, and a Significant Market Power framework for proportionate regulation.

The strategy also addresses the **Postal Services Ordinance**, which requires modernization to reflect Bangladesh Post's expanding role in logistics, financial services, and digital access. The chapter further examines the alignment of sectoral legislation with the Cyber Safety Ordinance (CSO), Personal Data Protection Ordinance (PDPO), and National Data Governance Ordinance (NDGO), which collectively form the legal triad governing data sovereignty, citizen privacy, and secure digital governance. The MOCSLA recruitment rules, pending since the authority's establishment in 2014, are identified as a critical gap requiring immediate resolution.

Telecom Infrastructure and Regulatory Capture (Chapter 3)

This chapter provides the most detailed publicly available assessment of Bangladesh's telecommunications infrastructure, market structure, spectrum governance, international connectivity, consumer protection regime, Social Obligation Fund management, and state-owned enterprise operations. The analysis exposes structural distortions that have constrained sectoral performance.

Regulatory Architecture: BTRC's operational autonomy was systematically dismantled following the 2010 amendments to the Bangladesh Telecommunication Act, which transferred effective decision-making authority to the executive branch. Commissioners were appointed through political patronage rather than merit-based selection, enforcement capacity was hollowed out, and the Research Division was never staffed. The result was a regulator captured by the interests it was supposed to oversee. The strategy proposes a Government Order formally codifying the separation of DoT's policy advisory role from BTRC's licensing and enforcement functions, reconstitution of the BTRC Research Division with independent analytical capacity, and the phased restoration of regulatory credibility through transparent decision-making and published reasoning.

Market Structure: The proliferation of over twenty-three licence categories created a layered market structure where intermediary operators extracted margins without adding proportionate value. The International Gateway (IGW) operators, International Internet Gateway (IIG) operators, and International Terrestrial Cable (ITC) operators collectively formed a toll-collection infrastructure between BSCPLC's international bandwidth and domestic end-users. Vertically integrated operators exploited this fragmentation through transfer pricing, with documented instances of bandwidth sold to affiliated entities at BDT 8 per Mbps when market rates were BDT 120 per Mbps. The TNL Policy's consolidation into four categories directly addresses this distortion.

Spectrum Governance: Spectrum pricing has prioritized revenue maximization over deployment efficiency. Auction designs have favoured incumbent operators, performance-linked conditions have been absent or unenforced, and secondary trading frameworks remain undeveloped. The strategy proposes a comprehensive Five-Year National Spectrum Plan

addressing 5G mid-band allocation (3.3 to 3.8 GHz), performance-linked pricing, technology-neutral assignments, and enterprise licensing provisions.

International Connectivity: BSCPLC operates three submarine cable systems: SEA-ME-WE 4 (operational since 2005, end of life 2030), SEA-ME-WE 5 (operational since 2016, end of life 2037), and SEA-ME-WE 6 (expected operational Q1 2026, adding 13,200 Gbps capacity, end of life 2045). Bandwidth has been historically overpriced at the wholesale level, constraining internet affordability. The private consortium BPCS (SIGMAR) cable provides limited competitive pressure. Starlink's entry in May 2025, with 880 Gbps gateway capacity and 1,800 active terminals, introduces the first meaningful alternative to terrestrial and submarine connectivity.

Social Obligation Fund: The SOF, intended as the primary instrument for universal digital inclusion, has been managed through discretionary allocation rather than systematic results-based financing. The strategy proposes restructuring the SOF to a results-based model that disburses funding only against verified delivery of measurable outcomes, confirmed by independent audits.

State-Owned Enterprises: BTCL, with capital assets exceeding BDT 579 billion and over 12,600 employees, operates legacy fixed-line infrastructure that generates declining revenue against rising operational costs. Teletalk holds approximately 6 percent mobile market share and reported a net loss of BDT 1.5 billion in FY2022-23. BSCPLC is one of the few profitable SOEs but its pricing policies have constrained downstream affordability. BSCL manages Bangabandhu-1 satellite with growing but still limited commercial revenue. BCSL manufactures copper and fibre optic cables under procurement frameworks that have not kept pace with market demand. TSS is repositioning as a domestic digital device manufacturer through strategic partnerships.

Sectoral Impacts Beyond Posts and Telecommunications (Chapter 4)

The strategy recognizes that the posts and telecommunications sector's significance extends far beyond its institutional boundaries. Telecommunications infrastructure and postal networks serve as enabling platforms for citizen-centric service delivery, e-commerce logistics, financial inclusion, disaster resilience, agricultural market connectivity, health and education access, and democratic processes.

Citizen-Centric Service Design: The strategy adopts a three-tier service design framework aligned with the National E-Service Guideline: digital-first (fully online services for digitally literate populations), assisted access (Nagorik Sheba centres with trained intermediaries for populations with limited digital literacy), and in-person services (retained for transactions requiring physical verification). Bangladesh Post's 9,800 offices represent the country's most

extensive physical service network and are being integrated with the Nagorik Sheba programme to create community-based digital service hubs offering NID, passport, tax filing, SIM registration, and connectivity services.

E-Commerce and Logistics: The strategy positions Bangladesh Post as the national last-mile logistics backbone for rural commerce. A reverse logistics middleware system enables two-way urban-rural e-commerce parcel delivery. API integration with the Domestic Mail Monitoring System supports real-time tracking. Cold chain infrastructure (14 chiller chambers, 84 deep freezers) enables agricultural and perishable goods logistics. The 2025 pilot that transported 65 tons of mangoes directly to consumers demonstrates the operational potential.

Financial Inclusion: Bangladesh Post's postal banking infrastructure, combined with the Nagad mobile financial services platform, positions the postal network as a vehicle for reaching financially excluded populations, particularly in rural areas. The strategy envisions expanded postal financial products including savings, micro-insurance, and remittance services by 2030.

Democratic Infrastructure: The postal ballot modernization programme, developed in coordination with the Election Commission, establishes a Standard Operating Procedure for ballot handling with digital tracking integration, ensuring chain-of-custody integrity for absentee and overseas voting. This positions Bangladesh Post as a contributor to democratic infrastructure.

Cross-Sectoral Coordination: The strategy establishes a coordination framework requiring PTD entities to engage systematically with the health, education, agriculture, social protection, disaster management, and justice sectors, ensuring that connectivity investments translate into measurable development outcomes rather than isolated infrastructure outputs.

Strategic Policies for Sovereignty and Innovation (Chapter 5)

This chapter translates the strategy's sovereignty principle into concrete policy frameworks governing data, technology, manufacturing, and environmental sustainability within the posts and telecommunications sector.

Digital Sovereignty: The strategy mandates vendor neutrality in all telecom infrastructure procurement, prohibiting single-vendor dependency in core network elements. The National Source Code Policy requires government ownership of all publicly funded software with mandatory escrow and audit provisions. The National Cloud Policy establishes a four-tier data classification framework (D0 Open through D4 Critical) with sovereign-by-design architecture and Zero Trust implementation for all government cloud deployments. Critical data must reside in nationally controlled data centres, with 100 percent sovereign cloud hosting targeted by 2030.

Cybersecurity and Critical Information Infrastructure: The strategy designates telecommunications networks, submarine cable landing stations, satellite ground stations, and data centres as Critical Information Infrastructure (CII) subject to enhanced security obligations. All major operators must achieve ISO 27001 certification by 2028, with the 72-hour mandatory breach reporting regime fully operational. A sectoral CERT establishment and biannual cybersecurity drills provide layered defence against evolving threats.

Local Manufacturing and Import Substitution: The strategy positions BCSL and TSS as anchors of a domestic digital manufacturing ecosystem. BCSL's optical fibre production capacity is being expanded to meet demand generated by the tower fiberization mandate. TSS has signed strategic partnership agreements with technology firms for CCTV equipment manufacturing, lithium-ion battery repair, 4G/5G dual-band WiFi router assembly, and computer accessories production. These initiatives contribute to import substitution, local employment, and technology transfer.

Environmental Sustainability: Green ICT practices are integrated across the strategy. The BTCL Hathazari Green Data Centre, developed as a PPP with ADB consultation, will pioneer renewable energy sourcing and modern containment cooling. Renewable energy adoption at tower sites is incentivized through spectrum fee rebates. A producer responsibility framework for electronic waste establishes recycling requirements and annual reporting obligations.

Emerging Technologies: The strategy establishes governance frameworks for 5G, IoT, AI, blockchain, and satellite technologies. A risk-based AI governance model requires algorithmic impact assessments for public-sector deployments. The National Blockchain Policy focuses on sovereign, permissioned networks for land records, identity validation, and supply chain integrity. The NGSO satellite framework, anchored by the Starlink operational precedent, enables multiple satellite operators with clear revenue sharing and local gateway requirements.

Institutional Reform (Chapter 6)

The institutional reform chapter addresses the human and organizational dimensions that legislative change alone cannot resolve. The analysis is grounded in documented patterns of governance failure across PTD entities, including politicized leadership appointments, irregular recruitment, unauthorized deputation extensions, absence of performance management systems, and board governance compromised by conflicts of interest.

The reform agenda operates across five interconnected tracks:

Human Resource Integrity: A unified digital Human Resource Information System (HRIS) will be deployed across all PTD entities, integrating personnel records, payroll, promotions, deputation tracking, and performance management into a single auditable platform. Merit-based

recruitment and promotion frameworks, standardized across entities, will replace the discretionary practices that enabled irregular appointments.

Board Governance: Boards of Directors across all SOEs will be reconstituted to include a majority of independent members with sectoral expertise. The PTD Secretary will be separated from the Board Chair position to prevent conflicts of interest between policy oversight and operational control. Independent mixed selection boards, including external experts and CAG observers, will conduct leadership appointments.

Accountability and Oversight: Comprehensive HR compliance audits by the Comptroller and Auditor General will be integrated into the regular audit cycle, with findings reported to Parliament. Secure whistleblower channels will guarantee anonymity and protection against retaliation. Fast-tracked disciplinary mechanisms will prevent accused officers from evading proceedings through retirement or transfer.

Structural Reform: A dedicated Telecommunication Cadre will be rebuilt under the Department of Telecommunications through competitive civil service recruitment, restoring technical capacity and institutional continuity lost since the corporatization of BTTB. Strict limits on deputation and ad-hoc extensions will rebuild professionalism across entities.

Entity-Specific Reform Programmes: Each of the ten PTD entities has a documented reform programme with specific done and pending actions. BTCL is pursuing fibre mapping digitization, MVNO technical integration with Teletalk, IPTV launch, ERP deployment, Green Data Centre development, and co-location service modernization. Teletalk has expanded 4G coverage to 85 percent of sites and signed tower sharing agreements for 2,300 additional sites. BSCPLC is commissioning SMW-6. Bangladesh Post has installed GPS trackers on 85 mail vans, established e-commerce API integration, initiated cold chain infrastructure, and is developing the National Address Management System. TSS has signed MoUs for CCTV manufacturing and battery repair. BCSL is expanding satellite services across South Asia. BCSL is modernizing optical fibre production lines. BTRC is preparing ERP specifications and the National Equipment Identity Register relaunch. MOCSLA is finalizing recruitment regulations.

Project Reform (Chapter 7)

The strategy includes the most comprehensive public assessment of PTD's development project portfolio, covering 42 projects with a combined investment exceeding BDT 20,000 crore. The Task Force review identified systemic governance failures including weak feasibility analysis, procurement irregularities, scope deviations from approved Development Project Proformas, implementation delays, dependency failures between projects, and falsified progress reporting.

The strategy documents specific interventions:

Cost Re-baselining: Inflated procurement costs for IT equipment have been slashed against international benchmarks, with a new value-for-money standard enforced for all future tenders. The cost per connection has been re-evaluated, eliminating the artificially inflated per-unit costs that characterized earlier procurement cycles.

Component Cancellation: Non-viable components have been identified and cancelled, saving hundreds of crores in public funds. The CRVS Software component (BDT 502.09 crore) and the Digital Village initiative (BDT 251.16 crore) were scrapped due to the failure of the Chinese G2G loan agreement and lack of feasibility. Funds have been reallocated or returned to the exchequer.

Third-Party Verification: A mandatory third-party verification mechanism has been established, with independent technical auditors physically inspecting random sites to validate contractor-submitted data before any bill is processed. The definition of connectivity has been shifted from physical installation to verified active usage, ensuring payments are released only for functional connections.

New Generation Projects: The strategy establishes a new generation of development projects designed with the governance safeguards absent from their predecessors. These include the Nagorik Sheba expansion, BTCL Green Data Centre, fourth submarine cable feasibility, Teletalk network modernization, Bangladesh Post digital logistics platform, and TSS manufacturing scale-up. Each project is subject to mandatory feasibility studies, third-party evaluations, centralized procurement databases, and ERP-driven monitoring.

Social Obligation Fund Reform: The SOF governance framework has been restructured from an input-based allocation model to a results-based financing system. Disbursements are linked to independently verified outcomes including broadband speeds, latency, active user adoption, and connectivity quality, rather than to input expenditure certificates.

Strategic and Governance Reforms (Chapter 8)

This chapter consolidates the entity-level and project-level reforms into eleven sector-wide governance frameworks that collectively define the operating environment for the posts and telecommunications sector through 2030.

Regulatory Architecture Reform: Formal separation of DoT's policy advisory function from BTRC's licensing and enforcement mandate, with published Government Orders, joint steering committees, and transparent decision-making protocols.

Market Design Modernization: Implementation of the TNL Policy's four-category licence structure, managed migration from legacy licences, the Significant Market Power framework, and open-access obligations to prevent anti-competitive concentration.

Spectrum Governance: The Five-Year National Spectrum Plan addressing 5G allocation, secondary trading, performance-linked pricing, and enterprise licensing, with public consultation and published reasoning for all spectrum assignments.

Digital Regulatory Systems: The BTRC ERP platform integrating licensing, revenue management, spectrum allocation, QoS monitoring, and enforcement. The National One-Stop Digital Permit Portal with 15-day and 30-day statutory SLA timelines for tower construction, fibre laying, and equipment import approvals.

Consumer Protection: District-level QoS performance scorecards published monthly, a unified digital complaint management system across web, mobile, and USSD channels, and modernized customer codes requiring plain-language tariff communication and explicit consent for value-added services.

Data Governance: ISO 27001/27701 adoption mandates for all operators, the 72-hour breach notification regime, sectoral CERT establishment, and Critical Information Infrastructure designation for core telecommunications assets.

Digital Inclusion: Results-based SOF financing, data and device affordability vouchers, zero-interest handset programmes for low-income populations, digital navigator helpdesks for rural and marginalized users, and measurable targets for closing the urban-rural connectivity divide.

Investment Climate: Clear, time-bound licensing processes, published fee schedules, independent dispute resolution mechanisms, intellectual property protections, and streamlined foreign investment approval pathways designed to restore investor confidence.

Emerging Technology Governance: 5G readiness including urban pilot deployment by 2028 and metropolitan commercial services by 2029, IoT interoperability standards, AI governance frameworks, satellite service licensing, and regulatory sandbox provisions.

Sustainability: The Green Data Centre at Hathazari, renewable energy incentives for tower sites, e-waste producer responsibility frameworks, and carbon footprint reporting requirements.

Evidence-Based Policymaking: The BTRC Research Division producing quarterly market analyses and regulatory impact evaluations, the Centre of Excellence for Digital Public Infrastructure conducting independent project evaluations, and the Reform Programme Management Office operating quarterly accountability dashboards.

Implementation Roadmap (Chapter 9)

The strategy is organized into a four-phase implementation roadmap with conditional sequencing, where each phase establishes the prerequisites for the next.

Phase	Period	Key Deliverables
Phase 1	2025 to 2026	Legal and institutional foundations: BTRC Autonomy Government Order gazetted; TNL Policy migration framework published; Postal Services Ordinance consultation draft circulated; Five-Year Spectrum Plan published; all SOE boards reconstituted; Digital HRIS deployed; BTRC ERP specified and procurement initiated; SMW-6 commissioned (13,200 Gbps); Teletalk tower sharing agreement (2,300 sites) operationalized; GPS fleet tracking integrated for Bangladesh Post; 500 post offices converted to Nagorik Sheba centres; TSS manufacturing pilots initiated; Reform PMO dashboards operational; CAG HR compliance audit completed.
Phase 2	2027 to 2028	Market restructuring and infrastructure scaling: All operators migrated to TNL Policy licence categories; MVNO commercially launched (BTCL-Teletalk); BTCL IPTV launched; SMP framework determinations published; Spectrum secondary trading operational; 80 percent tower fiberization achieved; Hathazari Green Data Centre PPP structured and construction commenced; BTRC ERP fully deployed; One-Stop Permit Portal operational; QoS Public Dashboard live; ISO 27001 achieved by major operators; National Address Management System implemented; Nagorik Sheba expanded to 3,000 centres; 5G urban pilot launched; NEIR fully integrated with customs databases.
Phase 3	2029 to 2030	Smart governance and maturity: 5G commercial deployment in metropolitan areas and all district headquarters; Research Division producing quarterly publications; Centre of Excellence conducting annual independent evaluations; Parliamentary Accountability Committee conducting regular oversight; Nagorik Sheba network reaches 8,000 centres; Bangladesh Post digital logistics platform fully operational; Postal financial services expanded; Hathazari Green Data Centre Tier III certified; ISO 27001 achieved by 100 percent of operators; Bangabandhu-2 procurement advanced.
Phase 4	2030 to 2035	Long-term vision: Bangabandhu-2 satellite commissioned; Fourth submarine cable operational (diversifying beyond SEA-ME-WE corridor); 6G research and spectrum preparedness; 95 percent household broadband availability; Comprehensive data sovereignty capabilities established; Bangladesh positioned as a regional digital

		connectivity hub with submarine cable landing, satellite gateway, and data centre hosting services.
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The roadmap includes a consolidated risk management framework addressing political discontinuity, institutional resistance, fiscal constraints, technology evolution, market concentration, and cybersecurity threats. Each risk category is matched with specific mitigation instruments embedded in the strategy's legal, institutional, and financial architecture.

Consolidated KPI Framework

The following master indicators provide the primary accountability reference for monitoring strategy implementation:

Indicator			Baseline (2025)	2026 Target	2028 Target	2030 Target
BTRC	Operational	Autonomy	Compromised	GO Published	Fully Operational	Institutionalized
Active Licence Categories			23+	In Migration	4 (Migrating)	4 (Complete)
Internet	Shutdowns	(Annual)	Multiple	Zero	Zero	Zero
Tower Fiberization			Approx. 35%	Approx. 45%	80%	95%+
International	Bandwidth	(Gbps)	Approx. 7,220	Approx. 15,000	20,000+	25,000+
Teletalk 4G Site Coverage			Approx. 3,400	5,100+	6,500+	7,500+
Nagorik	Sheba	Postal Centres	0	500	3,000	8,000
5G Commercial Service			None	Plan Published	Urban Pilot	District HQs
SOE Board Independence			Politicized	Reconstituted	Independent	Institutionalized

ISO 27001 Operator Coverage	Approx. 0%	Initiated	Major Operators	100%
QoS Public Dashboard	None	Specified	Operational	Mature
SOF Model	Input-Based	Framework Design	Pilots Running	Full Deployment
BD Post E-Commerce Platform	None	API Integrated	Full Network	Platform Mature
Research Division Output	None	Established	Quarterly Reports	Full Programme
Sovereign Cloud (Critical Data)	Low	40%	80%	100%
Green Data Centres	0	Under PPP	Construction	Tier III Certified

Strategic Imperative

Bangladesh's posts and telecommunications sector stands at an inflection point. The legislative, policy, and institutional reforms initiated in 2025 represent the most comprehensive restructuring of the sector in a generation. The Amended Ordinance's prohibition of internet shutdowns, with no precedent in South Asia, signals a fundamental commitment to digital rights and economic stability. The TNL Policy's elimination of intermediary licence layers addresses a structural distortion that has inflated connectivity costs for over a decade. The Starlink licensing demonstrated that new market entrants can be accommodated within weeks when regulatory processes are functional. The tower sharing agreements show that infrastructure expansion can be financed through commercial partnerships rather than public expenditure alone. The postal reforms demonstrate that legacy institutions can be repositioned for digital-age relevance.

These reforms have begun. The question is no longer whether transformation is possible but whether the institutions, legal frameworks, and accountability mechanisms established through this strategy can sustain reform momentum across electoral cycles, leadership transitions, fiscal pressures, and the resistance of entrenched interests. The strategy's architecture is deliberately designed to address this challenge. Legal commitments are gazetted in ordinances and government orders rather than administrative circulars. Institutional reforms are embedded in board reconstitution and digital systems rather than dependent on individual leadership. Financial discipline is enforced through third-party verification and results-based financing rather than

self-reported compliance. Performance accountability is published through dashboards and parliamentary oversight rather than internal reporting.

The ultimate measure of this strategy will not be the comprehensiveness of its analysis or the ambition of its targets. It will be whether, by 2030, every citizen in Bangladesh has access to affordable and reliable connectivity, whether the telecommunications market operates under transparent and predictable governance, whether the postal network has been transformed from a letter delivery service into a digital inclusion platform, whether public investment in the sector generates verified outcomes rather than procurement expenditure, and whether Bangladesh is recognized as a credible participant in regional and global digital governance. This strategy provides the analytical foundation, policy architecture, and implementation roadmap to achieve these outcomes. The responsibility for execution belongs to the institutions and public servants entrusted with the national interest.

1. Strategic Vision and Context

1.1 Background

Bangladesh's posts and telecommunications sector has, for over a decade, operated under conditions of structural fragmentation, regulatory capture, and institutional under-performance. The White Paper on Bangladesh's Telecom Crisis established, through extensive institutional audit, field visits, public hearings, and document analysis across ten entities, that the sector's difficulties are not primarily technological but governance-driven. Of the 42 projects executed under PTD totalling BDT 20,009.82 Crore, 18 high-value initiatives (BDT 16,744.03 Crore) were prioritised for scrutiny due to size or flagged irregularities, revealing widespread deficiencies in strategic planning, feasibility assessments, and sustainability metrics. None of the major entities—BTCL, Teletalk, BTRC, Postal Directorate, DoT, BSCPLC, TSS, or BCSL—maintained structured master plans.

This planning vacuum produced a cascade of consequences: misaligned infrastructure investments, underutilised assets, inflated expenditures, poor service quality, and limited digital access—disproportionately affecting students, small businesses, and marginalised communities. The International Long Distance Telecommunication Service (ILDTS) Policy of 2010, which governed the sector for fifteen years, created a layered licensing architecture of International Gateways (IGWs), International Internet Gateways (IIGs), Interconnection Exchanges (ICXs), and National Internet Exchanges (NIXs) that introduced unnecessary intermediary costs. As the Special Assistant to the Chief Adviser for Telecom and ICT characterised it, this structure created an “innovation-less bonsai structure” in which intermediaries extracted hundreds of crores through regulatory shielding while adding minimal infrastructure value.

The regulatory architecture itself suffered from compromised independence. The 2010 amendments to the Bangladesh Telecommunications Act centralised licensing authority within the Ministry, effectively curtailing BTRC's ability to independently regulate and develop the sector. Board-level interference in state-owned entities—exemplified by non-competitive appointments and the rule requiring the PTD Secretary to serve as chairman of entity boards—created persistent conflicts of interest. Structural weaknesses including dormant talent pipelines at DoT, reliance on deputation and contract staff at MOCSLA, and prolonged vacancy backlogs at the Directorate of Posts eroded institutional capacity from within.

Concurrently, the national policy environment has evolved decisively. The National Digital Transformation Strategy (2025–2030) established nine strategic pillars—from Unified Digital Public Infrastructure and Data Sovereignty to Future-Ready Human Capital—that collectively demand a telecommunications backbone capable of delivering interoperable, secure, and affordable connectivity to every citizen. The legal triad of the National Data Governance Ordinance (NDGO), Personal Data Protection Ordinance (PDPO), and Cyber Safety Ordinance (CSO), all enacted in 2025, creates enforceable obligations for data governance, privacy, and

cybersecurity that require every PTD entity to align its operations with new compliance standards. The PTD sector cannot merely digitise its existing processes; it must fundamentally restructure its institutions, modernise its regulatory framework, and reorient its operational priorities to serve as the “physical rail” upon which the nation’s digital ambitions travel.

1.2 Current Institutional Landscape and Challenges

The challenges confronting PTD's institutional ecosystem are systemic, interconnected, and documented. This section catalogues them across six dimensions that collectively define the reform imperative.

1.2.1 Absence of Strategic Planning and Master Plans

The single most consequential finding of the White Paper is that no major PTD entity possesses a structured master plan. Projects have been routinely launched without credible demand assessments or sustainability metrics. This has resulted in infrastructure that does not correspond to market needs, assets that remain underutilised after completion, and capital expenditures that fail to translate into citizen-facing service improvements. The Directorate of Posts, for example, operates one of the largest physical networks in the country—10,000 post offices—yet has no coherent digital roadmap to transform this network into a platform for e-commerce logistics, financial inclusion, or citizen service delivery. TSS, once positioned as a domestic telecom equipment manufacturer, has experienced a declining manufacturing base without a strategic plan for repositioning itself within the 5G and IoT supply chain. This planning vacuum is not a resource problem; it is a governance failure that this strategy must address as a first-order priority.

1.2.2 Regulatory Fragmentation and Capture

The licensing and regulatory architecture inherited from the ILDTS 2010 framework created concentrated ownership structures, cross-layered affiliations, and artificial monopolies. BTRC and MOCSLA both exhibited regulatory inertia, fragmented enforcement, and limited transparency. MOCSLA's largely manual, non-integrated processes and incomplete licensee database left many courier operators informal or under-regulated, limiting revenue capture and consumer protection in the rapidly growing e-commerce delivery ecosystem. Frequent policy reversals and ad hoc decisions heightened investor uncertainty and weakened long-term strategic planning.

The Telecommunications Network and Licensing Policy 2025 has begun to address these structural deficiencies by simplifying the licensing architecture from over twenty categories to four core types: Access Network Service Providers (ANSP), National Infrastructure and Connectivity Service Providers (NICSP), International Connectivity Service Providers (ICSP), and Non-Terrestrial Networks and Service Providers (NTNSP). Existing IGW, IIG, ICX, and NIX licences will expire naturally, with most lapsing by 2027. The policy introduces technology-neutral licensing, formal recognition of Mobile Virtual Network Operators (MVNOs), Significant Market Power (SMP) remedies, mandatory infrastructure sharing, and a National Quality of Service Dashboard with near-real-time performance data. However, policy approval is only the first step; the translation of policy into enforceable guidelines, licensing instruments, and institutional practice constitutes the core implementation challenge.



1.2.3 Governance Weaknesses in State-Owned Entities

Human resource and governance failures recur across PTD entities. The White Paper documented sector-wide patterns of non-competitive appointments, frequent departures from merit-based recruitment, credential-deficient promotions, and politically influenced board decisions that undermined institutional capability, accountability, and service delivery. The structural rule that the PTD Secretary serves as chairman of entity boards created an inherent conflict of interest between the regulatory and operational functions of the Division. At BTRC, non-competitive appointments and board-level interference compromised regulatory impartiality. At TSS, politically influenced recruitment weakened the organisation's technical capacity. At BTCL, the transition from the legacy Bangladesh Telegraph and Telephone Board left unresolved liabilities and an organisational culture resistant to commercial discipline.

The Bangladesh Telecommunication (Amendment) Ordinance 2025 has taken significant steps to remedy these governance deficiencies at the regulatory level by restoring BTRC's institutional independence, introducing an accountability committee chaired by the head of the relevant parliamentary standing committee, mandating quarterly public hearings, and establishing safeguards against conflicts of interest. The parallel challenge—reforming governance structures within the operational entities (BTCL, Teletalk, BSCCPLC, BSCL, TSS, BCSL, DoP)—requires equally systematic attention through this strategy.

1.2.4 Infrastructure Underutilisation and Connectivity Gaps

Bangladesh's bandwidth consumption stands at approximately 8,500 Gbps, of which 4,000 Gbps is supplied through submarine cables operated by BSCCPLC and the remainder through terrestrial links from India. Under the previous administration, more than 65 per cent of BSCCPLC's submarine cable capacity remained unused. Under the current reform dispensation, bandwidth supply has increased by over 2.2 terabits in one year—a growth rate of 105 per cent—and BTRC has mandated that all IIG operators use at least 50 per cent submarine bandwidth, reducing dependence on Indian terrestrial supply. The government has signed agreements to participate in the revised routes of SEA-ME-WE 4, SEA-ME-WE 5, and SEA-ME-WE 6 submarine cable systems.

Despite these advances, structural connectivity gaps persist. Only 30 per cent of mobile towers are connected via optical fibre, with most relying on low-capacity microwave links. Thousands of towers operate with only 300 Mbps of bandwidth shared among thousands of users. The Social Obligation Fund (SOF), intended to finance connectivity in underserved areas, has been characterised by dormant funds, poor coordination, absence of merit-based project selection, weak impact evaluation, and procurement irregularities. BTCL's fixed-line network, while extensive, requires fundamental modernisation to serve as a viable broadband access platform. The entry of Starlink—with four ground stations operational in Gazipur, Rajshahi, and Jashore and 880 Gbps of installed capacity—introduces a complementary non-terrestrial layer, but its integration into the broader connectivity architecture requires careful regulatory and commercial design.

1.2.5 Postal Sector Stagnation

The postal sector, governed by the Post Office Act of 1898 (as amended in 2010), operates one of the most geographically distributed institutional networks in Bangladesh. The 2010 amendment expanded the Directorate of Posts' mandate to include universal postal service provision, e-commerce facilitation, remittance transfer, postal banking, and data management. However, the operational reality has not kept pace with the legislative mandate. The Directorate of Posts faces prolonged vacancy backlogs, manual and paper-based processes, and an absence of digital infrastructure capable of supporting e-logistics, digital financial services, or interoperable citizen service delivery. Meanwhile, MOCSLA's inability to effectively regulate the courier and mailing sector—despite its rapid growth driven by e-commerce—represents both a governance failure and a significant foregone revenue opportunity.

1.2.6 Skills Deficit and Institutional Capacity Erosion

The sector suffers from a critical skills mismatch between institutional needs and available human capital. The abolition of the dedicated Telecom Cadre removed the institutional mechanism for developing and retaining specialised technical expertise within PTD and its entities. DoT, which should serve as the Division's policy and technical arm, operates with

dormant talent pipelines. Reliance on deputation and contract staff across multiple entities has produced a workforce without institutional continuity or deep sectoral expertise. TSS, which requires engineering capacity for its manufacturing and R&D mandate, has been particularly affected by recruitment practices that prioritised political considerations over technical qualifications. The reintroduction of a Telecom Cadre and the development of structured capacity-building programmes are essential preconditions for any sustainable reform.

1.3 Vision Statement

“To establish a transparent, accountable, and digitally sovereign posts and telecommunications ecosystem that delivers affordable, secure, and inclusive connectivity as the foundation for Bangladesh’s national digital transformation.”

This vision recognises that the posts and telecommunications sector is not an end in itself but the foundational infrastructure upon which every other dimension of digital transformation depends. The National Digital Transformation Strategy’s ambitions for interoperable Digital Public Infrastructure, 95 per cent end-to-end digital government services, and a \$10 billion ICT export economy by 2030 cannot materialise without a telecommunications backbone that is reliable, affordable, and available to every citizen. The vision is grounded in three foundational commitments: governance integrity as the precondition for infrastructure investment; digital sovereignty as the principle for technology and vendor decisions; and inclusive access as the measure of success.

1.4 Mission Statement

“To reform PTD’s institutional architecture, modernise its regulatory framework, and operationalise its entities as transparent, performance-driven organisations that deliver universal connectivity, competitive market structures, and citizen-centric postal and telecommunications services—aligned with the Bangladesh National Digital Architecture (BNDIA) and the legal requirements of the NDGO, PDPO, and CSO.”

The mission translates the vision into three actionable mandates. First, institutional reform: restructuring governance, recruitment, and accountability mechanisms across all ten entities to eliminate the conflicts of interest, planning vacuums, and procurement failures documented by the White Paper. Second, regulatory modernisation: completing the implementation of the Telecommunications Network and Licensing Policy 2025 and the Bangladesh Telecommunication (Amendment) Ordinance 2025 through enforceable guidelines, digital regulatory platforms, and independent oversight mechanisms. Third, operational transformation: deploying measurable programmes for infrastructure expansion, service quality improvement, postal sector digitisation, and workforce development that deliver citizen-facing outcomes tracked through published KPIs.

1.5 Guiding Principles

The following principles guide all reform initiatives, policy decisions, and institutional actions under this strategy. They are derived from the governance failures identified by the White Paper, the policy framework established by the National Digital Transformation Strategy, and the legal obligations created by the 2025 legislative triad.

Principle	Description
Governance-Led Reform	Connectivity outcomes depend fundamentally on governance quality rather than technology alone. Every infrastructure investment, licensing decision, and institutional restructuring must be preceded by governance safeguards: transparent procurement, independent feasibility assessment, published performance metrics, and accountability mechanisms.
Regulatory Independence and Accountability	BTRC must operate as an independent regulator, free from ministerial interference in licensing, tariff setting, and enforcement decisions. This independence must be matched by accountability through the parliamentary oversight committee, mandatory public hearings, and published regulatory impact assessments.
Digital Sovereignty	Government ownership of critical source code, vendor-neutral procurement, open standards, and control over national data assets. Technology decisions must prevent vendor lock-in and ensure that Bangladesh retains the capacity to operate, maintain, and evolve its telecommunications infrastructure independently.
Evidence-Based Planning	All projects must be grounded in rigorous, independent feasibility studies, credible demand assessments, and sustainability metrics. The absence of master plans across PTD entities must be replaced by structured planning frameworks that are publicly available and subject to third-party evaluation.
Citizen-Centric Service Design	All services—whether telecommunications, postal, or regulatory—must be designed from the citizen’s perspective. This means mobile-first interfaces, Bangla-first accessibility, end-to-end digital delivery, and elimination of the intermediary structures that compel citizens to rely on middlemen.
Competitive Neutrality	State-owned and private operators must be treated under consistent regulatory standards. Public entities are essential to guarantee coverage in underserved areas, but this obligation must not translate into regulatory favouritism that distorts market outcomes.
Fiscal Discipline and Transparency	Every project must embed cost-benefit analysis, milestone-based capital expenditure, and public financial reporting. The era of projects launched without sustainability metrics and completed without impact evaluation must end.
Interoperability and Alignment	All PTD systems, data repositories, and service platforms must be designed to interoperate with the Bangladesh National Digital Architecture (BNDIA) and comply with the data governance requirements of the NDGO. The

	“collect once, use many times” principle applies equally to telecommunications infrastructure and postal networks.
Inclusive Connectivity	Affordable, reliable connectivity is a foundational right, not a commercial luxury. Reform must specifically address the rural-urban digital divide, the affordability gap, and the exclusion of marginalised communities—students, women, persons with disabilities—from digital participation.



1.6 Global Benchmarks

The reform programme draws on international models where comparable governance and structural challenges have been addressed through systematic regulatory modernisation. These benchmarks are referenced not as aspirational targets to be copied wholesale, but as evidence-based reference points for the design of PTD’s specific reform interventions.

Country / Model	Key Achievements	Relevance to PTD Reform
Estonia	X-Road interoperability platform connecting all government and private sector systems; 99% of services delivered online; blockchain-secured digital identity; annual productivity savings equivalent to 2% of GDP.	Federated architecture model demonstrating that interoperability does not require centralisation. Estonia’s success was built on governance reform preceding technology deployment—directly relevant to PTD’s challenge.
India (TRAI / Aadhaar / UPI)	Independent Telecom Regulatory Authority (TRAI) with transparent tariff-setting and public consultation processes; Aadhaar digital identity for 1.4 billion citizens; UPI payment platform processing billions monthly.	Demonstrates that regulatory independence and scale-appropriate DPI can coexist. TRAI’s transparent consultation process is a model for BTRC’s reform. India’s spectrum auction mechanisms provide benchmarks for Bangladesh’s planned auctions.
United Kingdom (Ofcom / GDS)	Independent converged regulator (Ofcom) overseeing telecoms, broadcasting, and postal services; Government Digital Service (GDS) establishing user-centric design standards; agile procurement and skill-based recruitment.	Converged regulatory model relevant to PTD’s oversight of both telecommunications and postal sectors. Ofcom’s approach to SMP remedies and infrastructure sharing provides direct precedents for BTRC’s implementation of the 2025 Policy.
Singapore (IMDA)	Infocomm Media Development Authority (IMDA) integrating regulation with industry development; centralised tech procurement; rapid prototyping; AI integration in governance.	Demonstrates institutional model where regulator also drives innovation—relevant to DoT’s proposed evolution into a central policy integrator.
Rwanda	Rapid digitisation of postal services into e-logistics and financial inclusion platforms; mobile-first government services; structural transformation of state telecom operator.	Lower-middle-income country that achieved postal sector transformation through governance reform and digital platform deployment—directly comparable to Bangladesh’s Directorate of Posts challenge.

Global Benchmarks: Telecommunications & Digital Standards



Bangladesh's approach must be calibrated to its own institutional context. The critical lesson from all benchmarks is consistent: sustainable digital transformation requires governance reform to precede and accompany technology deployment. Countries that invested in infrastructure without first addressing regulatory capture, procurement discipline, and institutional accountability—as Bangladesh's telecom sector did over the past fifteen years—produced underutilised assets and eroded public trust. This strategy is designed to avoid repeating that sequence.

1.7 Reform Urgency

The urgency of comprehensive reform across PTD’s institutional ecosystem stems from multiple converging pressures, each of which compounds if reform is delayed.

National Digital Transformation Dependencies. The National Digital Transformation Strategy targets 95 per cent end-to-end digital government services, a \$10 billion ICT export economy, and 100 per cent fibre-to-tower connectivity by 2030. None of these targets can be achieved without a reformed telecommunications infrastructure. PTD entities control the physical layer—submarine cables, terrestrial fibre, mobile networks, satellite links, and postal delivery networks—upon which every digital service depends. Every month of governance paralysis in these entities delays the national programme.

Market Structure Transformation Window. The Telecommunications Network and Licensing Policy 2025 creates a narrow implementation window. Existing IGW, IIG, ICX, and NIX licences expire by 2027. The transition from the old licensing regime to the new four-tier framework must be managed through detailed guidelines, licensee migration plans, and infrastructure transfer protocols. Failure to execute this transition competently risks regulatory uncertainty, investment hesitation, and service disruption.



Satellite Connectivity Integration. The entry of Starlink—with operational ground stations and a proposal to establish Bangladesh as a regional internet transit hub serving neighbouring countries—introduces both an opportunity and an integration challenge. Starlink’s 880 Gbps installed capacity must be regulated and integrated in a manner that complements rather than

undermines submarine cable investments and terrestrial fibre deployment. This requires immediate regulatory clarity on bandwidth export, transit routing, and competitive neutrality between satellite and terrestrial providers.

5G Deployment Readiness. Teletalk holds 60 MHz of spectrum in the 3.5 GHz band for 5G services, but its experimental deployments have not progressed to commercial-scale operations. The Telecommunications Policy 2025 mandates that mobile operators connect 50 per cent of their towers to fibre within 18 months and 80 per cent within three years. Without accelerated fibre deployment and a credible 5G rollout strategy—potentially through spectrum auctions open to all qualified operators—Bangladesh risks falling further behind regional competitors who are already deploying commercial 5G networks.

Postal Sector Relevance. E-commerce transaction volumes continue to grow while the Directorate of Posts' market share in parcel and logistics services shrinks. The window for transforming post offices into platforms for e-logistics, financial inclusion, and citizen service delivery is narrowing as private courier and logistics operators consolidate market positions. Without rapid digitisation and institutional reform, the postal network's 10,000 physical touchpoints—the most distributed government infrastructure in the country—will become stranded assets rather than transformation enablers.

Fiscal Sustainability. Several PTD entities operate at a loss or with declining revenues. BTCL's legacy fixed-line business is structurally unprofitable; Teletalk's subscriber base and revenue lag far behind private operators; TSS's manufacturing revenues have diminished. Continued subsidisation without reform produces no return. This strategy's emphasis on fiscal discipline, milestone-based capital expenditure, and commercial orientation is essential to ensure that public investment translates into sustainable operations.

International Credibility. Bangladesh's ability to attract foreign direct investment in telecommunications, participate in international submarine cable consortia, and negotiate with global technology partners (including Starlink, Huawei, Nokia, and potential MVNO investors) depends on demonstrating a predictable, transparent, and rules-based regulatory environment. The reforms of 2025—the Ordinance amendment, the Licensing Policy, the ban on internet shutdowns—have begun to rebuild this credibility. The implementation phase must sustain it.

1.8 Strategic Objectives (2025–2030)

The following strategic objectives define the measurable outcomes that this strategy is designed to achieve. Each objective responds directly to the challenges documented in Section 1.2 and aligns with the corresponding pillars of the National Digital Transformation Strategy.

Strategic Objective	Description and Key Actions	Indicative KPIs
Governance and Institutional Reform	Mandate structured master plans for all ten PTD entities; implement open, competitive appointment processes for top management; establish independent board governance with balanced professional representation; reintroduce a dedicated Telecom Cadre; create a cross-agency steering committee with quarterly reporting to the Cabinet Division.	100% entity master plan completion; Open recruitment for all MD/Chairman positions; Telecom Cadre re-established; Quarterly dashboard publication
Regulatory Modernisation	Complete implementation of the Telecommunications Network and Licensing Policy 2025 through enforceable guidelines; operationalise BTRC's independence per the 2025 Ordinance; deploy the National Quality of Service Dashboard; implement SMP remedies and infrastructure sharing mandates; digitise and integrate MOCSLA's regulatory processes.	New licensing guidelines issued; QoS Dashboard operational; SMP determinations completed; MOCSLA online registration and tracking platform launched
Infrastructure Expansion and Optimisation	Achieve 50% fibre-to-tower connectivity within 18 months and 80% within 3 years; execute SEA-ME-WE 4/5/6 submarine cable upgrades; integrate Starlink capacity into the national bandwidth architecture; deploy BTCL MVNO service; complete BTCL network modernisation.	Fibre-to-tower percentage; Submarine cable capacity utilisation above 70%; BTCL MVNO subscribers; Bandwidth export volume
Postal Sector Digital Transformation	Digitise Directorate of Posts operations; establish e-logistics and parcel tracking platform; launch postal digital financial services; integrate post offices into the national e-service delivery framework; reform MOCSLA into an effective courier sector regulator.	E-logistics platform operational; Digital transactions per post office; MOCSLA registered operators as % of market
Domestic Manufacturing and Supply Chain	Reposition TSS as a domestic hardware and IoT device producer; establish strategic partnerships for 5G equipment, CCTV systems, and Wi-Fi router manufacturing; launch software incubation centre.	TSS revenue from new product lines; Local assembly/manufacturing output; Strategic partnership agreements executed
Workforce Development	Reintroduce Telecom Cadre; implement structured capacity-building programmes across all entities; establish technical competency frameworks aligned	Telecom Cadre positions filled; Staff trained per entity per year; Technical

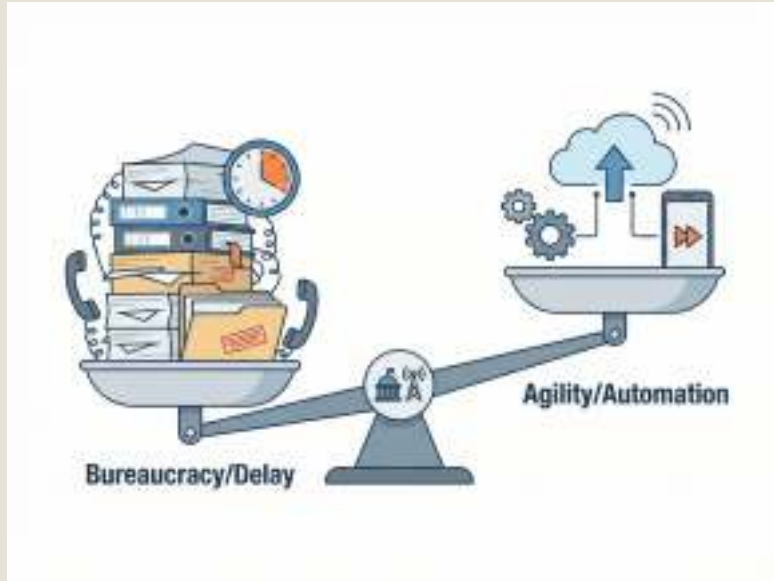
	with 5G, satellite, cybersecurity, and data governance requirements.	certification completion rates
Fiscal Sustainability	Implement milestone-based capital expenditure frameworks; establish cost-benefit analysis requirements for all projects above BDT 50 Crore; reform SOF governance; achieve revenue growth targets for BSCCPLC, BTCL, BSCL, and TSS.	SOF fund utilisation rate; Entity revenue growth; Project cost overrun reduction
Citizen-Centric Service Delivery	Ensure affordable broadband access benchmarked to ITU affordability targets; deploy mobile-first, Bangla-first interfaces for all PTD digital services; establish citizen feedback mechanisms for telecom and postal services.	Broadband affordability index; Citizen satisfaction survey scores; Complaint resolution time



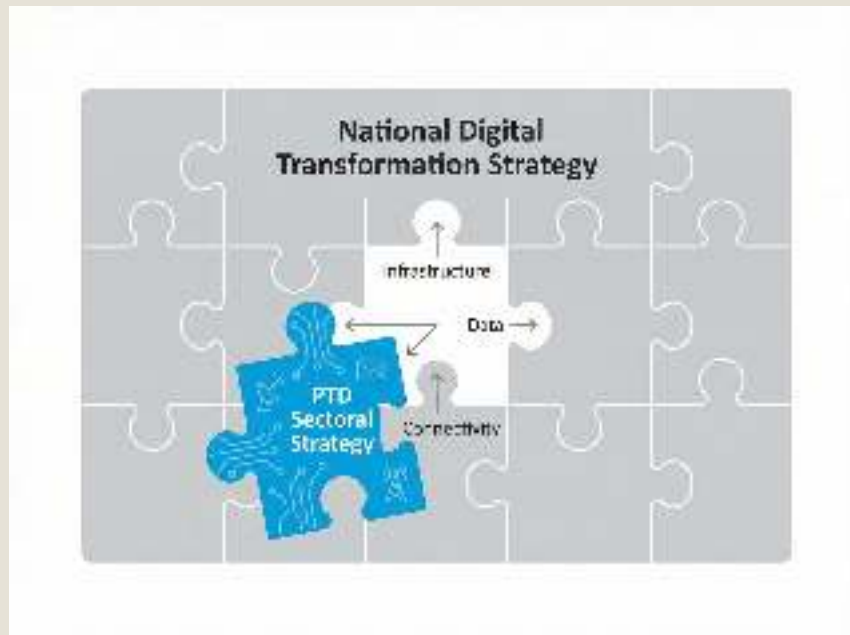
1.9 Alignment with the National Digital Transformation Strategy

This strategy is designed as the sectoral implementation framework for the telecommunications and postal dimensions of the National Digital Transformation Strategy (2025–2030). The alignment is structural, not nominal. Each PTD reform initiative maps directly to one or more of the nine strategic pillars established by the national strategy.

NDTS Strategic Pillar	PTD Alignment and Contribution
Unified Digital Public Infrastructure	BSCCPLC submarine cable upgrades, BTCL fibre network modernisation, Starlink integration, National Fiber Bank coordination—collectively delivering the physical connectivity layer upon which DPI operates.
Citizen-Centric Governance	Postal sector digitisation, BTCL service delivery improvements, QoS Dashboard, citizen feedback mechanisms—ensuring that telecom and postal services meet the “Digital by Default” and “Bangla First” mandates.
Data Sovereignty and Legal Certainty	Compliance of all PTD entities with NDGO, PDPO, and CSO requirements; digital sovereignty in procurement; government source code ownership; vendor-neutral technology choices.
National Connectivity Backbone	Fibre-to-tower targets, submarine cable capacity expansion, Starlink non-terrestrial layer, rural connectivity through reformed SOF—directly constituting the connectivity backbone referenced by the national strategy.
Autonomous Regulatory Oversight	BTRC independence under the 2025 Ordinance; MOCSLA reform; DoT evolution as policy integrator—establishing the regulatory architecture within PTD’s domain.
Economic Resilience and Innovation	MVNO market entry, competitive licensing framework, bandwidth export potential, TSS manufacturing repositioning—creating the conditions for investment, innovation, and export growth.
Future-Ready Human Capital	Telecom Cadre reintroduction, technical capacity building, TSS software incubation centre—developing the specialised workforce the sector requires.
Institutional Interoperability	Integration of PTD systems with BNDIA standards, postal network integration with e-service delivery framework, MOCSLA digitisation—eliminating data silos within the sector.
Proactive Security and Resilience	Compliance with CSO cybersecurity requirements, CIS (Centre for Information Support) establishment replacing NTMC, lawful interception under judicial oversight—ensuring telecommunications infrastructure supports national security without compromising civil liberties.



This alignment ensures that PTD’s reform programme is not an isolated sectoral exercise but an integral component of the national digital transformation. Progress on PTD’s strategic objectives directly advances the national strategy’s KPIs; conversely, failure within the telecommunications and postal sector cascades as delay across every other dimension of digital transformation.



— End of Chapter 1 —

2. Legal Landscape

2.1 Why Legal Reform Matters

The posts and telecommunications sector in Bangladesh has, for over a decade, operated under a legal architecture that simultaneously enabled regulatory capture, insulated intermediaries from competition, and failed to keep pace with the transformation of digital markets. Chapter 1 documented the governance failures that this architecture produced: the absence of master plans, procurement irregularities across BDT 20,009.82 Crore in projects, politically compromised regulatory independence, and a postal sector governed by a statute enacted in 1898. The legal reforms of 2025 represent the most comprehensive restructuring of the sector's legislative foundations since the enactment of the Bangladesh Telecommunication Act in 2001.

The importance of legal reform to the digital transformation programme cannot be overstated. Infrastructure investments, institutional restructuring, and service delivery improvements all operate within a framework of legal permissions and constraints. When the legal framework concentrates licensing authority within a Ministry, as the 2010 amendments did, independent regulation becomes structurally impossible regardless of the quality of the regulator's personnel. When a postal statute lacks provisions for digital logistics, electronic postage, or data protection compliance, no operational initiative can achieve sustainable transformation. When surveillance powers lack judicial oversight and internet shutdowns can be ordered without legal constraint, investor confidence erodes and citizen trust collapses.

This chapter examines the four principal legal instruments that collectively reshape PTD's operating environment: the Bangladesh Telecommunication (Amendment) Ordinance 2025, the Telecommunications Network and Licensing Policy 2025, the Postal Services Ordinance 2025 (draft), and the Mailing Operator and Courier Service Development and Regulatory Authority Ordinance 2025 (pending). It analyses each instrument's key provisions, situates them within the broader national legal reform programme—including the Cyber Safety Ordinance (CSO), Personal Data Protection Ordinance (PDPO), and National Data Governance Ordinance (NDGO)—and identifies the implementation imperatives that determine whether legislative intent translates into operational reality.

Internationally, the relationship between legal framework quality and telecommunications sector performance is well established. The International Telecommunication Union's ICT Regulatory Tracker consistently demonstrates that countries with independent regulatory authorities, technology-neutral licensing, and enforceable consumer protections achieve higher connectivity outcomes per capita investment. The European Electronic Communications Code (2018), India's Telecommunications Act (2023), and Singapore's Telecommunications Act (revised 2024) each represent recent legislative modernisations that Bangladesh's 2025 reforms parallel in structural intent, though adapted to the specific governance challenges documented by the White Paper.

2.2 Bangladesh Telecommunication (Amendment) Ordinance 2025

2.2.1 Background and Legislative Context

The Bangladesh Telecommunication Act 2001 (Act No. XVIII of 2001) established the Bangladesh Telecommunication Regulatory Commission (BTRC) as an independent statutory body with authority over licensing, tariff regulation, spectrum management, and market oversight. This foundational statute was designed to separate the regulatory function from the operational function—a principle recognised internationally as essential to effective sector governance.

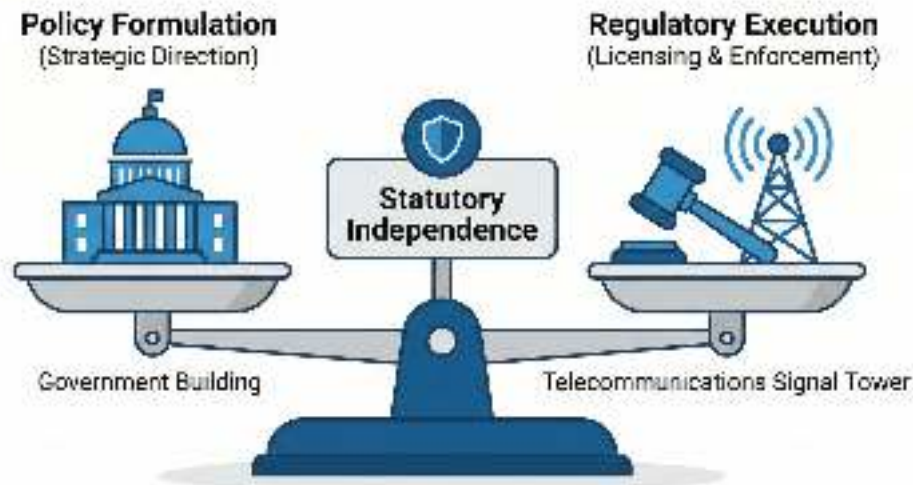
The Bangladesh Telecommunication (Amendment) Act 2010 fundamentally altered this architecture. The 2010 amendments transferred final authority over licence issuance, renewal, service approval, and tariff-setting from BTRC to the Ministry of Posts and Telecommunications. As the White Paper characterised it, this legislation marked the “Beginning of the End” for independent regulation. The Ministry’s assumption of licensing authority created a structural environment in which regulatory decisions could be influenced by political considerations rather than market analysis, consumer welfare, or evidence-based policy. The ILDTS Policy 2010, which followed shortly thereafter, was a direct product of this compromised regulatory environment—introducing a segmented licensing framework that lacked baseline demand analysis, failed to implement competitive safeguards, and enabled the formation of intermediary structures that extracted hundreds of crores in value while adding minimal infrastructure investment.

The Bangladesh Telecommunication (Amendment) Ordinance 2025, approved by the Advisory Council on 24 December 2025, reverses the most consequential elements of the 2010 amendments and introduces new provisions that did not exist in either the original 2001 Act or its 2010 iteration. It represents a legislative response to fifteen years of documented regulatory failure.

2.2.2 Key Provisions

Restoration of BTRC’s Regulatory Independence. The Ordinance restores to BTRC the authority over licensing, service approval, and tariff regulation that the 2010 amendments had transferred to the Ministry. This restoration is not merely administrative; it re-establishes the structural separation between policymaking (which remains with PTD) and regulation (which reverts to BTRC). International best practice, as codified in ITU recommendations and exemplified by regulators such as India’s Telecom Regulatory Authority (TRAI), the United Kingdom’s Ofcom, and Singapore’s Infocomm Media Development Authority (IMDA), consistently demonstrates that regulatory independence from the political executive is the single most important institutional precondition for competitive, investor-friendly telecommunications markets.

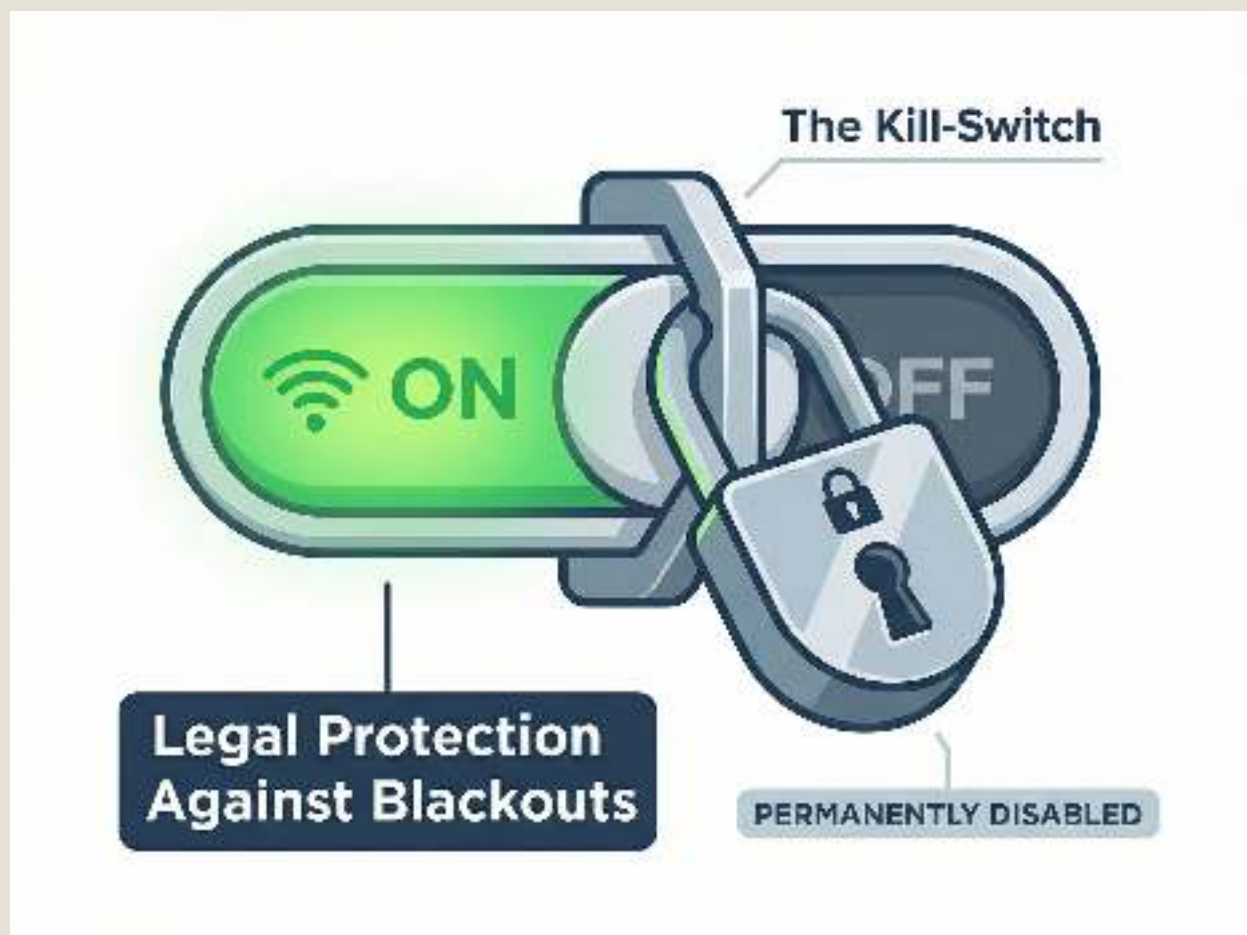
BTRC Independence Restoration: Separation of Powers



Permanent Prohibition on Internet Shutdowns (Section 97). The Ordinance introduces a permanent ban on state-ordered internet and telecommunications shutdowns, recognising continuous internet access as a citizen's right. This provision is significant both domestically and internationally. Between 2018 and 2024, Bangladesh experienced multiple instances of internet shutdowns, including nationwide mobile internet blackouts during the July 2024 protests, platform-specific blocks on social media and messaging applications, and connectivity restrictions in refugee camps. These shutdowns imposed measurable economic costs, disrupted essential digital services, and eroded Bangladesh's international credibility. The ban aligns Bangladesh with an emerging global norm: the United Nations Human Rights Council has repeatedly condemned internet shutdowns as violations of the right to freedom of expression, and the African Commission on Human and Peoples' Rights adopted a resolution in 2016 declaring that shutdowns violate the African Charter. Bangladesh is among the first countries in South Asia to enact a permanent statutory prohibition.

Abolition of the NTMC and Establishment of the Centre for Information Support (CIS). The National Telecommunications Monitoring Centre (NTMC), which operated under the Ministry of Home Affairs, is abolished. In its place, a Centre for Information Support (CIS) is established under the Ministry of Home Affairs with narrowly defined functions and explicit safeguards. The NTMC's operations had been characterised by opacity, absence of judicial oversight, and the potential for abuse of surveillance capabilities. The CIS is subject to the procedural requirements of the Cyber Safety Ordinance 2025, which mandates judicial authorisation for lawful interception.

Anti-Surveillance Protections (Section 71). The extraction of SIM registration data or device identification data for surveillance purposes is made a punishable offence. This provision addresses the documented practice of using telecommunications subscriber data for extrajudicial monitoring. It creates a legal boundary between lawful interception under judicial authorisation and the previously unconstrained access to subscriber metadata.



Accountability Committee and Public Hearings. An accountability committee, chaired by the head of the relevant parliamentary standing committee, is established to oversee BTRC's operations. Mandatory public hearings must be conducted every four months, with published follow-up actions. This dual mechanism—parliamentary oversight and mandatory public engagement—is designed to balance BTRC's restored independence with transparent accountability. The model draws on international precedents: TRAI in India conducts public consultations on all major regulatory decisions; Ofcom in the United Kingdom publishes impact assessments with every regulatory intervention; the Australian Communications and Media Authority (ACMA) operates under a legislative requirement for regular performance reviews by the Minister.

Revised Speech Offences. The Ordinance narrows speech-related offences to criminalise only incitement to violence, aligning the telecommunications law with the Cyber Safety Ordinance

2025. The overly broad provisions of prior legislation—which had been used to prosecute journalists, bloggers, and activists for online speech—are replaced by a narrower formulation that balances security interests with free expression. This alignment ensures that the telecommunications regulatory framework does not operate as a parallel censorship mechanism.

Reduced Licence Processing Timelines and Scaled Penalties. The Ordinance introduces streamlined timelines for licence processing and reduces penalty provisions that had been characterised as disproportionate. The objective is to lower barriers to market entry and reduce the regulatory burden on operators, consistent with the broader market liberalisation goals of the Telecommunications Network and Licensing Policy 2025.

2.2.3 Implementation Requirements

The Ordinance establishes the legislative framework; its translation into operational reality requires several implementation actions. BTRC must develop and publish new procedural guidelines for its restored licensing and tariff-setting functions within six months of the Ordinance’s commencement. The accountability committee must be constituted and its terms of reference gazetted. The first cycle of mandatory public hearings must be scheduled and publicised. The transition protocols for dismantling NTMC and establishing CIS must be executed within the Ministry of Home Affairs, with appropriate data handling and asset transfer procedures. Training programmes for BTRC staff on the exercise of restored regulatory functions must be designed and deployed. Without these implementation steps, the Ordinance’s intent remains aspirational rather than operational.

Implementation KPIs: Bangladesh Telecommunication (Amendment) Ordinance 2025

Indicator	Timeline	Responsible Entity
BTRC licensing guidelines published	Within 6 months of commencement	BTRC
Accountability committee constituted	Within 3 months	PTD / Parliamentary Standing Committee
First public hearing cycle completed	Within 4 months of committee formation	BTRC
CIS operational; NTMC dissolved	Within 6 months	Ministry of Home Affairs
QoS Dashboard operational with real-time data	Within 12 months	BTRC
Zero internet shutdowns post-commencement	Continuous	BTRC / CIS
Staff training on restored functions completed	Within 9 months	BTRC

2.3 Telecommunications Network and Licensing Policy 2025

2.3.1 Background and Rationale

The Telecommunications Network and Licensing Policy 2025 was gazetted on 22 September 2025, replacing the International Long Distance Telecommunication Service (ILDTS) Policy 2010. The gazette notification simultaneously repealed the ILDTS Policy 2010 upon the new policy's entry into force. The rationale for replacement was documented extensively in the White Paper: the ILDTS 2010 framework had created over twenty licence categories, imposed rigid segmentation between service layers, lacked demand-based calibration, and enabled cartel-like behaviour among intermediaries—particularly within the IGW Operators Forum (IOF), where operators coordinated on interconnection terms without effective regulatory oversight.

The policy's design reflects a global shift toward converged, technology-neutral licensing. The European Union's European Electronic Communications Code (2018) established the principle that service authorisation should not prescribe the technology through which services are delivered. India's Unified Licence regime consolidated multiple service-specific licences into a single authorisation framework. Singapore's Facilities-Based Operator (FBO) and Services-Based Operator (SBO) framework provides a simplified two-tier model. The Bangladesh policy adopts a comparable rationalisation, reducing complexity from over twenty categories to four core types while introducing provisions for infrastructure sharing, competitive safeguards, and quality of service transparency that were absent from the prior regime.

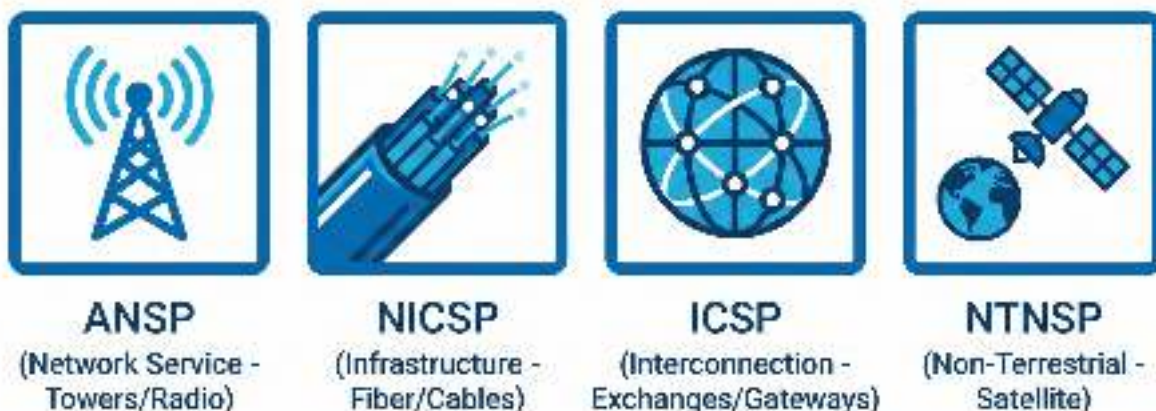


2.3.2 New Licensing Architecture

The policy establishes a three-tier network topology with four core licence categories, replacing the fragmented structure of IGWs, IIGs, ICXs, NIXs, and numerous specialised permits:

Licence Category	Scope of Operations	Key Conditions
Access Network Service Provider (ANSP)	Cellular mobile operators, fixed telecom operators, and telecom-enabled service providers. Delivers services directly to end users.	Minimum 15% domestic ownership for cellular operators (85% foreign cap). Must connect 50% of towers to fibre within 18 months; 80% within 3 years.
National Infrastructure and Connectivity Service Provider (NICSP)	National transmission, interconnection, and internet exchange functions. Backbone infrastructure layer between international connectivity and access networks.	45–55% foreign ownership cap. Mandatory infrastructure sharing obligations. Must provide non-discriminatory wholesale access to ANSPs.
International Connectivity Service Provider (ICSP)	International bandwidth, submarine cable landing, terrestrial cross-border links, and satellite earth station operations. Replaces IGW and IIG categories.	49% foreign ownership cap. Submarine cable operators and satellite gateway operators operate at this tier. Must ensure minimum 50% submarine bandwidth usage.
Non-Terrestrial Networks and Service Provider (NTNSP)	Satellite-based communications, including LEO, MEO, GEO, and HAPS services. First formal recognition in Bangladesh licensing.	Subject to specific terms per technology. Formal framework for Starlink-type satellite internet and future non-terrestrial entrants.

New Unified License Architecture (The 4 Pillars)



Existing licences issued under the ILDTS 2010 framework remain valid under their original terms and conditions, subject to the new policy’s provisions. IGW, IIG, ICX, and NIX licences will expire naturally, with most lapsing by 2027. This phased transition avoids abrupt disruption while ensuring that the market converges toward the simplified architecture within a defined timeframe.

2.3.3 Competitive Safeguards and Market Structure Reforms

The policy introduces several mechanisms to address the market concentration and anti-competitive behaviour documented in the White Paper:

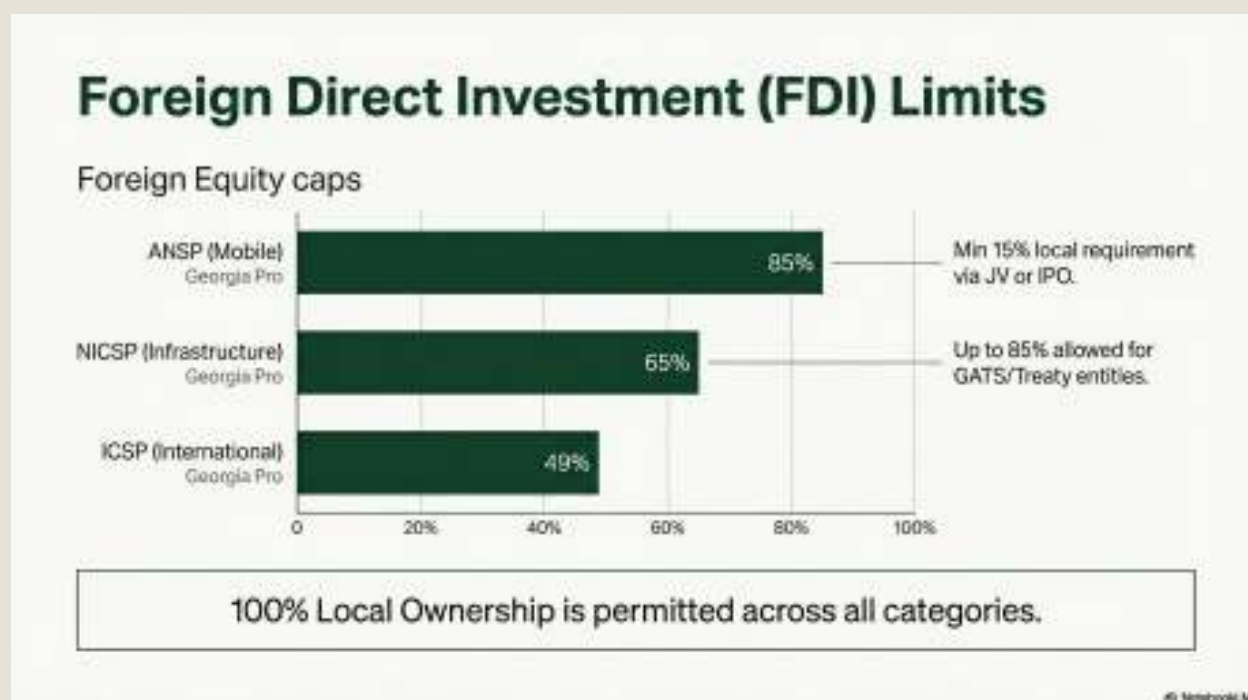
Significant Market Power (SMP) Remedies. BTRC is empowered to designate operators with significant market power and impose ex-ante remedies, including mandatory wholesale access, price controls, and accounting separation. This provision mirrors the SMP framework employed by Ofcom in the United Kingdom and the European Commission under the European Electronic Communications Code. The absence of SMP remedies under the ILDTS 2010 framework allowed dominant operators to leverage market position without regulatory constraint.

Mandatory Infrastructure Sharing. All licensees are required to share passive and active infrastructure—including fibre networks, towers, radio access networks, and core network elements—under guidelines to be issued by BTRC. Infrastructure sharing reduces duplication, lowers barriers to entry, and accelerates coverage expansion. India’s infrastructure sharing regime, governed by TRAI’s IP-I and UASL guidelines, is credited with enabling the rapid expansion of mobile broadband coverage to over 700 million users. The Bangladesh policy mandates that BTRC issue comprehensive infrastructure sharing guidelines, encompassing both passive elements (ducts, poles, towers) and active elements (spectrum, radio networks).

MVNO Recognition. The policy formally recognises Mobile Virtual Network Operators (MVNOs) for the first time in Bangladesh’s licensing framework. MVNOs—which operate on host network infrastructure without owning spectrum—have been instrumental in increasing market competition and consumer choice globally. BTCL’s approved MVNO pilot is a direct product of this policy enablement. The MVNO framework is expected to create additional market entry opportunities, particularly for niche service providers targeting underserved segments.

National Quality of Service Dashboard. The policy mandates the creation of a publicly accessible dashboard displaying near-real-time quality of service data for all licensed operators. This transparency mechanism enables consumers, regulators, and policymakers to monitor network performance against published benchmarks. Brazil’s Anatel publishes quarterly QoS reports with operator-specific metrics; India’s TRAI MySpeed portal crowdsources speed data from millions of users. Bangladesh’s dashboard is intended to provide a comparable level of public accountability.

Foreign Ownership Caps with Tiered Structure. The policy establishes differentiated foreign ownership limits across the network tiers: 85 per cent for cellular mobile operators (requiring a minimum 15 per cent domestic holding), 49 per cent for international connectivity, and 45–55 per cent for national infrastructure. This tiered approach balances the need for foreign investment and technology transfer with the objective of ensuring domestic stakeholder participation in critical national infrastructure. The design draws on comparative frameworks: India permits 100 per cent FDI in telecommunications but maintains security conditions; the Philippines recently raised its foreign ownership cap from 40 per cent to 100 per cent for telecommunications carriers.



2.3.4 Fibre Connectivity Targets

The policy establishes binding fibre connectivity milestones for mobile operators: 50 per cent of towers connected to optical fibre within 18 months of policy commencement, and 80 per cent within three years. This is a critical provision. As documented in Chapter 1, only approximately 30 per cent of Bangladesh’s mobile towers are currently connected via fibre, with the majority relying on low-capacity microwave links that constrain data throughput. The National Digital Transformation Strategy’s target of 100 per cent fibre-to-tower connectivity by 2030 depends on the accelerated execution of these intermediate milestones. Operators that fail to meet the targets face enforcement actions under BTRC’s licensing conditions, including potential licence amendments.

2.3.5 Implementation Requirements

The policy's effectiveness depends on the timely issuance of implementing guidelines by BTRC in several areas: detailed licensing guidelines for each of the four categories; infrastructure sharing guidelines encompassing both passive and active elements; SMP determination criteria and remedy procedures; MVNO operational guidelines; the National Quality of Service Dashboard technical specifications and data collection protocols; and cross-ownership restrictions to prevent concentration across network tiers. BTRC must also develop a migration roadmap for existing licensees, including timeline, fee structures, and transitional conditions. The policy explicitly reserves to the government the authority to grant exemptions for state-owned telecommunications entities, a provision that must be exercised transparently and subject to published criteria to avoid recreating the regulatory asymmetries of the prior regime.

Implementation KPIs: Telecommunications Network and Licensing Policy 2025

Indicator	Timeline	Responsible Entity
BTRC licensing guidelines for all 4 categories issued	Within 6 months of policy commencement	BTRC
Infrastructure sharing guidelines published	Within 9 months	BTRC
SMP determination framework operational	Within 12 months	BTRC
QoS Dashboard live with real-time operator data	Within 12 months	BTRC
50% fibre-to-tower connectivity achieved	Within 18 months	Licensed operators / BTRC oversight
MVNO operational guidelines issued	Within 6 months	BTRC
Legacy IGW/IIG/ICX/NIX licences expired or migrated	By 2027	BTRC / licensees
80% fibre-to-tower connectivity achieved	Within 3 years	Licensed operators / BTRC oversight

2.4 Postal Services Ordinance 2026

2.4.1 Context and Strategic Rationale

For over a century, the postal backbone of Bangladesh was governed by the *Post Office Act, 1898*—a colonial instrument designed primarily for the secure movement of royal mail and the maintenance of imperial communication lines. While effective in the pre-digital era, this archaic framework failed to address the complexities of the 21st-century digital economy, the explosion of e-commerce, the necessity of data sovereignty, and the imperative of financial inclusion.

The enactment of the **Postal Services Ordinance, 2026** marks a definitive paradigm shift. It repeals the 1898 Act and establishes a unified, time-befitting legal structure aimed at the modernization, standardization, and digital transformation of the postal, courier, and parcel sectors. This Ordinance is not merely a regulatory update; it is a strategic instrument that redefines the **Department of Post** (Bangladesh Post) not just as a mail carrier, but as the "Designated Operator" responsible for the nation's logistics sovereignty, a trustee of citizen wealth through postal banking, and the custodian of the National Address Infrastructure.

Crucially, the Ordinance recognizes the postal network as an **"Essential Service"**. It mandates that in times of national crisis, disaster, or emergency, the postal network must remain operational to ensure the continuity of state communication and the delivery of essential goods, granting it priority status equivalent to other emergency services.

2.4.2 The Designated Operator: Redefining the National Post

The Ordinance formally designates the **Department of Post** as the country's sole **"Designated Operator" (DO)**. This aligns Bangladesh with the Universal Postal Union (UPU) convention, establishing a "Single Postal Territory" where the state guarantees uninterrupted communication to every citizen, regardless of geography or economic status.

1. Mandate for Universal Postal Service (UPS): Unlike private operators driven by profit, the Designated Operator is legally bound to provide **"Universal Postal Service"**—a guaranteed minimum level of service at an affordable, uniform price to every address in Bangladesh. This includes the delivery of letters, personal documents, judicial notices, public examination papers, and safety net allowances. The Ordinance ensures that this mandate is not an unfunded liability; it obligates the Government to provide necessary budget allocations and administrative support to maintain this critical network.

2. The Distinction of Services: To ensure market fairness while protecting state interests, the Ordinance categorizes postal services into distinct tiers:

- **Basic Postal Service:** The core obligation of collecting, processing, and delivering mail under UPU quality standards.
- **Entitled Postal Service (Adhikari Seva):** Historic services provided by the state for financial and social inclusion, such as Postal Life Insurance and Savings Bank operations.
- **Agency Services:** The Post acting as a "proxy operator" for other government or private entities (e.g., passport delivery, allowance distribution).
- **Commercial & Premium Services:** High-speed "Express" and specialized "Premium" services where the Post competes with private players based on market rates.

2.4.3 The Financial Powerhouse: From Savings to "Postal Bank"

A cornerstone of the 2026 Ordinance is the revitalization of the Post Office as a primary engine for financial inclusion. Recognizing that the postal network reaches deep into rural areas where commercial banks often fear to tread, the law empowers the Department to expand its financial horizon significantly.

1. Digital Financial Services & Interoperability: The Ordinance authorizes the Department to operate Money Transfer Services (MTS), digital wallets, and electronic money transfers. Crucially, it mandates the **digital transformation** of all financial services, moving away from manual ledgers to a centralized, technology-driven core banking environment. This allows for "Money Distribution" services using the postal network as a cash-out/cash-in mesh for other financial institutions.

2. Treasury and Investment Management: The law integrates postal financial operations with the national treasury system. All savings and financial service accumulations are to be managed through a **Treasury Single Account (TSA)** mechanism. However, it grants the Department the strategic flexibility to invest these funds in Treasury Bonds or other government instruments, ensuring that citizen deposits generate returns that can be shared as profit.

3. The Vision of a Postal Bank: Perhaps the most transformative provision is the enabling clause for the future. The Government is empowered to establish a full-fledged **"Postal Bank"** to institutionalize rural and small-scale savings, graduating the current Post Office Savings Bank into a formal banking entity subject to appropriate banking regulations. This provision provides the legal roadmap for separating the financial arm of the Post into a specialized corporate entity.

2.4.4 Postal Life Insurance (PLI): A Sovereign Guarantee

The Ordinance reinforces the unique position of **Postal Life Insurance (PLI)** as the oldest and most secure insurer in the country, backed by a **sovereign state guarantee**.

- **Inclusivity:** PLI is mandated to be inclusive, offering policies not just to government employees but to any citizen with a regular income, including special provisions for risky professions like the defense forces.
- **Product Innovation:** The Department is authorized to introduce modern insurance products such as education insurance, marriage insurance, and health insurance, moving beyond simple endowment policies.
- **Fund Management:** A dedicated "**Post Office Insurance Fund**" will be maintained on a "no-profit, no-loss" basis, ensuring that surpluses are returned to policyholders as bonuses rather than siphoned off as corporate profit. The fund is to be valued regularly by a certified actuary to ensure solvency.

2.4.5 National Address Infrastructure: The Post as Data Fiduciary

In a digital economy, a physical address is as critical as a digital IP address. The Ordinance assigns the Department of Post the monumental task of establishing and maintaining the **National Address and Data Infrastructure**.

1. Data Fiduciary Role: The Department is designated as the "**Data Fiduciary**" for national address data. It is responsible for identifying every holding or structure in Bangladesh, assigning it a unique geolocation, and maintaining a dynamic database of these addresses. This is a strategic asset for the entire nation—vital for e-commerce delivery, emergency response, and urban planning.

2. Addressing the "Floating" Population: The law explicitly addresses the reality of climate change and migration. It mandates the preservation of address history for citizens displaced by river erosion or natural disasters ("Natural Migration"), ensuring that a citizen does not lose their identity simply because their land was washed away. The Post must develop methodologies to trace and serve floating populations who lack a fixed geographic abode.

3. Digital Codes and Interoperability: The Ordinance mandates the use of **Postcodes** and authorizes the introduction of innovative "Digital Location Codes". It also establishes a "**Central Logistics Tracking Platform**," a digital hub that ensures interoperability between the Post and private couriers. This platform will allow a user to track a package across different operators using a single portal, aiming to create a seamless national logistics grid.

2.4.6 Regulating the Private Sector: The Licensing Authority

The Ordinance brings discipline to the booming, yet chaotic, private courier and e-commerce delivery sector. It empowers the **Mailing Operator and Courier Service Licensing Authority** (hereinafter "The Authority") to regulate the market.

1. Licensing Regime: No person or entity can operate a courier or parcel service without a valid license. The Authority will categorize licenses based on service scope (e.g., international,

domestic, city-wide) and service nature (e.g., document, heavy parcel). This ends the era of unregulated "grey market" operators. Existing licensees under the 2013 rules are grandfathered in but must transition to the new framework.

2. Level Playing Field & Anti-Monopoly: The Ordinance includes strong anti-trust provisions. It prohibits predatory pricing, dumping, and exclusivity agreements that stifle competition. If a dominant player emerges, the Authority has the power to impose "**Price Caps**" to prevent exploitation. Crucially, to ensure fair competition between the state and private sectors, the Department of Post is required to maintain "**Account Separation**," ensuring that subsidies meant for Universal Service are not used to cross-subsidize its commercial competitive services.

3. Consumer Protection: Operators are now legally liable for the safety and secrecy of the mail. The law mandates compensation for lost or damaged goods based on declared value and service terms. It also forces operators to implement effective grievance redressal mechanisms.

2.4.7 Security, Sovereignty, and Customs Integration

As the volume of cross-border e-commerce grows, so does the risk of illicit trade. The Ordinance integrates postal operations with national security and customs frameworks.

1. Electronic Advance Data (EAD): To align with global security standards (WCO and UPU), the Ordinance mandates **Electronic Advance Data (EAD)**. Operators must transmit digital data about the sender, receiver, and contents of international parcels to customs and security agencies *before* the shipment physically arrives. This allows for risk-based screening and faster clearance of legitimate trade.

2. Know Your Customer (KYC): The law makes **KYC mandatory** for booking specific types of parcels, particularly international shipments or high-value domestic goods. Operators will be granted limited access to the National Identity (NID) server to verify sender identities in real-time, closing a major loophole used for trafficking contraband.

3. Prohibited Goods & Cyber Security: A strict schedule of prohibited items (explosives, narcotics, counterfeit currency) is established. Furthermore, the Ordinance mandates compliance with the *Cyber Safety Ordinance, 2025* and *National Data Management Ordinance, 2025*, ensuring that the massive amounts of personal data handled by logistics providers are stored and processed securely.

2.4.8 Commercialization: Post Shops and Franchisees

Recognizing that the Department cannot do everything alone, the Ordinance creates frameworks for commercial partnerships and asset utilization.

- **Post Shops:** The Department is authorized to convert its post offices into "**Post Shops**"—commercial hubs that sell not just stamps but also packaging materials, digital devices, and even third-party products, effectively becoming retail outlets for rural communities.
- **Franchise Post Offices:** To expand the network without bloating the government payroll, the Designated Operator can appoint "**Franchise Postmasters.**" These are private individuals or small businesses authorized to offer postal services in areas where a full-fledged post office is unviable. This fosters local entrepreneurship while ensuring service coverage.
- **Post E-Centers:** These are designated as digital inclusion hubs, offering IT training, e-commerce support, and government e-services to the rural youth, acting as a bridge to the "Smart Bangladesh" vision.

2.4.9 Offenses, Penalties, and Enforcement

The Ordinance replaces the minor fines of 1898 with substantial deterrents suitable for the modern economy.

- **Illegal Operation:** Operating a courier service without a license carries a penalty of up to **1 year imprisonment or 10 Lakh Taka fine.**
- **Mail Tampering & Theft:** Postal employees found guilty of stealing, opening, or destroying mail face up to **7 years imprisonment.** This high penalty reflects the "high responsibility" status of postal work.
- **Data Privacy Violations:** Unauthorized disclosure of mail contents or personal data is a criminal offense punishable by up to **2 years imprisonment.**
- **Administrative Fines:** For minor operational breaches, the Licensing Authority can impose administrative fines up to **10 Lakh Taka** without needing to go to court, streamlining enforcement.

2.4.10 Governance: The Postal Council

To ensure high-level inter-ministerial coordination and oversight, the Ordinance establishes the **Postal Council.** Chaired by the Secretary of the Posts and Telecommunications Division, this high-powered body includes representatives from Finance, Commerce, Home Affairs, and Bangladesh Bank.

The Council serves as the ultimate appellate body for grievances against the Department or the Licensing Authority. It has the powers of a civil court to summon records and witnesses, ensuring that the postal sector remains accountable, transparent, and aligned with national development goals.

2.4.11 Comparative Analysis: Old Framework versus New

Dimension	Prior Framework (1898/2010)	Postal Services Ordinance 2026
Governing statute	Post Office Act 1898 (as amended 2010)	Postal Services Ordinance 2026 (complete replacement)
Digital services	No provisions for digital postage, logistics tracking, or e-commerce integration	Digital postage stamps (QR/barcode), central logistics tracking platform, interoperability mandates
Data protection	No data protection requirements	Mandatory PDPO compliance: encryption, data minimisation, breach notification
Licensing authority	Limited authority under Sections 4B/4C; 9 of 23 staff positions filled	Empowered licensing authority with expanded regulatory mandate and enforcement powers
Penalties for unlicensed operations	BDT 50,000–200,000	Up to BDT 10 lakh
Universal service	Designated under Section 4A (2010 amendment); no accounting separation	Universal service obligations with mandatory separate accounting to prevent cross-subsidisation
Financial services	General authorisation for banking/remittance (2010 amendment); no specific framework	Postal Bank establishment (subject to banking regulations)
Competition oversight	Minimal provisions	Non-discrimination enforcement, competition oversight, licensing transparency

2.4.12 Implementation Requirements

The Postal Services Ordinance is in draft stage and must be approved by the Advisory Council, gazetted, and brought into force. Upon commencement, the Directorate of Posts and the licensing authority must develop implementing rules covering digital postage standards, logistics tracking platform specifications, PDPO compliance protocols, universal service accounting methodology, and Postal Bank governance and capitalisation. The draft's passage is a legislative priority: without it, the postal sector continues to operate under an 1898 statute that provides no legal basis for the digital services, data protection compliance, or institutional modernisation that this strategy requires.

2.4.13 Innovation and The Road Ahead

Finally, the Ordinance explicitly supports innovation through "**Regulatory Sandboxes.**" It allows the Department and private operators to test new technologies (like drone delivery or blockchain stamps) in a controlled environment with temporary regulatory relaxation. It also

legalizes "**Digital Stamps**" (QR codes, encrypted tokens), paving the way for a paperless postage system.

The *Postal Services Ordinance, 2026* is a comprehensive "Root and Branch" reform. It sheds the colonial baggage of the 1898 Act to construct a legal framework where the Post is a sovereign logistics guarantor, a financial inclusion engine, and a digital data steward. For the PTD Transformation Strategy, this Ordinance provides the statutory bedrock upon which the "Smart Logistics" and "Sovereign Network" pillars will be built.

2.6 Integration with the National Legal Reform Programme

The sector-specific legal instruments examined in Sections 2.2 through 2.5 operate within a broader national legal reform programme enacted in 2025. Three cross-cutting ordinances create enforceable obligations for every PTD entity:

2.6.1 Cyber Safety Ordinance 2025 (CSO)

The CSO, which repeals the Digital Security Act 2018 and the Cyber Security Act 2023, establishes the legal framework for cybersecurity governance, criminal offences, and institutional coordination. For PTD entities, the CSO's relevance is threefold. First, it recognises citizens' right to continuous internet access, reinforcing the Telecommunication Ordinance's shutdown ban with an additional statutory basis. Second, it establishes the National Cyber Emergency Response Team (N-CERT) and the National Security Operations Centre (NSOC), both of which require BTRC and licensed operators to participate in threat intelligence sharing and mandatory Critical Information Infrastructure (CII) incident reporting. Telecommunications networks constitute critical information infrastructure by any definition; the CSO's incident reporting and response requirements apply directly to all PTD-regulated operators. Third, the CSO introduces procedural guardrails for the filing of cybercrime cases—restricting the filing to police, victims, or legal representatives—that eliminate the previous regime's practice of using cybersecurity provisions for politically motivated prosecution of online speech.

2.6.2 Personal Data Protection Ordinance 2025 (PDPO)

The PDPO establishes a comprehensive data protection regime applicable to all entities that process personal data, including every PTD entity. BTRC processes subscriber registration data for all telecommunications operators. BTCL and Teletalk maintain extensive customer databases. The Directorate of Posts handles correspondence and is building address and financial service databases. BSCL manages satellite service user data. TSS and BCSL hold employee and contractor data. The PDPO's requirements—including consent-based processing, data minimisation, security measures, data breach notification, and Data Protection Impact Assessments for sensitive data processing—apply to all of these entities without exception. The appointment of Chief Data Officers by significant data fiduciaries, as required under Section 23, applies to the larger PTD entities. The 2026 Amendment to Section 29 further requires that any cloud-based processing of Critical Information Infrastructure or Restricted Personal Data must maintain at least one synchronised real-time copy within Bangladesh—a provision with direct implications for any PTD entity considering cloud-based service delivery.

2.6.3 National Data Governance Ordinance 2025 (NDGO)

The NDGO establishes the National Data Governance Authority (NDGA) and creates the institutional framework for data interoperability across the government. For PTD entities, compliance with the NDGO requires alignment of all data systems with the Bangladesh National Data Governance and Interoperability Architecture (BNDIA). This means that BTRC's licensing

databases, BTCL’s subscriber records, the Directorate of Posts’ address management and logistics tracking systems, and MOCSLA’s operator registry must all be designed or retrofitted to meet BNDIA interoperability standards. The NDGO’s data classification framework—which categorises government data into open, restricted, confidential, and classified tiers—applies to the data held by every PTD entity. Compliance is not optional; the NDGO imposes penalties for non-compliance and establishes dispute resolution mechanisms for inter-agency data sharing disagreements.

2.6.4 Cross-Legislative Framework Integration

The following table maps the key compliance obligations that the national legal reform programme creates for PTD entities:

PTD Entity	Cross-Legislative Compliance Obligations (CSO, PDPO, NDGO)
BTRC	CSO: CII incident reporting, threat intelligence sharing. PDPO: subscriber data protection, consent management, breach notification. NDGO: licensing database interoperability with BNDIA standards.
BTCL	CSO: CII compliance for network infrastructure. PDPO: customer data protection, Chief Data Officer appointment. NDGO: subscriber records aligned with BNDIA, data classification applied.
Teletalk	CSO: CII compliance, NSOC coordination. PDPO: subscriber and usage data protection. NDGO: data interoperability and classification compliance.
BSCCPLC	CSO: submarine cable infrastructure as CII, mandatory incident reporting. PDPO: operational and commercial data protection. NDGO: bandwidth and capacity data sharing per interoperability standards.
BSCL	CSO: satellite infrastructure security, CII compliance. PDPO: user data protection for Starlink resale operations. NDGO: data reporting to national infrastructure databases.
Directorate of Posts	CSO: digital postal platform security. PDPO: correspondence data, address data, and financial transaction data protection; Data Protection Impact Assessment for Postal Bank. NDGO: address management and logistics data interoperability.
MOCSLA	CSO: operator registry and compliance platform security. PDPO: licensee data handling. NDGO: operator registry aligned with BNDIA for cross-agency verification.
TSS / BCSL	PDPO: employee and contractor data protection. NDGO: production and operational data aligned with national standards.

2.7 International Benchmarks for Legal Reform

Bangladesh's 2026 legal reforms do not operate in isolation. Comparable legislative modernisations across the telecommunications and postal sectors provide reference points for implementation design and outcome expectations.

Legal Reform	Key Features	Relevance to Bangladesh
India: Telecommunications Act 2023	Replaced the Indian Telegraph Act 1885 and Indian Wireless Telegraphy Act 1933 with a unified framework. Introduced technology-neutral licensing, spectrum auctions, right of way provisions, regulatory sandbox, and enhanced consumer protection. Established TRAI's consultative mandate with mandatory public consultation on all major decisions.	Direct parallel to Bangladesh's replacement of the ILDTS 2010 regime. India's transition from legacy legislation to a unified framework provides implementation lessons: phased migration of existing licences, clear timelines, and preservation of existing operator rights during transition.
EU: European Electronic Communications Code 2018	Consolidated four directives into a single Code. Introduced technology-neutral authorisation, co-investment incentives for very high capacity networks, strengthened consumer rights, and harmonised spectrum management across 27 member states.	The Code's approach to SMP remedies, infrastructure sharing incentives, and co-investment frameworks directly informs BTRC's implementation of SMP provisions and infrastructure sharing mandates under the Licensing Policy 2025.
Singapore: Telecommunications Act (revised 2024)	IMDA operates as a converged regulator for telecom, media, and infocomm. Technology-neutral licensing with FBO/SBO two-tier framework. Strong consumer protection, mandatory QoS reporting, and integrated industry development mandate.	Singapore's converged regulatory model is relevant to PTD's oversight of both telecom and postal sectors. IMDA's integration of regulation with industry development informs DoT's proposed evolution.
Denmark: Postal Act Amendment 2023	Abolished the designation of a single universal service provider effective 1 January 2024. Levelled the regulatory playing field by applying identical rules to all postal operators. Reflects a market that can deliver services without a designated monopoly provider.	While Bangladesh's context differs (requiring continued universal service designation for the Directorate of Posts), Denmark's approach to competitive neutrality and its market-based assessment methodology provide a model for evaluating Bangladesh's postal market as it matures.
US: Postal Service Reform Act 2022	Addressed USPS's structural financial challenges by reforming retiree benefit obligations, requiring Medicare integration, mandating six-day delivery,	The Act's approach to coupling legislative reform with a strategic modernisation plan parallels this strategy's integration of the Postal

	and establishing real-time package tracking requirements. Enabled the “Delivering for America” ten-year modernisation plan.	Services Ordinance with the Directorate of Posts’ digital transformation programme.
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2.8 Strategic Impact of Legal Reforms

The legal reforms of 2026, taken collectively, produce five structural shifts that redefine the operating environment for PTD’s digital transformation programme:

From Regulatory Capture to Independent Oversight. The Telecommunication Ordinance’s restoration of BTRC’s independence, combined with the accountability committee and public hearing requirements, creates an institutional architecture in which regulatory decisions are made on the basis of evidence and market analysis rather than political direction. This shift is the single most important precondition for investor confidence, competitive market structures, and consumer welfare. The lesson from fifteen years of compromised regulation is that no amount of infrastructure investment produces optimal outcomes when the regulator serves the interests of the Ministry rather than the market.

From Fragmented Licensing to Simplified, Technology-Neutral Authorisation. The Licensing Policy’s reduction from over twenty licence categories to four core types eliminates the intermediary structures that extracted value without contributing proportionate infrastructure investment. The technology-neutral approach ensures that the licensing framework does not constrain innovation by prescribing specific technologies. The phased transition preserves existing operator rights while converging the market toward a simplified structure.

From Colonial Postal Law to Digital-Ready Framework. The replacement of the 1898 Act with a comprehensive Postal Services Ordinance that includes digital postage, logistics tracking, data protection compliance, and Postal Bank provisions creates the legal basis for transforming the Directorate of Posts from a letter carrier into a digital logistics and financial inclusion platform. Without this legislative foundation, operational initiatives—however well designed—lack the legal authority required for sustainable deployment.

From Unconstrained Surveillance to Rule-of-Law Governance. The combined effect of the Telecommunication Ordinance’s internet shutdown ban and anti-surveillance provisions, the CSO’s judicial oversight requirements for lawful interception, and the PDPO’s data protection obligations creates a legal framework in which telecommunications infrastructure serves citizens rather than surveilling them. This shift is essential for rebuilding the domestic trust and international credibility that documented abuses had eroded.

From Siloed Entities to Integrated Data Governance. The NDGO’s interoperability requirements, the PDPO’s data protection standards, and the sector-specific provisions of the Postal Services Ordinance collectively mandate that PTD entities design their digital systems for

integration rather than isolation. The era of each entity maintaining incompatible databases with no data sharing protocols must give way to a framework in which postal address data, telecommunications subscriber data, licensing records, and service quality metrics are governed by consistent standards and accessible through interoperable platforms—subject to appropriate privacy and security controls.

2.9 Key Highlights Summary

Legal Instrument	Status	Key Provisions
Bangladesh Telecommunication (Amendment) Ordinance 2025	Approved by Advisory Council, 24 December 2025	Restores BTRC independence; bans internet shutdowns permanently; abolishes NTMC; establishes CIS; creates parliamentary accountability committee; mandates quarterly public hearings; criminalises surveillance data abuse
Telecommunications Network and Licensing Policy 2025	Gazetted 22 September 2025	Reduces 20+ licence categories to 4 core types (ANSP, NICSP, ICSP, NTNSP); introduces SMP remedies, mandatory infrastructure sharing, MVNO recognition, QoS Dashboard, fibre-to-tower targets (50%/18 months, 80%/3 years); repeals ILDTS Policy 2010
Postal Services Ordinance 2026	Draft stage (pending Advisory Council approval)	Repeals Post Office Act 1898; introduces digital postage, central logistics tracking, PDPO compliance mandate, enhanced penalties (up to BDT 10 lakh), universal service with accounting separation, Postal Bank provisions, Establishes MOCSLA as statutory body with budgetary autonomy; expands regulatory mandate to digital commerce logistics; strengthens enforcement powers; creates merit-based leadership selection.
Cyber Safety Ordinance 2025	Enacted 2025 (cross-cutting)	Repeals DSA 2018 and CSA 2023; recognises right to continuous internet access; establishes N-CERT and NSOC; mandates CII incident reporting; restricts case filing to police/victims
Personal Data Protection Ordinance 2025	Enacted 2025 (cross-cutting)	Consent-based processing; data minimisation; breach notification; Chief Data Officer requirement for significant fiduciaries; cross-border data transfer conditions; cloud copy requirement for CII data (2026 amendment)
National Data Governance Ordinance 2025	Enacted 2025 (cross-cutting)	Establishes NDGA; data classification framework; BNDIA interoperability standards; mandatory compliance for all government entities; dispute resolution mechanism

The legal landscape of 2025 provides the Posts and Telecommunications Division with a comprehensive legislative foundation that, for the first time, aligns the sector’s legal framework with the ambitions of the National Digital Transformation Strategy. The implementation chapters

that follow—covering infrastructure, institutional reform, project reform, and governance—operate within and are enabled by this reformed legal architecture. The critical determinant of success is not the quality of the legislation, which is structurally sound, but the discipline, speed, and transparency of its implementation.

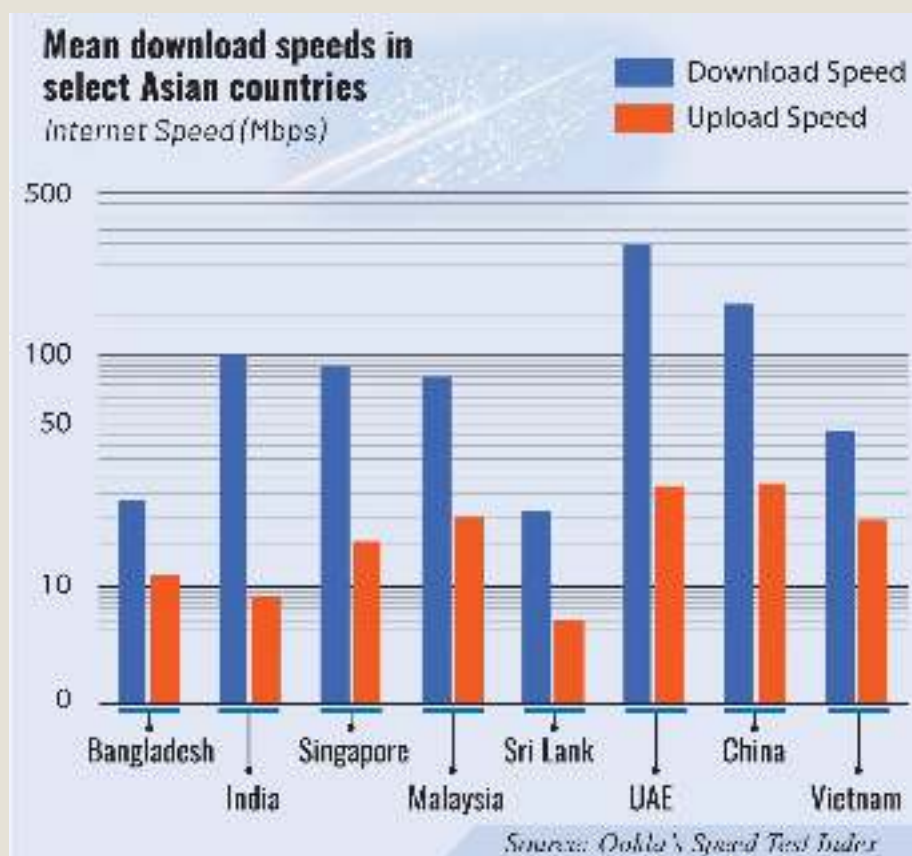
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3. Telecom Infrastructure and Regulatory Capture

3.1 The Infrastructure Imperative

Bangladesh's telecommunications sector has, by several measures, achieved remarkable scale. As of mid-2025, the country has approximately 188 million mobile phone subscriptions across four mobile network operators, 133.6 million internet subscribers, and 4G coverage reaching virtually 100 per cent of the population. The communications services market generates an estimated USD 4.9 billion in annual revenue. Mobile financial services process transactions exceeding BDT 13 lakh crore annually, making Bangladesh one of the most active mobile money markets in Asia. The sector directly employs over 30,000 people within PTD entities alone, with hundreds of thousands more across private operators, tower companies, ISPs, and the broader digital ecosystem.

Yet these aggregate figures conceal structural deficiencies that, left unaddressed, will constrain the sector's capacity to serve as the foundation of the national digital transformation. Internet penetration stands at only 44.5 per cent—significantly below regional peers such as India (52 per cent), Vietnam (79 per cent), and Malaysia (97 per cent). Between June 2024 and January 2025, the number of mobile internet users fell by 13.2 million, driven by a tax burden exceeding 39 per cent on mobile services (comprising supplementary duty, VAT, and surcharges), rising SIM card costs, and macroeconomic pressures that forced consumers to prioritise basic necessities over data packages. Only approximately 30 per cent of mobile towers are connected via fibre optic cables, with the remainder relying on low-capacity microwave links that constrain data throughput and network resilience. The median mobile download speed, at 28.3 Mbps (January 2025, Ookla data), places Bangladesh behind India (94.5 Mbps), Vietnam (50.7 Mbps), and significantly behind regional leaders.



The White Paper's analysis reveals that these deficiencies are not merely the product of underinvestment or market immaturity. They are, in substantial part, the consequence of regulatory capture—a condition in which the regulatory authority, designed to serve the public interest, instead serves the interests of the entities it is meant to regulate, or of the political executive that controls it. The erosion of BTRC's independence through the 2010 amendments, the issuance of licences without baseline demand analysis, the tolerance of ownership structures that breached BTRC's own guidelines, and the maintenance of artificially high submarine cable bandwidth pricing all reflect a regulatory environment in which the structures of oversight were subverted to benefit concentrated interests at the expense of competitive markets and consumer welfare.

This chapter examines the state of Bangladesh's telecommunications infrastructure and the regulatory architecture that governs it. It documents the governance failures that produced the current conditions, analyses the reform measures initiated in 2025, identifies the gaps that remain, and establishes the implementation imperatives and performance indicators that will determine whether the sector achieves the connectivity outcomes that the National Digital Transformation Strategy requires.

Bangladesh Telecom Sector Reset

Building a Foundation for Digital Rights and Connectivity

PROBLEM: Regulatory Capture & Stagnation



IF IT RESET: Foundational Sector Overhaul



SO WHAT? Foundation for Digital Rights and Connectivity

Structural & Operational Modernization

5G, 4G, 3G Networks

Provision of 5G and 4G networks is essential for the growth of digital economy and connectivity.

Integration of satellite services is essential for connectivity in remote areas and disaster response.

Governance, Rights, and Accountability

Ensuring the security and integrity of digital infrastructure is critical for digital rights and accountability.

Surveillance and data collection must be done in a transparent and accountable manner to protect digital rights.

Accountability is a key factor in ensuring digital rights and connectivity.

42%

KEY PERFORMANCE INDICATORS

INDICATOR	STATUS
5G Network Coverage	40% Coverage
4G Network Coverage	80% Coverage
3G Network Coverage	100% Coverage
Surveillance and Data Collection	Transparency and Accountability

3.2 The Regulatory Commission: BTRC

3.2.1 Institutional Architecture

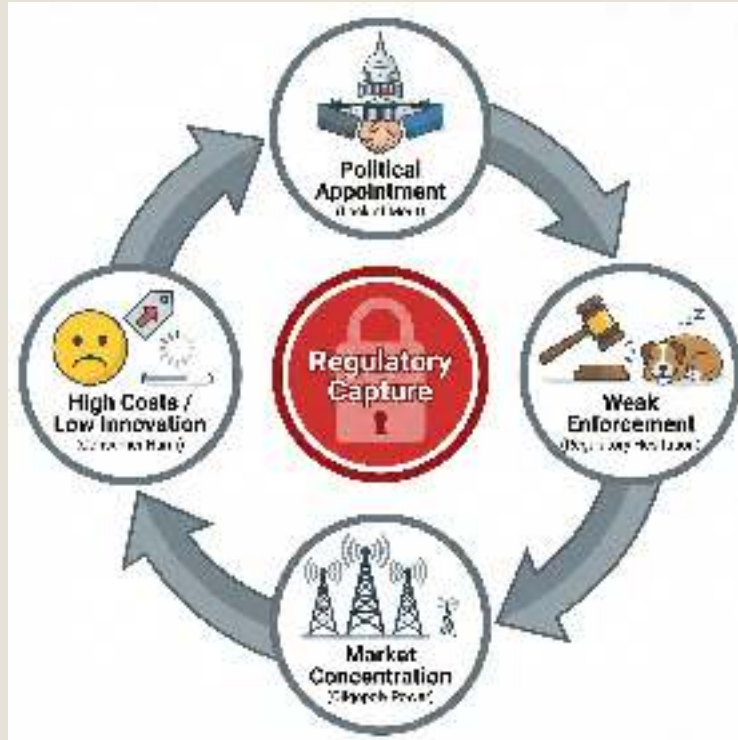
The Bangladesh Telecommunication Regulatory Commission (BTRC) was established under the Bangladesh Telecommunication Act 2001 (Act No. XVIII of 2001) as an independent statutory body responsible for licensing, tariff regulation, spectrum management, quality of service enforcement, and market oversight. The Commission structure comprises a Chairman, a Vice-Chairman, and three Commissioners, supported by eight functional divisions: Engineering and Operations, Administration and Human Resources, Spectrum Management, Systems and Services, Legal and Licensing, Enforcement and Inspection, Finance Accounts and Revenue, and Public Relations. Five divisions are headed by Director Generals, with two Directors and one Deputy Director reporting directly to the Commission.

BTRC's revenue generation is substantial. Its licensing and regulatory fees constitute a major source of non-tax revenue for the national exchequer, with revenue peaking around 2017–2018. The development budget for fiscal year 2023–24 allocated over BDT 4,773 crore to BTRC, reflecting its central role in the sector's development financing. The Commission oversees a market comprising four mobile network operators (Grameenphone, Robi Axiata, Banglalink, and Teletalk), two NTTN licensees, 24 IGW operators, 22 ICX operators, 36 IIG operators, approximately 2,500 ISPs, and multiple tower companies, satellite operators, and specialised service providers.

3.2.2 The Erosion of Independence: 2010–2024

The Bangladesh Telecommunication (Amendment) Act 2010 fundamentally altered BTRC's institutional character. As documented in the White Paper, the 2010 amendments transferred final authority over licence issuance, renewal, service approval, and tariff-setting from BTRC to the Ministry of Posts and Telecommunications. This transfer converted BTRC from an independent regulator into, effectively, an implementing agency of the Ministry. Key decisions on spectrum management, licensing, and tariff approval required ministerial endorsement, weakening BTRC's operational discretion. Regulatory decisions became subject to ad hoc instructions, ministerial intervention, and political influence rather than rule-based governance.

The consequences were systemic. The ILDTS Policy 2010, which followed shortly after the amendments, was issued under the Ministry's assumed licensing authority and introduced a licensing regime that the White Paper characterises as having “enabled cross-layer ownership and expanded licence issuance” without baseline demand analysis. This created “fertile ground for collusion, rent-seeking, and monopolistic behavior.” The over-fragmented market that resulted—ICX (22 licensees), IGW (24), IIG (36), approximately 2,500 ISPs—was misaligned to actual demand. Legal international voice minutes collapsed tenfold since 2015. Chronic under-utilisation and arrears demonstrated that the economic model for the majority of licensees was unsustainable.



Internationally, the relationship between regulatory independence and sector performance is well-established. The ITU’s Global ICT Regulatory Tracker consistently demonstrates that countries with “fourth generation” regulatory frameworks—characterised by independent, multi-stakeholder regulatory bodies—achieve higher connectivity outcomes per unit of investment than those with first- or second-generation frameworks characterised by ministerial control. The OECD’s Principles for the Governance of Regulators (2019) underscore the need for clear separation between policymaking and regulation to prevent political or commercial interference. The European Electronic Communications Code prohibits external directives that could compromise objective decision-making by National Regulatory Authorities. The FCC in the United States and Ofcom in the United Kingdom both operate under statutory protections for institutional autonomy. TRAI in India, while advisory in function, conducts mandatory public consultations on all major regulatory decisions, creating procedural accountability even where formal independence is limited.

Bangladesh’s experience between 2010 and 2024 provides a case study in what the absence of these protections produces: a regulatory environment characterised by unpredictability, investor uncertainty, delayed spectrum auctions, slow deployment of next-generation technologies, and a licensing regime that served concentrated interests rather than competitive market development.

3.2.3 Governance Failures: Recruitment and Appointments

The White Paper documents extensive governance failures within BTRC’s own institutional processes. Since 2002, BTRC has had eleven Chairmen in twenty-one years, with many replaced

when political administrations changed. Commissioner positions were filled by executive decision without transparent search processes or external scrutiny. Director Generals were commonly drawn from government cadres or promoted internally with Ministry clearance, reinforcing BTRC's dependence on the parent Ministry.

The recruitment and promotion irregularities identified between 2009 and 2022 included: the reclassification of contract and project staff as “departmental candidates” to bypass competitive examination requirements; selective age relaxations for preferred candidates; cross-stream promotions that violated branch-specific criteria in the BTRC Service Regulations; appointment of a Commissioner in 2017 whose siblings retained telecom business licences, creating a conflict of interest inconsistent with OECD, Ofcom, and US federal impartiality standards; and the appointment in 2022 of a Commissioner found to lack a formal service contract. These irregularities stemmed from institutional weakness, administrative negligence, nepotism-linked preferential practices, abuse of discretionary authority, and alleged political interference.

The White Paper's root-cause assessment identifies seven contributory factors: institutional weakness and incapacity (absence of a centralised HR oversight cell); negligence and procedural lapses (failure to maintain applicant lists, examination scripts, and result archives); nepotism and preferential favouritism; abuse of discretionary powers (Commission-level age relaxations and qualification waivers without documented justification); political and external influence on appointments at Director General, Commissioner, and Chairman levels; corruption-prone governance gaps (lack of transparent competitive examinations and weak conflict-of-interest safeguards); and regulatory and policy gaps (ambiguous definitions enabling procedural arbitrage).

3.2.4 Reform Agenda

The Bangladesh Telecommunication (Amendment) Ordinance 2025, analysed in Chapter 2, restores BTRC's regulatory independence as a matter of law. However, institutional reform requires more than legislative restoration. The White Paper recommends a comprehensive reform programme comprising:

Competitive, rules-based recruitment: All vacancies publicly advertised with clear eligibility criteria, transparent PSC-style examinations, published mark weightings, and mandatory document retention for at least ten years. This curbs non-competitive entry and ensures equal opportunity consistent with Article 29 of the Constitution.

Merit- and ACR-based promotions with time-in-grade rules: All promotions linked to documented Annual Confidential Report/KPI performance, minimum time-in-grade, and prescribed qualifications, embedded in a central Human Resource Information System (HRIS) that auto-enforces rules and blocks promotions of officers under investigation or lacking required tenure.

Branch-locked career ladders with bridging pathways: Technical roles within technical streams, administrative within administrative, financial within finance, and legal within legal, with cross-stream promotions permitted only after completion of accredited bridging programmes and minimum qualifying tenure in the target stream.

Independent search-cum-selection committee for senior appointments: An amendment to the Telecommunication Act 2001 establishing an independent committee for the selection of Chairman, Commissioners, and Director Generals, replacing the current executive-driven model that has produced politicised, non-transparent appointments.

Strengthened conflict-of-interest framework: A mandatory declaration and recusal regime aligned with OECD and Ofcom standards, ensuring that no Commissioner or senior officer has personal or family financial interests in regulated entities.

Reform KPIs: BTRC Institutional Governance

Indicator	Timeline	Responsible
HRIS deployed with automated rule enforcement	Within 12 months	BTRC
Search-cum-selection committee constituted for senior appointments	Within 6 months of Ordinance commencement	PTD / BTRC
All vacancies advertised with published eligibility criteria	Immediate (continuous)	BTRC
Conflict-of-interest declarations filed by all Commissioners and DGs	Within 3 months	BTRC
First independent external audit of HR processes completed	Within 18 months	BTRC / external auditor
Zero cross-stream promotions without accredited bridging qualification	Continuous from HRIS deployment	BTRC

3.3 Market Structure and Competition

3.3.1 Mobile Network Operators

Bangladesh’s mobile market is served by four operators. Grameenphone (a subsidiary of Norway’s Telenor Group) leads with approximately 86.5 million subscribers, representing nearly 45 per cent of the market. Robi Axiata follows with approximately 57.4 million subscribers (30 per cent), having consolidated its position through its 2016 merger with Airtel Bangladesh. Banglalink (part of the VEON group) holds approximately 37.9 million subscribers, while Teletalk, the sole state-owned mobile operator, serves approximately 6.6 million subscribers. Collectively, these operators generate the majority of the sector’s revenue and drive the consumer experience of telecommunications in Bangladesh.



Operator	Ownership	Subscribers (Mid-2025)	Market Share	Key Characteristics
Grameenphone	Telenor (55.8%), Grameen Telecom (34.2%)	~86.5 million	~45%	Market leader; SMP designated; highest QoS scores (video experience 57.6, download 23.3 Mbps per OpenSignal Feb 2025)

Robi Axiata	Axiata Group (61.82%), Bharti Airtel (28.18%)	~57.4 million	~30%	Merged with Airtel 2016; parent Axiata also controls edotco tower company
Banglalink	VEON Group	~37.9 million	~20%	Strong youth demographic; digital lifestyle focus
Teletalk	100% Government of Bangladesh	~6.6 million	~3.5%	Persistent losses (Tk 196.97 crore FY23); declining subscriber base; MVNO pilot with BTCL approved

The market exhibits significant concentration. Grameenphone has been designated as an operator with Significant Market Power (SMP), a designation that in principle triggers ex-ante remedies including wholesale access obligations and price controls. In practice, as the White Paper documents, SMP remedies have existed “largely on paper but are rarely enforced.” The Telecommunications Network and Licensing Policy 2025 empowers BTRC to establish a functioning SMP framework; the critical implementation task is translating this authority into operational enforcement—including periodic market reviews, published determinations, and enforceable remedies with transparent compliance monitoring.

The decline in mobile subscribers between mid-2024 and early 2025—approximately 6 million mobile subscriptions and 13.2 million mobile internet users lost—warrants analysis. The decline is attributable to multiple reinforcing factors: a tax burden exceeding 39 per cent on mobile services following the 2024–25 budget’s increase of supplementary duty from 15 to 20 per cent; rising SIM card prices (from subsidised levels to BDT 100–300); macroeconomic pressures that reduced discretionary spending on data packages; and the deactivation of dormant multi-SIM subscriptions. While broadband internet subscriptions increased slightly to approximately 14 million during the same period (suggesting substitution toward Wi-Fi-based access), the net decline represents a significant setback to the digitisation trajectory and underscores the policy imperative of reducing the tax burden on mobile services to levels consistent with regional norms.

3.3.2 Tower Infrastructure Concentration

The tower infrastructure market in Bangladesh is dominated by two companies: edotco Bangladesh Company Limited (EBCL) and Summit Tower Limited, which together control approximately 90 per cent of the passive tower infrastructure. This duopoly raises significant competition and sovereignty concerns.

edotco Bangladesh Company Limited (EBCL): Established in 2013 as a subsidiary of edotco Group Sdn Bhd (itself a subsidiary of Malaysia's Axiata Group Berhad), EBCL's formation involved the extraction of Robi Axiata's tower assets into a separate entity. The White Paper documents a troubling pattern of ownership changes. At formation, Robi Axiata held 51 per cent and edotco Group 49 per cent. In 2015, Robi sold 5,258 towers to EBCL for USD 250 million. In November 2016, BTRC approved edotco's increase of its stake to 80 per cent—directly contravening BTRC's own guideline limiting foreign partner shareholding to 70 per cent. In September 2018, Robi transferred its remaining 20 per cent to edotco Group. BRAC subsequently acquired 26 per cent and Greencon Tower/GETCO 4 per cent, bringing the current structure to edotco Group (70 per cent), BRAC (26 per cent), and Greencon/GETCO (4 per cent). The practical effect was that the Axiata Group re-rated tower assets to higher valuation multiples while depleting Robi's asset base, with Malaysia's sovereign wealth fund Khazanah Nasional Berhad holding significant stakes in both Axiata Group (36.7 per cent) and edotco Sdn Bhd (approximately 32 per cent).

The regulatory questions raised by the White Paper remain pertinent: how did BTRC approve an 80 per cent foreign stake that exceeded its own 70 per cent guideline? Why was the ILDS Policy 2010's prohibition on foreign ownership of new telecom entities not applied to the tower asset transfer? These questions illustrate the regulatory capture dynamic: ownership restrictions were enforced inconsistently, with the effect of enabling one corporate group to consolidate control over critical national infrastructure.

Reform Imperatives for Tower Infrastructure: The Licensing Policy 2025's mandatory infrastructure sharing provisions, if enforced, create the framework to address tower market concentration. BTRC must establish open-access pricing guidelines for shared tower infrastructure, conduct transparent periodic SMP reviews of the tower market, enforce non-discrimination obligations on dominant tower providers, and ensure that the infrastructure sharing guidelines—which must cover both passive elements (towers, poles, ducts) and active elements (radio access networks)—create a genuinely competitive wholesale tower market. India's tower sharing experience, where Indus Towers (a joint venture of Airtel and Vodafone Idea) and American Tower Corporation operate under a framework regulated by TRAI's infrastructure provision (IP-I) guidelines, provides a comparative reference. Indonesia's tower regulation, which mandates co-location and sharing with regulated pricing, offers another model for markets with concentrated tower ownership.

3.3.3 NTTN Duopoly and Vertical Integration

The Nationwide Telecommunications Transmission Network (NTTN) market in Bangladesh is served by two operators: Fiber@Home and Summit Communications. This duopoly controls the fibre backbone infrastructure that interconnects access networks, international gateways, and end users. The concentration of national transmission capacity in two entities creates the conditions for market power that the White Paper identifies as inadequately addressed: “The duopoly in

NTTN provision by Fiber@Home and Summit... has not been counterbalanced by robust open-access price controls, audited non-discrimination, or transparent periodic SMP reviews.”

Compounding the duopoly concern is vertical integration across network layers. Cross-ownership arrangements combining NTTN, ITC (International Terrestrial Cable), and IIG (International Internet Gateway) licences have enabled integrated groups to influence upstream and downstream pricing and access. The White Paper observes that regulatory enforcement of accounting separation has been weak, allowing conflicts of interest to persist. Stand-alone operators—particularly smaller ISPs—pay more for the same wholesale services, creating structural disadvantage that undermines competitive entry and market diversity.

The Licensing Policy 2025’s rationalisation of licence categories addresses this concern by design: the National Infrastructure and Connectivity Service Provider (NICSP) licence integrates transmission, interconnection, and internet exchange functions under a single category with mandatory non-discriminatory wholesale access obligations. However, the transition from the current duopoly to a genuinely competitive NICSP market requires BTRC to implement SMP determinations, establish transparent wholesale pricing methodologies, mandate accounting separation with independent audits, and create monitoring mechanisms that detect discriminatory conduct. The European Commission’s regulatory treatment of national network operators under the European Electronic Communications Code—including the Significant Market Power framework, access and interconnection obligations, and cost-oriented pricing—provides a well-developed institutional model.

3.3.4 Licensing Over-Fragmentation and the ILDTS Legacy

The licensing regime established under the ILDTS Policies of 2007 and 2010 created over twenty distinct licence categories, producing a fragmented market structure that the White Paper characterises as the product of regulatory failure rather than market-driven evolution.

The ILDTS Policy 2007 introduced segmented licensing for International Gateways (IGW), Interconnection Exchanges (ICX), International Internet Gateways (IIG), and Internet Protocol Telephony Service Providers (IPTSP), with fixed licence caps established without any documented baseline demand analysis. The policy arbitrarily restricted the number of licensees: three IGWs (in addition to BTTB), and six ICXs (in addition to BTTB). The White Paper concludes that “allocating only 3 IGW licenses in 2008 was a flawed decision.”

The ILDTS Policy 2010 compounded the 2007 regime’s deficiencies by removing licence caps entirely without implementing competitive safeguards. The resulting flood of sub-scale operators stretched BTRC’s oversight capacity, diluted investment, and produced a market of “survival-mode firms, price wars, leakage, and weak QoS—without matching consumer benefit.” The multi-layered infrastructure stack—the rigid ITC→IIG/IGW→ICX→ANS chain—added cost and latency at each hop, forced additional audits and approvals, and imposed disproportionate costs on smaller ISPs.

Legacy Licence Categories and Market Oversupply

Category	Active Licensees	Key Issue
IGW	24	Severe oversupply; legal international voice minutes collapsed 10x since 2015; chronic under-utilisation; formation of IGW Operators Forum (IOF) enabling coordinated behaviour
ICX	22	Oversupply relative to declining voice interconnection demand; quality and monitoring challenges
IIG	36	Multiple IIG licences concentrated in vertically integrated groups; limited independent competition
ISP	~2,500	Vast majority sub-scale; many dormant or operating with expired licences; disproportionate compliance burden from multi-desk NOC processes
NTTN	2	Duopoly (Fiber@Home and Summit) controlling national fibre backbone; insufficient SMP enforcement
Tower Company	Multiple (duopoly)	edotco + Summit control ~90% of market; open-access pricing not enforced

The Licensing Policy 2025 addresses this legacy by consolidating the entire licence landscape into four categories (ANSP, NICSP, ICSP, NTNSP), with existing licences permitted to run to natural expiry—most lapsing by 2027. The transition period must be managed carefully: BTRC must publish a clear migration roadmap specifying timelines, fee structures, and transitional conditions for each legacy category; ensure that operators understand their obligations under the new framework; and provide a transparent appeal mechanism for disputes arising during transition.

3.4 International Connectivity: Submarine Cables and Bandwidth Pricing

3.4.1 BSCPLC and the Bandwidth Pricing Challenge

Bangladesh Submarine Cable Company PLC (BSCPLC)—the state-owned operator responsible for international submarine cable connectivity—has, in financial terms, been among the most consistently profitable PTD entities since its inception. Its 2012 IPO was successful, and its stock has performed steadily in the market. Yet this financial performance, as the White Paper documents in detail, came at a significant cost to the national interest: the price of submarine cable bandwidth was maintained at artificially high levels, keeping a large portion of available capacity unsold while Bangladesh’s internet users paid more than they should have for international connectivity.

The data is stark. In FY 2023–24, BSCPLC’s lit-up capacity stood at 6,850 Gbps, but actual utilisation was only 2,696 Gbps—39 per cent of available capacity. This consistent and remarkably large gap between capacity and utilisation is, as the White Paper notes, “a clear outcome of highly priced BSCPLC bandwidth” that has “had a negative impact on the overall expansion of the Internet in Bangladesh.” International bandwidth has zero shelf-life, and submarine cables have limited operational lifetimes of approximately twenty-five years; holding up unused bandwidth represents a direct loss of national economic value. While International Terrestrial Cable (ITC) operators may have benefited from BSCPLC’s pricing, the average internet user did not.

By 2023, the total downstream bandwidth brought in by ITC operators equaled BSCPLC’s bandwidth, and has since surpassed it. BSCPLC’s market share in international bandwidth is declining as operators seek lower-cost terrestrial alternatives. This trend will accelerate with the entry of the Bangladesh Private Cable Consortium (BPCS) and the commissioning of SEA-ME-WE 6.

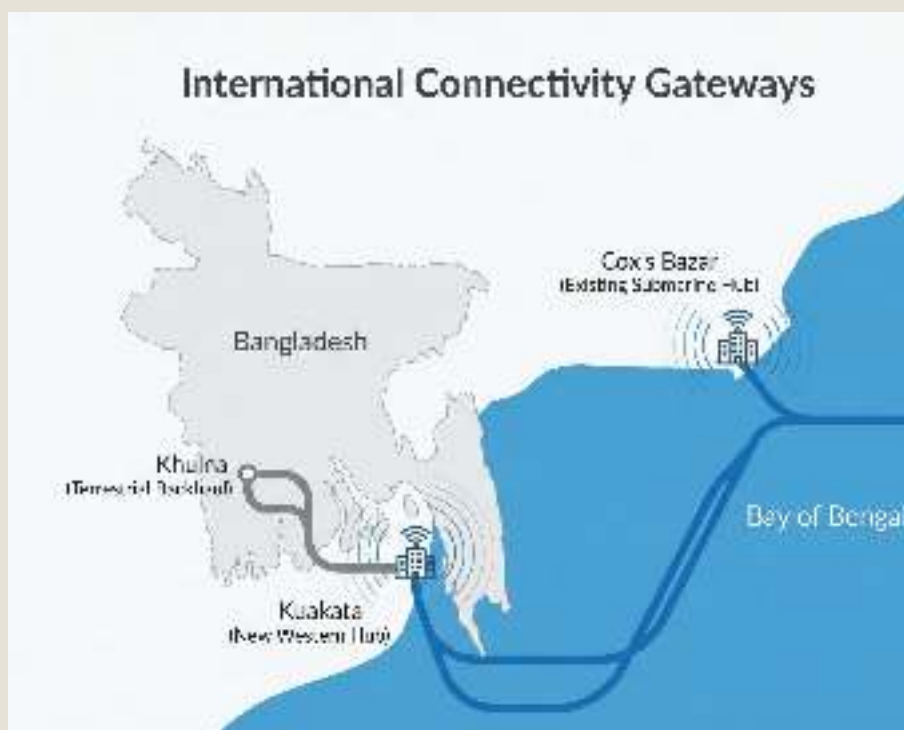
Internationally, state-owned submarine cable operators face similar challenges. Brazil’s Telebras and India’s BSNL have both faced the tension between maintaining profitability as state enterprises and pricing bandwidth at levels that maximise national connectivity. The resolution, invariably, involves pricing reform that reduces wholesale bandwidth costs to competitive levels, coupled with commercial diversification into value-added services (managed wavelength services, data centre interconnection, co-location at cable landing stations) that generate margins from services rather than artificial scarcity. BSCPLC must undertake a comparable strategic pivot: price reductions to maximise utilisation, combined with new service offerings that generate sustainable revenue.

3.4.2 SEA-ME-WE 6, BPCS, and Redundancy

Bangladesh is investing in SEA-ME-WE 6 (SMW-6), a high-capacity international submarine cable expected to become operational in 2026. SMW-6 will significantly increase Bangladesh’s international bandwidth capacity and provide redundancy for the existing SEA-ME-WE 4 and

SEA-ME-WE 5 connections. The cable's design capacity of approximately 100 Tbps per fibre pair places it among the highest-capacity submarine cable systems globally.

The Bangladesh Private Cable Consortium (BPCS), licensed to invest in the SIGMAR cable (a 1,400 km stretch of UMO), introduces private sector participation in international submarine connectivity for the first time. BPCS offers dark fibre Indefeasible Right of Use (IRU) ownership, enabling operators to light their own capacity and bypass BSCPLC's wholesale bandwidth model. This structural change may further divert traffic and revenue from BSCPLC unless it substantially revises its pricing and service models.



The White Paper raises legitimate concerns about BPCS's licensing structure: three separate private bidders formed a consortium after the licensing process, raising questions about whether BTRC should have encouraged consortium formation earlier to direct investment into a redundancy cable rather than a parallel capacity addition. The timing of BPCS licensing during SMW-6's rollout risks diluting national investment in BSCPLC and creating market fragmentation rather than strategically expanding Bangladesh's cable landing diversity—a critical resilience concern given that Bangladesh currently has only two cable landing points (both in Cox's Bazar), creating a single geographic point of failure.

Reform Imperatives: Bangladesh's international connectivity strategy must address four priorities: pricing reform (BSCPLC bandwidth pricing reduced to competitive levels to maximise utilisation); redundancy (diversification of cable landing points beyond Cox's Bazar, potentially through a Chittagong or Mongla alternative landing station); competitive framework (BTRC enforcement of fair-play rules ensuring BPCS entry stimulates competition rather than

fragmentation); and strategic coordination (alignment of SMW-6 commissioning, BPCS operations, and Starlink satellite connectivity into a coherent national bandwidth strategy that optimises cost, capacity, and resilience).

3.4.3 Satellite Internet: The Starlink Entry

The arrival of Starlink in Bangladesh added a new dimension to the connectivity landscape in 2025. SpaceX's LEO satellite constellation, offering broadband internet with low latency, received regulatory approval to operate in Bangladesh, with Bangladesh Satellite Company Limited (BSCL) designated as the national resale partner. The entry of satellite internet—formalised through the Non-Terrestrial Networks and Service Provider (NTNSP) licence category in the Licensing Policy 2025—is the first time Bangladesh's licensing framework has provided explicit provisions for satellite broadband.

Starlink's potential impact is significant in three dimensions: geographic coverage (LEO satellites can provide broadband connectivity to the approximately 55 per cent of Bangladesh's population in rural areas, the Chittagong Hill Tracts, and coastal regions where terrestrial infrastructure is limited or absent); disaster resilience (satellite connectivity is independent of terrestrial infrastructure and can maintain communications during cyclones, floods, and other natural disasters that regularly affect Bangladesh); and competitive pressure (satellite broadband creates an alternative pathway to international connectivity that bypasses the entire terrestrial value chain, potentially disciplining pricing by ITC operators, BSCPLC, and the NTTN providers).

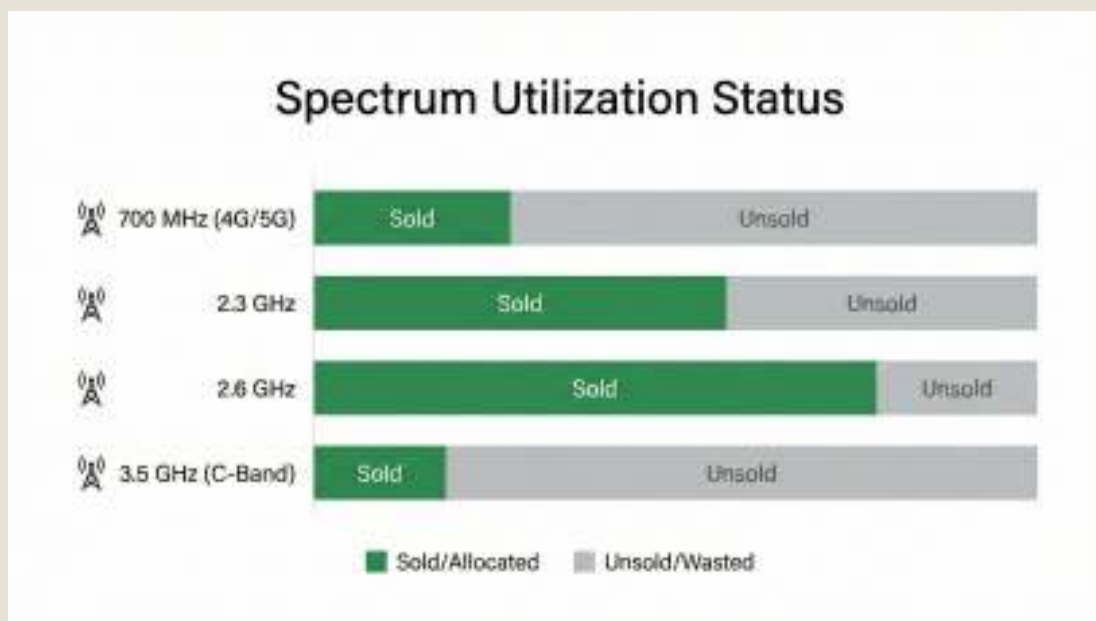
However, challenges remain: Starlink's consumer pricing must be affordable for the Bangladeshi market (current US pricing of approximately USD 120 per month for equipment lease plus USD 120 per month for service is well beyond the reach of most Bangladeshi consumers); regulatory conditions must ensure that Starlink's operations comply with PDPO data protection requirements and the CSO's CII provisions; and BSCL's operational capacity to serve as an effective resale partner must be strengthened.

3.5 Spectrum Management and 5G Readiness

3.5.1 Current Holdings and Allocation History

Spectrum management in Bangladesh has been characterised by the White Paper as “price-heavy and flexibility-light.” Auctions have carried rigid rules and inflated reserve prices, while the absence of secondary spectrum trading, dynamic spectrum management, spectrum sharing, refarming, and technology-neutral rights has constrained efficient utilisation and delayed the rollout of next-generation services.

The 2022 spectrum auction allocated 220 MHz across the 2.3 GHz and 2.6 GHz bands: Grameenphone and Robi each secured 60 MHz from the 2.6 GHz band for approximately BDT 3,360 crore each; Banglalink acquired 40 MHz from the 2.3 GHz band for BDT 2,241 crore; and Teletalk obtained 30 MHz for BDT 1,680 crore. Notably, 30 MHz from the 2.3 GHz band remained unsold in a second phase, with neither Teletalk nor Banglalink showing interest—an indication that reserve pricing exceeded operators’ willingness to pay at the margin. In August 2024, BTRC allocated an additional 20 MHz in the 2.6 GHz band to Grameenphone and Robi (10 MHz each) at a negotiated price of BDT 580 million per MHz, with payment terms structured over ten annual instalments.



3.5.2 The 700 MHz Band and Rural Broadband

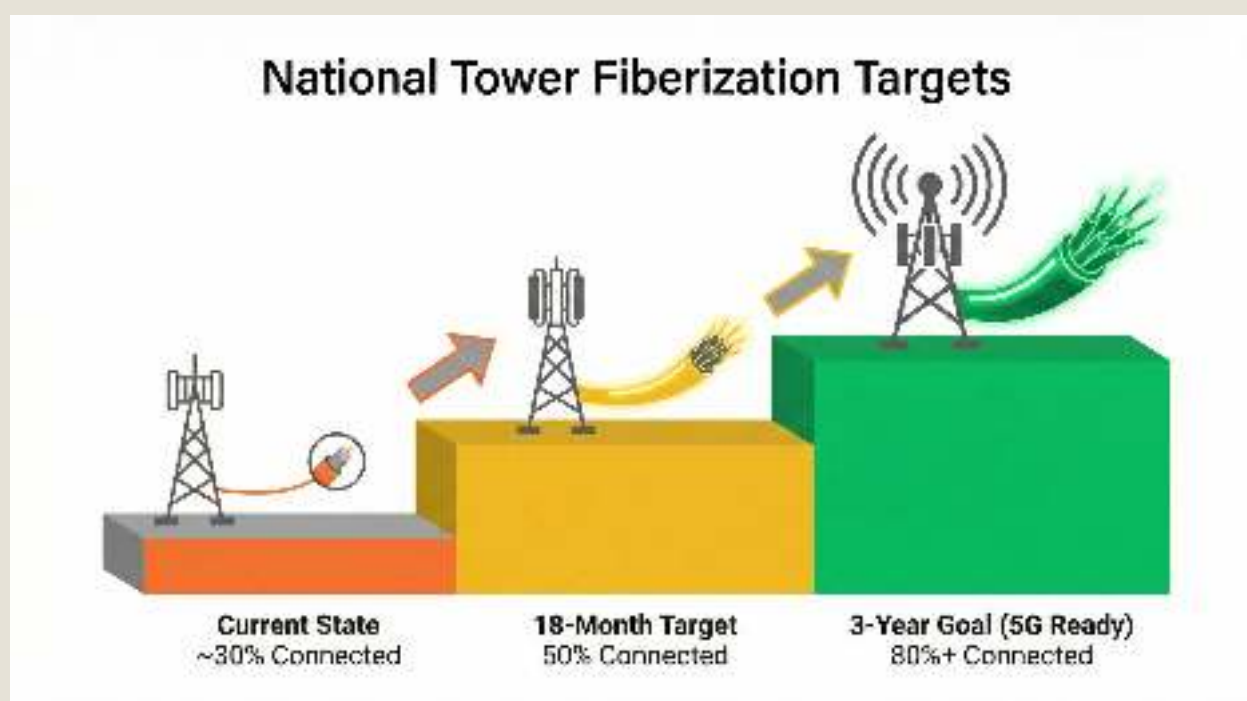
BTRC planned to auction the 700 MHz band in 2025, a frequency band that is critical for mobile broadband expansion in rural areas due to its superior propagation characteristics (wider coverage and better building penetration compared to higher frequencies). However, a legal dispute with an ISP over a portion of this band has delayed the auction. The proposed base price of BDT 263 crore per MHz was identified by the White Paper as being far above international

benchmarks. At this price level, spectrum costs would reach 14–16 per cent of operator revenue (compared to the current 9–12 per cent range), discouraging investment and potentially leaving the band unsold.

Internationally, spectrum pricing reform has been a critical enabler of rural broadband expansion. India's approach to the 700 MHz band is instructive: after initial auctions in 2016 saw the band go unsold at reserve prices that exceeded operators' valuations, TRAI recommended substantial reductions. The 2024 Indian auction attracted bids at significantly lower reserve prices, enabling operators to deploy rural broadband infrastructure. The United Kingdom's Ofcom set 700 MHz reserve prices at levels designed to ensure allocation rather than maximise auction revenue, recognising that the economic value of spectrum is generated through its deployment, not its auction price. Bangladesh must resolve the legal dispute and reset the 700 MHz reserve price to levels that ensure allocation and deployment, particularly in the approximately 2,600 unions identified by the National Digital Transformation Strategy as requiring enhanced connectivity.

3.5.3 5G Readiness and Roadmap

Bangladesh has not yet commercially deployed 5G services, placing it behind regional peers including India (launched June 2022), Vietnam (commercial trials ongoing since 2024), Thailand (launched March 2020), and the Philippines (launched March 2021). The prerequisites for 5G deployment include: adequate spectrum allocation in mid-band (3.3–3.8 GHz) and high-band (millimetre wave) frequencies; fibre backhaul to at least 80 per cent of tower sites (current estimate: approximately 30 per cent); a regulatory framework for small cell deployment (including rights of way, permitting, and municipal coordination); and an investment environment that provides operators with confidence in the return on 5G capital expenditure.



The Licensing Policy 2025’s provisions for technology-neutral licensing, infrastructure sharing, and NTNSP licences create the regulatory preconditions. The fibre-to-tower targets (50 per cent within 18 months, 80 per cent within three years) directly address the backhaul constraint. However, 5G spectrum allocation—particularly in the 3.5 GHz band that is the global anchor for 5G mid-band deployment—must be scheduled, priced competitively, and allocated through a transparent process. BTRC must develop and publish a 5G spectrum roadmap within twelve months, identifying candidate bands, indicative timelines, and the technical conditions (including electromagnetic compatibility assessments and satellite coordination) required for allocation.

The RAN (Radio Access Network) and spectrum sharing rules that the White Paper identifies as “years late”—MORAN (Multi-Operator Radio Access Network) and MOCN (Multi-Operator Core Network) frameworks—must be finalised and published by BTRC. Active infrastructure sharing reduces duplicative capital expenditure, enables more efficient spectrum utilisation, and accelerates rural coverage. Without these rules, operators continue to invest in parallel infrastructure that increases costs and slows deployment—a pattern that is particularly damaging for Teletalk, whose limited capital base makes shared infrastructure essential to its viability.

Reform KPIs: Spectrum Management and 5G Readiness

Indicator	Timeline	Responsible
700 MHz legal dispute resolved; auction completed	Within 12 months	BTRC / Ministry of Law
700 MHz reserve price reset to internationally benchmarked level	Pre-auction	BTRC
5G spectrum roadmap published (3.5 GHz, mmWave bands)	Within 12 months	BTRC
MORAN/MOCN active sharing guidelines published	Within 9 months	BTRC
Spectrum trading and sharing framework established	Within 18 months	BTRC
First commercial 5G pilot deployment	By 2027	Licensed operators / BTRC
Technology-neutral spectrum rights implemented for all bands	Within 12 months of Licensing Policy guidelines	BTRC

3.6 Consumer Protection and Quality of Service

3.6.1 The QoS Gap

Consumer experience of telecommunications in Bangladesh falls substantially short of both the standards that regulation prescribes and the outcomes that peer countries achieve. Bangladesh ranks 93rd among 154 countries in Ookla's global index of mobile internet quality, behind Vietnam (13th), India (28th), Indonesia (82nd), and Sri Lanka (85th). The median mobile download speed of 28.3 Mbps, while improved from prior years, is less than a third of India's 94.5 Mbps and roughly half of Vietnam's 50.7 Mbps. Fixed broadband performance, at a median of 48.9 Mbps, similarly trails regional comparators.

BTRC has established QoS regulations and conducts periodic drive tests, but the White Paper identifies critical weaknesses in the enforcement framework. Drive tests in 2024 revealed non-compliance with standards: excessive call drops, poor 4G coverage in areas nominally covered, and high latency. Despite these findings, penalties have been rare. Methodologies lack transparency and standardisation; disputes between operators and BTRC regarding test procedures remain unresolved; and public QoS reports are irregularly published and lack detail on test criteria and compliance levels. The new QoS benchmarks proposed in early 2025—requiring ISPs to deliver at least 75 per cent of subscribed download speed and 50 per cent of upload speed—represent an improvement, but their effectiveness depends entirely on the enforcement machinery.

Certain regulatory decisions have, counterintuitively, degraded the consumer experience. In 2021, BTRC directed the removal of cache servers from small and medium-sized ISPs, which increased operational costs and reduced service quality. Although this restriction was lifted for most ISPs in December 2024, the intervening three years imposed significant financial and technical strain on smaller operators and their subscribers. Similarly, restrictions on ISPs sharing physical network resources (equipment and fibre-optic infrastructure) have contributed to higher transmission costs and consumer prices, with the impact most pronounced in rural and semi-urban areas. A recent study found that rural users pay higher rates than urban users while receiving comparatively lower internet speeds—a regressive outcome that deepens the digital divide.

3.6.2 Consumer Rights and Complaint Resolution

There is currently no dedicated consumer protection unit within BTRC and no centralised complaint portal. The existing complaint mechanisms are fragmented: BTRC maintains a call centre (100), individual operators have customer service desks, but escalation paths are unclear and independent oversight is absent. Public hearings reveal that fewer than 2 per cent of complaints receive formal responses, reflecting BTRC's limited grievance-handling capacity.

The absence of a unified national consumer protection code for telecommunications is a significant institutional gap. Global peers provide instructive models: India's TRAI has established an Ombudsman for telecom complaints with binding dispute resolution powers; the United Kingdom's Ofcom operates the Communications and Internet Services Adjudication Scheme (CISAS) and Ombudsman Services: Communications as independent Alternative

Dispute Resolution (ADR) bodies; and the US FCC’s Consumer Complaint Center provides an online portal for filing, tracking, and resolving complaints with mandatory operator response timelines. These models share common features: an online portal for complaint submission and tracking, mandatory response timelines for operators, escalation to an independent adjudication body, and periodic publication of complaint resolution data.

Reform Imperatives: The Licensing Policy 2025’s QoS Dashboard requirement provides the foundation for transparent quality monitoring. BTRC must implement: a national digital complaint management system (a single online and mobile platform for submitting, tracking, and escalating grievances with monthly public reporting); a modernised customer code requiring plain-language tariff communication, explicit consent for value-added services (VAS), standardised billing formats, and transparent fair-usage policies; district-level independent performance scorecards combining drive-test and crowdsourced data on speed, latency, and complaint resolution; and an independent telecom ombudsman or ADR mechanism for disputes that operators’ internal processes cannot resolve within prescribed timelines.

3.6.3 Affordability and Taxation

The total tax burden on mobile services in Bangladesh exceeds 39 per cent—comprising supplementary duty (increased from 15 to 20 per cent in the 2024–25 budget), VAT, and surcharges. This is among the highest effective tax rates on telecommunications in Asia and substantially above the levels that international evidence suggests are consistent with maximising digital inclusion. The GSMA’s annual Mobile Tax Gap analysis estimates that sector-specific taxes above approximately 25–30 per cent of the total cost of mobile ownership begin to reduce adoption among lower-income consumers, with each additional percentage point of taxation producing a disproportionate reduction in mobile internet penetration among the bottom two income quintiles.

The impact is evident in Bangladesh’s subscriber data: the 13.2 million decline in mobile internet users between June 2024 and January 2025 coincides with the supplementary duty increase and the consequent rise in retail prices. The decline is concentrated among pre-paid users, who represent the most price-sensitive segment of the market. Internationally, countries that have reduced telecommunications taxation to stimulate adoption—including India (which reduced its GST rate on mobile services from 18 to 12 per cent in certain categories) and Kenya (which reversed a proposed excise duty increase after evidence of subscriber decline)—have observed demonstrable recovery in subscriber growth and data consumption.

The strategy must advocate for: a phased reduction of the supplementary duty on mobile services toward 15 per cent; alignment of total effective taxation on telecommunications with the GSMA’s recommended ceiling; and demand-side affordability interventions including data and device vouchers for low-income populations, zero-interest handset financing programmes, and targeted subsidies for educational and healthcare connectivity—reforms that fall within the

broader fiscal policy domain but must be articulated as essential preconditions for the digital transformation agenda.

Reform KPIs: Consumer Protection and Quality of Service

Indicator	Timeline / Target	Responsible
QoS Dashboard operational with quarterly publication	Within 12 months	BTRC
National digital complaint management system launched	Within 12 months	BTRC
Modernised customer code gazetted and enforced	Within 9 months	BTRC
Telecom ombudsman or ADR mechanism established	Within 18 months	PTD / BTRC
Complaints resolved within 15 days (%)	Target: 80% by 2027	All operators
Bangladesh Ookla ranking improvement	Target: Top 70 by 2028	Market-wide
New QoS benchmarks (75% download, 50% upload) enforced	Within 6 months	BTRC

3.7 Social Obligation Fund: Governance Failures and Reform

3.7.1 Background and Scale

The Social Obligation Fund (SOF) was established in November 2011 under the Telecommunication Act (Amendment) 2010, mandating telecom operators to contribute 1 per cent of their audited gross annual revenue to finance connectivity in underserved areas. The fund is intended to universalise internet access by supporting broadband infrastructure in remote coastal zones, river island (char) areas, hilly regions, and areas affected by river erosion—locations where low commercial returns inhibit private investment.

Over fourteen years, the SOF accumulated BDT 3,558.98 crore in total contributions and interest. Seven SOF-funded projects were approved between 2018 and 2021, implemented by five different agencies, with a combined estimated cost exceeding BDT 2,000 crore. These projects encompassed school digitisation, rural broadband rollout, mobile internet deployment in haor and island areas, and infrastructure installation in underserved regions.

3.7.2 Governance Failures

The White Paper’s analysis of the SOF identifies three categories of systemic failure that have rendered the fund substantially less effective than its potential:

Non-compliance and revenue shortfalls: The statutory 1 per cent levy was unevenly enforced. While major mobile operators contributed with reasonable consistency, the majority of ISPs, divisional and district-level service providers, and value-added service licensees either remitted irregularly or not at all. Only recently, in 2023, were approximately 3,000 additional licensees (ISPs, IIGs, VAS providers) considered for inclusion—raising questions about the reasons for their prior exclusion. Discretionary waivers further eroded the contribution base: for certain internet services, BTRC selectively waived the 1 per cent SOF alongside a 5.5 per cent revenue-sharing requirement.

Idle funds and delayed deployment: Despite accumulating substantial sums, approximately 35 per cent of total funds raised (approximately BDT 1,500 crore) remained idle due to regulatory and institutional bottlenecks. From its establishment in 2011, the SOF saw no project activity until early 2018—seven years of institutional inertia during which collected public monies were not deployed for their intended purpose. The prolonged non-deployment represents a significant opportunity cost: backbone and last-mile investments that could have been financed when costs were lower were foregone, and subsequent spending was hurriedly approved under political pressure, creating conditions for poor procurement and weak project design.

Governance failures in project implementation: The White Paper documents fund mismanagement, multiple cases of Public Procurement Rules (PPR) violation in contract and tender awards, instances of political interference in project site selection and installation placement, lack of transparency in project selection, and deviations from SOF guideline

principles in terms of impact and sustainability plans. SOF projects were excluded from the Annual Development Programme (ADP), bypassing standard national planning scrutiny. The SOF Management Committee’s governance structure—with the Minister serving simultaneously as Committee Chairperson and the power to withdraw funds on the Minister’s signature alone—created vulnerability to political direction that the White Paper characterises as a “key reason for their vulnerability in terms of conception, implementation and impact framework.”

Sustainability risks were pervasive. The “Connected Bangladesh” project was to be handed to entities that either lacked necessary licences or had unclear operational budgets. BTCL hotspot installations lacked ring-fenced operation and maintenance funding. VSAT terminals, solar systems, and classroom tablets rarely budgeted for spare parts, batteries, or refresh cycles beyond limited warranties. Backbone deployments without guaranteed last-mile service provision risked underutilisation. The White Paper notes that some SOF project approvals appeared “designed to generate demand for TSS or other politically connected vendors rather than being grounded in competitive need assessment.”

3.7.3 International Comparators

SOF-related governance challenges are not unique to Bangladesh. India’s Universal Service Obligation Fund (USOF) has faced allegations of misallocation, incomplete work, and manufactured cost inflation; some projects were abandoned post-funding. Malaysia’s Universal Service Provision (USP) has been criticised for misaligned priorities and procurement bias. However, both countries have developed stronger institutional frameworks: India’s USOF operates under the Department of Telecommunications with published project evaluation criteria, competitive bidding, and independent completion audits; Malaysia restructured its USP through the Malaysian Communications and Multimedia Commission (MCMC) with transparent annual reporting and outcome-based disbursement. Bangladesh’s SOF lacks these institutional safeguards.

3.7.4 Reform Imperatives

The SOF must be restructured to operate on a results-based financing model. Disbursements should be awarded only for verified delivery of measurable outcomes—broadband speeds, latency, active user adoption—confirmed by independent audits. Specific reforms include:

Institutional restructuring: Separate the SOF governance body from the ministerial chain by establishing an independent SOF Board with representation from BTRC, PTD, civil society, industry, and the Planning Commission. Remove the single-signatory withdrawal authority.

Harmonisation with national planning: Integrate SOF project appraisal with the ADP process, requiring Planning Commission review for projects above a defined threshold (recommended: BDT 50 crore).

Competitive and transparent procurement: Mandate open competitive bidding for all SOF projects with published evaluation criteria, and include private operators as eligible implementing entities.

Outcome-based disbursement: Tie funding tranches to independently verified milestones (connectivity delivered, active users registered, speed benchmarks met). Withhold final payment until post-completion audit.

Sustainability requirements: Require every SOF project proposal to include a credible exit plan with identified successor operator, ring-fenced O&M funding for at least five years, and total cost of ownership (TCO) budgeting that includes equipment refresh cycles.

Impact evaluation: Establish a centre of excellence for impact evaluation (as originally proposed in the inaugural SOF meeting of March 2018 but never followed through), independent of BTRC, to conduct third-party assessments of SOF project outcomes.

Reduced NTTN dependency: Avoid dependence on private NTTN entities for future SOF project awards. Where upstream connectivity flows through private NTTN licensees, PTD should finance expansion of government-owned optical fibre as an alternative, or BTRC should revisit NTTN licence conditions to empower government NTTN operators.



Reform KPIs: Social Obligation Fund

Indicator	Timeline / Target	Responsible
Independent SOF Board constituted	Within 6 months	PTD
SOF projects above BDT 50 crore submitted to Planning Commission	Immediate (continuous)	SOF Board / Planning Commission
Automated contribution tracking system operational	Within 12 months	BTRC
Contribution compliance rate from all licensees	Target: 90% by 2027	BTRC

Percentage of SOF funds deployed (vs. idle)	Target: 85% by 2028	SOF Board
Independent completion audit for all active SOF projects	Within 18 months	SOF Board / external auditor
Impact evaluation centre established	Within 12 months	PTD / SOF Board

3.8 State-Owned Telecom Entities: Institutional Challenges and Strategic Direction

3.8.1 BTCL: From Legacy Operator to National Infrastructure Provider

Bangladesh Telecommunications Company Limited (BTCL) was created through the 2008 corporatisation of the legacy fixed-line operations of the Bangladesh Telegraph and Telephone Board (BTTB). BTCL inherited an extensive but ageing copper network, a bureaucratic management culture, high maintenance costs, and a workforce structure designed for a monopoly era. Its revenue contribution to PTD stands at over BDT 300 crore annually, but this figure is misleading: in FY 2023–24, BTCL reported a small profit driven primarily by non-operating income (fixed deposit interest) rather than core telephony and data operations, which continue to decline. The fixed-line subscriber base has contracted sharply as mobile and broadband services have substituted for traditional telephony.

The White Paper identifies BTCL's challenges as structural and financial. Without regulatory incentives or modernisation policies following the 2008 separation, BTCL struggled to transition toward fibre-based broadband and digital services. Policy gaps in performance monitoring and public-sector reform meant that inefficiencies persisted, reducing BTCL's ability to compete with private ISPs and mobile operators. The 2008 separation resulted in institutional fragmentation rather than functional specialisation—the lack of a unified governance model and performance-linked regulatory oversight undermined the goal of creating a dynamic, competitive public telecom ecosystem.

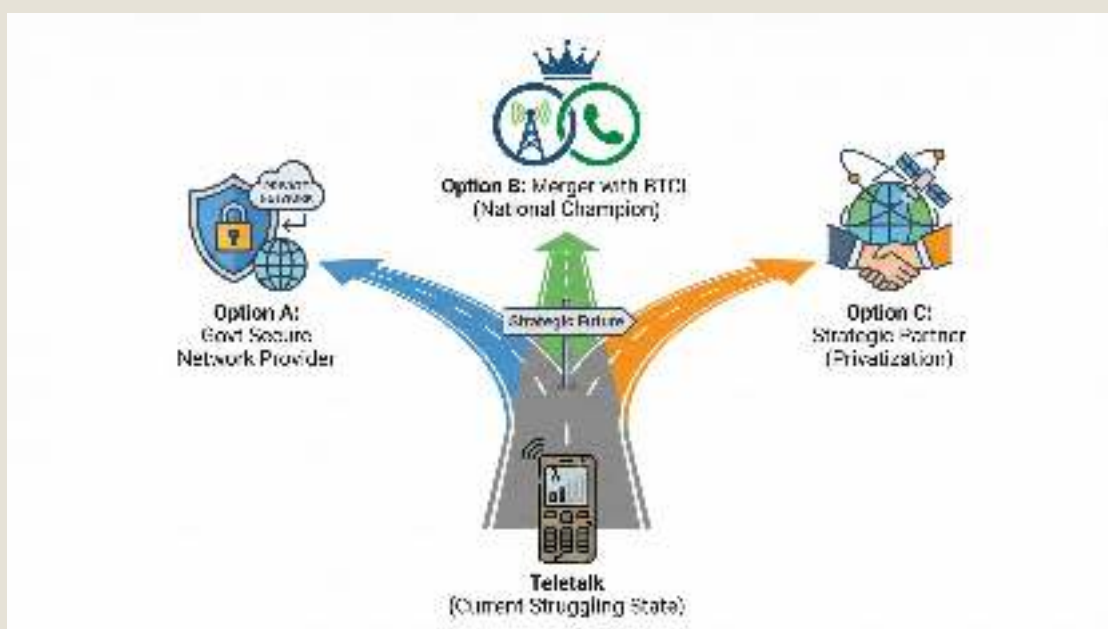
Strategic Repositioning: The White Paper identifies a transformative opportunity for BTCL to redefine its role from a legacy fixed-line utility into a strategic national infrastructure provider. With one of the largest and most valuable infrastructure portfolios in the country—including extensive duct networks, exchange buildings, and right-of-way assets—BTCL is uniquely positioned to lead the national infrastructure-sharing framework mandated by the Licensing Policy 2025. By offering wholesale backhaul and interconnect capacity to both public and private operators, it can become the backbone of Bangladesh's digital ecosystem. Key strategic directions include: leading fibre-to-tower deployment (BTCL's duct network provides the most cost-effective pathway to meet the 80 per cent fibre-to-tower target within three years); becoming the primary wholesale broadband provider to ISPs in underserved areas; and underpinning smart grid, smart city, and digital public goods infrastructure.

An MVNO initiative with Teletalk has been approved, enabling BTCL to offer mobile services using Teletalk's network capacity. This represents an important step toward diversifying BTCL's revenue base beyond declining fixed-line operations. However, the initiative's success depends on BTCL developing commercial capabilities (marketing, customer acquisition, digital service delivery) that its current institutional culture has not prioritised.

3.8.2 Teletalk: The State-Owned Mobile Operator

Teletalk Bangladesh Limited, the only state-owned mobile network operator, has struggled with persistent financial losses throughout most of its operational history. Total losses reached BDT 196.97 crore in FY 2022–23, driven by a declining subscriber base (approximately 6.6 million, representing just 3.5 per cent of the market), deteriorating service quality, and substantial debts owed to regulatory authorities. The White Paper notes that “Teletalk’s losses totaled Tk 196.97 crore in FY23, driven by a declining subscriber base, revenue, and deteriorating service quality, along with substantial debts owed to regulatory authorities. Its revival prospects remain uncertain due to weak financial controls and operational inefficiencies.”

Recent operational initiatives include: negotiating tower sharing agreements with tower companies for 2,300 sites (with agreements for 1,720 sites completed by late 2025 and 15 new sites activated from shared towers); expanding 4G coverage by adding 1,699 new 4G BTS sites (increasing daily data consumption from 354 TB to 464 TB, a 20 per cent increase); upgrading battery capacity at 1,420 sites to improve network availability; and planning further 4G expansion to 3,600 additional sites by June 2027, targeting 85 per cent 4G coverage of existing Teletalk sites.



Strategic Assessment: Teletalk’s viability as an independent mobile network operator must be honestly assessed. With 3.5 per cent market share in a four-operator market, persistent losses, and a capital base insufficient to fund 5G deployment, Teletalk cannot sustain competitive operations on its current trajectory. The strategy must consider three options: first, restructuring Teletalk as a government network services provider (maintaining its spectrum assets and network for government communications, emergency services, and rural connectivity while partnering with private operators through MVNO and infrastructure-sharing arrangements); second, merging Teletalk’s operations with BTCL to create a single state-owned telecommunications entity with sufficient scale to compete in wholesale infrastructure and government services; third,

strategic partnership with a private operator (either through a joint venture or management contract) that brings commercial capabilities while preserving government's strategic interests. Each option involves trade-offs between financial sustainability, employment, sovereign control over spectrum, and the political economy of public-sector reform. What is not sustainable is the status quo of continuous losses funded by public resources.

3.8.3 BSCPLC: The Submarine Cable Company

Bangladesh Submarine Cable Company PLC (BSCPLC) has been discussed in Section 3.4 in the context of international connectivity. From a state-owned entity management perspective, BSCPLC presents a paradox: it is the most consistently profitable PTD entity, yet its profitability has been achieved through pricing practices that the White Paper documents as having constrained national internet adoption. Revenue from BSCPLC stands at approximately BDT 60 crore annually, and it is one of the few PTD entities listed on the Dhaka Stock Exchange (with general public investors holding 7.75 per cent).

With BPCS entry, SMW-6 commissioning, and ITC operators now collectively exceeding BSCPLC's bandwidth, the company's business model requires fundamental restructuring. The strategic direction must shift from high-margin, low-volume bandwidth sales to competitive pricing that maximises capacity utilisation, complemented by value-added services (managed wavelength, co-location, data centre interconnection at cable landing stations). BSCPLC must also lead the diversification of cable landing infrastructure beyond Cox's Bazar to ensure national resilience.

3.8.4 Department of Telecommunications (DoT)

The Department of Telecommunications (DoT) faces several structural and operational challenges. The White Paper identifies its authority as narrow and its coordination with both BTRC and the Ministry as weak, leading to duplication of responsibilities and slow, unpredictable decision-making. Many policy and oversight functions originally assigned to DoT have drifted toward BTRC, while BTRC has expanded beyond its traditional compliance role into policy agenda-setting—a role inversion that has left DoT without a clearly defined operational mandate.

The White Paper recommends reintroducing a dedicated Telecom Cadre to institutionalise technical specialisation within DoT. This would create a permanent pool of technical expertise capable of formulating evidence-based telecommunications policy, monitoring sector performance, and providing the Ministry with informed advice on spectrum, infrastructure, and technology evolution—functions that require domain expertise distinct from the generalist administrative cadre that currently staffs the department. Revenue collection functions should be transferred from BTRC to appropriate government financial authorities, clarifying the boundary between regulatory and fiscal functions.

3.8.5 TSS, BSCL, and Other PTD Entities

Telephone Shilpa Sangstha (TSS), the state-owned telecom equipment manufacturer, and Bangladesh Satellite Company Limited (BSCL) both face existential questions about their relevance and sustainability. TSS was identified in the White Paper’s analysis of SOF projects as a vendor whose demand was, in certain instances, artificially generated through project approvals. BSCL’s role has become more significant with the Starlink partnership and the NTNSP licensing framework, but its operational capacity to manage satellite resale operations must be strengthened substantially. Bangladesh Cable Shilpa Limited (BCSL), the state-owned cable manufacturer, occupies a niche in the domestic cable supply chain but faces competition from imports.

The common thread across these entities is the absence of performance-linked governance. The comprehensive reform of state-owned telecom entities—bringing BTCL, Teletalk, BSCPLC, TSS, BSCL, BCSL, and DoT under a coordinated portfolio with shared performance targets (network coverage, service reliability, customer satisfaction, financial sustainability)—would create synergy among public operators and instill a culture of accountability. Formal service contracts for public mandates (rural connectivity, emergency communications, disaster-ready networks), backed by clear funding frameworks, measurable service levels, and independent audits, would redefine each entity’s role from an unfunded obligation model to a transparent, performance-based public service delivery model.

Summary: State-Owned Telecom Entities – Status and Strategic Direction

Entity	Current Status	Key Challenge	Strategic Direction
BTCL	~BDT 300 cr revenue; small profit from non-operating income; declining fixed-line base	Legacy copper infrastructure; bureaucratic culture; lack of commercial orientation	Reposition as national wholesale infrastructure provider; lead fibre-to-tower; MVNO with Teletalk
Teletalk	~6.6M subscribers (3.5%); persistent losses (BDT 197 cr FY23); declining base	Insufficient scale; weak financials; cannot self-fund 5G; regulatory debts	Three-option assessment: government network provider, merge with BTCL, or strategic partnership
BSCPLC	~BDT 60 cr revenue; consistently profitable; 39% capacity utilisation	Artificially high bandwidth pricing; declining market share vs. ITCs and BPCS	Price reduction to maximise utilisation; diversify into value-added services; diversify cable landing points

DoT	Narrow mandate; weak coordination with BTRC; drifted functions	No defined operational role; staffed by generalist cadre; policy vacuum	Reintroduce Telecom Cadre; define policy formulation mandate; transfer revenue collection from BTRC
TSS / BSCL / BCSL	Marginal revenue; sustainability questions; TSS linked to SOF procurement concerns	Relevance; operational capacity; competitive imports; vendor capture risk	Performance audit; role redefinition; BSCL to lead satellite resale partnership; BCSL to supply fibre rollout

3.9 Chapter Summary: The Regulatory Capture Diagnosis and the Reform Response

The analysis presented across both parts of this chapter establishes a central finding: Bangladesh's telecommunications infrastructure—despite achieving impressive aggregate scale—has been shaped by regulatory capture that has produced outcomes misaligned with the public interest. The erosion of BTRC's independence through the 2010 amendments, the issuance of licences without demand analysis, the tolerance of ownership structures that breached regulatory guidelines, the maintenance of artificially high bandwidth pricing, the mismanagement of the Social Obligation Fund, and the failure to enforce consumer protection standards collectively constitute a pattern in which the structures of oversight served concentrated interests rather than competitive markets and consumer welfare.

The reform programme initiated in 2025—comprising the Telecommunication (Amendment) Ordinance, the Licensing Policy, and the institutional measures recommended by the White Paper—addresses the structural roots of these failures. However, legislative and policy reform creates possibility, not certainty. The transformation of possibility into outcome depends on implementation: the capacity, will, and accountability of the institutions responsible for executing the reform agenda.

3.9.1 Five Structural Shifts

The reforms documented in this chapter, if implemented, will produce five structural shifts in Bangladesh's telecommunications landscape:

From regulatory capture to independent oversight. The restoration of BTRC's independence, the establishment of a search-cum-selection committee for senior appointments, and the HRIS-enforced merit-based HR processes create the institutional preconditions for regulatory decisions that serve the public interest rather than political or commercial interests.

From fragmented licensing to simplified, technology-neutral authorisation. The consolidation of 20+ licence categories into four (ANSP, NICSP, ICSP, NTNSP) reduces regulatory complexity, lowers barriers to entry, and eliminates the multi-layered value chain that inflated costs and created opportunities for cartel behaviour.

From concentrated infrastructure control to mandated sharing and open access. The infrastructure sharing mandates, SMP framework, and mandatory non-discrimination obligations address the tower duopoly, the NTTN duopoly, and the vertical integration that have constrained competition. Implementation requires BTRC to develop and enforce open-access pricing, conduct transparent SMP reviews, and mandate accounting separation with independent audits.

From high-price, low-utilisation bandwidth to competitive international connectivity. BSCPLC's pricing reform, the entry of BPCS, SMW-6 commissioning, and Starlink's satellite

broadband collectively diversify Bangladesh’s international connectivity options and create competitive pressure that should drive bandwidth prices toward cost-reflective levels.

From opaque, politically directed universal access spending to results-based digital inclusion financing. The SOF’s restructuring—independent governance, ADP integration, outcome-based disbursement, competitive procurement, and independent impact evaluation—transforms a fund that has been characterised by idle balances, governance failures, and sustainability risks into an instrument of verifiable digital inclusion.

3.9.2 Consolidated Reform Timeline

Timeframe	Priority Actions
0–6 Months	BTRC accountability committee constituted; search-cum-selection committee for senior appointments established; SOF Board constituted; conflict-of-interest declarations filed; licensing migration roadmap published; new QoS benchmarks enforcement begins; customer code gazetted
6–12 Months	HRIS deployed at BTRC; QoS Dashboard operational; national digital complaint management system launched; BTRC guidelines for all 4 licence categories published; MORAN/MOCN sharing rules published; SOF contribution tracking system operational; 5G spectrum roadmap published; 700 MHz auction completed; BSCPLC pricing reform initiated; impact evaluation centre established
12–18 Months	Infrastructure sharing guidelines published; SMP framework operational with first market review; first independent HR audit of BTRC completed; telecom ombudsman established; spectrum trading framework established; independent SOF project completion audits
18–36 Months	50% fibre-to-tower target achieved; legacy licences migrated to new framework; BSCPLC capacity utilisation above 60%; first 5G commercial pilot; Teletalk strategic options assessment completed and implemented; cable landing diversification study initiated; SOF fund deployment exceeds 85%
By 2030	80% fibre-to-tower achieved; 5G commercially available in metropolitan areas; internet penetration exceeds 60%; Bangladesh in top 70 Ookla mobile quality ranking; SOF-financed connectivity in all 2,600 underserved unions; state-owned entity portfolio operating under performance contracts with independent audit

3.9.3 Interdependencies with Other Strategy Chapters

The telecommunications infrastructure and regulatory reforms analysed in this chapter are foundational to—but not sufficient for—the broader digital transformation. Chapter 4 (Sectoral Impacts Outside PTD) depends on the connectivity outcomes produced here: citizen-centric digital service delivery requires reliable broadband reach; digital commerce requires trusted, affordable connectivity; and educational and health applications require quality-of-service standards that are enforced rather than aspirational. Chapter 5 (Strategic Policies for Sovereignty and Innovation) builds on the spectrum management and satellite connectivity frameworks

established here. Chapter 6 (Institutional Reform) addresses the organisational restructuring of PTD entities whose current conditions are documented in Section 3.8. Chapter 7 (Project Reform) directly addresses the SOF project governance failures analysed in Section 3.7. And the implementation roadmap in Chapter 9 integrates the timelines established in Section 3.9.2 with the reform sequences of all other chapters.

The central argument of this chapter is that connectivity outcomes are shaped not by technology alone, but by the quality of governance. When licensing frameworks reflect network realities, spectrum flows to its highest-value use, essential facilities are accessible on fair terms, consumers enjoy enforceable rights, and public funds drive measurable inclusion—and under these conditions, even modest investment produces substantial connectivity gains. Conversely, when regulatory processes are fragmented, captured, or slow, even the most advanced technologies underperform, and costs cascade across the entire value chain. The reforms articulated here are designed to shift Bangladesh’s telecommunications sector decisively from the latter condition to the former.

— End of Chapter 3—

4. Sectoral Impacts Beyond Posts and Telecommunications Division

4.1 Introduction: PTD as a Cross-Sectoral Enabler

The preceding chapters have examined the legislative framework and infrastructure architecture of the Posts and Telecommunications Division. This chapter shifts focus from supply-side reforms to demand-side outcomes: how the infrastructure, institutions, and regulatory frameworks under PTD translate into tangible improvements for citizens, enterprises, and sectors that fall outside the Division's direct mandate. The central proposition is that postal and telecommunications assets are not ends in themselves but enabling platforms whose value is realised through the services they make possible in health, education, agriculture, financial inclusion, e-commerce, emergency response, and citizen-centric governance.

Bangladesh's 188 million mobile subscriptions, 133 million internet connections, and 9,886 post offices constitute the country's most extensive physical and digital service delivery network. No other government institution matches this combined geographic reach. Yet the sectoral potential of this network remains substantially unrealised. The Post e-Centre project, which invested BDT 541 crore to convert 8,500 rural post offices into digital centres, stands as a cautionary example: a Comptroller and Auditor General (CAG) audit found that the project had comprehensively failed to achieve its objectives, with government funds effectively wasted. Mobile Financial Services, while transformative in their reach, have developed largely outside the institutional control of the Postal Directorate, as demonstrated by the Nagad experience documented in the PTD White Paper. Telemedicine platforms depend on reliable broadband that the telecom infrastructure does not yet consistently deliver to rural health facilities.

The National Digital Transformation Strategy 2025 to 2030, adopted by the ICT Division, establishes a comprehensive citizen-centric service design framework anchored in nine strategic actions: a national definition of digital public services, annual digital maturity assessments, common design and portal standards, the once-only principle for data reuse, procurement reform for outcomes, digital-first legislative reform, a digital service workforce framework, reusable BanglaGovStack building blocks, and public digital literacy programmes. PTD's contribution to this whole-of-government agenda is not to replicate these strategic actions but to ensure that its infrastructure and institutions provide the reliable, affordable, and inclusive connectivity without which the broader digital transformation cannot reach the populations that need it most.

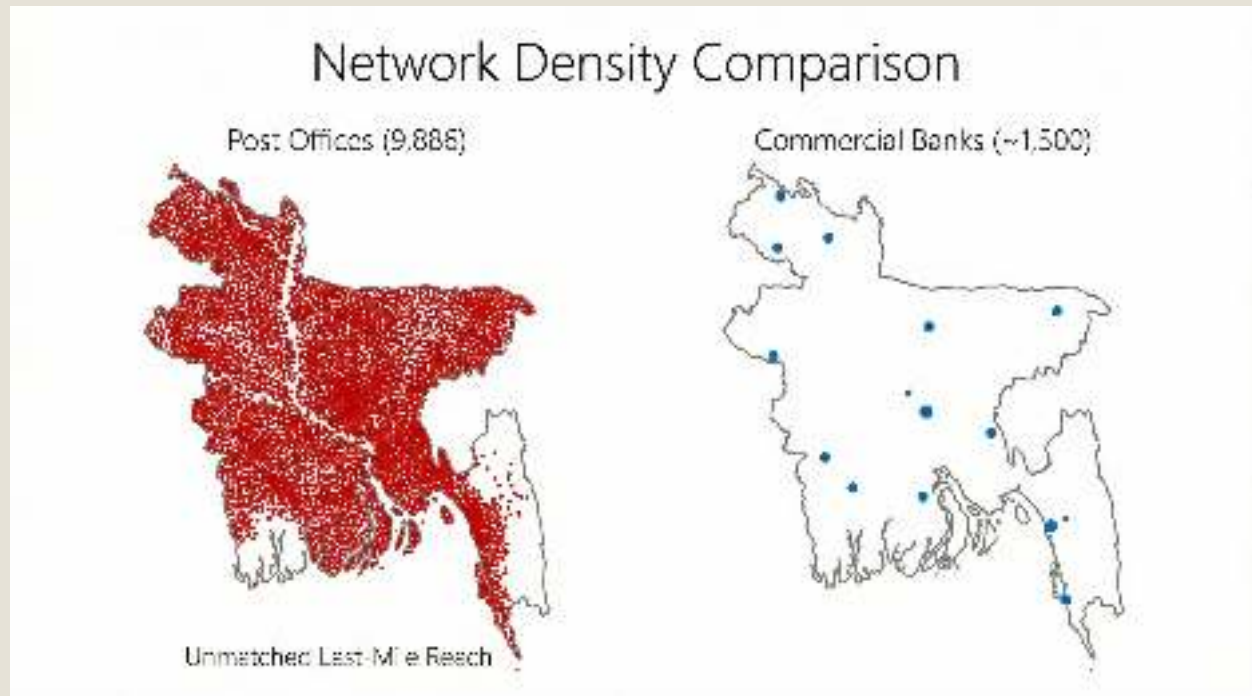
This chapter therefore examines five domains where PTD's reform agenda intersects with sectoral outcomes: the transformation of the postal network into a citizen service delivery platform; financial inclusion through postal and mobile channels; e-commerce and logistics modernisation; the enabling role of telecom infrastructure across health, education, agriculture, and social protection; and emergency communications and disaster resilience. For each domain, the analysis identifies the current state, the specific reforms required from PTD entities, the

interdependencies with other government agencies, and the measurable outcomes that should be achieved by 2030.

4.2 The Postal Network as a Citizen Service Delivery Platform

4.2.1 The 9,886-Office Opportunity

The Bangladesh Post Office operates 9,886 offices across the country, supported by approximately 40,000 personnel, making it one of the largest public service networks in the country. This physical infrastructure reaches communities that no other government agency consistently serves, particularly in remote char lands, haor areas, and hill districts where Union Parishad offices may be the only other institutional presence. The Universal Postal Union's Integrated Index for Postal Development (2IPD) assessed Bangladesh at a composite score of approximately 54 in early 2025, ranking the country 68th among 174 nations, with a reliability sub-score of 72 in the 2024 assessment. While these scores represent improvement from a reliability score of 16.3 in 2021 and an overall IPD of 13.9, they continue to lag behind higher-performing operators and confirm that substantial operational gaps persist.



The operational reality is that the Bangladesh Post Office functions primarily as a traditional mail carrier, even as traditional mail volumes have declined precipitously worldwide. The general delivery standard of two to three days for domestic mail is frequently not met due to inadequate sorting infrastructure, a transport fleet without modern route optimisation, and a workforce whose skills have not kept pace with the shift from paper mail to parcels and digital services. The 14 Automated Mail Processing Centres established across districts at a cost of approximately BDT 365 crore have remained substantially dormant, with sophisticated sorting equipment accumulating dust due to inadequate training, maintenance failures, and integration gaps. Meanwhile, private courier and logistics firms, including Sundarban, eCourier, Paperfly,

Pathao, and RedX, have captured the e-commerce logistics market, offering real-time tracking, same-day delivery in urban areas, and digital payment integration that the postal network cannot match.

Internationally, postal operators that have successfully transformed their operations share a common trajectory: India Post's IT 2.0 programme, backed by an investment of INR 5,800 crore (approximately USD 700 million), deployed the Advanced Postal Technology platform across 1.70 lakh post offices by August 2025, introducing QR-code payments, OTP-based delivery authentication, real-time tracking, and integration with the Open Network for Digital Commerce. India Post now processes 3.2 million bookings and 3.7 million deliveries daily through a microservices-based architecture hosted on government cloud. It delivered 43 lakh passport applications, processed 6.94 lakh export shipments worth INR 148.19 crore through Dak Ghar Niryat Kendras, and opened 3.5 crore India Post Payments Bank accounts, 99 per cent at the doorstep and 59 per cent for women. Singapore's SingPost and South Korea's Korea Post have similarly repositioned themselves as logistics-technology platforms integrated with national e-commerce ecosystems.

The lesson for Bangladesh is clear: without a comparable technology investment and institutional transformation, the postal network will continue to lose relevance. The Postal Services Ordinance 2026, approved by the Council of Advisers on 5 February 2026, provides the legislative foundation by replacing the 1898 Post Office Act, introducing digital postage stamps and e-stamping, establishing e-commerce compliance requirements, mandating a modern address management system, and increasing the maximum administrative fine for unlicensed postal, courier, or parcel operations from BDT 50,000 to BDT 10 lakh. What remains is execution: converting legal authority into operational capability.

4.2.2 Post Offices as Nagorik Sheba Hubs

The National Digital Transformation Strategy identifies Nagorik Sheba Bangladesh as its flagship last-mile service delivery initiative: a nationwide programme to transform unused government premises, including post offices and union facilities, into community-based digital service centres operated by trained local entrepreneurs delivering over 500 government services through a human-assisted digital service model. The programme was inaugurated on 26 May 2025 by the Chief Advisor, with three pilot centres in Gulshan, Uttara, and Nilkhet, 60 trained entrepreneurs (42 per cent women), and a target to scale to 8,000 centres serving 80 million active users by 2030.

Post offices are natural host sites for Nagorik Sheba centres. They provide physical premises in locations where citizens already visit for postal, savings, and insurance services. Their nationwide distribution ensures geographic coverage that no purpose-built network could replicate at comparable cost. The Nagorik Sheba model operates as a zero-cost interoperability hub: centres are managed by licensed entrepreneurs under a sustainable social business

framework, with private financing replacing direct government subsidies. For the Postal Directorate, hosting Nagorik Sheba centres converts underutilised postal premises into multi-service hubs that generate footfall, introduce new revenue streams, and create local employment without burdening the postal budget.



The integration pathway requires specific actions from PTD. First, the Postal Directorate must conduct a nationwide premises audit to identify post offices with adequate space, power supply, and internet connectivity to host Nagorik Sheba operations. The BD Post progress report indicates that six Nagorik Sheba centres have been established in postal premises at Gulshan, Uttara, and four additional locations, with a target of transforming 1,000 post offices into such centres. Second, BTCL broadband connectivity must be provisioned to all designated postal Nagorik Sheba sites, ensuring minimum speeds of 10 Mbps for service delivery operations. Third, the Postal Directorate must establish formal service-level agreements with the Nagorik Sheba programme authority to define premises standards, revenue sharing, operational boundaries, and dispute resolution mechanisms.

4.2.3 Lessons from the Post e-Centre Project

Any strategy for transforming the postal network into a digital service platform must reckon honestly with the failure of the Post e-Centre for Rural Community project, which invested BDT 541 crore between 2012 and 2017 to equip 8,500 rural post offices with computers, internet connections, and point-of-sale machines for delivering e-governance, e-commerce, e-education, and financial services. The CAG audit, conducted through visits to 900 centres over five and a

half months, found that the project had comprehensively failed. Services listed in the Development Project Proposal, including internet access, e-governance, e-commerce, e-agriculture, and accelerated remittance disbursement, were included, in the CAG's words, only to secure project approval. Government funds had been wasted.

The Implementation Monitoring and Evaluation Division (IMED) corroborated these findings: IT services were not fully activated in many offices; essential equipment such as photocopiers, POS machines, webcams, and computers was missing or non-functional; operators lacked ICT training; and electricity problems in rural areas went unaddressed. The infrastructure underutilisation was staggering: of 17,000 POS machines procured, only 704 were in use; of 10,000 finger vein machines, only 643 were operational; of 500 servers purchased, only 33 served their intended purpose while 467 were misapplied to unrelated functions. Equipment purchased under the Mail Processing and Logistics Services project was non-functional. The Postal Headquarters building itself remained unused for two years after completion.

The post-mortem reveals systemic failures that any successor initiative must explicitly address. The project lacked a viable operating model: equipment was procured without corresponding revenue streams, maintenance budgets, or trained operators. There was no demand-side assessment to determine which digital services rural populations actually needed and would use. No exit strategy ensured sustainability beyond the project period. Procurement was supply-driven, prioritising hardware acquisition over service design. And oversight was absent: the Project Completion Report was submitted a year late, and implementing agency responses to IMED observations were never reported.

Structural Safeguards for Future Initiatives: Any postal digitalisation programme must incorporate the following non-negotiable design principles. First, demand-validated service portfolios: before any technology procurement, the Postal Directorate must survey actual citizen demand at the upazila level and design service offerings accordingly. Second, sustainable operating models: each digital post office must have a business plan demonstrating revenue sufficiency for at least five years of operations, maintenance, and equipment refresh. Third, outcome-based procurement: technology contracts must be structured around service delivery milestones, not hardware delivery, with payments linked to verified operational status. Fourth, independent monitoring: quarterly audits by IMED or an independent evaluator, with findings published and mandatory management responses. Fifth, mandatory connectivity prerequisites: no equipment should be installed until broadband connectivity of at least 5 Mbps is confirmed and operational.

4.3 Financial Inclusion through the Postal and Telecom Network

4.3.1 Postal Financial Services

The Bangladesh Post Office has historically been a significant channel for financial inclusion in underserved communities. The Post Office Savings Bank and Postal Life Insurance provide savings and insurance products to populations that have limited access to commercial banking. The Electronic Money Transfer Service (EMTS), launched on 26 March 2010 and expanded to over 2,750 post offices, enables inter-office fund transfers using mobile phones, providing a basic remittance channel for rural households. These legacy financial services remain important: in a country where only 53 per cent of adults hold a bank account (Global Findex 2021), the post office network's physical presence in every upazila offers a distribution advantage that no private financial institution can replicate.

However, the existing postal financial infrastructure operates on manual or semi-digital processes. Savings bank transactions are recorded in paper passbooks with limited electronic reconciliation. Insurance claim processing is paper-intensive with extended settlement periods. The EMTS, while functional, lacks the speed, convenience, and feature set of modern Mobile Financial Service platforms. The contrast with India Post Payments Bank is instructive: India's postal banking arm, backed by its IT 2.0 digital platform and a Core Banking Solution linking 25,022 departmental post offices, opened 3.5 crore accounts with biometric authentication and doorstep service, of which 59 per cent were opened for women. Bangladesh's postal financial services have no comparable digital backbone.

Reform Direction: The Postal Services Ordinance 2026 explicitly incorporates provisions for postal life insurance, savings banks, and expanded financial services. The Postal Directorate should digitise all savings bank transactions through a core banking platform integrated with the national payment switch (Bangladesh Electronic Fund Transfer Network). Postal Life Insurance should be modernised with digital policy issuance, premium collection through MFS channels, and electronic claims processing. The target should be to bring all postal financial transactions onto a digital platform by 2028, with interoperability with the national financial system ensured through Bangladesh Bank coordination.

4.3.2 The Nagad Experience: Governance Lessons

The most consequential financial service to emerge from the postal ecosystem is Nagad, the digital financial service launched by the Bangladesh Post Office on 26 March 2019 under the authority of the Post Office (Amendment) Act 2010, which permits the Postal Directorate to appoint agents for financial service delivery and to partner with institutions for remittance and banking services. Nagad recorded its highest monthly transaction value of BDT 34,000 crore in March 2025, surpassing private competitors in certain categories. Its Digital Know Your Customer (DKYC) system, which combines Bangla OCR, automated identity verification, and localised data to reduce customer onboarding time to one-tenth of market practice, represents a genuine technical innovation.

Yet the PTD White Paper's Task Force findings reveal profound governance failures surrounding Nagad that must be documented for institutional learning. The Memorandum of Understanding signed between the Postal Directorate and the operating company on 26 April 2017 was for a pilot programme to test agency operations of the Postal Cash Card on a profit-sharing basis, raising the question of whether the Postal Directorate was legally permitted to launch a full-scale financial service through an MoU rather than a formal licensing process. The Postal Directorate's institutional control over Nagad's operations progressively eroded: the operating company changed its name to Nagad Limited based on a no-objection letter whose authenticity the Task Force found questionable, as the referenced source documents were missing from official files. Bangladesh Bank's role was ambiguous, with the institution applying Mobile Financial Services Regulations 2018 to what the Postal Directorate consistently termed a Digital Financial Service, without amending the underlying legislation. Excess e-money creation raised serious accountability concerns.

The governance lessons are threefold. First, public digital financial services operated through the postal network must be subject to dual regulatory oversight by both Bangladesh Bank and the PTD, with clear delineation of prudential supervision (Bangladesh Bank) and institutional governance (PTD). Second, any agency arrangement for financial service delivery must be established through formal contracts approved by the relevant ministry, not through MoUs or pilot agreements that lack the transparency and accountability mechanisms of standard public procurement. Third, the Postal Directorate must retain demonstrable operational control over financial services bearing its institutional name, including control over customer data, transaction records, brand usage, and agent network management. The new Postal Services Ordinance provides the legal framework; the challenge is ensuring that implementing regulations enforce these governance standards.

4.3.3 Mobile Financial Services and Telecom's Enabling Role

Mobile Financial Services have become the primary channel for digital financial inclusion in Bangladesh, with active accounts exceeding 60 million and monthly transaction volumes surpassing BDT 1 lakh crore. This ecosystem operates entirely on the telecommunications infrastructure regulated by BTRC and provided through mobile network operators. The MFS sector's continued growth depends on three telecom-related conditions: universal mobile network coverage, including in rural areas where financial exclusion is most acute; affordable mobile data pricing that does not deter low-income users from transacting digitally; and network reliability that ensures transactions complete without failure, a critical trust factor for populations transitioning from cash.

PTD's contribution to financial inclusion through MFS is therefore primarily infrastructural. The reforms outlined in Chapter 3, particularly the reduction of the telecommunications tax burden from 39 per cent, the expansion of 4G coverage through the Social Obligation Fund, the

introduction of Teletalk's strengthened rural network (targeting 85 per cent 4G coverage by June 2027 through 3,600 additional BTS sites), and BTCL's broadband reach, all directly affect MFS penetration. Additionally, BTRC's regulatory authority over USSD access pricing and short code allocation directly influences MFS operational costs and accessibility. BTRC should ensure that USSD access pricing for financial services remains regulated at levels that do not create barriers for low-income users, and that mobile operators provide non-discriminatory network access to all licensed MFS providers.

4.4 E-Commerce and Logistics Modernisation

4.4.1 Bangladesh's E-Commerce Landscape

Bangladesh's e-commerce sector has emerged as a significant economic force, with market revenue projected at approximately USD 7 billion in 2025, growing at a compound annual growth rate of 8 to 13 per cent through 2029 when the market is expected to reach USD 9.6 to 12.4 billion. Despite this growth, online retail accounts for only 3 to 5 per cent of total retail transactions, indicating that the market remains substantially under-penetrated compared to regional peers. The sector is led by Daraz (owned by Alibaba Group), with niche platforms including Chaldal (grocery), Pickaboo (electronics), and a growing ecosystem of social commerce operators leveraging Facebook and Instagram. Private logistics providers, including Pathao, RedX, Paperfly, eCourier, and Sundarban Courier, dominate last-mile delivery, with the Courier, Express, and Parcel segment projected to grow at a CAGR of 3.6 per cent through 2030.

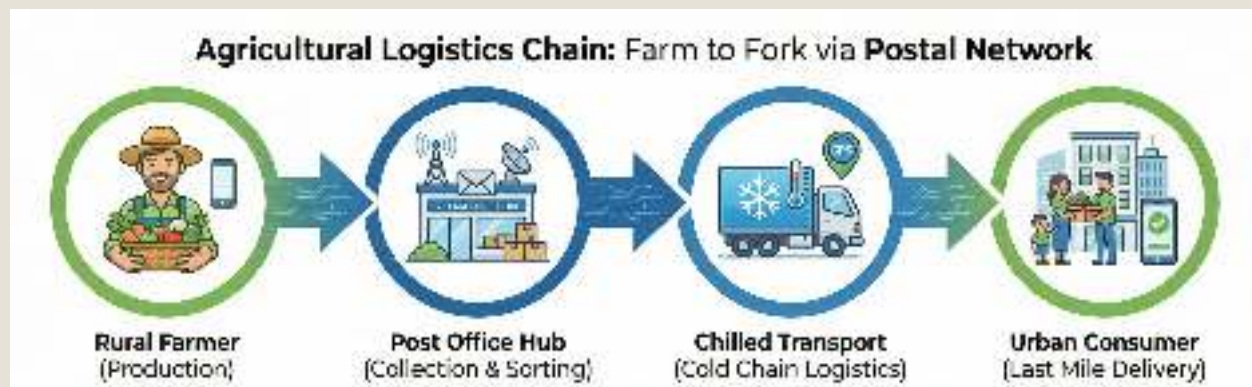
The e-commerce sector's growth faces two structural constraints that fall within PTD's sphere of influence. First, the absence of a standardised national address system renders last-mile delivery inefficient and expensive, particularly outside major metropolitan areas. Delivery riders rely on phone calls to locate recipients, adding time and cost to each delivery. Over 90 per cent of e-commerce transactions use cash-on-delivery, reflecting both consumer distrust of digital payments and the logistical complexity of payment collection at unstandardised addresses. Second, the logistics infrastructure outside Dhaka and Chittagong is fragmented and underdeveloped, with limited cold chain capacity, inconsistent delivery timelines, and no integrated parcel tracking standards.

4.4.2 Postal Logistics Transformation

The Bangladesh Post Office has the physical infrastructure to participate meaningfully in the e-commerce logistics market, but it must fundamentally redefine its operating model from a mail carrier to a logistics platform. The BD Post progress report documents several promising but early-stage initiatives: a Central Logistics Tracking Platform (CLTP) is being developed; 85 mail vans have been equipped with GPS trackers for route transparency; 14 chiller chambers and 84 deep freezers have been installed for perishable goods; a reverse logistics middleware system for e-commerce returns is in the planning stage; and the domestic mail monitoring software (DMS) is being integrated through APIs with e-commerce and courier platforms.

Perhaps the most compelling proof of concept is the postal department's mango delivery service, which connects farmers in Rajshahi and Chapainawabganj directly with consumers in Dhaka, cutting out intermediaries and securing fairer prices. In its most recent season, the service transported 65 tonnes of mango through the postal network. This agricultural market linkage model demonstrates the postal network's potential as a rural logistics backbone connecting

producers to urban and digital markets. If systematised and expanded to other seasonal products, it could establish the Post Office as a key node in the agricultural supply chain.



Strategic Direction: The postal logistics transformation should prioritise three capabilities. First, parcel and e-commerce fulfilment: the postal network must offer real-time tracking, digital proof of delivery, OTP-based authentication for high-value parcels, and integration with major e-commerce platforms through open APIs, replicating the capabilities that India Post's IT 2.0 platform has demonstrated. Second, agricultural logistics: the mango delivery model should be expanded into a year-round agricultural market linkage service connecting rural producers with urban and export markets, utilising cold chain infrastructure and GPS-tracked transport. Third, reverse logistics: the planned middleware system for processing e-commerce returns should be developed to enable the postal network to compete for the growing reverse logistics market, which private operators find commercially challenging in semi-urban and rural areas.

Data-Driven Management: Implement a centralized platform for customer complaint management and data-analytics-based performance analysis.

Digital Tracking & Security: Deploy a National Address Management System (NAMS) and domestic mail tracking integrated with a secure source code repository to ensure data sovereignty.

Logistics Team Reform: Form specialized teams for effective mailing and courier service reforms to ensure KPI-based monitoring and the prevention of illegal/prohibited item transport.

4.4.3 National Address Management System

The National Address Management System (NAMS) represents one of the most consequential infrastructure initiatives for e-commerce, logistics, and citizen service delivery. The BD Post progress report indicates that the Project Evaluation Committee has provided conditional approval for NAMS, which will divide Bangladesh into structured micro-regions with unique identifiers, introduce GPS-based geofencing for accurate deliveries, and align with international postal standards to facilitate cross-border trade.

The absence of standardised addresses imposes costs across the economy. E-commerce firms report that 10 to 15 per cent of delivery attempts fail on the first try due to address ambiguity, adding approximately BDT 50 to 100 per delivery in re-attempt costs. Emergency services cannot geocode incident locations for efficient dispatch. Social protection payments cannot be reliably delivered to verified addresses. Land records, utility connections, and tax administration all suffer from the lack of a universal address register. Countries that have invested in postal code and address standardisation, including India (PIN code system with six-digit codes covering all 1.4 billion citizens), the United Kingdom (Royal Mail's Postcode Address File covering 30 million addresses), and Rwanda (which implemented a comprehensive addressing system in partnership with the UPU in under three years), have demonstrated measurable improvements in logistics efficiency, tax collection, and emergency response times.

Implementation Pathway: NAMS should be implemented in three phases. Phase 1 (2026 to 2027): establish the addressing framework, including the coding structure, database architecture, and institutional governance, in partnership with the Survey of Bangladesh and the Local Government Division, covering all 64 districts and 495 upazilas with unique geo-referenced codes. Phase 2 (2027 to 2028): integrate NAMS with the National Portal, e-commerce platforms, and logistics providers through standardised APIs, and digitise all post office delivery zones using the new addressing system. Phase 3 (2028 to 2030): achieve 70 per cent GPS-geotag address coverage nationwide, as targeted by the National Digital Transformation Strategy, and establish cross-border address interoperability aligned with UPU S42 standards for international e-commerce fulfilment.

4.4.4 MOCSLA and Courier Sector Regulation

The Mailing Operator and Courier Services Licensing Authority (MOCSLA), established in 2014, is responsible for licensing, monitoring, and enforcing compliance among private courier and mailing operators. It maintains a public registry of licensed operators and oversees reforms to modernise postal logistics. The Postal Services Ordinance 2026 significantly strengthens MOCSLA's enforcement capacity by increasing penalties for unlicensed operations and establishing a modern regulatory framework for the courier and parcel sector.

The courier sector in Bangladesh is characterised by a small number of established operators alongside a large informal segment of unregistered delivery services, particularly those serving social commerce transactions on Facebook and Instagram. MOCSLA's regulatory challenge is to bring this informal segment into a licensed framework that ensures consumer protection, including parcel tracking, insurance for high-value items, and complaint resolution, without imposing compliance costs that stifle the sector's growth. The regulatory model should be risk-proportionate: lighter registration requirements for micro-operators handling fewer than 100 parcels per day, with progressive compliance obligations as operators scale.

Table 4.1: E-Commerce and Postal Logistics KPI Framework

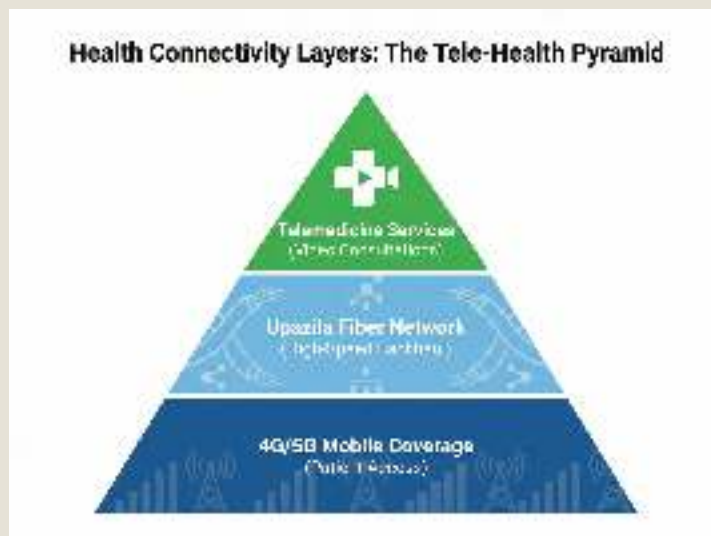
KPI	2025	2026	2027	2028	2030
Post offices with real-time parcel tracking	50 (pilot)	500	2,000	5,000	8,000+
NAMS coverage (districts with geocoded addresses)	Framework design	16 districts	32 districts	64 districts	100% coverage
Postal parcel volume (daily average)	Baseline TBD	+20%	+50%	+100%	+200%
MOCSLA licensed operators	Current registry	+30%	+60%	+80%	95% formalised
Agricultural products delivered via postal network (tonnes/year)	65	200	500	1,500	5,000
E-commerce platforms integrated with postal API	0	3	8	15	All major

4.5 Telecom Infrastructure as Enabler of Sectoral Transformation

The telecommunications network regulated by BTRC and operated by both private and state-owned providers serves as the enabling layer for digital transformation across every sector of the economy. The National Digital Transformation Strategy sets ambitious targets: 50 to 75 per cent of health facilities digitised by 2027 to 2028, 40 to 60 per cent of schools with digital classrooms, 5 to 10 million farmers using agricultural applications, and 50 to 75 million active e-service users. None of these targets can be met without reliable, affordable, and geographically comprehensive telecommunications. This section examines four sectors where the connection between PTD's reforms and citizen outcomes is most direct.

4.5.1 Health: Telemedicine and Health Information Systems

Telemedicine has emerged as a critical healthcare delivery channel in Bangladesh, with over 1 million annual teleconsultations delivered through platforms supported by the Ministry of Health and Family Welfare. The National Digital Transformation Strategy targets 8 to 15 million teleconsultations annually by 2027 to 2028, rising to 25 to 40 million by 2030, alongside 25 to 35 million citizens with unique Health IDs by 2028 and 50 million by 2030. These targets require broadband connectivity at every community health centre, upazila health complex, and district hospital, with minimum bandwidth sufficient to support video consultations, medical image transmission, and electronic health record synchronisation.



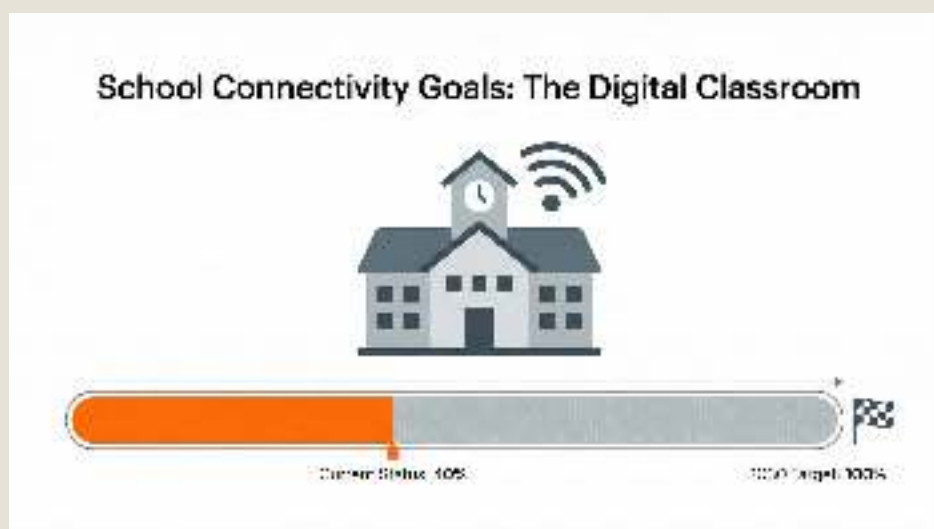
The infrastructure gap is significant. Many rural health facilities rely on mobile data connections with inconsistent speeds, making video-based teleconsultation unreliable. The Social Obligation Fund's Connected Bangladesh project, examined in Chapter 3, has deployed internet connectivity in selected community clinics, but coverage remains incomplete and the sustainability of these connections beyond the project period is uncertain. BTCL's fibre network, currently underutilised, could provide dedicated broadband links to health facilities in areas

where it passes, but no systematic health connectivity programme exists to convert this infrastructure potential into actual service.

PTD Action Required: BTRC should collaborate with the Ministry of Health to establish a Health Connectivity Programme that maps all public health facilities against existing fibre and mobile coverage, identifies connectivity gaps, and prioritises infrastructure investment accordingly. BTCL should be mandated to provide wholesale broadband to health facilities within 2 kilometres of its fibre route at concessional rates funded through the Social Obligation Fund. For facilities beyond fibre reach, Teletalk’s 4G network expansion and the proposed satellite connectivity through BSCL’s Starlink resale partnership should be designated as alternative delivery channels. The target should be broadband connectivity at 100 per cent of upazila health complexes by 2027 and 75 per cent of community clinics by 2029.

4.5.2 Education: Digital Classrooms and E-Learning

The education sector’s digital transformation depends fundamentally on school connectivity. The National Digital Transformation Strategy targets 40 to 60 per cent of schools with digital classrooms by 2027 to 2028, 10 to 15 million students on e-learning platforms, and 200,000 to 300,000 teachers trained in ICT. By 2030, the targets rise to 80 to 100 per cent of schools with digital classrooms and 20 to 25 million students on digital learning platforms. Bangladesh has approximately 150,000 primary and secondary educational institutions, of which a substantial proportion, particularly government primary schools in rural areas, currently lack internet connectivity.



The ITU’s Giga initiative, which aims to connect every school to the internet globally, estimates that school connectivity in low and lower-middle-income countries requires a combination of fibre, fixed wireless, and satellite technologies, with the choice depending on school location, population density, and existing infrastructure. For Bangladesh, BTCL’s fibre network and the

NTTN infrastructure of Fiber@Home and Summit can serve schools in urban and peri-urban areas. Teletalk's expanded 4G network can provide connectivity to schools in areas with mobile coverage. Satellite solutions, including the proposed NTNSP licensing framework for providers like Starlink, will be necessary for schools in remote haor, char, and hill-tract areas.

PTD Action Required: BTRC should work with the Ministry of Education to develop a School Connectivity Plan following the Giga methodology, mapping all schools against available infrastructure and producing a cost deployment roadmap. The Social Obligation Fund should allocate a dedicated window for school connectivity, with clear per-school cost benchmarks and outcome-based disbursement linked to verified operational connectivity. Spectrum allocation decisions, particularly the assignment of 700 MHz band for rural broadband, should incorporate school connectivity targets as a condition of spectrum licence obligations for successful bidders.

4.5.3 Agriculture: Connectivity for Rural Enterprise

Agriculture employs approximately 38 per cent of Bangladesh's workforce and contributes around 12 per cent of GDP, but the sector's productivity remains constrained by information asymmetries: farmers lack timely access to market prices, weather forecasts, pest management guidance, and input supply information. The National Digital Transformation Strategy targets 5 to 10 million farmers using agricultural applications by 2028 and 15 to 20 million by 2030, alongside 30 to 50 per cent of villages with agricultural information hubs and 2,000 to 5,000 drone-assisted crop surveys by 2028.

The postal network's role in agricultural transformation extends beyond connectivity. The mango delivery service demonstrates that post offices can function as aggregation and distribution points for agricultural produce. The BD Post progress report notes agricultural market linkage activity connecting farmers directly with consumers. The potential to expand this model into a systematic rural e-commerce logistics service is significant: post offices could serve as collection centres where farmers deposit produce for onward distribution, receive payments through postal financial services or MFS, and access price information through digital kiosks. This model is particularly relevant for perishable products where the cold chain infrastructure (chiller chambers and deep freezers) that the Postal Directorate has begun installing could ensure product quality.

PTD Action Required: The agricultural logistics model requires coordination between the Postal Directorate, the Department of Agricultural Extension, and e-commerce platforms. PTD should establish an Agricultural Postal Logistics Unit within the Postal Directorate to manage collection, cold chain, and distribution operations for agricultural products. BTRC should ensure that mobile network quality-of-service standards cover agricultural zones, including char and haor areas, where connectivity gaps directly constrain agricultural advisory services. Teletalk, as a state-owned operator, should be given specific coverage obligations for agricultural production zones identified by the Department of Agricultural Extension.

4.5.4 Social Protection: Digital Payment Delivery

Bangladesh's social safety net programmes serve over 9 million beneficiaries through allowances for the elderly, persons with disabilities, widows, and other marginalised groups. The shift from manual cash disbursement to digital payment, primarily through Mobile Financial Services, has reduced leakage and improved payment timeliness, but significant coverage gaps persist, particularly in areas where MFS agent networks are thin. The post office network offers an alternative and complementary disbursement channel: its presence in every upazila provides a physical payment point where beneficiaries who cannot use MFS (due to age, disability, or lack of a mobile phone) can collect payments.

The post office's role in social protection payment delivery was recognised as early as 2020, when the Prime Minister directed that Nagad be used for social safety net allowance disbursement. However, the Nagad governance concerns documented in the White Paper have complicated this channel. The reform pathway is to decouple the payment delivery function from any single MFS provider and instead establish the postal network as a multi-channel payment disbursement platform that can work with any licensed MFS provider, bank, or payment system to deliver social protection payments to beneficiaries at post office counters using biometric authentication.

PTD Action Required: The Postal Directorate should develop a Social Protection Payment Gateway that integrates with the Ministry of Social Welfare's beneficiary database and supports disbursement through multiple channels: direct post office counter payment with biometric verification, MFS credit to beneficiary mobile wallets, and bank account transfer. The platform should be channel-agnostic, ensuring that the choice of payment method rests with the beneficiary, not the provider. The target should be to make post office payment disbursement available in all 64 district headquarters by 2027 and all 495 upazilas by 2029.

Table 4.2: Sectoral Enablement KPI Framework

Sector / KPI	2026	2027	2028	2030
Health facilities with broadband (upazila level)	50%	100% upazila	50% community clinics	75% community clinics
Schools with internet connectivity	20%	40%	60%	80%

Post offices with social protection payment capability	64 districts	200 upazilas	350 upazilas	495 upazilas
Agricultural zones with verified 4G coverage	60%	75%	85%	95%
Post offices hosting Nagorik Sheba centres	100	500	1,000	2,000+
Postal financial transactions fully digital (%)	20%	50%	75%	100%

4.6 Disaster Resilience and Emergency Communications

Bangladesh's exposure to climate-related disasters, including cyclones, floods, riverbank erosion, and sea-level rise, makes telecommunications resilience a matter of life safety. The National Digital Transformation Strategy identifies disaster recovery and smart national infrastructure safety as a strategic priority, proposing AI-enabled surveillance, satellite and drone coverage for disaster zones, IoT-based sensors for critical infrastructure monitoring, and smart highway systems. The telecommunications network's role in disaster resilience operates at three levels: early warning dissemination, emergency response coordination, and post-disaster service restoration.

At the first level, cell broadcast technology enables mass dissemination of disaster warnings to all mobile phones within affected areas without requiring recipients to have data connections or specific applications. BTRC should mandate the deployment of cell broadcast capability across all mobile networks by 2027, following the model of the EU's Public Warning Systems Directive and Japan's Earthquake Early Warning System, which broadcasts alerts within seconds of seismic detection. At the second level, the integration of National Helplines 333 (citizen services) and 999 (emergency services) through the technical and operational framework proposed in the National Digital Transformation Strategy requires telecommunications backhaul capacity and call routing infrastructure that BTRC must ensure operators provide. At the third level, network resilience requires physical infrastructure hardening, including tower reinforcement in cyclone-prone zones, battery backup extending beyond current four-hour standards, and geographic redundancy in core network routing to ensure that the loss of a single node does not isolate an entire region.



The submarine cable landing station at Cox's Bazar represents a specific vulnerability: as documented in Chapter 3, all of Bangladesh's international fibre connectivity lands at a single coastal location highly exposed to cyclone damage. The diversification of cable landing points, whether through the proposed Khulna or Payra landing station or through satellite backup via BSCL's Starlink partnership and the proposed NTNSP licensing framework, is a disaster

resilience imperative that extends well beyond the telecommunications sector to every digital service that depends on international connectivity.

PTD Action Required: BTRC should develop a Telecom Disaster Resilience Framework, in coordination with the Ministry of Disaster Management and Relief, that establishes minimum infrastructure resilience standards for telecommunications facilities in each disaster zone category; mandates cell broadcast deployment on all mobile networks by 2027; requires operators to maintain pre-positioned mobile cell-on-wheels (COWs) for rapid post-disaster coverage restoration; and ensures that the 333/999 integration has geographically redundant call processing centres. The Social Obligation Fund should include disaster-resilient connectivity as an eligible expenditure category, funding tower hardening, backup power, and emergency satellite terminals in the most vulnerable districts.

4.7 Chapter Summary: From Infrastructure to Impact

This chapter has demonstrated that the value of PTD’s infrastructure and institutions is ultimately measured by the sectoral outcomes they enable. The postal network’s 9,886 offices can become citizen service delivery platforms, financial inclusion channels, agricultural logistics hubs, and social protection payment points, but only if the institutional failures exemplified by the Post e-Centre project and the Nagad governance lapses are addressed through the structural safeguards identified in this chapter. The telecommunications network’s 188 million mobile connections can enable telemedicine, digital classrooms, agricultural advisory services, and disaster early warning, but only if the quality-of-service, affordability, and coverage reforms outlined in Chapter 3 are implemented.

The cross-cutting theme is that PTD’s sectoral impact requires sustained coordination with line ministries and agencies that it has historically not maintained. The Health Connectivity Programme requires joint planning with the Ministry of Health. The School Connectivity Plan requires coordination with the Ministry of Education. The National Address Management System requires partnership with the Local Government Division and the Survey of Bangladesh. The Social Protection Payment Gateway requires integration with the Ministry of Social Welfare’s beneficiary database. Agricultural logistics require collaboration with the Department of Agricultural Extension. Disaster resilience requires coordination with the Ministry of Disaster Management and Relief.

The institutional mechanism for this coordination is addressed in Chapter 6 (Institutional Reform). The project governance framework is addressed in Chapter 7 (Project Reform). The implementation timeline integrating all sectoral targets is consolidated in Chapter 9 (Roadmap). The immediate priority for PTD is to recognise that its performance will increasingly be judged not by the volume of spectrum auctioned or licences issued, but by the measurable improvement in citizens’ access to health, education, financial, and government services that its infrastructure makes possible.

Table 4.3: Consolidated Reform Actions and Responsible Entities

Reform Action	Lead Entity	Partner	Timeline	Expected Outcome
Postal premises audit for Nagorik Sheba hosting	Postal Directorate	Nagorik Sheba Authority	Q3 2026	1,000 post offices identified for conversion

Postal Logistics Technology Platform (tracking, API integration, digital proof of delivery)	Postal Directorate	ICT Division, BCC	2026-2027	Real-time tracking at 2,000+ post offices
NAMS framework design and pilot deployment	Postal Directorate	Survey of Bangladesh, LGD	2026-2028	64-district geocoded address coverage
Postal Savings Bank digital platform	Postal Directorate	Bangladesh Bank, BEFTN	2026-2028	100% digital postal financial transactions
Health Connectivity Programme	BTRC	MoHFW, BTCL, Teletalk	2026-2029	100% upazila, 75% community clinic broadband
School Connectivity Plan	BTRC	MoE, BTCL, SOF	2026-2030	80% schools with internet by 2030
Social Protection Payment Gateway	Postal Directorate	MoSW, Bangladesh Bank	2026-2029	Payment disbursement in 495 upazilas
Telecom Disaster Resilience Framework	BTRC	MoDMR, MNOs	2026-2027	Cell broadcast on all networks; COW pre-positioning
MOCSLA courier sector formalisation	MOCSLA	BTRC, MoC	2026-2028	95% courier market formalised
Agricultural Postal Logistics Unit	Postal Directorate	DAE, e-commerce platforms	2026-2028	5,000 tonnes agricultural products annually by 2030

— End of Chapter 4 —

5. Strategic Policies for Sovereignty and Innovation

5.1 Introduction: Policy Imperatives for the Posts and Telecommunications Sector

The preceding chapters have established the legal foundations (Chapter 2), infrastructure and regulatory landscape (Chapter 3), and cross-sectoral service delivery ambitions (Chapter 4) of the Posts and Telecommunications Division. This chapter addresses the strategic policies that will govern **how** PTD entities operate, procure, secure, and innovate within the broader National Digital Transformation Strategy 2025-30 framework. Where the National Strategy sets whole-of-government policy direction across areas such as artificial intelligence, cloud computing, source code governance, cybersecurity, green ICT, and cross-border digital trade, this chapter translates those policies into specific, actionable obligations for the posts and telecommunications sector.

The policy translation is not theoretical. PTD entities collectively manage some of the most critical digital infrastructure in Bangladesh: 188 million mobile subscriptions traverse networks regulated by BTRC; BTCL operates the nation's fixed broadband backbone; BSCPLC provides over 95 per cent of international bandwidth through submarine cables; Teletalk serves as the state-owned mobile operator for government-sensitive communications; and Bangladesh Post handles financial transactions through 9,886 offices reaching every union in the country. A failure in cybersecurity at any of these entities does not merely affect a single organisation; it threatens the entire national digital ecosystem. Similarly, vendor lock-in at BTCL or the Postal Directorate does not merely waste public funds; it undermines the digital sovereignty that the 2025 Legal Trinity (Cyber Safety Ordinance, Personal Data Protection Ordinance, and National Data Governance Ordinance) was designed to protect.

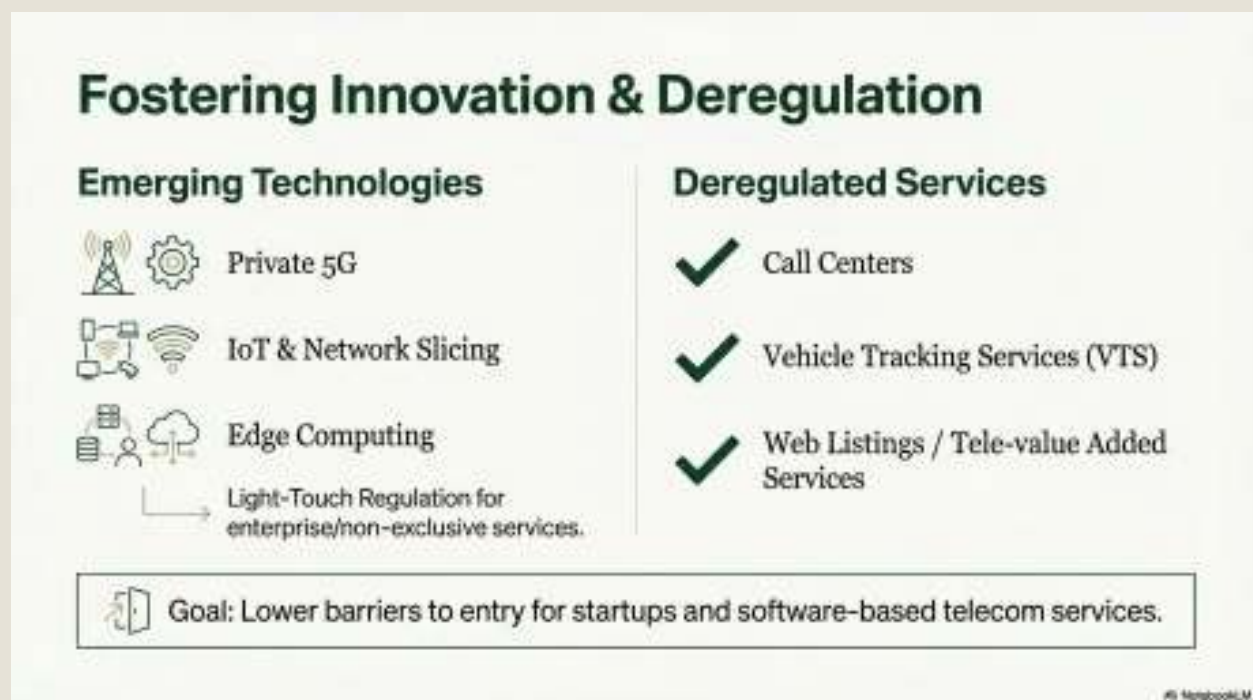
The chapter is organised around six policy domains that carry the greatest strategic weight for the posts and telecommunications sector: digital sovereignty and vendor independence; cybersecurity and critical information infrastructure protection; data protection and privacy compliance; local manufacturing and technology self-reliance; environmental sustainability in telecom operations; and emerging technology adoption. Each section identifies the national policy anchor, diagnoses the sector-specific gap, prescribes the PTD action, and defines measurable outcomes with timelines.

5.2 Telecommunications Network and Licensing Policy, 2025

5.2.1 The Strategic Shift: From Voice-Centric to Data-Sovereign

For fifteen years, Bangladesh's telecommunications sector operated under the *International Long Distance Telecommunication Service (ILDTS) Policy, 2010*. While that policy successfully navigated the early days of mobile voice proliferation, it was fundamentally architected for a world that no longer exists—a world where voice minutes were the primary commodity, and the internet was an auxiliary service. The rigid hierarchy of the 2010 framework (separating IGWs, ICXs, and IIGs) created a "siloed" ecosystem that, by 2024, had become a bottleneck for innovation, inflating latency and creating artificial layers of cost.

The enactment of the **Telecommunications Network and Licensing Policy, 2025**, gazetted on September 22, 2025, marks the definitive end of that era. By repealing the ILDTS Policy 2010, the Government has not merely updated a regulation; it has re-engineered the foundational logic of Bangladesh's connectivity.



This new policy serves as the bedrock for the PTD's 2030 vision. It shifts the regulatory focus from "gatekeeping voice traffic" to "enabling data sovereignty." It recognizes that in an AI-driven economy, the network must be a seamless, technology-neutral utility that empowers services rather than restricting them. The 2025 Policy introduces a layered, horizontal licensing architecture designed to achieve three strategic imperatives: **Decoupling**, **De-regulation**, and **Resilience**.

5.2.2 Decoupling Infrastructure from Service: The TIP and TESP Model

The most significant structural reform in the 2025 Policy is the decoupling of physical network ownership from service delivery. Historically, regulatory mandates often forced service providers to build their own heavy infrastructure, leading to capital inefficiency, road digging redundancies, and high barriers to entry for startups.

The new policy resolves this by establishing two distinct functional categories:

1. Telecom Infrastructure Providers (TIP): These entities are the "builders." They are licensed specifically to deploy, own, and maintain the passive and active infrastructure—fiber optic backbones, towers, and transmission networks. By consolidating infrastructure rights under TIPs, the policy encourages a "build once, share many" philosophy. This reduces national CAPEX waste and ensures that fiber resources are treated as a shared national utility rather than a proprietary weapon of market dominance.

2. Telecom Enabled Service Providers (TESP): These entities are the "innovators." A TESP does not need to lay cables or build towers. Instead, they utilize the infrastructure provided by TIPs to deliver value-added services to the end-user. This category is critical for the "Innovation" pillar of this chapter. It lowers the barrier to entry for Bangladeshi tech companies, allowing a local startup to offer specialized communication services, OTT platforms, or enterprise data solutions without needing the massive capital required for physical infrastructure.

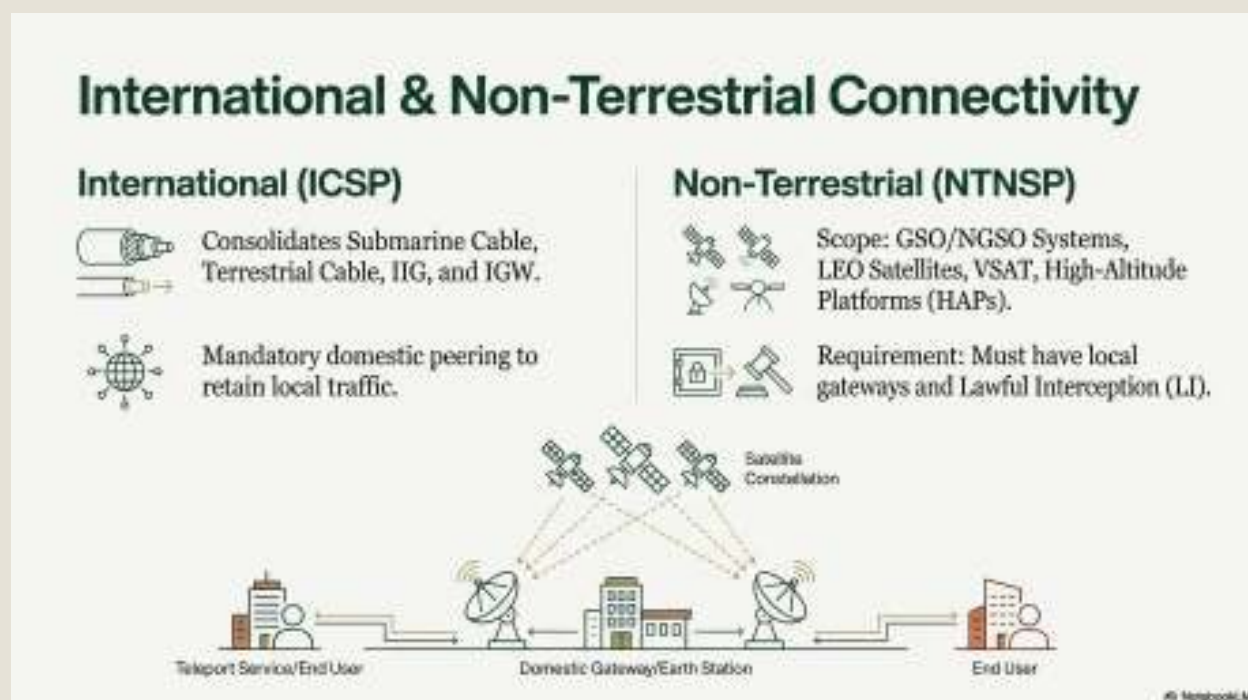
This separation mirrors the successful "Layered Regulation" models seen in advanced digital economies (such as the EU's distinction between Electronic Communications Networks and Services). For Bangladesh, this means the regulatory burden is heavy on the physical layer (where safety and order are paramount) but light on the service layer (where speed and innovation are paramount).

5.2.3 Democratizing Access: The Licensing Tiers

To ensure that the benefits of this new architecture reach every corner of the nation—and not just the urban centers of Dhaka and Chattogram—the 2025 Policy introduces a nuanced tier of service provider licenses.

Cellular Mobile Service Providers: The policy maintains a robust framework for Mobile Network Operators (MNOs), but shifts the expectation from simple coverage to "quality of service" and high-speed data throughput. MNOs are now viewed as the primary "last mile" access point for the majority of the population, with renewed mandates to integrate with the TIP ecosystem for backhaul efficiency.

Fixed Telecom Service Providers: Recognizing that fiber-to-the-home (FTTH) is the gold standard for bandwidth, this category consolidates wired and fixed-wireless services. However, the true strategic innovation lies in the sub-category described below.



District Fixed Telecommunication Service Providers: This is a pro-poor, pro-rural policy instrument. By creating a specific license category for *District-level* providers, the Government is decentralizing the internet economy. Previously, large national ISPs often neglected remote districts due to lower immediate ROI. The District license empowers local entrepreneurs—who understand the specific needs and geography of their home districts (e.g., Kurigram or Bhola)—to become formal connectivity providers. This creates a grassroots mesh of connectivity that is more responsive to local demand than a centralized Dhaka-based monolith could ever be.

5.2.4 Sovereignty in the Skies: Non-Terrestrial Networks (NTN)

Perhaps the most forward-looking aspect of the 2025 Policy—and the one most relevant to "Sovereignty"—is the formal recognition of **Non-Terrestrial Networks and Service Providers (NTNSP)**.

For decades, Bangladesh's connectivity was terrestrial-bound, relying on submarine cables (SEA-ME-WE) and cross-border terrestrial links. While efficient, this created single points of failure. A cable cut in the Bay of Bengal or political instability at a land border could theoretically isolate the nation.

The NTNSP category authorizes the operation of:

- **Low Earth Orbit (LEO) Satellite Constellations:** allowing high-speed, low-latency internet anywhere in the country, regardless of terrain.
- **High-Altitude Platforms (HAPs):** utilizing stratospheric balloons or drones for temporary or permanent coverage enhancement.

Strategic Implication: This is a sovereignty assurance mechanism. By licensing NTN providers, Bangladesh ensures that its critical command, control, and emergency communication systems can survive catastrophic floods, earthquakes, or submarine cable severances. It transforms the definition of "national network" from a 2D map of cables to a 3D dome of connectivity. Furthermore, by bringing these global satellite giants under a licensing regime, the PTD ensures that they operate within Bangladeshi law, adhering to lawful interception and data security requirements, rather than operating in a "grey market" vacuum.

5.2.5 Data Sovereignty: Regulating Hyperscalers

In the digital age, sovereignty is defined by where data lives. The 2010 policy was silent on cloud computing; the 2025 Policy explicitly defines and regulates **Hyperscalers**.

Hyperscalers are the large-scale cloud service providers (global giants operating extensive data center infrastructures). Their inclusion in the policy framework is a decisive move to:

1. **Encourage Local Hosting:** By creating a regulatory home for Hyperscalers, Bangladesh signals its readiness to host "Availability Zones" and physical data centers within its borders.
2. **Ensure Data Governance:** It brings the storage and processing of Bangladeshi citizen data under the purview of national law, moving away from the "wild west" era where national data was stored offshore without oversight.
3. **Catalyze the Software Industry:** Local hosting by Hyperscalers drastically reduces latency for local developers, making Bangladesh a viable hub for real-time applications, AI processing, and fintech hubs.

5.2.6 The Innovation Engine: De-Regulation

A distinctive feature of the 2025 Policy is the formal introduction of **De-regulation**.

In the past, the regulator often attempted to license every new technology, stifling innovation (e.g., the early restrictions on VoIP). The 2025 Policy explicitly states that "Services under the de-regulated category may be offered without a license."

This "Negative List" approach—where everything is permitted unless specifically restricted—is a paradigm shift for the BTRC. It allows Bangladesh's youth and developers to experiment with emerging technologies (like IoT meshes, localized ad-hoc networks, or experimental protocols)

without facing bureaucratic headwinds. It signals that the PTD views regulation as a tool for safety and equity, not as a tool for control and revenue extraction.

5.2.7 Implementation and the Road Ahead

The transition from the ILDTS 2010 to the Network and Licensing Policy 2025 is not merely administrative; it is operational. The BTRC is tasked with the immediate migration of existing licensees into this new tiered structure. This involves:

- **Grandfathering and Migration:** Ensuring current operators can transition to TIP/TESP or Provider licenses without service disruption.
- **Audit of Infrastructure:** Mapping the nation's fiber assets to enforce the TIP shared-infrastructure mandate effectively.
- **New Licensing Windows:** Opening applications for NTN and District providers to immediately jumpstart rural and satellite connectivity.

Conclusion The *Telecommunications Network and Licensing Policy, 2025* is the legislative engine of the PTD Transformation Strategy. It replaces the "Scarcity Model" of 2010 with the "Abundance Model" of 2030. By recognizing the sky (NTN), the cloud (Hyperscalers), and the grassroots (District Providers), it constructs a telecom sector that is resilient enough to protect our sovereignty and flexible enough to fuel our innovation.

5.3 Digital Sovereignty and Vendor Independence

The National Digital Transformation Strategy identifies vendor neutrality and digital sovereignty as a foundational policy pillar. It observes that over the past decade, certain government ministries, agencies, and state-owned enterprises have procured vendor-locked or closed software and hardware systems that create long-term strategic vulnerabilities, including restrictive licensing models, lack of interoperability, and prohibitively high renewal or change fees. This assessment applies with particular force to the telecommunications sector, where network management systems, billing platforms, spectrum monitoring tools, and postal automation equipment represent critical national infrastructure that must remain under sovereign operational control.

5.3.1 Vendor Neutrality in Telecom Procurement

The telecommunications sector is among the most hardware-intensive and vendor-dependent segments of government. BTCL's Huawei-supplied core network, BTRC's spectrum monitoring systems, BSCPLC's submarine cable terminal equipment, and the Postal Directorate's automated mail processing centres all represent significant vendor dependencies. In several cases, even routine maintenance or configuration changes require vendor authorisation and payment, effectively transferring operational sovereignty to foreign suppliers.

The National Strategy mandates that public funds must not be used to procure software or hardware that creates dependency on a single vendor. For the posts and telecommunications sector, this translates into three operational requirements. First, all new telecom system procurements by BTCL, Teletalk, BSCPLC, and the Postal Directorate must include open interface specifications, allowing integration with equipment from multiple vendors and ensuring compatibility with the Bangladesh National Digital Architecture (BNDIA). Second, existing vendor-locked systems must be subject to phased migration plans, beginning with an audit of all current contracts, license expiry dates, renewal costs, and migration feasibility. Third, BTRC's licensing conditions must require that telecom operators deploy open-standard network interfaces, particularly for 5G core and radio access networks, following the model of the O-RAN Alliance specifications that countries such as Japan, Germany, and India are increasingly mandating to reduce dependence on any single equipment supplier.

India's Department of Telecommunications (DoT) has been particularly active in promoting Open RAN through its Trusted Telecom Portal and Trusted Sources mandate, which requires that telecom equipment used in Indian networks comes from sources deemed trustworthy by a designated committee. Japan's NTT DOCOMO has invested heavily in Open RAN deployment, while Germany's O-RAN Town project in Neubrandenburg demonstrated multi-vendor 5G deployment in a real-world environment. These international precedents validate the feasibility and strategic importance of vendor-neutral telecom procurement for a country like Bangladesh

that seeks to maintain sovereign control over its digital infrastructure without isolating itself from global technology markets.

5.3.2 Source Code Ownership for PTD Entities

The National Source Code Policy establishes the principle that all government-funded software is public property and must be registered in the National Source Code Repository (NSCR) before deployment. This principle addresses a recurring failure across PTD entities where software developed with public funds has been controlled by vendors rather than by the government.

The Post e-Centre project, discussed in Chapter 4, exemplifies this failure. BDT 541 crore was spent on a system that equipped 8,500 post offices with digital equipment, yet when the project was audited, the software could not be independently maintained, modified, or redeployed because the government did not retain effective control over the source code and system architecture. Similarly, BTCL's billing and customer management systems, Teletalk's value-added services platform, and BTRC's Electronic Regulatory Platform (ERP) all rely on proprietary software where the government's ability to modify, audit, or migrate is constrained by vendor contracts.

To implement the National Source Code Policy across PTD, the following directives apply. BTCL must register all existing government-developed or government-funded software in the NSCR by Q4 2027, beginning with its billing platform, network management system, and customer service applications. BTRC must require, as a licensing condition under the Telecommunications Network and Licensing Policy 2025, that all regulatory reporting interfaces and compliance systems deployed by licensees use open APIs conforming to BNDIA standards. The Postal Directorate must ensure that all software deployed under the Comprehensive Logistics Transformation Programme (CLTP) and the Document Management System (DMS) includes mandatory source code escrow with the government retaining full modification rights. Teletalk must adopt open-source core network solutions where technically feasible, beginning with its OSS/BSS stack, and contribute improvements to the national repository.

The European Union's interoperability framework, through the European Commission's Open Source Software Strategy 2020-2023, provides a relevant benchmark. The EU mandated that all new software developed by or for European institutions be open source by default, with exceptions requiring formal justification. France's policy through DINUM (Direction interministerielle du numerique) has created a national catalogue of recommended open-source solutions for government. Bangladesh's National Source Code Policy follows this direction, and PTD entities, as managers of critical national infrastructure, must be among the first to comply.

5.3.3 Bandwidth Sovereignty and Strategic Reserve

Bangladesh's international connectivity depends on submarine cable infrastructure managed primarily by BSCPLC through the SEA-ME-WE-4 and the forthcoming SEA-ME-WE-6 systems. As documented in Chapter 3, BSCPLC's lit-up capacity of 6,850 Gbps in FY 2023-24 far exceeded actual utilisation of 2,696 Gbps (39 per cent), yet the entry of Bangladesh Private Cable System (BPCS) with its dark fibre Indefeasible Right of Use (IRU) model threatens to divert traffic and revenue from the state-owned operator. The PTD White Paper identified several concerns about this licensing decision, including the potential dilution of national investment in BSCPLC and the fragmentation of market control over international bandwidth.

The Telecommunications Network and Licensing Policy 2025 provides the regulatory foundation for addressing these concerns through its provisions on International Connectivity Service Providers (ICSP), infrastructure sharing mandates, and Significant Market Power (SMP) monitoring. Building on these provisions, this strategy proposes the establishment of a National Bandwidth Reserve, a policy mechanism that requires a defined percentage of total international bandwidth capacity to be held under state control through BSCPLC at all times, regardless of the number of private ICSP licensees. This reserve ensures that during geopolitical disruptions, natural disasters, or commercial disputes, Bangladesh retains sovereign access to sufficient international connectivity for essential government, financial, and emergency communications.

International precedent supports this approach. Kenya's experience with the TEAMS and SEACOM submarine cables demonstrated that a mixed public-private ownership model can sustain competition while protecting national interests. Singapore's open cable landing station policy, which requires non-discriminatory access to landing infrastructure, has made it a regional connectivity hub without compromising sovereignty. Australia's Subsea Cable Protection Zone legislation protects critical cable infrastructure as a matter of national security. Bangladesh should draw from each of these models: the bandwidth reserve from the strategic perspective, the open landing station principle for competitive access, and the infrastructure protection framework for physical security of the Cox's Bazar cable landing station.

Table 5.1: Digital Sovereignty Policy Actions and KPIs

Policy Action	Lead Entity	Timeline	KPI / Target
Vendor lock-in audit of all PTD entity systems	DoT	Q4 2026	100% of systems audited; migration plans for locked systems
Open interface mandate for new telecom procurements	BTRC / DoT	Q2 2027	All new RFPs require open API, multi-vendor compatibility
PTD entity source code registration in NSCR	BTCL, Teletalk, Postal Directorate	Q4 2027	All government-funded software registered
BTRC open API requirement for licensee regulatory reporting	BTRC	Q2 2027	100% licensee systems using BNDIA-compliant APIs

National Bandwidth Reserve policy	PTD / BSCPLC	Q4 2026	Minimum 30% of total international capacity under state control
Open cable landing station access framework	BTRC / BSCPLC	Q2 2027	Non-discriminatory access rules published and enforced

5.4 Cybersecurity and Critical Information Infrastructure Protection

The Cyber Safety Ordinance 2025 establishes the National Cyber Safety Agency (NCSA) and the National Security Operations Center (NSOC) as the apex institutions for national cyber defence. It mandates that entities designated as Critical Information Infrastructure (CII) must maintain secure connectivity with NSOC for real-time threat monitoring, report cyber incidents to the National Cyber Emergency Response Team (NCERT), and establish sector-specific Computer Incident Response Teams (CIRTs). The Telecommunications Network and Licensing Policy 2025, in Section 11.10, explicitly requires that all licensees operating critical service platforms be designated as CII by the NCSA, and comply with regular security audits including penetration testing and vulnerability assessments.

For the posts and telecommunications sector, these requirements carry extraordinary weight. Telecom networks are not merely CII in their own right; they are the infrastructure upon which every other CII sector depends. A compromise of BTCL's backbone affects banking systems, power grid management, and government communications simultaneously. A breach of BTRC's spectrum management system could enable interference with aviation and emergency frequencies. An attack on BSCPLC's submarine cable terminal could sever Bangladesh's international connectivity entirely. The interdependent nature of telecom CII demands a layered, sector-specific cybersecurity framework that exceeds the baseline requirements of the CSO.

5.4.1 Telecom Sector CII Designation and CIRT Establishment

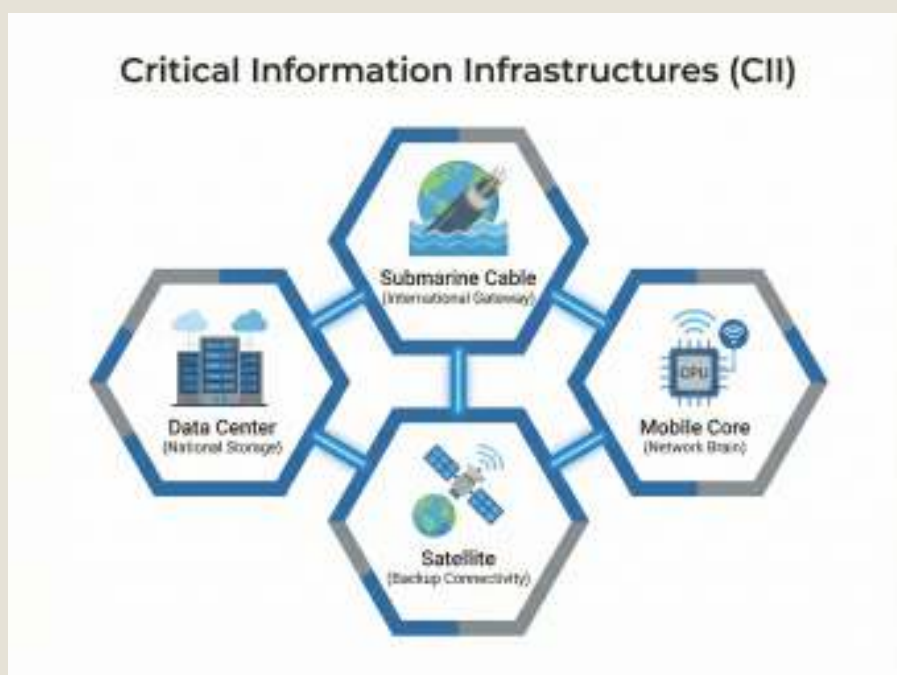
The NCSA has already initiated the process of updating the CII designation list, with the Directorate of Posts being processed for inclusion as of early 2026. This strategy recommends that the following PTD entities and infrastructure be formally designated as CII within the first year of implementation: BTCL's core switching and transmission network; BSCPLC's submarine cable terminal and international gateway systems; BTRC's spectrum management and monitoring infrastructure; Teletalk's mobile core network; and the Postal Directorate's financial transaction platforms including the Electronic Money Transfer Service (EMTS) and any successor platforms handling postal savings and insurance data.

Each designated CII entity must establish a dedicated Computer Incident Response Team (CIRT) with minimum staffing of five cybersecurity professionals, 24/7 monitoring capability, and direct secure connectivity to the NSOC. India's approach provides a relevant model: the Department of Telecommunications operates a Telecom Security Operations Centre (TSOC) that monitors threats specific to the telecom sector, while the NCIIPC (National Critical Information Infrastructure Protection Centre) coordinates across sectors. Singapore designates telecom and infocomm systems as CII under its Cybersecurity Act, requiring operators to implement the Cybersecurity Code of Practice (CCoP 2.0), conduct regular audits, and report incidents

promptly. Bangladesh's telecom CIRT framework should combine India's sector-specific monitoring approach with Singapore's compliance and audit rigour.

5.4.2 Telecom Cybersecurity Standards and Compliance

Beyond CII designation, the telecom sector requires a comprehensive cybersecurity standards framework that addresses network-layer, application-layer, and supply-chain security. The National Digital Transformation Strategy identifies the need for Vulnerability Assessment and Penetration Testing (VAPT) audits, ISO 27001 compliance, and zero-trust architecture as baseline requirements. For the telecom sector, these translate into specific obligations.



First, all mobile network operators, including Teletalk, must implement SS7 and Diameter protocol security measures to prevent signalling-layer attacks that can enable location tracking, call interception, and denial of service. These vulnerabilities, well documented since the 2014 demonstrations by researchers at the Chaos Communication Congress, remain exploitable in networks that have not deployed filtering and monitoring tools. Second, BTRC must issue a Telecom Cybersecurity Compliance Directive requiring all licensees to achieve ISO 27001 certification for their core network operations by 2028, with annual VAPT audits conducted by NCSA-accredited auditors. Third, all PTD entities must implement the National Cloud Policy's zero-trust architecture requirements, ensuring that network location does not confer trust and that all access requests are authenticated, authorised, and continuously verified regardless of origin.

The supply-chain dimension is equally critical. The National Strategy warns against vendor dependencies, but in cybersecurity the risk extends to the integrity of network equipment itself.

The European Union's 5G Toolbox and the United States' Secure and Trusted Communications Networks Act both address the risk of compromised network equipment from untrusted suppliers. For Bangladesh, where network infrastructure relies heavily on equipment from a limited number of manufacturers, BTRC should develop a Trusted Telecom Equipment Framework that requires vendors to submit to security evaluations before their equipment can be deployed in CII-designated networks. This does not require excluding any specific vendor, but it does require that equipment deployed in critical national infrastructure meets independently verified security standards.

5.4.3 Submarine Cable and Landing Station Security

The physical security of submarine cable infrastructure represents a unique cybersecurity concern that sits at the intersection of digital and physical defence. Bangladesh's international connectivity depends overwhelmingly on the Cox's Bazar cable landing station, a single point of geographic concentration that is vulnerable to both natural disasters (cyclones, earthquakes) and deliberate interference. Chapter 4 identified this vulnerability in the context of disaster resilience; here, the concern is specifically cybersecurity and national security.

BSCPLC must implement multi-layered security for the cable landing station, including physical perimeter security compliant with international standards, cybersecurity monitoring of all terminal equipment with NSOC connectivity, and encrypted management channels for all remote operations. Australia's Subsea Cable Protection Zone legislation, which creates exclusion zones around critical cable infrastructure and criminalises unauthorised interference, provides a model for physical protection. The diversification imperative identified in Chapter 4, including the need for alternative landing sites at Khulna or Payra, carries cybersecurity as well as resilience justification: geographic concentration creates a single point of failure that a determined adversary could exploit.

Table 5.2: Cybersecurity Policy Actions and KPIs

Policy Action	Lead Entity	Timeline	KPI / Target
CII designation for all PTD critical systems	NCSA / PTD	Q2 2027	BTCL, BSCPLC, BTRC, Teletalk, Postal Directorate designated
Sector CIRT establishment at each PTD entity	Each PTD entity	Q4 2027	5+ cybersecurity professionals per CIRT; 24/7 monitoring
ISO 27001 certification for telecom licensees	BTRC	2028	100% of Tier 1 licensees (MNOs, ICSPs) certified
SS7/Diameter signalling security deployment	MNOs / BTRC	Q4 2027	100% of MNOs deploy signalling firewalls
Telecom VAPT audit regime	BTRC / NCSA	Q2 2027	Annual audits for all CII-designated entities

Trusted Telecom Equipment Framework	BTRC	Q4 2027	Security evaluation mandatory for CII network equipment
Cable landing station security upgrade	BSCPLC	Q2 2028	Physical and cyber security to international standards

5.5 Data Protection and Privacy in the Telecom Sector

The Personal Data Protection Ordinance 2025 (PDPO) grants citizens legal ownership of their digital footprint and establishes the Data Protection Authority as the enforcement body. The National Data Governance Ordinance (NDGO) mandates interoperability and creates the National Data Governance Authority (NDGA) with powers to regulate data flows across government and the private sector. For the telecommunications sector, these ordinances create specific and substantial compliance obligations.

5.5.1 Telecom Data as Personal Data

Telecommunications operators hold among the most extensive personal data sets in the country. Call Detail Records (CDRs) contain information about who communicated with whom, when, for how long, and from where. Mobile network data reveals location patterns, movement habits, and social relationships. Mobile Financial Services (MFS) data encompasses financial transactions, account balances, and spending patterns. Internet browsing data, though not typically stored by operators for extended periods, passes through their infrastructure and can be intercepted if not properly protected. The PDPO classifies much of this data as sensitive personal data requiring enhanced protection.



The Nagad experience documented in Chapter 4 illustrates the consequences of inadequate data governance in the telecom-financial intersection. As a mobile financial service built on the postal network, Nagad handles BDT 34,000 crore in monthly transactions as of March 2025, yet the PTD White Paper documented governance failures including reliance on informal memoranda of understanding rather than formal licensing arrangements, loss of Postal Directorate operational control, and questionable documentation around no-objection letters. These governance failures carry direct data protection implications: without formal institutional control, the ability to enforce data protection standards, conduct audits, and respond to breaches is fundamentally compromised.

BTRC must issue a Telecom Data Protection Directive that operationalises the PDPO for all telecom licensees. This directive must address, at minimum, the following requirements: lawful basis for data processing (consent, contractual necessity, legal obligation); data minimisation (operators may not collect or retain data beyond what is necessary for service provision and lawful interception obligations); purpose limitation (CDR and location data may not be used for commercial profiling without explicit, informed, and freely given consent); data retention periods (aligned with lawful interception requirements but with mandatory deletion after the prescribed period); breach notification (immediate reporting to the Data Protection Authority and affected individuals within 72 hours, consistent with GDPR standards); and cross-border data transfer restrictions (subscriber data must not leave Bangladesh without NDGA approval, consistent with the PDPO's cross-border transfer provisions).

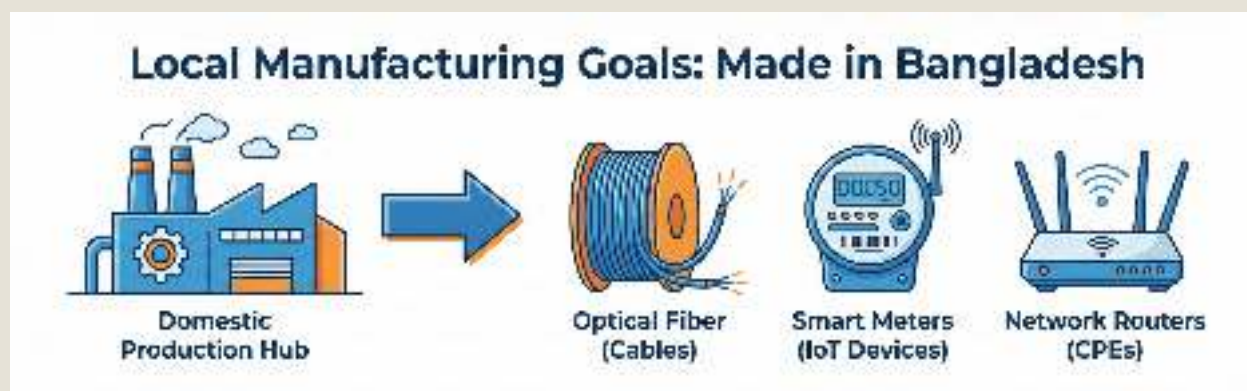
5.5.2 Mobile Financial Services Data Governance

The convergence of telecom and financial services through MFS platforms creates a regulatory intersection that demands coordinated oversight between BTRC and Bangladesh Bank. The National Strategy identifies the need for dual regulatory oversight for services that span sectoral boundaries. For MFS data, this means that Bangladesh Bank's payment system regulations and the PDPO's personal data protection requirements must both apply, with BTRC responsible for ensuring that the underlying telecom infrastructure meets the necessary security and privacy standards.

Specific policy actions include: mandatory data segregation between telecom and financial services data within MFS platforms; regular data protection impact assessments (DPIAs) for all MFS services, conducted jointly by BTRC and Bangladesh Bank auditors; customer consent management systems that meet PDPO requirements for informed, specific, and freely withdrawable consent; and binding service level agreements between MFS providers and the Postal Directorate or telecom operators that host their services, replacing the informal MoU arrangements that characterised the Nagad relationship. The three governance lessons identified in Chapter 4 apply directly: dual regulatory oversight from BTRC and Bangladesh Bank; formal contracts rather than memoranda; and institutional operational control by the hosting organisation.

5.6 Local Manufacturing and Technology Self-Reliance

The National Digital Transformation Strategy identifies technology transfer, domestic manufacturing capability, and reduced import dependency as strategic priorities. The National Semiconductor Policy and the broader industrial policy framework aim to position Bangladesh in the global technology supply chain. For the posts and telecommunications sector, these ambitions intersect with existing manufacturing assets in two state-owned entities: Bangladesh Cable Shilpa Limited (BCSL), the country's only domestic fibre optic cable manufacturer, and Telephone Shilpa Sangstha Limited (TSS), the state-owned digital device manufacturing and integration company.



5.6.1 Bangladesh Cable Shilpa Limited: Fibre Optic Manufacturing

BCSL, established in 1967 in Khulna and operational since 1973, has evolved from copper telecom cable production to become Bangladesh's sole domestic manufacturer of optical fibre cable, with commercial OFC production beginning in 2011. The company operates from a 600,000 square metre facility with annual production capacity of 20,000 kilometres of optical fibre cable, along with silicon-core HDPE ducts (added in 2016) and power cables (added in 2019). In FY 2024-25, BCSL earned a profit of BDT 31 crore, with cumulative profits over the preceding five years reaching BDT 148 crore. The company currently supplies approximately 60 per cent of the optical fibre telecom and copper cables used across the country.

The strategic opportunity is substantial. Industry estimates indicate that Bangladesh requires a five lakh kilometre fibre optic cable network for comprehensive connectivity, while only approximately 1.5 lakh kilometres are currently deployed. Moreover, 75 per cent of the existing network uses overhead deployment, which is vulnerable during cyclones and floods. The transition to underground fibre deployment, essential for disaster resilience, will generate massive domestic demand for both fibre cable and HDPE ducts that BCSL is positioned to supply. BCSL has also announced plans to export fibre optic cables to Nepal and select African countries by 2028, potentially transforming a domestic manufacturer into a regional supplier.

However, BCSL faces significant constraints. Its machinery, while upgraded over time, is aging compared to competitors in neighbouring countries. The company's workforce of 33 officers, 163 staff, and 125 outsourced workers is operating with significant vacancies: 29 out of 63 officer positions and 176 out of 341 regular employee positions remain unfilled. Quality testing relies on external partners (BUET, KUET, CUET, and MIST) rather than in-house capabilities, and the company imports approximately 99 per cent of its copper rods, over 60 per cent of aluminium rods, and all optical fibre preforms from countries including Australia, Korea, Malaysia, China, and India.

The policy response must address both expansion and supply chain vulnerability. First, PTD should mandate a domestic content requirement for all government-funded fibre deployment projects, requiring that a minimum percentage (initially 40 per cent, rising to 60 per cent by 2029) of fibre cable be sourced from BCSL, provided quality standards are met. Second, BCSL's capacity must be expanded through machinery modernisation, supported by a capital investment plan funded through retained earnings and PTD allocations, targeting an increase in annual OFC capacity from 20,000 to 50,000 kilometres by 2030. Third, BCSL should pursue a joint venture or technology licensing arrangement for optical fibre preform manufacturing, the critical upstream input that the company currently imports entirely. India's Sterlite Technologies and Finolex Cables provide examples of successful domestic fibre preform production that reduced import dependency. The BCSL directive from PTD also requires the company to obtain internationally recognised environmental certification (the 'Green Certificate' initiative) by July 2026, positioning its products for export markets that demand environmental compliance.

Fibre Migration & Production: BCSL will prioritize the production of next-generation fiber optic cables to facilitate the national transition from legacy copper to high-speed fiber infrastructure.

Strategic Utility Supplier: Position BCSL as the primary transmission and distribution cable supplier for all national power agencies, including REB, DESA, DESCO, and NESCO.

Standardization: In coordination with BTRC and TSS, BCSL will enforce strict national quality standards (aligned with ITU/EU) for all imported and domestic cables to eliminate substandard imports and reduce electronic waste through "Unified Charging" and auxiliary cable standards.

5.6.2 Telephone Shilpa Sangstha: From Assembly to Manufacturing

TSS, founded in 1967 as a joint venture with Siemens AG of West Germany and reconstituted as a limited company in 1973, has a complex history. In its post-independence phase, TSS produced EMD telephone exchanges, analog PABX systems, fax machines, and telephone sets. It later transitioned to digital device assembly under the DOEL brand, producing laptops, notebooks, tablets, desktops, biometric devices, multimedia projectors, sound boxes, and smart prepaid

energy meters. The DOEL laptop, launched in 2011 as the first laptop assembled in Bangladesh with components primarily from Malaysia and China, represented an important symbolic milestone but did not achieve commercial sustainability. The project assembled over 40,000 units at prices starting from BDT 10,000, but the majority of components (approximately 90 per cent) were imported, and the initiative was unable to develop meaningful local manufacturing depth.

An initiative was announced in 2025 to transform TSS into a high-tech park, with the Special Assistant to the Chief Adviser responsible for the Ministry of Posts, Telecommunications, and Information Technology confirming that the approach would focus initially on developing TSS as an effective mid-tech or low-tech manufacturing institution rather than an immediate high-tech park. Discussions were reported about aligning TSS's existing infrastructure with efforts to attract Chinese manufacturing plants to Bangladesh. The TSS facility in Tongi, Gazipur, has available land and industrial infrastructure, though reports indicate that land allocation for non-industrial purposes and encroachments have constrained the available space.

The strategy for TSS must be realistic about what can be achieved. Rather than attempting to compete in finished consumer electronics where global supply chains are deeply established, TSS should focus on three areas where domestic manufacturing serves clear national security or policy objectives. First, telecom network components: manufacturing of cable assemblies, patch panels, fibre distribution frames, outdoor cabinets, and tower-mounting hardware that are currently imported but do not require advanced semiconductor fabrication. Second, IoT and smart devices for government programmes: smart energy meters for utilities, environmental monitoring sensors, GPS tracking devices for postal logistics (building on the 85 GPS-tracked postal vans already deployed), and biometric devices for government authentication systems. Third, assembly and integration services for telecom equipment: providing controlled-environment facilities where imported components are assembled into finished products under local oversight, ensuring supply chain visibility and enabling gradual import substitution as domestic capabilities mature.

Prohibition of Substandard Imports and National Standardization: To protect national network integrity and encourage local industry, the PTD shall implement a strict ban on the import of low-quality cables from non-verified foreign vendors. The **Bangladesh Telecommunication Regulatory Commission (BTRC)**, in coordination with **Telephone Shilpa Sangstha (TSS)** and **Bangladesh Cable Shilpa Limited (BCSL)**, will establish a mandatory national certification and standardization framework for all telecommunications and power cables used in public infrastructure. This framework will ensure that all deployed cables meet rigorous durability and signal-integrity benchmarks, reducing maintenance costs and improving overall Quality of Service (QoS).

Consumer Electronics Expansion: TSS will scale domestic production of International Standard (ITU/EU) mobile devices, laptops, auxiliary cables, adapters, and chargers.

Infrastructure Components: Through partnerships with entities like **edotco**, TSS will manufacture specialized hardware for telecom towers and power units, ensuring a local supply chain for the 5-year network evolution plan.

Vietnam's experience offers a relevant benchmark. Through targeted FDI policies and industrial zone development, Vietnam attracted Samsung, Intel, and other major manufacturers to establish production facilities, progressing from basic assembly to increasingly sophisticated manufacturing over two decades. Bangladesh's approach through TSS cannot replicate Vietnam's scale, but it can follow the same trajectory of starting with assembly, building worker skills, and gradually deepening local content. The key enablers are: designation of TSS as a Hi-Tech Park to access existing tax exemptions and customs duty waivers; establishment of a joint venture with an established manufacturer to bring technology and management expertise; and integration of TSS into PTD procurement processes so that government demand provides a baseline revenue stream.

Table 5.3: Local Manufacturing Policy Actions and KPIs

Policy Action	Lead Entity	Timeline	KPI / Target
Domestic content requirement for government fibre projects	PTD	Q2 2027	40% by 2027, 60% by 2029 from BCSL
BCSL capacity expansion to 50,000 km/year OFC	BCSL	2030	2.5x increase from current 20,000 km capacity
Optical fibre preform JV or technology licensing	BCSL / PTD	Q4 2028	Domestic preform production initiated
BCSL Green Certificate international certification	BCSL	Jul 2026	Certification obtained; export market access
BCSL vacancy recruitment drive	BCSL	Q4 2026	80% of sanctioned positions filled
TSS Hi-Tech Park designation and JV partnership	TSS / PTD	Q4 2027	JV agreement signed; manufacturing commenced
TSS integration into PTD procurement	DoT / TSS	Q2 2027	GPS trackers, meters, cable assemblies sourced from TSS
BCSL export to Nepal and Africa	BCSL	2028	First export shipments completed

5.7 Environmental Sustainability in Telecom Operations

The National Digital Transformation Strategy includes a National Green ICT Policy that recognises the environmental footprint of digital infrastructure and mandates the integration of sustainability into all technology procurement, deployment, and operations. The

Telecommunications Network and Licensing Policy 2025, in Section 11.11, requires all licensed operators to adopt environmentally friendly technologies, promote renewable energy solutions, implement e-waste management practices, and reduce their carbon footprint. Section 11.12 commits the government to promoting and incentivising environmentally sustainable practices in alignment with the Sustainable Development Goals and the Paris Agreement.

5.7.1 Telecom Tower Energy Transition

Bangladesh's telecom tower infrastructure, numbering approximately 35,000 to 40,000 towers serving both mobile operators and tower companies, represents one of the largest sources of energy consumption in the sector. Many towers, particularly in rural and off-grid areas, rely on diesel generators for primary or backup power, creating significant carbon emissions, fuel cost burdens, and logistical challenges for fuel delivery to remote sites.

The policy mandate from the Telecommunications Network and Licensing Policy is clear: operators must integrate renewable energy sources, such as solar, wind, or hybrid systems, into telecom tower operations wherever technically and economically feasible. To operationalise this mandate, BTRC should issue a Green Tower Directive with the following elements. First, a mandatory target of 50 per cent renewable energy share for all telecom towers by 2028 and 75 per cent by 2030, measured by energy consumption rather than merely by the number of towers with any renewable component. Second, priority deployment of solar-hybrid systems at the approximately 10,000 off-grid or unreliable-grid towers where diesel dependency is highest and the economic case for solar is strongest. Third, inclusion of green energy obligations in new tower company licensing conditions and in 5G spectrum allocation requirements.

India's telecom tower companies provide a relevant benchmark. Indus Towers (the largest tower company globally with over 220,000 towers) has committed to converting 85 per cent of its towers to renewable energy. American Tower Corporation's India operations have deployed solar and lithium-ion battery solutions across thousands of sites, reducing diesel consumption by up to 80 per cent at converted towers. In Bangladesh, the tower companies Edotco, Summit Towers, and BSCCL Tower can draw on these precedents, while SOF allocations could support renewable conversion at towers serving Universal Service Obligation areas.



5.7.2 E-Waste Management in the Telecom Sector

The rapid evolution of telecom technology generates substantial electronic waste: decommissioned base station equipment, replaced network hardware, retired handsets, and obsolete postal automation equipment. The National Green ICT Policy and the Telecommunications Network and Licensing Policy both mandate effective e-waste management, yet Bangladesh currently lacks a comprehensive e-waste regulatory framework specific to the telecom sector.

BTRC should develop a Telecom E-Waste Management Guideline that requires operators to: maintain inventories of all electronic equipment with projected end-of-life dates; ensure that decommissioned network equipment is processed through certified recycling facilities rather than entering informal waste streams; partner with BCSL for recovery of valuable materials (copper, aluminium, optical fibre) from decommissioned cables; and submit annual e-waste reports to BTRC. TSS could play a role in this ecosystem by establishing a refurbishment and recycling line at its Tongi facility, creating economic value from waste while building technical capabilities in electronics processing. The European Union's WEEE (Waste Electrical and Electronic Equipment) Directive provides the most comprehensive regulatory model, requiring producers to finance collection, treatment, and recovery of electronic waste.

Standardization for E-Waste Reduction: The PTD shall enforce a Universal/Single Charging and Data Transfer Cable standard for all mobile devices and handheld electronics sold in Bangladesh. This initiative, led by the BTRC in coordination with the Ministry of Commerce, aims to drastically reduce electronic waste (e-waste) by eliminating the need for multiple proprietary cables and chargers for every new device purchase.

Import Restrictions and Market Integrity: To ensure the success of this standard and protect the national digital infrastructure, the following regulatory measures will be implemented:

- **Ban on Substandard Imports:** The Ministry of Commerce and BTRC will implement a strict prohibition on the import of low-quality, non-compliant cables and auxiliary accessories from foreign vendors that do not meet **ITU/EU safety and signal-integrity standards**.
- **Digital Equipment Import Registry (DEIR):** All imported cables and adapters must be registered in the **National Equipment Identity Register (NEIR)** or a dedicated Digital Import Registry to ensure they meet the universal port specifications.
- **Auxiliary Cable Oversight:** Regulation will extend to all auxiliary cables, adapters, and chargers to prevent the dumping of obsolete or low-grade electronics in the domestic market, which contributes to rapid equipment failure and increased environmental toxicity.
- **Local Manufacturing Alignment:** Telephone Shilpa Sangstha (TSS) and Bangladesh Cable Shilpa Limited (BCSL) will serve as the domestic anchors for this standard, manufacturing high-quality, certified universal cables and adapters to provide a reliable local alternative to imports.

5.7.3 Green Manufacturing at BCSL

BCSL's Khulna manufacturing facility has been directed by PTD to pursue international environmental certification through the 'Green Certificate' initiative. The steps include applying to the Bangladesh Small and Cottage Industries Corporation (BSCIC) for certification, which requires meeting environmental standards for emissions, waste disposal, and energy efficiency. Additionally, the installation of solar panels at the BCSL facility has been identified as a priority for reducing grid dependency and qualifying for green certification.

The strategic benefit extends beyond environmental compliance. Green certification positions BCSL's products for export markets where environmental standards are increasingly a condition of procurement. The European Union, for example, requires compliance with environmental regulations for imported infrastructure components. BCSL's planned exports to Nepal and African countries would benefit from internationally recognised environmental certification. The target of obtaining green certification by July 2026, as directed by PTD, should be treated as a firm commitment with regular progress monitoring.

Table 5.4: Environmental Sustainability Policy Actions and KPIs

Policy Action	Lead Entity	Timeline	KPI / Target
Green Tower Directive: renewable energy targets	BTRC	Q2 2027	50% renewable share by 2028; 75% by 2030

Solar-hybrid deployment at off-grid towers	Tower companies / SOF	2028	10,000 off-grid towers converted
Telecom E-Waste Management Guideline	BTRC	Q4 2027	Annual e-waste reporting by all licensees
TSS e-waste recycling and refurbishment facility	TSS	2028	Facility operational; processing 500 tonnes/year
BCSL Green Certificate	BCSL	Jul 2026	International environmental certification obtained
BCSL solar panel installation	BCSL	Q4 2026	On-site renewable energy for factory operations
Green energy obligations in 5G spectrum allocation	BTRC	2027	Mandatory green provisions in 5G license conditions

5.8 Emerging Technology Adoption in Posts and Telecommunications

The National Digital Transformation Strategy addresses emerging technologies through a series of draft policies covering artificial intelligence, blockchain, drones, cloud computing, and semiconductors. For the posts and telecommunications sector, several of these technologies have specific applications that can improve operational efficiency, enhance regulatory effectiveness, and create new service capabilities. The key principle, drawn from the National Strategy, is that emerging technology adoption should be guided by risk-based assessment, aligned with the Legal Trinity of 2025, and focused on augmenting rather than replacing human capabilities.

5.8.1 Artificial Intelligence in Telecom Operations and Regulation

The National AI Policy establishes a risk-based classification framework with four tiers (prohibited, high-risk, limited-risk, low-risk) and designates the NDGA as the central regulatory body for AI governance. For the telecom sector, AI applications span both operational and regulatory domains.

On the operational side, AI can enhance network management through predictive maintenance (identifying equipment likely to fail before actual failure), dynamic spectrum allocation (optimising frequency usage in real time based on demand patterns), automated quality of service monitoring (detecting degradation before user complaints), and fraud detection in mobile financial services (identifying suspicious transaction patterns in real time). These applications fall into the limited-risk or low-risk categories under the National AI Policy and can be deployed under standard compliance requirements.

On the regulatory side, BTRC can leverage AI for automated spectrum monitoring (detecting unauthorised transmissions), market analysis (identifying anti-competitive behaviour through automated analysis of pricing and traffic data), and consumer complaint triage (classifying and routing complaints for faster resolution). These applications require transparency and human oversight but do not fall into the high-risk category unless they involve enforcement decisions affecting licensee rights, in which case the National AI Policy's mandatory Algorithmic Impact Assessment (AIA) requirement would apply.

For the postal sector, AI applications include intelligent parcel routing (optimising delivery sequences based on address data, traffic patterns, and delivery priorities), demand forecasting for financial services (predicting transaction volumes at individual post offices to optimise cash management), and automated mail sorting at the 14 Automated Mail Processing Centres that have been underutilised since the Post e-Centre project. These applications can revitalise dormant infrastructure by adding software intelligence to existing physical equipment.

5.8.2 Blockchain for Telecom and Postal Trust Infrastructure

The National Blockchain Policy 2026 focuses on sovereign, permissioned blockchain networks for public service delivery, explicitly distinguishing government blockchain applications from speculative cryptocurrency markets. For the posts and telecommunications sector, blockchain applications address specific trust and verification requirements.

First, the Postal Services Ordinance 2026 introduces digital postage stamps and digital postal marks. Blockchain provides the ideal verification infrastructure for these instruments: each digital stamp can be minted as a unique token on a permissioned blockchain, preventing counterfeiting and enabling real-time verification of postage validity. Second, the NAMS (National Address Management System) described in Chapter 4 could use blockchain to create an immutable registry of verified addresses, preventing unauthorised modifications and providing a trusted foundation for e-commerce logistics. Third, BTRC's licensing and compliance records could benefit from blockchain-based documentation, creating an auditable, tamper-resistant record of licensing decisions, spectrum assignments, and compliance certifications that addresses the transparency concerns identified in the PTD White Paper regarding inconsistent regulatory documentation.

5.8.3 Satellite and Non-Terrestrial Network Strategy

Bangladesh Satellite Company Limited (BSCL) manages the country's first geostationary satellite, Bangabandhu-1, launched in 2018 for television broadcasting, telecommunications, and emergency services. The National Digital Transformation Strategy identifies the need for satellite connectivity to complement terrestrial networks, particularly for disaster resilience, maritime communications, and connecting areas beyond the reach of terrestrial infrastructure.

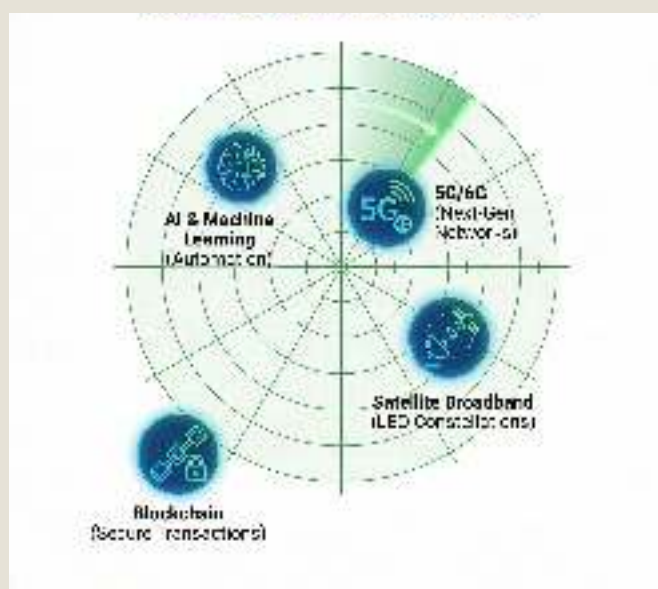
The rapid evolution of Low Earth Orbit (LEO) satellite constellations, particularly Starlink (SpaceX), OneWeb, and Amazon's Project Kuiper, creates both opportunity and regulatory challenge. These services can provide broadband connectivity to rural and maritime areas that terrestrial networks cannot economically serve. However, they also raise sovereignty concerns regarding data routing (traffic may transit through foreign ground stations), spectrum coordination (LEO constellations require careful management to avoid interference with existing satellite and terrestrial services), and market impact (unregulated entry could undermine investment in domestic infrastructure including BSCL's Bangabandhu-1).

The policy approach should balance access with sovereignty. BTRC should develop a Non-Terrestrial Network Service Provider (NTNSP) licensing category within the Telecommunications Network and Licensing Policy framework, with the following conditions: mandatory partnership with a local entity (BSCL being the preferred partner for government-related services); data routing through domestic gateways to ensure traffic visibility and lawful interception compliance; spectrum coordination through BTRC's existing frequency management framework; and a Universal Service contribution requirement that supports the cha

SOF's connectivity objectives. BSCL's strategic role should evolve from managing a single geostationary satellite to becoming the national coordinator for all non-terrestrial connectivity, including potential ground station operations for LEO services.

5.8.4 5G Readiness and Spectrum Strategy

The successful auction of the 700 MHz band in January 2026, where Grameenphone acquired 10 MHz of spectrum for approximately BDT 2,370 crore, demonstrated BTRC's capacity to execute complex spectrum transactions and resolve longstanding regulatory disputes. The 700 MHz band, with its long-range coverage and indoor penetration characteristics, is ideal for extending 4G/LTE coverage to underserved areas and laying the foundation for 5G deployment.



The 5G readiness strategy for PTD must address three dimensions. First, spectrum allocation: BTRC should develop a phased spectrum roadmap identifying mid-band (3.3-3.8 GHz) and high-band (millimetre wave) frequencies for 5G, with timeline, pricing methodology, and coverage obligations. Second, infrastructure preparation: the National Fiber Bank and BTCL's backbone network must be expanded to provide the dense fibre backhaul that 5G requires, particularly in urban areas where small cell deployment will supplement macro towers. Third, use-case development: rather than treating 5G as merely faster mobile broadband, PTD should identify specific applications where 5G's low latency and massive machine connectivity capabilities enable transformative services, including remote healthcare (supporting the telemedicine targets in Chapter 4), smart port operations at Chittagong and Payra (supporting logistics efficiency), and industrial IoT in export processing zones (supporting economic competitiveness).

Teletalk's role in 5G strategy is particularly important. As the state-owned mobile operator, Teletalk can serve as the 5G testbed for government applications, deploying pilot networks in controlled environments such as the Tongi TSS facility, government office complexes, and port areas before commercial-scale deployment. This approach, similar to South Korea's strategy where Korea Telecom (KT) deployed 5G first for industrial and government applications before consumer services, allows the technology to be proven in practical settings while building domestic expertise.

Table 5.5: Emerging Technology Policy Actions and KPIs

Policy Action	Lead Entity	Timeline	KPI / Target
AI-based network management pilot at BTCL/Teletalk	BTCL / Teletalk	Q4 2027	Predictive maintenance reducing downtime by 25%
AI spectrum monitoring at BTRC	BTRC	2028	Automated detection covering 90% of assigned bands
Blockchain digital stamp verification	Postal Directorate	Q4 2027	Digital stamp system operational on permissioned chain
NTNSP licensing framework	BTRC	Q2 2027	Licensing category published; BSCL partnership mandate
5G spectrum roadmap (mid-band and mmWave)	BTRC	Q4 2027	Roadmap published with timeline, pricing, obligations
Teletalk 5G testbed deployment	Teletalk	2028	Pilot networks at 3+ government/industrial sites
AI-based postal parcel routing	Postal Directorate	2028	Intelligent routing at 5+ major sorting centres
BSCL LEO ground station feasibility	BSCL	Q4 2027	Feasibility study completed; partnership framework

5.9 Digital Skills and Workforce Development for the Telecom Sector

The National Digital Transformation Strategy targets a workforce of 7 to 8 million ICT professionals by 2030 and identifies the BEAR framework (Biotech, Electronics, AI, Robotics) as the priority skills direction. For the posts and telecommunications sector, workforce development addresses two distinct challenges: upgrading the skills of existing employees within PTD entities, and building the broader pool of telecom professionals that the sector requires.

5.9.1 PTD Entity Workforce Modernisation

The workforce challenges across PTD entities are well documented. BCSL operates with only 34 of 63 sanctioned officer positions filled and 165 of 341 regular employee positions occupied. TSS faces similar constraints. BTCL, managing a legacy network that requires both traditional telecom engineering and modern IP networking skills, needs to retrain its existing workforce while recruiting new talent in areas such as fibre optic network design, IP core engineering, and cybersecurity. The Postal Directorate's 40,000-person workforce must develop digital literacy, logistics technology skills, and financial services competencies to operate the transformed post offices envisioned in Chapter 4.

The strategy requires a Sectoral Skills Development Plan coordinated by DoT with the following components. First, an immediate vacancy filling programme for BCSL, TSS, and other PTD entities where critical positions remain unfilled, targeting 80 per cent occupancy of sanctioned positions by Q4 2026. Second, a mandatory digital competency certification programme for all PTD employees, delivered in partnership with the Bangladesh Telecommunications Academy (if revitalised as proposed in Chapter 6) or through external training providers, with completion targets of 50 per cent by 2027 and 100 per cent by 2029. Third, a specialised cybersecurity training programme for the estimated 30 to 50 professionals needed to staff the CIRTs at each PTD entity, potentially supported by SOF allocations for cybersecurity capacity building. Fourth, a postal workforce digital literacy programme aligned with the Nagorik Sheba hub model, ensuring that staff at the 1,000 post offices targeted for transformation can operate digital platforms, provide citizen assistance, and handle MFS transactions securely.

5.9.2 Telecom Sector R&D and Innovation Ecosystem

The PTD White Paper recommends creating an in-house BTRC R&D capability and funding external R&D through SOF allocations for technologies relevant to digital public services. This recommendation addresses a significant gap: Bangladesh currently has minimal telecom R&D capacity, with most innovation arriving embedded in imported equipment rather than being developed domestically.

The strategy proposes a Telecom Innovation Fund, financed from SOF allocations (a dedicated percentage of annual SOF collections), to support applied research in areas directly relevant to PTD's mission. Priority research areas include: rural broadband solutions optimised for Bangladesh's geography (riverine terrain, flood-prone areas, dispersed settlements); Bangla natural language processing for automated customer service and postal applications; low-cost smart device design for the Nagorik Sheba ecosystem; environmental monitoring sensors using telecom infrastructure (leveraging tower sites for weather, air quality, and flood sensors); and postal logistics optimisation algorithms. The fund should operate through competitive grants to universities, startups, and research institutions, with mandatory partnership with a PTD entity to ensure practical relevance and technology transfer.

Additionally, the establishment of a nationally accredited SAR (Specific Absorption Rate) testing and certification laboratory under BTRC, as recommended in the PTD White Paper, would serve dual purposes: enabling local verification of mobile handset compliance (reducing reliance on foreign labs and preventing falsified reports) and building domestic expertise in radio frequency engineering and electromagnetic compatibility testing. This laboratory could eventually expand to provide type-approval testing services for the broader South Asian market, generating revenue while building regional technical leadership.

5.10 Chapter Summary: From National Policy to Sectoral Action

This chapter has translated the National Digital Transformation Strategy's policy framework into specific, measurable obligations for the posts and telecommunications sector. Six policy domains have been addressed, each with identified gaps, prescribed actions, lead entities, timelines, and performance indicators. The policies are not standalone; they form an integrated framework where digital sovereignty enables effective cybersecurity, data protection builds citizen trust that supports service adoption, local manufacturing reduces dependency that threatens sovereignty, environmental sustainability ensures long-term viability, and emerging technologies provide the tools to achieve each objective more effectively.

Three cross-cutting themes emerge from the analysis. First, the principle of sovereign control: from bandwidth reserves to source code ownership to trusted equipment frameworks, the recurring requirement is that Bangladesh maintains effective control over the infrastructure, data, and systems on which its digital future depends. This does not mean isolation from global technology markets; it means ensuring that engagement with global markets occurs on terms that protect national interests. Second, the principle of governance maturity: the Nagad experience, the Post e-Centre failure, and the PTD White Paper's documentation of regulatory inconsistencies all demonstrate that policy ambition without institutional capacity and governance discipline produces waste rather than progress. Every policy proposed in this chapter requires the institutional reforms described in Chapter 6 and the project governance framework described in Chapter 7 to be implemented effectively. Third, the principle of measurable accountability: every policy action has been assigned a specific KPI, timeline, and lead entity, creating a framework for monitoring that previous reform efforts lacked.

Table 5.6: Consolidated Chapter 5 Reform Actions

Policy Domain	Number of Actions	Key Milestones	Chapter Dependencies
Digital Sovereignty	6 actions	Vendor audit Q4 2026; Bandwidth Reserve Q4 2026; Open APIs Q2 2027; NSCR registration Q4 2027	Ch. 3 (infrastructure), Ch. 6 (institutional capacity)
Cybersecurity and CII	7 actions	CII designation Q2 2027; CIRTs Q4 2027; ISO 27001 2028; Trusted Equipment Q4 2027	Ch. 2 (CSO legal basis), Ch. 6 (NCSA coordination)
Data Protection	PDPO compliance + MFS governance	Telecom Data Protection Directive Q2 2027; MFS DPIA regime 2027; Formal service contracts 2027	Ch. 2 (PDPO), Ch. 4 (Nagad governance lessons)
Local Manufacturing	8 actions	Domestic content 40% by 2027; BCSL 50K km by 2030; TSS JV Q4 2027; BCSL export 2028	Ch. 3 (fibre deployment), Ch. 7 (project governance)

Environmental Sustainability	7 actions	Green Tower Directive Q2 2027; 50% renewable 2028; BCSL Green Certificate Jul 2026	Ch. 3 (tower infrastructure), Ch. 9 (roadmap)
Emerging Technology	8 actions	AI pilots Q4 2027; 5G roadmap Q4 2027; NTN5P licensing Q2 2027; Blockchain stamps Q4 2027	Ch. 3 (spectrum), Ch. 4 (postal services)
Skills and Workforce	Vacancy filling + training + R&D	80% vacancies filled Q4 2026; Digital certification 100% by 2029; SAR lab 2028; Telecom Innovation Fund 2027	Ch. 6 (institutional reform), Ch. 9 (timeline)

The implementation of these policies requires coordinated action across all PTD entities under the strategic direction of the Division and the operational coordination of the Department of Telecommunications. Chapter 6 addresses the institutional reforms necessary to build this coordination capacity, Chapter 7 provides the project governance framework to prevent repetition of past implementation failures, and Chapter 9 integrates all policy actions into a consolidated implementation roadmap with quarterly milestones through 2030.

— End of Chapter 5 —

6. Institutional Reform

6.0 Introduction: The Imperative for Institutional Transformation

The Posts and Telecommunications Division (PTD) oversees an institutional ecosystem comprising ten affiliated entities, each carrying distinct mandates that span regulation, infrastructure operation, manufacturing, satellite services, and postal logistics. These entities were established at different historical junctures, often inheriting organizational cultures, staffing patterns, and operational frameworks from predecessor bodies such as the Bangladesh Telegraph and Telephone Board (BTTB). Over time, many have been subject to governance fragmentation, political patronage in appointments, overlapping mandates, and a persistent disconnect between policy commitments and operational execution.

The institutional reform agenda articulated in this chapter is grounded in a simple but urgent premise: the digital transformation of Bangladesh's posts and telecommunications sector cannot advance through legislative or regulatory reform alone. Without corresponding institutional modernization at the entity level, new laws, policies, and infrastructure investments risk being absorbed into structures that lack the capacity, accountability, or commercial orientation to deliver results. The PTD Whitepaper Task Force findings of April 2025 confirm that recurring patterns of misgovernance, accountability gaps, and strategic neglect have impaired the ability of state owned operators to compete effectively, eroded public trust, and depleted institutional memory.



This chapter presents a structured, entity by entity reform programme that translates the strategic policy directions of Chapters 2 through 5 into concrete institutional actions. Each entity section follows a consistent analytical framework: a brief diagnosis of institutional challenges, a catalogue of reform initiatives currently underway or recently completed (drawn from official progress reports dated through January 2026), a forward looking improvement agenda, and a set of measurable Key Performance Indicators (KPIs) with defined timelines. Where relevant, international benchmarks from comparable jurisdictions are referenced to contextualize the reform trajectory.

The overarching institutional reform philosophy rests on three pillars. First, commercial viability: state owned entities must transition from subsidy dependent operations toward self sustaining, revenue generating models that justify continued public ownership. Second, governance integrity: board composition, appointment processes, audit compliance, and oversight mechanisms must be restructured to eliminate patronage, enforce transparency, and restore meritocratic progression. Third, strategic complementarity: rather than competing with private sector operators on unfavourable terms, PTD entities should position themselves as essential infrastructure partners, niche service providers, or sovereign capability anchors in domains where private provision alone is insufficient. Countries such as Singapore (with Singtel's transformation from a state monopoly to a globally competitive operator), South Korea (with Korea Telecom's post privatization modernization), and Rwanda (with the Rwanda Utilities Regulatory Authority's performance based governance model) offer instructive precedents for the kind of institutional evolution envisioned here.

6.1 Bangladesh Telecommunication Regulatory Commission (BTRC)

6.1.1 Rationale for Reform

BTRC occupies the pivotal position in Bangladesh's telecommunications governance architecture as the sole sectoral regulator. Its effectiveness determines the quality of spectrum management, the integrity of licensing processes, the enforcement of quality of service standards, and the credibility of consumer protection mechanisms. The Telecom Ordinance 2025 has significantly expanded BTRC's regulatory perimeter to encompass OTT platforms, artificial intelligence governance, and social media oversight, while simultaneously introducing technology neutral licensing and strengthened enforcement provisions. This expanded mandate requires a corresponding institutional upgrade in BTRC's technical capacity, operational independence, and data driven decision making capabilities.

Between March 2025 and January 2026, BTRC has undertaken several consequential reforms under PTD direction. These include the successful completion of the 700 MHz spectrum auction on 14 January 2026, which generated approximately BDT 2,370 crore in revenue through Grameenphone's acquisition of 10 MHz and resolved a legal impasse spanning two decades. The Commission has also advanced the Automated Spectrum Management System (ASMS) and Telecom Monitoring System (TMS) for real time spectrum and quality monitoring, implemented a reformed licensing framework with one stop service provisions, and operationalized the National Equipment Identity Register (NEIR) to combat illegal and cloned handsets. These achievements represent a marked departure from the regulatory inertia that characterized earlier periods.

6.1.2 Reform Initiatives and Current Status

Initiative	Current Status	Timeline	Expected Outcome
Spectrum Management (ASMS) and Spectrum Pricing Rationalization	700 MHz band 25 MHz auction completed; ASMS and TMS enabling real time monitoring; global benchmarking of spectrum pricing underway	2027	Re farming of unused spectrum for additional auction revenue; transparent, formula based pricing
Telecom Ordinance 2025 Implementation	Ordinance enacted; OTT, AI, and social media brought under regulatory framework; enforcement mechanisms being operationalized	2025 to 2026	Legal foundation for digital sovereignty and cybersecurity enforcement
New Telecom Licensing Policy and Guidelines 2025	One stop service launched; technology neutrality principle embedded; entry barriers reduced for ISPs and district level operators	2025	Increased FDI attraction; local entrepreneurship in fixed telecom services

Social Obligation Fund (SOF) Reform	SOF financing connectivity in underserved areas and digital literacy; smart device subsidy programme under consideration	2026 to 2027	Targeted digital inclusion for marginal communities
NEIR Operationalization	Fully operational; illegal and clone handset control ongoing; revenue leakage prevention active	Completed	Handset market integrity; national security strengthening; local manufacturing protection
6G Research Cell and NGSO Policy	ITU guideline alignment initiated; NGSO satellite internet policy framework under development	2027 to 2028	Future readiness for next generation connectivity; satellite internet access for remote populations
Quality of Service (QoS) Enforcement	TMS based regular audits being institutionalized; call drop monitoring and broadband speed verification ongoing	Ongoing	Measurable improvement in consumer experience; operator accountability

6.1.3 Institutional Strengthening Agenda

BTRC's institutional reform trajectory must address four interconnected dimensions. First, regulatory autonomy must be reinforced through legal amendments that reduce ministerial pre clearance requirements and align BTRC's governance model with international independent regulators such as India's TRAI, the United Kingdom's Ofcom, or Australia's ACMA. This includes reconstituting the Commission's board to include a majority of independent members with demonstrated sectoral expertise and establishing fixed, non renewable terms for Commissioners.

Second, BTRC must develop specialized technical units for emerging regulatory domains. The expansion of the regulatory perimeter to OTT, AI, and platform governance requires recruitment of professionals with backgrounds in data science, competition economics, and digital rights, not merely traditional telecommunications engineering. Singapore's Infocomm Media Development Authority (IMDA) offers a useful model, having established dedicated units for platform regulation, data governance, and AI ethics within a converged regulatory structure.

Third, the SOF must be fundamentally reformed from a largely dormant pool of unspent resources into an active, outcomes driven investment vehicle. International experience with universal service funds in Malaysia (USP Fund), Colombia (Fondo de Tecnologías de la Información y las Comunicaciones), and India (USOF) demonstrates that transparent project selection criteria, independent evaluation, and published impact assessments are essential to fund credibility. SOF resources should be redirected toward smart device subsidies for students and marginal farmers, community broadband in underserved areas, and digital literacy programmes with measurable completion rates.

Fourth, BTRC should establish a formal 6G Research Cell in coordination with ITU guidelines, engaging universities, research institutions, and private operators in anticipatory regulatory research. South Korea's early establishment of a 6G research fund in 2020, well before any commercial deployment, positioned the country to lead international standards development. Bangladesh need not match that scale of investment but should ensure that regulatory awareness of emerging technologies does not lag behind commercial deployment by a decade, as occurred with 3G and 4G.

6.1.4 KPI Framework

Reform Area	KPI	Baseline (2025)	Target	Timeline
Spectrum Revenue	Annual spectrum auction and re farming revenue	BDT 2,370 crore (2026)	BDT 5,000 crore cumulative	2028
Licensing Efficiency	Average licensing processing time	60+ days	15 business days	2027
QoS Compliance	Operator compliance rate with QoS benchmarks	Below 70%	90%	2028
SOF Deployment	SOF fund utilization rate	Below 40%	85%	2027
NEIR Coverage	Percentage of handsets registered	70%	98%	2027
Digital Inclusion	Underserved areas with broadband access via SOF	Limited	500 union parishads	2028

6.2 Bangladesh Telecommunications Company Limited (BTCL)

6.2.1 Rationale for Reform

BTCL represents both the most significant institutional challenge and the most substantial reform opportunity within the PTD ecosystem. Inherited from the corporatization of the Bangladesh Telegraph and Telephone Board (BTTB), BTCL possesses a nationwide fixed line network, extensive duct and fiber optic infrastructure, valuable urban real estate including prime locations in Mohakhali and across 34 district headquarters, and an established brand in the .bd domain name space. Yet for over a decade, these assets have been dramatically underutilized. Legacy dependence on declining landline revenues, institutional inertia, and governance weaknesses identified in the PTD Whitepaper Task Force findings left BTCL operationally stagnant while private operators captured the growth segments of mobile voice, data, and broadband.

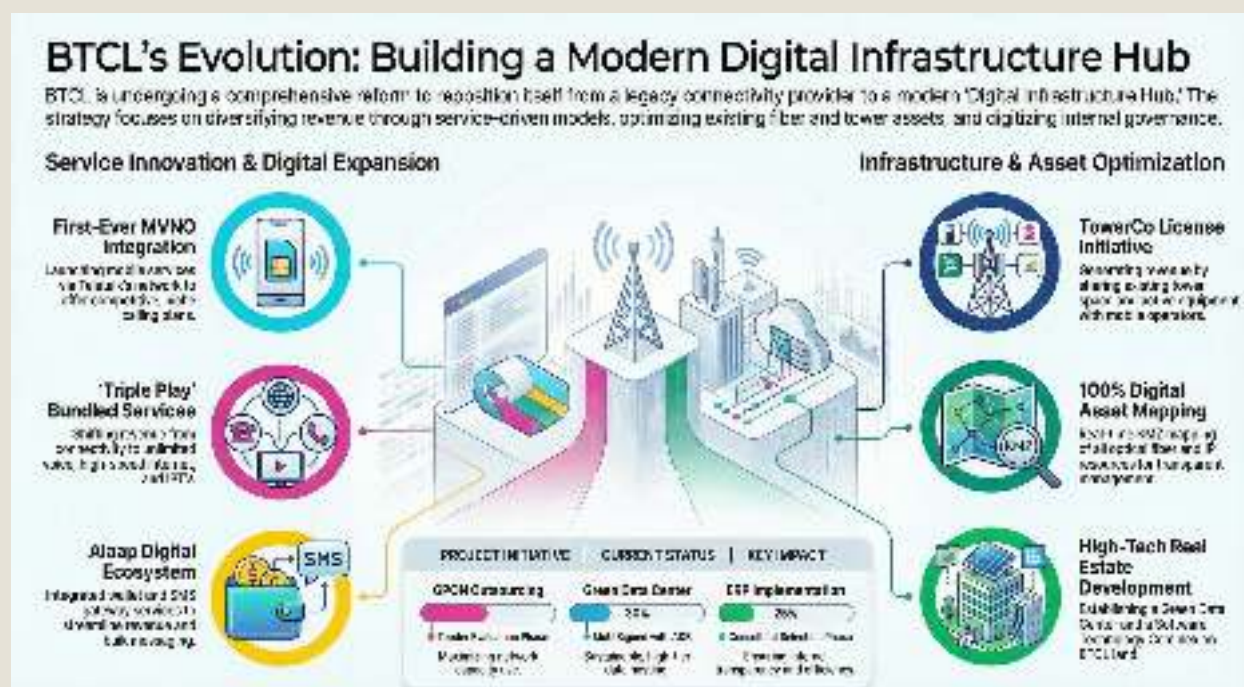
The reform programme initiated in 2025 is designed to transform BTCL from a legacy landline operator into a commercially viable Digital Infrastructure Hub. The strategic logic is straightforward: BTCL should not attempt to compete head to head with private mobile operators in consumer services, where it lacks scale and brand equity. Instead, it should leverage its unique physical assets, particularly its fiber network, duct infrastructure, tower sites, and real estate, to become an essential wholesale partner and infrastructure enabler for the broader telecommunications ecosystem. This mirrors the trajectory of entities such as Chorus Limited in New Zealand, which was structurally separated from Spark and repositioned as a wholesale fiber network provider, or TelkomSA's Openserve division in South Africa.

6.2.2 Reform Initiatives and Current Status

Initiative	Current Status	Timeline	Expected Outcome
GPON Outsourcing for Capacity Monetization	Service provider selection tender under evaluation; target for completion by November 2025; outsourced model to maximize utilization of existing GPON capacity	November 2025	Revenue generation from previously idle fiber capacity; reduced operational burden through outsourcing
TowerCo License Application	Application submitted to BTRC on 16 February 2025; enabling tower space and active sharing with mobile operators	Pending BTRC approval	New recurring revenue stream; strengthened mobile backbone infrastructure; reduced operator CAPEX
IP Phone and Call Centre Services	Session Border Controller and Digital Services Platform procurement completed; testing ongoing with target launch by January 2026	January 2026	Hosted IP phone and cloud call centre services for enterprise clients

Nagorik Sheba (Citizen Service) Centres	Operational at Gulshan, Uttara, and Nilkhet exchanges since September 2025; repurposing underutilized exchange buildings for public services	Operational	Citizen services access; entrepreneurship support; revenue from facility utilization
Domain Reseller Programme	Six resellers registered as of October 2025; DNS migration to new servers ongoing; domain pricing reduced	Ongoing	Expanded .bd domain ecosystem; reduced per domain cost for end users
Alaap Application Enhancement	Wallet service launched October 2024; SMS gateway launched October 2025; enabling bill payment and bulk SMS services	Operational	Digital payment integration; new revenue from bulk SMS; improved customer convenience
Infrastructure Sharing MoUs	Draft MoU prepared for execution with PGCB, Bangladesh Railway, and BIWTA for fiber resource sharing; target by July 2025	July 2025	Shared fiber corridors; reduced duplication of infrastructure investment
MVNO with Teletalk	Technical integration underway; billing and provisioning links established; 500 SIM pilot prepared	Awaiting launch	Bangladesh's first MVNO service; youth market capture; new mobile revenue for BTCL
IPTV Service Launch	Tender process determination referred to BPPA for approval guidance; expected launch alongside GPON	January 2026	Triple play bundled service offering (voice, data, television)
Green Data Centre (Hathazari, Chattogram)	MoU signed on 26 January 2025 among BTCL, PTD, PPP Authority, and Asian Development Bank for PPP based facility	January 2025 MoU signed	Tier III data centre; cloud hosting revenue; disaster recovery services
Digital Mapping and Inventory Portal	Transmission digital map live at btcltip.net.bd; immovable property digital inventory portal prepared	Completed	Transparency; asset protection; simplified maintenance planning
Fiber Optic Expansion (30 Districts)	Multiple phases covering 30 districts; feasibility studies ongoing	Ongoing	Extended fiber backbone; last mile connectivity enablement

Mohakhali Telecom and Software Technology Complex (BTSTC)	Feasibility study PFS restructured per Planning Commission review (28 October 2025); RADP inclusion for FY 2025 to 2026 submitted	FY 2025 to 2026	Technology innovation hub; software industry cluster; commercial real estate revenue
District Telecom Infrastructure Modernization (Phase 1, 34 Districts)	Included in ADP green list for FY 2025 to 2026; Finance Division clearance on GOB financing nature sought	FY 2025 to 2026	Co location services; modernized district level connectivity infrastructure
ERP Software Implementation	Consultant appointment tender issued for specification development; customized ERP to follow	January 2026	Internal process digitization; financial transparency; operational efficiency



6.2.3 Strategic Improvement Agenda

The current reform trajectory is commendable in its ambition and scope, but several areas require accelerated attention to ensure that BTCL's transformation is irreversible and commercially sustainable. The first priority is completing the TowerCo licensing process. The global tower company model, exemplified by entities such as American Tower, Helios Towers in Africa, or edotco in Southeast Asia, generates stable, long term rental income with high margins. BTCL's existing tower portfolio, once licensed and professionally managed, can generate predictable revenue that reduces dependence on volatile retail services.

The second priority is ensuring that the Hathazari Green Data Centre proceeds on the PPP timeline without the procurement delays that have afflicted previous BTCL projects. The ADB's involvement provides technical assurance, but BTCL must invest in developing internal data centre management competencies through structured training programmes and, potentially, operational partnerships with established data centre operators such as Equinix, Digital Realty, or regional providers with emerging market experience.



Third, the MVNO partnership with Teletalk must be launched expeditiously. The concept of BTCL offering mobile services through Teletalk's network is strategically sound, but extended delays risk diminishing the market window. International MVNO models, such as Lebara or Lycamobile in Europe, demonstrate that MVNOs succeed through differentiated pricing and targeted demographic appeal rather than network scale. BTCL's MVNO should be positioned specifically for enterprise clients, government users, and the diaspora community, segments where brand trust in a state owned entity is an asset rather than a liability.

Fourth, the digital mapping and property inventory portal, now operationally live, must be protected through legal mechanisms that prevent encroachment, unauthorized occupation, or disposal of BTCL's immovable assets. International precedents in Nigeria and India, where state owned telecom land was progressively appropriated during periods of institutional weakness, underscore the importance of proactive asset protection. BTCL should pursue formal land title registration, GIS based boundary demarcation, and periodic physical verification aligned with Comptroller and Auditor General audit cycles.

Finally, BTCL will transition to an AI-assisted performance management system to monitor network health in real-time. A unified Customer Complaint Management System will be integrated with data analytics to identify recurring failure points, enabling proactive resolution of customer issues. This shift will be governed by strict Key Quality Indicators (KQI) and Quality of Service (QoS) benchmarks, with customer satisfaction directly linked to departmental performance evaluations.

6.2.4 KPI Framework

Reform Area	KPI	Baseline (2025)	Target	Timeline
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Revenue Diversification	Non landline revenue as percentage of total	Below 20%	50%	2028
GPON Utilization	Active GPON connections via outsourced model	Minimal	100,000 connections	2027
Tower Revenue	Annual TowerCo rental income	Zero	BDT 200 crore	2028
Data Centre	Hathazari Green Data Centre operational status	MoU stage	Tier III operational	2028
MVNO Subscribers	Active MVNO SIM connections	Zero	50,000	2027
Digital Inventory	Percentage of assets digitally mapped and verified	Partial	100%	2026
ERP Deployment	Modules operational across BTCL offices	Zero	All core modules live	2027
District Infrastructure	Districts with modernized co location facilities	Zero	34 districts	2028

6.3 Teletalk Bangladesh Limited

6.3.1 Rationale for Reform

Teletalk Bangladesh Limited occupies a unique but precarious position as the sole state owned mobile operator in a market dominated by four private operators (Grameenphone, Robi, Banglalink, and newly licensed entrants). With a subscriber base representing a modest fraction of the total mobile market, Teletalk's commercial viability has long been questioned. The PTD Whitepaper documents persistent challenges including delayed 4G rollout, inadequate network coverage, dependence on government and foreign funded projects for capital expenditure, and governance weaknesses in procurement and human resource management.

Nevertheless, Teletalk serves critical sovereign functions that private operators cannot be relied upon to fulfil. These include secure government communications, biometric SIM verification infrastructure (the National Identification Database interface), coverage obligations in commercially unviable areas such as the Chittagong Hill Tracts and border regions, and the provision of mobile connectivity for government employees and defence forces. The reform agenda for Teletalk must therefore balance two objectives: improving commercial

competitiveness to reduce fiscal dependence, while strengthening Teletalk's role as a sovereign communications backbone. Norway's Telenor (prior to full privatization) and India's BSNL restructuring (2019 revival package) offer relevant precedents for reforming state owned mobile operators.

6.3.2 Reform Initiatives and Current Status

Initiative	Current Status	Timeline	Expected Outcome
VoLTE Service Launch	Limited rollout completed; preparation for full subscriber base rollout underway	February 2026	Improved voice call quality for LTE enabled devices; enhanced customer satisfaction
VoWiFi Service Launch	Limited rollout completed; full deployment preparation ongoing	February 2026	Extended indoor coverage; reduced call drops in weak signal areas
Tower Sharing (2,300 Sites)	Lease agreement signed November 2025; 1,720 site surveys ongoing; 95 sites under equipment installation; 15 sites live	December 2026	Reduced CAPEX for new site deployment; accelerated network expansion; operational cost reduction
Battery Capacity Expansion	1,420 sites modernized under various projects; additional 350 sites under modernization	March 2026	Improved network availability; reduced downtime from power disruptions
4G Coverage Expansion	1,699 new 4G BTS sites activated; daily data usage increased from 354 TB to 464 TB (20% increase); 3,600 additional sites planned	June 2027	85% of existing sites 4G enabled; 20% increase in 4G subscriber capacity
BTCL MVNO SIM	Billing and provisioning links established; three packages designed; 500 SIM pilot supply prepared	Awaiting launch	BTCL mobile market entry; revenue sharing between state entities
5G BTS Launch	Four 5G sites successfully launched in November 2025	Completed	Technology readiness demonstration; regulatory and technical validation for 5G
New ADP Projects (Remote Areas)	Three project DPPs prepared for ADP green list: remote area coverage, public university connectivity, and highway corridor 4G	April 2026	Digital inclusion in underserved areas; educational connectivity; highway safety communications

Sales and Marketing Acceleration	Cloud Phone service launched December 2025; MFS recharge partnerships with bKash and others; SIM distribution via Post Offices launched	Completed	Revenue growth; diversified distribution channels; improved customer accessibility
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6.3.3 Strategic Improvement Agenda

Teletalk's most critical near term challenge is the tower sharing programme. The agreement to lease 2,300 sites from a tower company represents a fundamental shift from BTCL-era asset ownership models toward the shared infrastructure approach that has proven cost effective across emerging markets in Africa and Southeast Asia. The conversion of lease agreements into operational sites must be accelerated, with clear monthly targets, a dedicated project management unit, and penalty clauses for delayed handover. edotco Group's tower sharing operations in Malaysia and Myanmar provide a relevant regional model for the operational framework Teletalk should adopt.

The 5G demonstration at four sites is symbolically important but must be followed by a realistic 5G deployment strategy that does not repeat the delays of 4G rollout. Given Teletalk's financial constraints, 5G deployment should initially target high value use cases (enterprise customers, industrial IoT zones, university campuses) rather than competing for mass consumer 5G coverage, which private operators with larger spectrum holdings and deeper capital reserves will inevitably provide first. India's BSNL adopted a similar phased approach after its 2019 revival package, prioritizing 4G network completion before selective 5G deployment.

The distribution partnership with Bangladesh Post for SIM sales through post offices is an innovative cross entity synergy that should be expanded. With over 8,500 post offices nationwide, this channel provides unmatched rural reach. However, success depends on training postal staff in SIM activation procedures, maintaining adequate SIM inventory at post offices, and integrating the distribution data into Teletalk's customer management systems. Japan Post's partnership with mobile operators for device and plan distribution at post offices offers a proven model for this approach.

6.3.4 KPI Framework

Reform Area	KPI	Baseline (2025)	Target	Timeline
Network Expansion	4G population coverage percentage	Approximately 60%	85%	2027
Tower Sharing	Operational shared tower sites	15 sites	2,300 sites	December 2026
Revenue Growth	Year on year revenue increase	Modest growth	15% annual growth	2027

Data Usage	Daily data consumption on network	464 TB	700 TB	2027
Distribution Channels	Post offices actively selling Teletalk SIMs	Pilot stage	1,000 post offices	2027
5G Readiness	5G enabled sites in priority zones	4 sites	50 sites	2028
Customer Satisfaction	VoLTE and VoWiFi adoption rate	Limited	40% of active subscribers	2027

6.4 Bangladesh Submarine Cable Company PLC (BSCPLC)

6.4.1 Rationale for Reform

BSCPLC operates Bangladesh's submarine cable landing stations and manages the country's international internet bandwidth connectivity. As the sole state owned international connectivity service provider, BSCPLC's pricing, service quality, and capacity planning directly affect the cost and reliability of internet access for every citizen and enterprise in Bangladesh. The company manages connections through the SEA ME WE 4 and SEA ME WE 5 submarine cable systems and provides IPLC, IP transit, co location, port services, and NIX (National Internet Exchange) services to domestic operators and international carriers.

The reform programme has focused on two immediate objectives: reducing bandwidth prices to lower end user internet costs, and strengthening financial discipline through aggressive dues recovery. Between March 2025 and January 2026, BSCPLC has reduced bandwidth prices by 30 to 50 percent effective 1 April 2025, introduced customer friendly packages for IPLC and IP transit clients, reduced port charges by at least 10 percent, and launched a systematic dues recovery programme targeting major defaulters. These price reductions, while commercially significant, bring Bangladesh closer to regional parity with India and Vietnam, though further reductions remain necessary to achieve the national target of affordable broadband for all.

6.4.2 Reform Initiatives and Current Status

Initiative	Current Status	Timeline	Expected Outcome
Tariff and Package Rationalization	New tariff effective 1 April 2025; bandwidth price reduced 30 to 50%; promotional packages for IPLC customers including three free 10G circuits per 100G purchase on SMW4	Completed	Consumer level internet cost reduction; increased competitiveness with ITC providers
Service Simplification and Client Engagement	Responsibility matrix developed; customer consultation held 27 February 2025; 24 hour upgrade commitment on demand note payment; 7 day wet segment activation promise	Ongoing	Faster service delivery; improved client relationships; reduced service complaints
Dues Recovery and Financial Discipline	Major defaulter list prepared (including Amra Technologies, Mango Teleservices); BTRC notified; active pursuit of outstanding payments	Ongoing	Strengthened balance sheet; improved cash flow; reduced bad debt exposure

Bandwidth Usage Ratio Enforcement	BTRC directed 50:50 submarine to ITC bandwidth ratio; Windstream Communications Ltd obtained High Court stay on 31 July 2025; BSCPLC pursuing ratio reinstatement	Under legal review	Level playing field between submarine and ITC bandwidth; market discipline
Port Charge Reduction	Port charges reduced minimum 10% effective 1 April 2025	Completed	Lower interconnection costs for operators; downstream savings for consumers
ITC Monitoring and Licensing Compliance	Recommendations submitted for accurate ITC bandwidth import measurement and strict telecom licensing policy enforcement	Pending regulatory action	Market transparency; fair competition; revenue protection for licensed operators

6.4.3 Strategic Improvement Agenda

BSCPLC's most pressing institutional challenge is the bandwidth usage ratio dispute. The principle that a balanced ratio between submarine cable and International Terrestrial Cable (ITC) bandwidth protects the viability of the state owned submarine cable investment is strategically sound, but the legal challenge from Windstream Communications demonstrates the need for a regulatory framework that can withstand judicial scrutiny. BTRC should formalize the ratio policy through a transparent regulatory determination process, supported by published cost studies and public consultation, rather than relying solely on administrative directives. International precedents from the European Union's Significant Market Power (SMP) framework and India's interconnection usage charge regime demonstrate how such access obligations can be legally robust when grounded in transparent cost modelling.

Second, BSCPLC must prepare for the anticipated growth in bandwidth demand driven by 5G deployment, expanding OTT consumption, cloud migration, and enterprise digitization. Capacity planning for SEA ME WE 6 participation and potential alternative routing through additional cable systems (such as proposed Bay of Bengal cable projects) should be initiated to ensure that Bangladesh is not constrained by single point of failure submarine cable dependencies. Japan's NICT and Singapore's IMDA have both invested in submarine cable diversity as a matter of national digital resilience.

Third, the tax and duty burden on telecommunications, including supplementary duty, VAT, and surcharges, is creating downstream pricing pressure that undermines the government's own objective of affordable internet. BSCPLC, as the primary wholesale bandwidth provider, is well positioned to provide evidence-based analysis to the Ministry of Finance and National Board of Revenue on the impact of fiscal policies on bandwidth pricing. A sector specific tax rationalization study, benchmarked against regional competitors, should be commissioned jointly by BSCPLC and BTRC.

Additionally, the following reforms must be introduced:

Landing Station Modernization: Introduce data centers, cloud hosting, cache storage, and edge routers at cable landing stations to increase domestic digital services.

Infrastructure Redundancy: Ensure land connectivity redundancy between Kuakata and Cox's Bazar through BTCL/PGCB fiber to divert capacity if a landing station fails.

Operational Governance: Conduct biannual tariff coordination and stakeholder meetings. Ensure 100% payment security and zero-default protocols for all customers.

Vendor Independence: Implement a technology transfer protocol with vendors/OEMs to reduce long-term dependency and enhance local technical handover.

6.4.4 KPI Framework

Reform Area	KPI	Baseline (2025)	Target	Timeline
Bandwidth Pricing	Average per Mbps wholesale price reduction	30 to 50% reduction achieved	Additional 15% reduction	2027
Client Satisfaction	Service activation turnaround time	Variable	24 hours post payment	2026
Dues Recovery	Outstanding dues recovered as percentage of total	Recovery initiated	80% of identified defaults	2027
Capacity Planning	SEA ME WE 6 or alternative cable investment decision	Not initiated	Investment commitment	2027
Revenue Growth	Annual revenue from bandwidth services	Baseline 2025	20% cumulative increase	2028

6.5 Bangladesh Post Office

6.5.1 Rationale for Reform

Bangladesh Post Office is arguably the PTD entity with the greatest untapped potential for transformative impact on citizen services. With a network of over 8,500 post offices reaching into the most remote villages, a workforce with deep community trust, and physical infrastructure in every upazila, the postal network represents a pre existing national service delivery platform that no private logistics company can replicate at equivalent scale. Globally, the postal reform movement initiated by the Universal Postal Union (UPU) in the early 2000s has demonstrated that modernized postal services can serve as platforms for financial inclusion, e commerce logistics, government service delivery, and rural entrepreneurship. Countries as diverse as Brazil (Correios), India (India Post), Kenya (Posta Kenya), and China (China Post) have successfully repositioned their postal systems from letter delivery monopolies into diversified logistics and financial service platforms.

The reform programme underway since May 2025 has been remarkably multifaceted, addressing logistics modernization, digital service delivery, agricultural market linkages, cold chain development, legal reform, and the ambitious programme to convert 1,000 Digital Centres into full service Nagorik Sheba (Citizen Service) Centres. The draft Postal Services Ordinance, currently awaiting Advisory Council approval, represents the most significant legal reform in the postal sector in decades, providing the statutory foundation for a commercially oriented, digitally enabled postal service.

6.5.2 Reform Initiatives and Current Status

Initiative	Current Status	Timeline	Expected Outcome
Central Logistics Tracking Platform (CLTP)	Onboarding phase active since March 2025; stakeholder meetings completed with e commerce, courier, payment gateway, and regulatory bodies; integrated digital logistics platform foundation established	Ongoing	Unified logistics tracking; interoperability with private couriers and e commerce platforms
Domestic Mail Monitoring Software (DMS) with API Sharing	API sharing initiative with courier and e commerce platforms under development	In progress	Real time parcel tracking; seamless integration with commercial logistics networks
Reverse Logistics Middleware System	Planning initiated for bidirectional urban rural e commerce network	Planning stage	Rural to urban and urban to rural e commerce enabling; digital divide reduction in commerce

GPS Tracker Installation on Mail Vans	85 mail vans equipped with GPS trackers during May to June 2025	Completed	Vehicle monitoring; route optimization; transportation transparency and security
Cold Chain Logistics Strengthening	14 chiller chambers and 84 deep freezers installed under ongoing project (2024 to 2025)	Ongoing	Perishable goods transport capability; agricultural value chain strengthening
Courier Delivery Simplification at Secretariat	Alternative mechanism established for courier mail delivery through postal system due to courier representative access restrictions	In progress	Resolution of secretariat courier delivery bottleneck
E Commerce Service Hour Extension	Committee formed to evaluate office hour extension for entrepreneurs and field officers	Planning stage	Improved accessibility for e commerce entrepreneurs; extended service availability
Financial Transaction Modernization (EMS and EMTS)	EMS and EMTS service modernization and expansion ongoing	In progress	Faster merchant payments; improved e commerce money order processing
National Address Management System (NAMS)	Project entitled Expanding and Facilitating Existing Postal Services at Citizen Level; PEC meeting completed; conditional approval provided	Conditional approval	Digital delivery capability; geocoded addressing; last mile delivery accuracy
Agricultural Market Linkage (Mango Transport)	65 tonnes of mangoes transported from Rajshahi, Chapainawabganj, Naogaon, and Satkhira to Dhaka during May to July 2025 season; direct farmer to consumer delivery	Completed (seasonal)	Profitable market access for small farmers; intermediary bypass; rural income enhancement
Commemorative Stamp Programme	Five commemorative stamps released: Independence Day (26 March), World Telecom Day (17 May), July Uprising Day (5 August), World Post Day (9 October), and Victory Day (15 December 2025)	Completed	National identity promotion; philatelic revenue; cultural and institutional branding
Nagorik Sheba (Citizen Service) Centres	Three pilot centres launched 26 May 2025 at Gulshan, Uttara, Nilkhet by Chief Adviser; three additional centres at Banasree, Mohammadpur, Gulistan opened 22 October 2025; phased conversion of 1,000 Digital Centres ongoing	Ongoing (target December 2026)	End to end government service delivery at single location; one stop citizen access

Postal Services Ordinance	Draft finalized by PTD; awaiting Advisory Council approval for policy endorsement	Pending	Legal foundation for modernized postal services; regulatory reform enabling commercial operations
Postal Ballot Modernization	SOP development in coordination with Election Commission; digital tracking system integration for ballot paper security and confidentiality	In progress	Democratic participation for election duty officials, overseas citizens, and remote area residents
ED Employee Management Reform	ED employee policy prepared and submitted to Ministry for approval; ED allowance increase proposal also submitted	Pending approval	Workforce motivation; socioeconomic security for Extra Departmental employees

6.5.3 Strategic Improvement Agenda

The most transformative initiative in the postal reform programme is the conversion of 1,000 Digital Centres into Nagorik Sheba Centres. This programme has the potential to create the largest citizen service delivery network in Bangladesh, offering government services, financial transactions, e commerce facilitation, and digital literacy training at a single physical location in every upazila. The model draws inspiration from India Post's transformation under the India Post Payments Bank (IPPB) initiative, which converted post offices into banking access points serving over 60 million accounts in underserved areas, and Brazil's Correios, which provides over 200 government services through its post office network.

However, the success of this programme depends critically on three factors. First, the digital infrastructure at each centre must be standardized and maintained, including reliable internet connectivity, biometric verification devices, secure payment terminals, and trained staff. Second, the service catalogue must be progressively expanded through interoperability agreements with other government agencies, moving beyond basic services toward passport applications, land records, health insurance enrolment, and social safety net disbursements. Third, the Nagorik Sheba Centre model must be financially sustainable through a combination of government service delivery fees, commission on financial transactions, and e commerce facilitation charges that generate sufficient revenue to cover operational costs.

The National Address Management System (NAMS) represents a foundational infrastructure investment that will benefit not only the postal sector but the entire logistics and service delivery ecosystem. Without a standardized, geocoded addressing system, last mile delivery remains inefficient, emergency response is hampered, and digital service delivery (including drone based delivery, which the National Digital Transformation Strategy envisions) cannot be reliably scaled. The United Kingdom's Postcode Address File and India's PIN Code system are long established examples; more recent initiatives such as what3words and Estonia's address data

infrastructure demonstrate how digital addressing can be implemented rapidly in emerging markets.

The cold chain logistics initiative, while modest in current scale (14 chiller chambers and 84 deep freezers), addresses a critical gap in Bangladesh's agricultural value chain. The loss of perishable produce between farm gate and consumer is estimated at 30 to 40 percent in Bangladesh, representing both a food security challenge and an economic waste. Expanding the postal cold chain, particularly during seasonal harvests of mangoes, lychees, and vegetables, can position Bangladesh Post as a trusted agricultural logistics partner. Thailand Post's farm to consumer cold chain service and China Post's rural e commerce logistics network provide relevant models.

6.5.4 KPI Framework

Reform Area	KPI	Baseline (2025)	Target	Timeline
Nagorik Sheba Centres	Operational citizen service centres	6 centres	1,000 centres	December 2026
Logistics Digitization	Parcels tracked through CLTP	Onboarding phase	100% of parcels	2027
Agricultural Logistics	Seasonal agricultural produce transported (tonnes)	65 tonnes (mango season)	500 tonnes annually	2027
Cold Chain Capacity	Cold storage equipped post offices	14 chiller sites	50 sites	2028
SIM Distribution	Teletalk SIMs sold through post offices monthly	Pilot	10,000 monthly	2027
NAMS Coverage	Addresses geocoded in NAMS database	Nil	All urban addresses	2028
Postal Ballot	Election coverage with postal ballot SOP	Under development	Nationwide SOP operational	2027

6.6 Bangladesh Satellite Company Limited (BSCL)

6.6.1 Rationale for Reform

Bangladesh Satellite Company Limited (BSCL) operates the nation's first geostationary communications satellite, Bangabandhu Satellite 1 (BS 1), and is positioning itself as the institutional anchor for Bangladesh's space economy ambitions. BS 1, launched in May 2018, provides C band and Ku band transponder capacity for broadcasting, telecommunications backhaul, and VSAT connectivity. However, capacity utilization has remained below commercial

expectations, partly due to limited domestic demand development and constrained international marketing capability. BSCL's mandate has expanded significantly in 2025 to encompass Starlink reselling, maritime satellite services, cable television digitalization, and earth observation satellite planning, representing a strategic pivot from single satellite operation toward a diversified space services portfolio.

6.6.2 Reform Initiatives and Current Status

Initiative	Current Status	Timeline	Expected Outcome
BS 1 Bandwidth Sales (Domestic)	Contracts signed with Bangladesh Navy, DGFI, and Police Special Branch for BS 1 bandwidth utilization; security sector adoption progressing	Ongoing	Internal security communications strengthening; sovereign satellite utilization increase
BS 1 Bandwidth Sales (International)	Preliminary discussions completed with Nepal; proof of concept preparation ongoing; discussions with Philippines and Indonesia underway	Ongoing	Foreign exchange earnings; regional market presence; satellite capacity monetization
Starlink Reselling in Bangladesh	BSCL operating as Starlink reseller; kits and data plans being sold; services deployed to BGB, WFP, Chittagong Hill Tracts schools (approximately 10 institutions); Grameenphone partnership for service expansion underway	Operational	High speed internet in remote and underserved areas; institutional connectivity in difficult terrain
Starlink at State Guest House Jamuna	Starlink internet deployed at all locations used by Chief Adviser at Jamuna state guest house	Completed	Secure, high bandwidth connectivity for state leadership; technical capability demonstration
Maritime AIS 2.0 System	Initial specifications prepared; stakeholder consultations completed; funding proposal submitted to BTRC SOF Committee	In progress (target June 2026)	Maritime security; vessel tracking in territorial waters; blue economy enablement
Cable Television Network Digitalization	Concept paper presented to Information and Broadcasting Ministry; verbal policy approval received; BSCL to develop central platform, set top box specifications, and TRP measurement framework	In progress (target June 2026)	Broadcasting transparency; TRP based advertising market discipline; content distribution modernization

Earth Observation (EO) Satellite Planning	Draft writeup on environmental impact of EO satellites under review by Managing Director	In progress (target December 2026)	Climate monitoring; disaster preparedness; agricultural planning; sovereign earth observation capability
TRP Device Deployment	Awaiting minutes of Information and Broadcasting Ministry meeting for next steps	Pending	Transparent television viewership data; advertising market rationalization

6.6.3 Strategic Improvement Agenda

BSCL's institutional transformation must be evaluated against the backdrop of a rapidly evolving global satellite industry. The emergence of low earth orbit (LEO) constellation providers, primarily Starlink but also OneWeb, Amazon Kuiper, and Telesat Lightspeed, has fundamentally altered the economics of satellite connectivity. Rather than viewing LEO as a competitive threat to BS 1, BSCL's strategy of becoming a Starlink reseller while continuing to market BS 1's geostationary capacity for specific use cases (broadcasting, maritime VSAT, government communications) represents a pragmatic adaptation. The reseller model ensures that Bangladesh participates in the LEO connectivity revolution while maintaining sovereign oversight and local distribution control.

The cable television digitalization initiative presents a major institutional opportunity. Bangladesh's cable TV market, estimated at over 20 million households, currently operates largely through analogue distribution by local cable operators with minimal transparency in subscriber counts, content licensing, or viewership measurement. BSCL's proposal to serve as the central digital platform, supported by standardized set top boxes and TRP measurement, mirrors the digital migration mandated by India's Telecom Regulatory Authority (TRAI) beginning in 2012, which transformed India's cable industry from an opaque, tax evading sector into a digitized, auditable industry. The implementation must, however, be carefully sequenced to avoid the disruptions India experienced during its initial digitalization phases.

BS 2 planning must commence immediately given BS 1's finite orbital life. The procurement and launch cycle for a geostationary communications satellite typically requires four to six years from initial planning to orbit. BSCL should engage with potential satellite manufacturers, evaluate the relative merits of traditional GEO capacity versus hybrid GEO MEO constellations, and develop a business case that reflects realistic demand forecasts rather than the aspirational projections that characterized the BS 1 procurement process.

6.6.4 KPI Framework

Reform Area	KPI	Baseline (2025)	Target	Timeline
BS 1 Utilization	Transponder capacity sold as percentage of total	Below 30%	60%	2028

International Sales	Countries with active BS 1 bandwidth contracts	Zero (discussions only)	3 countries	2027
Starlink Reach	Institutions and locations served via Starlink	Approximately 10	500 locations	2027
Maritime AIS	AIS 2.0 system operational status	Specification stage	Fully operational	2027
Cable TV Digitalization	Households on digital cable via BSCL platform	Zero	5 million	2028
BS 2 Planning	BS 2 procurement process stage	Not initiated	RFP issued	2027

6.7 Bangladesh Telephone Shilpa Sangstha (TSS)

6.7.1 Rationale for Reform

Bangladesh Telephone Shilpa Sangstha (TSS), the state owned telecom equipment manufacturing entity, has experienced the most dramatic institutional pivot among PTD entities. Historically focused on telephone handset and exchange equipment production, TSS found its traditional product lines obsolete as digital switching, mobile communications, and IP based networks replaced analogue telephony. Rather than allowing the entity to continue a slow decline, the reform programme has reimagined TSS as an innovation enabling state owned platform, leveraging its manufacturing infrastructure, Hi Tech Park designation, and strategic partnerships to produce a diversified portfolio of digital products.

On 6 April 2025, TSS was designated a Hi Tech Park by the ICT Division of the Ministry of Posts, Telecommunications and Information Technology. This designation provides regulatory benefits including tax incentives, simplified licensing, and access to the Hi Tech Park ecosystem of investors, technology transfer partners, and skilled workforce pipelines. The designation is being operationalized through multiple strategic partnership agreements and manufacturing pilot programmes that aim to reduce Bangladesh's dependence on imported digital equipment.

6.7.2 Reform Initiatives and Current Status

Initiative	Current Status	Timeline	Expected Outcome
Software Incubation Centre	Established with strategic partner support; ICT related innovative ideas being converted into practical products; operations ongoing	Operational	Local software entrepreneurship; product development pipeline for government digital services
Edotco MoU for Battery Refurbishment	MoU signed 4 December 2025 between TSS and edotco Bangladesh Ltd; technician training commenced for lithium ion battery and rectifier repair and refurbishment centre	December 2026 (pilot year)	Circular economy revenue; reduced battery import dependency; skilled workforce development
Strategic Partnership Agreements (IT and IoT)	Two strategic partnership agreements signed 15 September 2025 with technology firms; production planning for computers, accessories, network devices, telecom devices, and IoT equipment underway	Ongoing (8 to 12 months)	Local assembly and production of low to mid tech digital products; import substitution

CCTV and Surveillance Equipment Manufacturing	Korean OEM partner (Ticon Systems Ltd) technical and investment support secured; project proposal approved by TSS board	December 2026	Domestic surveillance equipment production; reduced import dependency for security infrastructure
4G and 5G Dual Band WiFi Router Production	SKD and CKD assembly from ZTE and Huawei components; supply to PTD affiliated entities planned	December 2026	Local router assembly; supply chain security for government telecom entities
Low Tech Product Assembly (Set Top Boxes, Chargers, Electric Poles)	Strategic partners engaged for set top box, single charging port, mobile charger, and multipurpose electric pole production	8 to 12 months	Diversified product portfolio; employment generation; domestic market supply
Smart Electric Meter Production	Expatriate Bangladeshi scientists developing local IoT and LTE SIM based smart meter technology; production at existing TSS meter plant	December 2026	Import substitution for smart meters; Made in Bangladesh IoT product; energy sector modernization support
Hi Tech Park and Electric Bike Manufacturing	Hi Tech Park designation received 6 April 2025; Liban Industries Ltd application approved by HTPA for electric bike production at TSS premises	December 2026	Green mobility manufacturing; Hi Tech Park ecosystem activation; new revenue from facility leasing

6.7.3 Strategic Improvement Agenda

TSS's transformation from a defunct telephone manufacturing unit into an innovation enabling platform is conceptually compelling but faces significant execution risks. The primary risk is overextension: attempting to simultaneously launch software incubation, battery refurbishment, CCTV manufacturing, router assembly, smart meter production, and electric bike manufacturing may exceed the entity's management capacity and financial resources. A phased approach, prioritizing initiatives with the clearest revenue potential and lowest technology risk, is advisable.

TSS is pivoting toward high-demand digital accessories to capture the retail market and support tower infrastructure. This includes the domestic manufacturing of high-quality charging cables and power adapters. Furthermore, through strategic partnerships with infrastructure providers like edotco, TSS will manufacture and supply specialized hardware such as multipurpose electric poles and power management units for telecom towers, ensuring a "Made in Bangladesh" supply chain for critical network sites.

The battery refurbishment centre, developed through the edotco partnership, deserves particular emphasis. The telecom tower industry in Bangladesh and across South Asia faces an escalating

challenge of lithium ion battery lifecycle management. With tens of thousands of tower sites requiring periodic battery replacement, a certified refurbishment facility reduces costs, addresses environmental disposal concerns, and generates recurring revenue. South Korea's battery recycling industry and Japan's circular economy policies for electronics provide frameworks for how this can evolve into a significant business line.

The smart electric meter initiative, leveraging IoT technology developed by expatriate Bangladeshi scientists, aligns directly with the national energy sector's modernization needs. The Bangladesh Energy Regulatory Commission (BERC) and electricity distribution companies have identified smart metering as a priority for reducing system losses and improving billing accuracy. If TSS can produce smart meters at competitive prices with adequate quality certification, it could capture a meaningful share of the domestic smart meter procurement market, currently dominated by imports from China and India. However, this requires rigorous testing, IEC standard compliance, and pilot deployments with distribution utilities before large scale procurement orders can be secured.



6.7.4 KPI Framework

Reform Area	KPI	Baseline (2025)	Target	Timeline
Product Diversification	Number of distinct products in commercial production	Zero (legacy only)	6 products	2027
Battery Refurbishment	Batteries refurbished annually (after pilot year)	Zero	10,000 units	2027
Router Assembly	4G and 5G routers assembled and supplied	Zero	20,000 units	2027

Smart Meters	Smart meters produced and deployed	Zero	50,000 units	2028
Hi Tech Park Tenants	Companies operating from TSS Hi Tech Park	1 (Liban Industries)	5 companies	2027
Revenue Recovery	Annual revenue from new product lines	Negligible	BDT 50 crore	2028

6.8 Bangladesh Cable Shilpa Limited (BCSL)

6.8.1 Rationale for Reform

Bangladesh Cable Shilpa Limited (BCSL, commonly referred to as BAKESHI) is the state owned cable manufacturing entity producing optical fiber cables, power cables, and related products. BCSL occupies a strategic position in the national infrastructure supply chain, providing the physical medium through which digital connectivity and power distribution are realized. In a context where fiber optic deployment is accelerating under national broadband expansion plans and power grid modernization is a priority under Bangladesh's energy strategy, BCSL's relevance has arguably never been greater.

The reform programme has focused on production capacity enhancement, product diversification, and market expansion beyond the traditional government procurement channel. The inter ministerial initiative to prioritize BCSL manufactured products in government infrastructure projects has already generated visible results in order volumes.

6.8.2 Reform Initiatives and Current Status

Initiative	Current Status	Timeline	Expected Outcome
Super Enamelled Copper Wire Plant	300 metric tonne annual capacity plant established; commercial production commenced	Operational	New product line; reduced import dependency for transformer and motor winding wire
Power Cable Capacity Expansion	International tender for 5 new technology machines opened 29 October 2025; technical evaluation completed; capacity to increase from 600 to 1,500 metric tonnes	October 2026	2.5 times capacity increase; enhanced ability to serve power distribution companies
LAN and Network Cable (CAT6 and CAT6A) Production	International tender for plant procurement; technical evaluation completed after tender opening 29 October 2025	October 2026	Domestic network cable manufacturing; import substitution for enterprise networking

Copper and Aluminium Rod Production Plant	Feasibility study documentation under preparation and review	June 2027	Vertical integration of raw material production; cost reduction in downstream cable manufacturing
Power Sector Market Expansion (WZPDCO)	Correspondence and continuous engagement with West Zone Power Distribution Company; bilateral supply agreement under negotiation	June 2026	Diversified customer base beyond traditional telecom procurement; stable revenue from power sector
Green Certification and Solar Panel Installation	Application submitted to BSCIC for Green Certificate; solar panel installation underway as prerequisite	July 2026	International brand value; ESG compliance; reduced energy costs from solar generation
11 and 33 kV Underground Power Cable Production	Feasibility study and draft tender for production plant underway; represents significant product diversification into high voltage cable segment	2028	Entry into high value underground power cable market; national electrification infrastructure support
Recycling Plant Activation	Scrap duct recycling plant fully activated; maximum waste reuse being ensured	Operational	Production cost reduction; waste minimization; circular economy contribution

6.8.3 Strategic Improvement Agenda

BCSL's production capacity expansion, particularly the planned increase from 600 to 1,500 metric tonnes in power cable output, positions the entity to capture a larger share of domestic infrastructure procurement. However, three strategic gaps merit attention. First, quality certification to international standards (IEC, ISO 9001, and product specific certifications) is essential to compete not only in domestic government procurement but also in potential export markets. Vietnam's cable manufacturing industry, which has successfully entered regional export markets through aggressive quality certification and competitive pricing, offers a relevant benchmark.

Second, the Green Certificate initiative and solar panel installation represent progressive environmental positioning, but should be accompanied by a comprehensive sustainability strategy including energy efficiency improvements in manufacturing processes, waste heat recovery, and water recycling. European cable manufacturers such as Prysmian Group and Nexans have established ESG reporting and sustainability targets that increasingly influence procurement decisions by multilateral development banks and international buyers.

Third, the 11 and 33 kV underground power cable production capability, once operational, will represent a qualitative leap in BCSL's product sophistication. Underground power distribution is expanding in Dhaka, Chattogram, and other major cities as urban density increases and overhead

line hazards become less acceptable. BCSL should engage with DPDC, DESCO, BPDB, and other distribution utilities early to understand technical specifications and secure advance procurement commitments before the production facility is completed.

6.8.4 KPI Framework

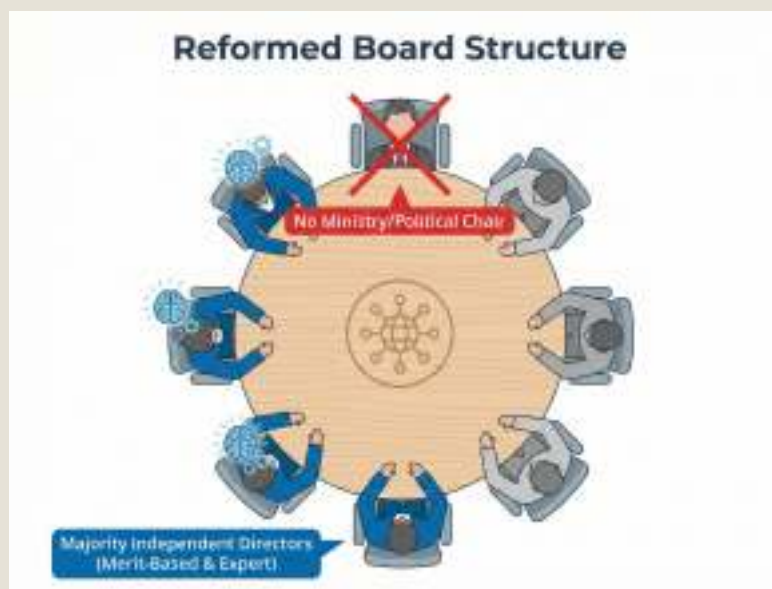
Reform Area	KPI	Baseline (2025)	Target	Timeline
Power Cable Production	Annual production volume (metric tonnes)	600 MT	1,500 MT	October 2026
Product Lines	Distinct cable products in commercial production	3 to 4 products	8 products	2028
Green Certification	Green Certificate obtained from BSCIC	Applied	Certified	July 2026
Market Diversification	Revenue from non telecom sectors as percentage	Low	40%	2028
Raw Material Integration	Copper and aluminium rod plant operational	Feasibility stage	Operational	June 2027
Order Volume	Annual order book value	Baseline 2025	50% increase	2027

6.9 Cross Cutting Institutional Reform Priorities

Beyond the entity specific reforms detailed above, several cross cutting institutional priorities apply across all PTD entities and require coordinated action at the divisional level.

6.9.1 Board Governance and Appointment Reform

The PTD Whitepaper Task Force findings documented a pervasive pattern of governance capture, where political patronage influenced board appointments, executive selection, and procurement decisions across multiple entities. Restoring governance integrity requires that Boards of Directors across all SOEs be reconstituted to include a majority of independent members with demonstrated sectoral expertise. The PTD Secretary should not serve as Board Chair of any SOE, separating policy oversight from operational control. Selection committees for executive positions should include external experts and observers from the Comptroller and Auditor General (CAG) or independent professional bodies. These measures align with the OECD Guidelines on Corporate Governance of State Owned Enterprises (2015) and the World Bank's Corporate Governance of State Owned Enterprises Toolkit.



6.9.2 Human Resource Digitization and Merit Based Progression

All PTD entities should implement centralized, digital Human Resource Management Information Systems (HRMIS) that ensure recruitment, promotion, and transfer decisions are transparently documented, rule based, and auditable. The post BTTB fragmentation of HR functions across BTCL, Teletalk, DoT, and other entities created overlapping mandates and enabled irregular appointments, ad hoc hiring, and post retirement reappointments without competitive processes. A unified digital HR framework, accessible to both internal management and external auditors, will institutionalize fairness and remove procedural ambiguity. Estonia's

public sector HRMIS and Singapore's Whole of Government HR portal provide models for digital HR governance in the public sector.



6.9.3 Accountability, Oversight, and Audit Compliance

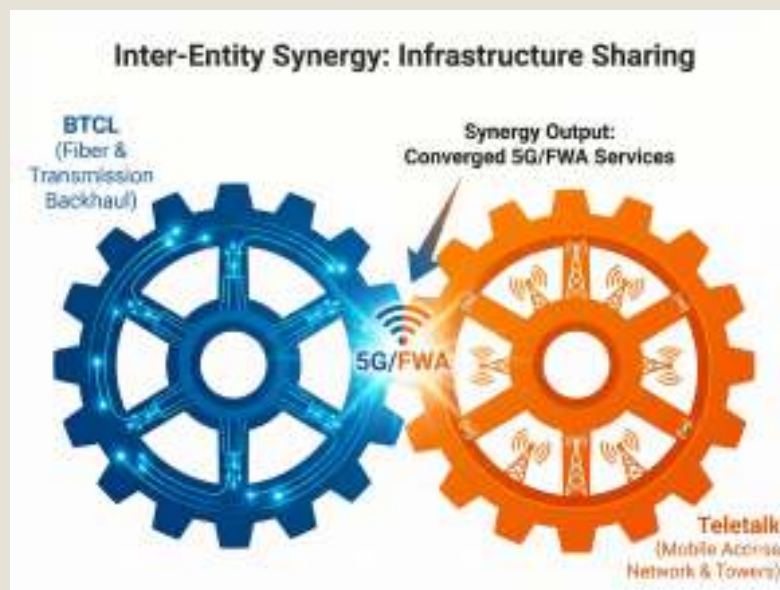
Comprehensive HR compliance audits by the CAG should be integrated into the regular audit cycle for all PTD entities, with findings submitted to Parliament to reinforce public accountability. An ex post regularization programme must review and rectify past irregular appointments through transparent procedures. Legal and disciplinary mechanisms should be fast tracked, with accused officers barred from retirement or transfer to evade proceedings. Secure whistleblower channels guaranteeing anonymity and protection against retaliation should be introduced across all entities. The completion of pending legal instruments, including the MOCSLA recruitment regulations and the clear definition of DoT BTRC functional boundaries, will eliminate structural ambiguity and prevent overlapping mandates.

6.9.4 Rebuilding the Telecommunication Cadre

The corporatization of BTTB dismantled the dedicated Telecommunication Cadre without creating an adequate replacement mechanism. The Department of Telecommunications (DoT), originally envisioned as the strategic anchor for sectoral leadership, has been structurally marginalized and rendered practically nonfunctional. Rebuilding a dedicated Telecommunication Cadre under DoT through competitive civil service recruitment will restore technical capacity and institutional continuity essential for managing emerging technologies including 5G, IoT, cybersecurity, and AI governance. This cadre will also ensure that PTD retains in-house technical expertise for policy formulation, project evaluation, and regulatory oversight, reducing dependence on external consultants and deputation based staffing.

6.9.5 Inter Entity Synergy and Resource Sharing

Several reform initiatives described in this chapter involve deliberate inter entity cooperation: BTCL's MVNO partnership with Teletalk, Teletalk's SIM distribution through Bangladesh Post, BCSL's cable supply to BTCL and power utilities, TSS's router production for PTD entities, and BSCL's Starlink reselling in partnership with telecom operators. These synergies should be formalized through structured inter entity service agreements with defined service levels, pricing mechanisms, dispute resolution procedures, and performance monitoring. A PTD level coordination body, meeting quarterly and reporting to the Secretary, should be established to oversee inter entity cooperation, resolve bottlenecks, and ensure that the collective reform agenda advances coherently rather than in institutional silos.



6.10 Chapter Summary

Chapter 6 presents a structured, entity by entity institutional reform programme for the ten organizations under the Posts and Telecommunications Division. The chapter diagnoses the institutional legacy of governance fragmentation, political patronage in appointments, and the disconnect between policy ambition and operational execution that the PTD Whitepaper Task Force documented in April 2025. Each entity section follows a consistent analytical framework: rationale for reform, current status of initiatives (drawn from official progress reports through January 2026), a strategic improvement agenda with international benchmarks, and measurable KPIs with defined timelines. BTRC's reform centres on restoring regulatory autonomy, operationalizing the Telecom Ordinance 2025, and transforming the Social Obligation Fund into an outcomes driven vehicle. BTCL's programme repositions the company from a legacy landline operator into a Digital Infrastructure Hub through TowerCo licensing, GPON outsourcing, the Hathazari Green Data Centre PPP, and the MVNO partnership with Teletalk. Teletalk's reform balances commercial competitiveness with sovereign communications obligations, anchored by the 2,300 site tower sharing agreement and phased 5G readiness. BSCPLC has reduced bandwidth prices by 30 to 50 percent and is pursuing dues recovery and capacity planning for next generation submarine cable systems. Bangladesh Post is undergoing the most multifaceted transformation, converting 1,000 Digital Centres into Nagorik Sheba Centres, building cold chain logistics capability, developing the National Address Management System, and modernizing postal ballot procedures. BSCL is pivoting from single satellite operation toward a diversified space services portfolio encompassing Starlink reselling, maritime AIS, cable television digitalization, and earth observation satellite planning. TSS has been redesignated as a Hi Tech Park and is pursuing battery refurbishment, router assembly, CCTV manufacturing, and smart meter production through strategic partnerships. BCSL is expanding power cable production capacity from 600 to 1,500 metric tonnes and diversifying into network cables and underground power cables. Five cross cutting priorities apply across all entities: board governance reconstitution with independent majorities, unified digital HRMIS implementation, CAG integrated accountability mechanisms, rebuilding the Telecommunication Cadre under the Department of Telecommunications, and formalizing inter entity synergies through structured service agreements overseen by a PTD level coordination body. The overarching institutional reform philosophy rests on three pillars: commercial viability to justify continued public ownership, governance integrity to eliminate patronage and restore meritocratic progression, and strategic complementarity to position state entities as essential infrastructure partners rather than ineffective competitors to private operators.

— End of Chapter 6 —

7. Project Reform

7.0 Introduction: The Project Governance Crisis

The Posts and Telecommunications Division (PTD) has, between January 2009 and August 2024, implemented or initiated 42 development projects across its affiliated entities, with a combined estimated value exceeding BDT 20,000 crore. These projects span fiber optic network expansion, submarine cable connectivity, satellite infrastructure, postal modernization, equipment manufacturing, and mobile network development. They have been funded through the Annual Development Programme (ADP), entity self financing, and the Social Obligation Fund (SOF). Yet the PTD Whitepaper Task Force, upon reviewing 18 priority projects representing 83 percent of total project value, found pervasive patterns of procurement irregularity, specification manipulation, cost inflation, inter project dependency failures, delayed completion, and in several cases, active Anti Corruption Commission (ACC) investigations.

These findings are not isolated incidents of individual misconduct. They reveal systemic weaknesses in the project governance architecture: the absence of mandatory feasibility studies for projects below BDT 50 crore, the lack of entity level master plans or strategic frameworks to guide project selection, chronic non submission or delayed submission of Project Completion Reports (PCRs), inadequate IMED (Implementation Monitoring and Evaluation Division) evaluation, and most critically, the structural exemption of SOF funded projects from ADP inclusion, IMED monitoring, and CAG audit scrutiny. The consequence has been a project pipeline that serves procurement cycles rather than strategic outcomes, where success is measured by expenditure rates rather than service delivery impact.

This chapter presents a comprehensive project reform framework designed to transform the PTD's project governance from its current state of fragmented accountability and opportunistic procurement into a disciplined, outcomes oriented, and strategically coherent system. The reform framework addresses five interconnected dimensions: the legacy project portfolio and its remediation, SOF governance reform, the new generation of strategically aligned projects initiated in 2025, the introduction of project governance safeguards, and the establishment of a project monitoring and evaluation architecture that ensures continuous accountability. International best practices from the United Kingdom's Infrastructure and Projects Authority (IPA), South Korea's Public Investment Management System (PIMAC), and Australia's Infrastructure Australia assessment framework inform the reform design.

7.1 Legacy Project Portfolio: Diagnosis and Remediation

7.1.1 Systemic Findings from the Task Force Review

The PTD Whitepaper Task Force's examination of 18 priority projects identified recurring governance failures that transcend individual entities. These systemic patterns must be understood as structural vulnerabilities requiring institutional remedies rather than merely disciplinary action against individual officers. The principal findings, drawn from the Task Force's documented analysis, are as follows.

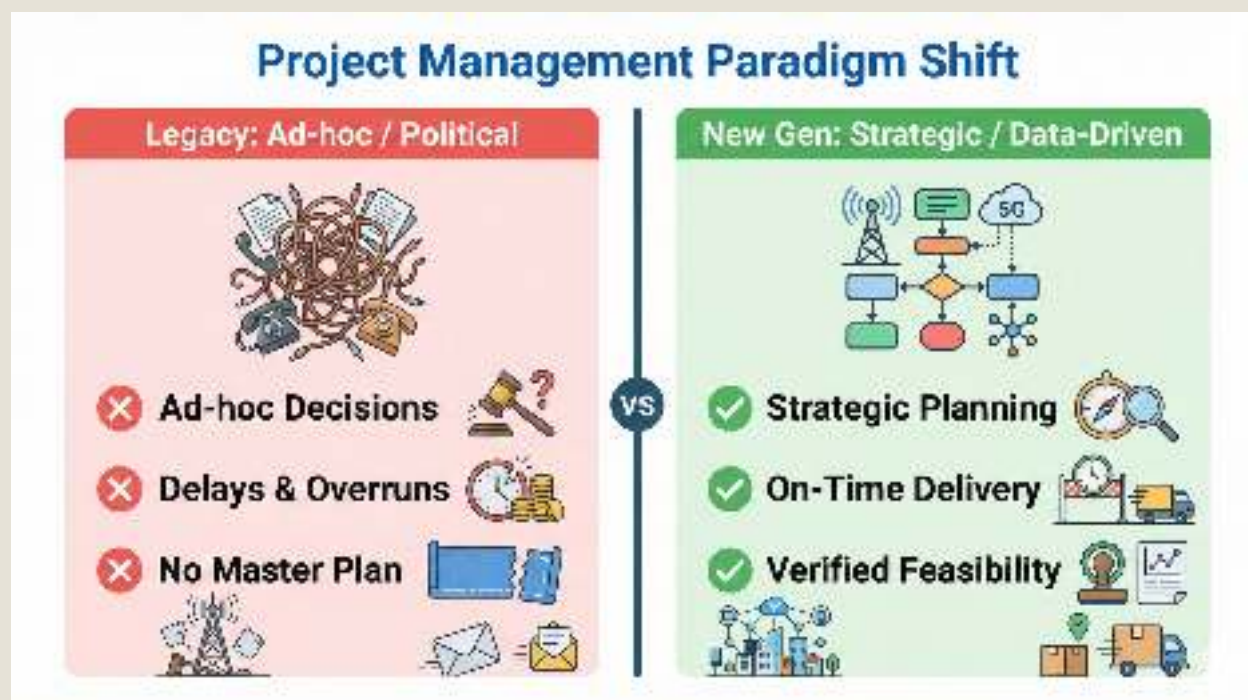
Absence of Strategic Framework and Master Plans. Multiple implementing entities, including BTCL, lacked articulated vision statements, mission definitions, or master plans at the time projects were conceived. Projects were therefore designed in a strategic vacuum, without reference to long term institutional objectives or sectoral coordination. The consequence was that projects addressed immediate procurement opportunities rather than coherent infrastructure development sequences. The Task Force recommended that all entities formulate clearly articulated visions and missions accompanied by comprehensive master plans spanning short, medium, and long term horizons, grounded in an understanding of the evolving global telecom landscape.

Feasibility Study Failures. The Ministry of Planning's June 2022 guideline mandates comprehensive feasibility studies for all investment projects exceeding BDT 50 crore. Despite this requirement, the Task Force found instances where feasibility studies were either not conducted, conducted superficially, or their findings disregarded during project implementation. In the case of BTCL's Optical Fiber Transmission Network Development for 5G Readiness Project (approved at BDT 1,059.10 crore by ECNEC in February 2022), the tender specifications for the GD 1 package were altered to require 400G/100G equipment, deviating from the 100G/10G specifications recommended in the approved feasibility study and Development Project Proposal (DPP). This unauthorized modification constituted a violation of Public Procurement Rules (PPR) and was subsequently flagged by the ACC for investigation.

Inter Project Dependency Failures. The 5G Readiness Project and BTCL's Internet Protocol (IP) Network Development and Expansion Project were technically interdependent: the IP network's servers required the DWDM transmission network to operate. Yet neither project's DPP formally documented this dependency, and the GD 1 specification alteration in one project effectively handicapped the other. The IP network project was reported to be 70 to 80 percent complete but rendered non functional because the DWDM network it depended upon was stalled by procurement disputes and ACC investigation. This situation, where substantial public expenditure on one project is rendered valueless by the failure of a linked project, exemplifies the absence of a centralized coordination mechanism under the Head of the Procuring Entity (HOPE).

Procurement Irregularity and Political Interference. The Task Force documented instances of political interference in procurement processes across multiple entities. In the 5G Readiness

Project, documentary evidence indicated that both a former Minister and a former Secretary sought to influence the tender process in favour of different bidding companies (ZTE Corporation and Huawei Technologies respectively). The BTCL board's 213th meeting revealed that the proposed 126 terabyte infrastructure capacity far exceeded projected demand of 26 terabytes by 2030, with only half the capacity expected to be utilized even by 2040. Industry experts described this capacity over specification as indicative of deliberate cost inflation. The project, valued at BDT 463 crore for the GD 1 tender package alone, is now under ACC investigation.



Delayed Project Completion Reports and IMED Evaluation. The Task Force found that Project Completion Reports were frequently submitted late (in one case, 18 months after completion) or not submitted at all for projects still nominally in implementation phase years after their original completion dates. IMED impact evaluations were either pending or, in the case of self financed projects, not required under existing rules. This evaluation gap means that the government has no systematic evidence base for assessing whether project expenditure achieved its intended outcomes.

7.1.2 Remediation Framework for Legacy Projects

Legacy projects under active ACC investigation must be allowed to proceed through the investigative and, where applicable, judicial process without executive interference. The ACC's June 2025 communication explicitly warned that proceeding with procurement activities in projects where preliminary evidence of PPA 2006 and PPR 2008 violations has been found may constitute a legal violation and that related expenditure would likely be deemed unlawful.

Regardless of pressures to advance projects citing national interest or technological urgency, the integrity of the anti corruption process must be safeguarded.

For legacy projects not under investigation but suffering from delays or scope problems, a structured Project Health Assessment should be conducted by an independent technical team, reviewing each project against its original DPP, feasibility study recommendations, procurement compliance, physical progress, and financial expenditure patterns. Projects that are salvageable should receive revised DPPs with realistic timelines, updated cost estimates reflecting currency depreciation and technology evolution, and clearly defined completion milestones. Projects that have been rendered technically obsolete by the passage of time or overtaken by alternative solutions should be formally cancelled, with Project Completion Reports submitted and lessons learned documented.

The United Kingdom's Infrastructure and Projects Authority conducts annual reviews of major government projects, assigning a Delivery Confidence Assessment (rated green, amber, or red) that is published publicly. Bangladesh should adopt a similar mechanism for PTD projects, with the PTD Secretary commissioning quarterly project health assessments and publishing summary findings to restore public confidence in project governance.

7.2 Social Obligation Fund (SOF) Governance Reform

7.2.1 Diagnosis of SOF Governance Failures

The Social Obligation Fund, established in November 2011 under the Telecommunication Act (Amendment) 2010, requires telecom operators to contribute 1 percent of their audited gross annual revenue. By January 2025, cumulative contributions and interest had reached approximately BDT 3,559 crore. The SOF's intended purpose, financing telecommunications and ICT development in underserved areas including remote coastal zones, river islands, hilly regions, and communities with limited commercial viability, is critically important for digital inclusion. However, the Task Force's review revealed that the SOF has been undermined by five structural governance failures.

First, non compliance with mandatory contributions. While major mobile operators contributed more consistently, the majority of ISPs, district level operators, and telecom value added service (TVAS) licensees either remitted irregularly or did not remit at all. BTRC's weak enforcement and ad hoc waiver practices eroded compliance norms, reducing the effective funding base and creating an uneven distribution of the contribution burden.

Second, large idle balances and delayed deployment. Approximately 35 percent of total funds raised remained idle due to regulatory and institutional bottlenecks. Large tranches of money were left undeployed for years, violating the fiduciary expectation that collected public monies be actively applied for their intended purpose. When spending was eventually hurried under political pressure, it created space for poor procurement and weak project design.

Third, exclusion from national oversight frameworks. SOF projects are not included in the Annual Development Programme (ADP). Consequently, their progress is not discussed in ADP coordination meetings, they are not monitored by IMED, and they are not audited by the Audit Directorate. The approval powers of the SOF Council, chaired by the Minister/State Minister in charge, are not harmonized with ECNEC's authority, and no upper cost limit is set for ministerial approvals. This structural exemption created an accountability vacuum that facilitated procurement shortcuts and ad hoc decisions.

Fourth, fragmented accountability architecture. The SOF design separates revenue collection (BTRC) from project selection and implementation (SOF Management Committee and diverse PTD agencies). No single entity carries the full chain of accountability for impact. BTRC can collect but not compel performance, while implementing agencies execute projects without sustained BTRC oversight. Ministerial approval signatures bypass routine cross checks, creating a fragmented accountability trail.

Fifth, absence of central monitoring and evaluation. BTRC maintains no dedicated M&E unit for SOF nor a unified asset and performance dashboard. Field inspections reported routers and hotspots as installed but not operational, with no central mechanism to capture status changes in

real time. The institutional culture privileged procurement milestones over sustained operations and actual service delivery.

7.2.2 SOF Reform Architecture

The reform of SOF governance requires structural changes across five dimensions, drawing on international best practices from India's Universal Service Obligation Fund (USOF), Malaysia's Universal Service Provision (USP) Fund, and Colombia's Fondo de Tecnologías de la Información y las Comunicaciones (FONTIC).

Inclusion in National Oversight Frameworks. All SOF funded projects must be included in a dedicated annex to the ADP, subjecting them to the same IMED monitoring, ADP coordination review, and CAG audit requirements as government funded development projects. The SOF Council's approval powers must be harmonized with ECNEC's authority, with projects exceeding BDT 50 crore requiring ECNEC clearance regardless of funding source. India's USOF, while administered by the Department of Telecommunications, operates within the framework of the national budget process and is subject to CAG audit, providing a precedent for this integration.

Independent Project Selection and Prioritization. A formal project invitation and evaluation process must replace the current ad hoc system where neither a dedicated proposal invitation committee nor clear prioritization criteria exist. SOF project proposals should be evaluated by an independent technical committee against published criteria including: geographic targeting accuracy (underserved area verification), cost effectiveness analysis, sustainability planning (maintenance, operational continuity), and alignment with PTD strategic priorities. Colombia's FONTIC employs a competitive project selection process with published evaluation criteria and independent technical assessment, which has significantly improved fund deployment effectiveness.

Unified Digital SOF Management Platform. BTRC must establish a dedicated M&E unit for SOF and commission a digital SOF management platform that provides real time asset tracking, functionality monitoring, usage analytics, and public reporting. This platform should integrate with BTRC's existing Automated Spectrum Management System (ASMS) and Telecom Monitoring System (TMS) to verify that SOF funded connectivity infrastructure is actually operational. Asset registers, functionality reports, and usage metrics must be consolidated into a single dashboard accessible to auditors, policymakers, and the public.

Enforcement of Operator Contributions. BTRC must implement automated revenue verification linked to operator audit reports, with graduated penalties for late or incomplete remittance. Licensees in persistent default should face licensing consequences, including suspension of spectrum assignment or license renewal conditionality. Malaysia's USP Fund, governed under Section 204 of the Communications and Multimedia Act 1998, imposes a mandatory 6 percent levy on licensees with revenue exceeding RM 2 million, with enforcement backed by licensing sanctions.

Performance Based Project Monitoring. SOF project contracts must include service level agreements (SLAs) with measurable performance indicators: connectivity uptime, user registration rates, training completion rates, and operational sustainability over a defined maintenance period. Payment tranches should be linked to verified performance milestones rather than procurement completion alone. Projects that fail to meet SLA benchmarks within a grace period should trigger remedial action, including contract termination with recovery provisions.

7.2.3 SOF Reform KPI Framework

Reform Area	KPI	Baseline (2025)	Target	Timeline
Contribution Compliance	Operator remittance rate against statutory 1%	Below 70% for non MNOs	95% across all licensees	2027
Fund Utilization	SOF funds deployed vs collected	65% deployed	90% deployed annually	2027
ADP Integration	SOF projects included in ADP annex	Zero	100%	2026
Asset Functionality	SOF funded installations verified operational	Unknown	90% verified operational	2027
M&E Platform	Digital SOF management dashboard operational	None	Fully operational	2026
Project Evaluation	SOF projects with published impact assessment	None	All completed projects	2027

7.3 New Generation Projects: Strategically Aligned Initiatives (2025 Onwards)

The reform programme initiated in 2025 has introduced a new generation of projects that, unlike many of their predecessors, are embedded within a coherent strategic framework defined by the PTD's digital transformation agenda. These projects are distinguished by their alignment with entity transformation plans, emphasis on commercial sustainability, cross entity coordination, and citizen facing service delivery outcomes. This section catalogues the principal new projects across PTD entities, assessing their strategic coherence and implementation readiness.

7.3.1 Nagorik Sheba Bangladesh: The Flagship Citizen Service Platform

Nagorik Sheba Bangladesh represents the most ambitious cross institutional project in the PTD reform portfolio. Launched on 26 May 2025 by Chief Adviser Professor Muhammad Yunus at

the State Guest House Jamuna, the initiative transforms unused government facilities, including post offices and exchange buildings, into modern citizen service centres operated by trained local entrepreneurs. The programme operates on a zero cost to public money model, with centres managed under a sustainable social business framework through public private community partnership, where licensed entrepreneurs provide services for a fee while paying rent for government space.



The pilot phase established centres at Gulshan, Uttara, and Nilkhet in Dhaka, with subsequent expansion to Banasree, Mohammadpur, and Gulistan by October 2025. By November 2025, a total of 1,122 Nagorik Sheba Centres were reported operational, integrating over 41 services with 453 services available through the MyGov platform. More than 2,000 entrepreneurs nationwide were trained, with a rigorous selection process initially certifying 100 entrepreneurs (50 women and 50 men) from an initial pool of 200 applicants. A daylong training workshop held on 13 November 2025 at Dak Bhaban brought together representative entrepreneurs from all 64 districts for standardized training on the platform, digital security, cyber security, and data protection.

Nagorik Sheba is structurally significant for PTD because it operationalizes three simultaneous transformation objectives. For Bangladesh Post, it converts the postal network from a declining mail delivery service into a citizen service delivery platform, generating revenue from service facilitation fees and revitalizing physical infrastructure. For BTCL, the Nagorik Sheba centres established in former telephone exchange buildings (Gulshan, Uttara, Nilkhet) demonstrate the commercial repurposing of underutilized telecom real estate. For the Digital Transformation Strategy as a whole, Nagorik Sheba serves as the last mile physical manifestation of the Bangladesh National Digital Architecture (BNDIA), acting as a decentralized interoperability

hub that bridges the digital divide through trained human intermediaries who navigate complex digital services on behalf of citizens with limited literacy or technical skills.

Milestone	Status (January 2026)	Target	KPI
Pilot centres operational	6 centres in Dhaka	Complete	Service transactions per centre per month
National centre deployment	1,122 centres operational as of November 2025	8,000+ centres by 2030	Centres per 100,000 population
Entrepreneur training	2,000+ trained; 64 district workshop completed	20,000+ by 2030	Women entrepreneur percentage (target 50%)
Service catalogue integration	41 services integrated; 453 via MyGov	1,500+ services by 2030	End to end digital service percentage
Post office conversion	Pilot at select post offices	All suitable post offices by 2028	Post offices with Nagorik Sheba status
Zero budget sustainability	Demonstrated in pilot	Sustained nationally	Revenue per centre covering operational costs

7.3.2 BTCL Infrastructure Modernization Projects

BTCL's new project pipeline reflects the entity's strategic repositioning from a legacy landline operator to a digital infrastructure hub. Three major projects are in various stages of approval and implementation.

Mohakhali Telecom and Software Technology Complex (BTSTC)

The BTSTC project proposes to develop BTCL's prime Mohakhali real estate into a technology innovation hub and software industry cluster. The Pre Feasibility Study (PFS) was restructured following Planning Commission review on 28 October 2025, with inputs provided for RADP inclusion in FY 2025 to 2026. A verification committee meeting was held on 10 November 2025 to advance the PFS toward full DPP preparation. The project represents a significant departure from BTCL's historical project approach: rather than procuring telecommunications equipment, it seeks to develop BTCL's most valuable urban asset into a revenue generating technology complex. The strategic logic mirrors the Nairobi Innovation Hub development in Kenya and Cyberjaya in Malaysia, where state owned land was leveraged for technology cluster development.

District Telecom Infrastructure Modernization (34 Districts, Phase 1)

This project aims to modernize BTCL infrastructure across 34 districts to provide co location services and updated connectivity infrastructure. The project was included in the ADP green list for FY 2025 to 2026, with a Finance Division letter dispatched on 3 November 2025 seeking clarification on the GOB financing modality. The Tejgaon Telecom Tower and Innovation

Complex, a related initiative, received BTCL Board approval at the 240th meeting on 25 November 2025 for BTCL's own financing. The feasibility study for the 34 district Phase 1 programme is progressing, and the project is designed to complement the Nagorik Sheba programme by ensuring that district level facilities have the infrastructure required to serve as technology service hubs.

Green Data Centre (Hathazari, Chattogram)

The Hathazari Green Data Centre project, anchored by an MoU signed on 26 January 2025 among BTCL, PTD, the PPP Authority, and the Asian Development Bank, represents BTCL's entry into the data centre market. The project is structured as a public private partnership, with ADB providing technical advisory support for a Tier III specification facility. ERP consultant procurement is ongoing to support BTCL's internal readiness for data centre operations. Given Bangladesh's rapidly growing data hosting demand, driven by financial sector compliance requirements, e government expansion, and private cloud adoption, the timing is strategically sound. However, execution risk remains elevated given BTCL's limited experience with PPP structures and data centre operations, necessitating the operational partnerships and capacity building measures identified in Chapter 6.

7.3.3 Teletalk Network Expansion Projects

Teletalk has three new Development Project Proposals (DPPs) prepared and submitted for ADP green list consideration, targeting April 2026 Planning Commission approval. These projects address three strategic gaps in Teletalk's coverage: remote area connectivity in the Chittagong Hill Tracts and river island communities, 4G coverage extension at public universities, and highway corridor 4G deployment for safety communications. The projects are complemented by the ongoing 4G expansion programme, which has already activated 1,699 new 4G BTS sites and increased daily data consumption by approximately 20 percent (from 354 TB to 464 TB). The tower sharing agreement for 2,300 sites, signed in November 2025, provides the infrastructure foundation for accelerated network expansion without the prohibitive capital expenditure of own tower construction.

Critically, these projects reflect lessons learned from Teletalk's previous ADP project history. The DPPs include detailed demand justification, phased implementation schedules, and defined handover criteria that address the Task Force's concerns about project completion reporting and impact evaluation. The integration of SIM distribution through post offices and the MVNO partnership with BTCL provides cross entity revenue sharing models that strengthen the business case for continued public investment in Teletalk's infrastructure.

7.3.4 Bangladesh Post Modernization Projects

Bangladesh Post's project pipeline encompasses three transformative initiatives, each grounded in the broader postal reform strategy.

National Address Management System (NAMS)

The NAMS project, conditionally approved following a Project Evaluation Committee (PEC) meeting in 2025, establishes a standardized, geocoded addressing system for Bangladesh. Under the proposed design, NAMS introduces a National Address Identifier (NAID), a 12 character alphanumeric code that serves as a permanent digital key for any location in Bangladesh. The system operates as a foundational layer of Digital Public Infrastructure, with Bangladesh Post serving as the primary custodian of the address registry, leveraging its nationwide field force for verification and data maintenance. The expected outcomes by 2030 include 100 percent household and business NAID assignment, elimination of addressing ambiguity in emergency response routing, and a projected 30 percent reduction in e commerce logistics costs through elimination of failed deliveries. NAMS directly operationalizes the National Digital Transformation Strategy's vision of a standardized delivery infrastructure for digital commerce and government service delivery.

Central Logistics Tracking Platform (CLTP)

The CLTP, in its onboarding phase since March 2025, creates an integrated digital logistics platform connecting Bangladesh Post with private courier companies, e commerce platforms, payment gateways, and regulatory bodies. Stakeholder consultations have been completed, and the platform's architecture is being finalized to support real time parcel tracking, interoperability with commercial logistics networks, and API based data sharing. The CLTP, combined with the Domestic Mail Monitoring Software (DMS) API sharing initiative and the planned Reverse Logistics Middleware System, positions Bangladesh Post as a national logistics backbone capable of serving the growing e commerce market while leveraging its unmatched physical reach into rural areas.

Cold Chain Logistics Expansion

The installation of 14 chiller chambers and 84 deep freezers under the ongoing postal development project (2024 to 2025) establishes Bangladesh Post's entry into cold chain logistics. The successful pilot transport of 65 tonnes of mangoes from Rajshahi, Chapainawabganj, Naogaon, and Satkhira to Dhaka during the May to July 2025 season demonstrates commercial viability and social impact. The expansion of cold chain capacity to additional post offices, particularly during seasonal fruit and vegetable harvests, addresses both the 30 to 40 percent post harvest loss problem in Bangladesh's agricultural sector and creates a sustainable revenue stream for the postal service. Thailand Post's farm to consumer cold chain model and India Post's Kisan Rin Portal (connecting farmers directly to urban markets) offer operational benchmarks for scaling this initiative.

7.3.5 BSCL Space and Satellite Projects

BSCL's project pipeline encompasses the Maritime Automatic Identification System (AIS 2.0) project, with preliminary specifications prepared and a financing proposal submitted to BTRC's

SOF Committee (target completion June 2026), the cable television network digitalization initiative approved in principle by the Ministry of Information and Broadcasting, and the Earth Observation (EO) satellite planning initiative currently under Managing Director review. The AIS 2.0 project, designed to provide satellite based vessel tracking in Bangladesh's territorial waters, has direct implications for maritime safety, Blue Economy development, and border security. Its financing through the SOF represents an appropriate use of the fund's mandate for infrastructure development in underserved or strategically critical domains.

The cable television digitalization project, proposing BCSL as the central digital platform with standardized set top box specifications and transparent TRP (Television Rating Points) measurement, carries implications beyond BCSL's institutional boundaries. If implemented effectively, it will modernize a cable industry currently estimated at over 20 million households, establish auditable subscriber and viewership data, and create a transparent advertising market. India's TRAI mandated digital migration, implemented in phases from 2012 onwards, provides both positive lessons (dramatic increase in revenue transparency and tax compliance) and cautionary examples (consumer disruption during rapid forced migration) that BCSL should study carefully.

7.3.6 BCSL and TSS Manufacturing Projects

The manufacturing entities' project pipelines, while smaller in individual scale, collectively address a critical strategic gap: Bangladesh's near total dependence on imported telecommunications and digital equipment. BCSL's power cable capacity expansion (from 600 to 1,500 metric tonnes), LAN/Network cable (CAT6/CAT6A) production facility, and 11/33 kV underground power cable feasibility study represent progressive capacity building in physical infrastructure manufacturing. TSS's diversification into CCTV and surveillance equipment (with Korean OEM support), 4G/5G WiFi router assembly (from ZTE/Huawei components), and smart electric meter production (using IoT technology from expatriate Bangladeshi scientists) represents a technology transfer model that, if successfully implemented, reduces import dependency while building domestic manufacturing capability.

These manufacturing projects are strategically important precisely because they are modest in ambition. Unlike the legacy mega projects that suffered from over specification and procurement capture, the manufacturing initiatives are structured as phased expansions with identifiable revenue streams, specific customer bases (power utilities, government procurement, telecom operators), and manageable technology risk. The critical success factor is quality certification: without IEC, ISO, and product specific certifications, domestically manufactured products will face resistance in both government procurement and commercial markets. Vietnam's experience in building a cable and electronics manufacturing export industry through systematic quality certification and competitive pricing provides a roadmap that BCSL and TSS should follow.

7.4 Project Governance Safeguards: Preventing Recurrence

The lessons from legacy project failures must be institutionalized through binding project governance safeguards that apply to all future PTD projects, regardless of funding source. The following safeguards are proposed for adoption through PTD administrative order and, where necessary, through amendments to the SOF Guidelines and entity procurement manuals.

7.4.1 Mandatory Feasibility Study and Independent Review



All PTD projects exceeding BDT 25 crore must undergo a comprehensive third party feasibility study by a qualified, independent team selected through competitive procurement. The current BDT 50 crore threshold established by the Ministry of Planning is insufficient for the telecom sector, where technology cycles are shorter and cost inflation risk is higher. Feasibility studies must include demand analysis validated by independent market data, technology options appraisal with life cycle costing, risk assessment (including technology obsolescence, currency fluctuation, and procurement timeline risks), and a clear statement of assumptions that can be verified during implementation. South Korea's PIMAC (Public Investment Management Center), which conducts independent feasibility studies for all public projects exceeding 50 billion won (approximately BDT 340 crore), provides a rigorous model. PIMAC's studies include benefit cost analysis, policy relevance assessment, and regional balance considerations, with a rejection rate of approximately 50 percent, demonstrating genuine gatekeeping rather than rubber stamping.

7.4.2 DPP Integrity and Change Control

Once a Development Project Proposal (DPP) is approved, any material change in project scope, specifications, cost, or timeline must undergo formal revision through the same approval channel that authorized the original DPP. The Task Force's finding that tender specifications were altered without DPP correction, as occurred in the 5G Readiness Project's GD 1 package, constitutes a fundamental governance breach. A formal Change Control Board, chaired by the HOPE and including representation from the Ministry of Planning and the relevant sector division of the Planning Commission, must authorize any deviation from approved specifications. Unauthorized

specification changes should be treated as prima facie procurement rule violations, triggering automatic referral to the CAG and, where warranted, to the ACC.

7.4.3 Inter Project Coordination Mechanism

All projects within a single entity or with technical dependencies across entities must be mapped in a Project Dependency Register maintained by the PTD's project monitoring cell. The IP Network and 5G Readiness fiasco demonstrates the consequences of implementing interdependent projects without formal coordination. Before any project tender is floated, the HOPE must certify that all technical dependencies have been identified, that linked projects are on track, and that specifications are mutually compatible. Where dependencies involve different entities (for example, BTCL fiber expansion feeding Teletalk coverage deployment), formal inter entity coordination protocols must be established, including shared milestone reporting and escalation mechanisms for dependency failures.



7.4.4 Procurement Transparency and Anti Interference Protocols

The documented political interference in BTCL's 5G Readiness tender, where both a Minister and a Secretary sought to influence the process in favour of different bidders, represents a governance failure that must be addressed through institutional design rather than merely individual accountability. The following protocols are proposed. All communications between political office holders (Ministers, State Ministers, Special Assistants) and procuring entities regarding specific tender processes must be formally documented and disclosed to the relevant review authority. The Head of the Procuring Entity (HOPE) must submit a sworn declaration at each procurement milestone confirming that no external direction has been received that conflicts with PPR requirements. Tender Evaluation Committee (TEC) membership must include at least one external expert nominated from a panel maintained by the Institution of Engineers,

Bangladesh or an equivalent professional body, ensuring independent technical judgment. Technical evaluation reports must be made available to all Board members in advance of any board meeting where procurement decisions are discussed, addressing the practice identified by the Task Force where external board members were denied access to evaluation reports.

7.4.5 Mandatory Project Completion Reporting and Impact Evaluation

Project Completion Reports (PCRs) must be submitted within 90 days of project completion or termination, with the responsible Project Director personally accountable for timely submission. Late PCR submission should be reflected in the Project Director's Annual Performance Assessment, with delays exceeding six months triggering administrative inquiry. IMED impact evaluations must be commissioned within 12 months of project completion, regardless of funding source. For self financed projects that currently fall outside IMED's mandate, the entity's own internal audit function must conduct an equivalent evaluation, with findings submitted to the PTD Secretary and the CAG. Australia's Infrastructure Australia mandates post completion reviews for all nationally significant infrastructure projects, with published assessment reports that inform future project design. This practice of transparent, published evaluation should be adopted for all PTD projects exceeding BDT 100 crore.

7.5 Project Monitoring and Evaluation Architecture

7.5.1 PTD Project Monitoring Cell

A dedicated PTD Project Monitoring Cell should be established within the Division, staffed by officers with project management qualifications and reporting directly to the PTD Secretary. The Cell's mandate would encompass maintaining a consolidated project portfolio dashboard covering all ADP, self financed, and SOF funded projects; conducting quarterly project health assessments using a standardized delivery confidence methodology; maintaining the Project Dependency Register for inter project coordination; tracking PCR submission compliance and IMED evaluation scheduling; and preparing semi annual project portfolio reports for submission to the Secretary and, through the Secretary, to the Chief Adviser's Office. The United Kingdom's Infrastructure and Projects Authority's Government Major Projects Portfolio (GMPP) reporting system provides a proven template for this function. Singapore's Ministry of Finance also maintains a centralized project monitoring function through its Centre for Strategic Futures, which conducts prospective evaluations of major public investments.



7.5.2 Digital Project Management Information System

The project monitoring function must be supported by a digital Project Management Information System (PMIS) that integrates project planning, procurement tracking, financial management, milestone monitoring, and evaluation data in a single platform. The PMIS should be accessible to implementing entities, PTD, the Ministry of Planning, IMED, and the CAG, with role based access controls ensuring appropriate information sharing while maintaining procurement confidentiality where legally required. India's Pragati (Pro Active Governance and Timely Implementation) platform, which enables the Prime Minister's Office to monitor infrastructure project progress across all central government ministries through a unified digital dashboard, demonstrates the transformative potential of centralized digital project monitoring.

7.5.3 Public Accountability and Transparency

Project information, including DPP summaries, procurement timelines, physical and financial progress, and completed evaluation reports, should be published on the PTD website in a machine readable format. This transparency measure serves multiple purposes: it enables parliamentary oversight, facilitates civil society monitoring (including by organizations such as Transparency International Bangladesh that have previously flagged PTD project irregularities), provides data for academic research on project governance, and creates reputational incentives for implementing entities to maintain performance standards. The Philippines' Department of Budget and Management publishes project level ADP data through its Open Data Portal, and Rwanda's RPPA (Rwanda Public Procurement Authority) publishes procurement data in real time through its e Procurement system. Bangladesh should adopt similar practices, beginning with PTD as a pilot division.

7.5.4 Consolidated Project Reform KPI Framework

Reform Area	KPI	Baseline (2025)	Target	Timeline
Feasibility Studies	Projects with independent FS completed before DPP	Partial compliance	100% compliance	2026
PCR Compliance	PCR submitted within 90 days of completion	Significant delays	100% on time	2027
IMED Evaluation	Completed projects with IMED impact evaluation	Below 50%	100%	2027
SOF ADP Integration	SOF projects in ADP annex	Zero	100%	2026
Procurement Transparency	Projects with published TEC reports	Zero	All projects above BDT 50 crore	2027
Project Monitoring	PTD PMIS operational with real time data	None	Full deployment	2027
Nagorik Sheba Scale	Operational citizen service centres	1,122 (Nov 2025)	8,000+	2030
Legacy Remediation	Legacy projects with health assessment completed	None	All active projects	2026
Inter Entity Projects	Cross entity projects with formal coordination protocol	Informal	100% formalized	2026
Public Transparency	Projects with published progress data	Minimal	All ADP and SOF projects	2027

7.6 Chapter Summary

Chapter 7 addresses the systemic project governance crisis within the Posts and Telecommunications Division, where 42 development projects valued at over BDT 20,000 crore were implemented between 2009 and 2024 with pervasive patterns of procurement irregularity, specification manipulation, cost inflation, and delayed completion. The chapter is organized across five reform dimensions. First, the legacy project portfolio diagnosis documents the Task Force's principal findings: absence of entity level master plans, feasibility study failures (including the unauthorized specification alteration in BTCL's 5G Readiness Project from 100G/10G to 400G/100G equipment), inter project dependency breakdowns that rendered substantial expenditure valueless, documented political interference in procurement by both a former Minister and a former Secretary, and chronic non submission of Project Completion Reports. Second, the Social Obligation Fund governance reform addresses five structural failures including non compliance with mandatory contributions, large idle balances, exclusion from ADP and IMED oversight, fragmented accountability, and absence of monitoring and evaluation, proposing ADP annex integration, independent project selection, a unified digital SOF management platform, automated contribution enforcement, and performance based project monitoring. Third, the new generation of strategically aligned projects initiated in 2025 is catalogued, led by the Nagorik Sheba Bangladesh programme (1,122 centres operational by November 2025, targeting 8,000 by 2030), BTCL's Mohakhali Technology Complex and Hathazari Green Data Centre, Teletalk's remote area and highway corridor coverage projects, Bangladesh Post's NAMS and CLTP platforms, BSCL's maritime AIS and cable TV digitalization initiatives, and BCSL and TSS manufacturing expansions. Fourth, binding project governance safeguards are proposed: mandatory independent feasibility studies for all projects above BDT 25 crore, formal DPP change control boards, a Project Dependency Register for inter project coordination, procurement anti interference protocols requiring sworn HOPE declarations and external TEC membership, and mandatory PCR submission within 90 days with IMED impact evaluation within 12 months. Fifth, the project monitoring and evaluation architecture establishes a dedicated PTD Project Monitoring Cell, a digital Project Management Information System accessible to all oversight bodies, and public transparency through machine readable project data publication. International frameworks from the United Kingdom's Infrastructure and Projects Authority, South Korea's PIMAC, Australia's Infrastructure Australia, India's Pragati platform, and the Philippines' Open Data Portal inform the reform design throughout.

— End of Chapter 7 —

8. Strategic and Governance Reforms

8.0 Introduction

The preceding chapters have diagnosed the specific institutional, project, and infrastructure failures that have constrained the posts and telecommunications sector. This chapter turns from diagnosis to architecture. It sets out the strategic and governance reforms necessary to create a regulatory and policy environment capable of sustaining the transformation agenda across electoral cycles, leadership transitions, and technological shifts. While Chapter 6 addressed the internal restructuring of PTD entities and Chapter 7 reformed the project management ecosystem, this chapter operates at a higher level of abstraction: the rules, institutions, and accountability frameworks that govern the entire sector.

The reform imperatives are substantial. The Task Force review documented an ecosystem in which regulatory independence existed in statute but not in practice, where spectrum policy functioned as a revenue extraction mechanism rather than an enabler of connectivity, where over 3,000 licences across 23 categories created artificial barriers to competition and innovation, and where consumer protection, data governance, and cybersecurity frameworks remained either absent or unenforceable. The July to August 2024 internet shutdown, in which BTRC was made an instrument of executive overreach in ordering a complete nationwide connectivity blackout, crystallized the governance deficit at the highest level.



Significant reform has already commenced. The Bangladesh Telecommunication (Amendment) Ordinance 2025, approved by the Advisory Council in December 2025, permanently prohibits internet and telecom service shutdowns, restores BTRC operational autonomy, and establishes a

parliamentary accountability committee to oversee regulatory performance. The Telecommunications Network and Licensing Policy 2025, gazetted on 22 September 2025, replaces the International Long Distance Telecommunication Service (ILDTS) Policy 2010 and consolidates over a dozen fragmented licence categories into four technology neutral categories. Starlink received its operating licence in April 2025 and launched commercial service on 20 May 2025, representing the fastest Non-Geostationary Satellite Orbit (NGSO) licence issuance in Bangladesh's regulatory history.

These are foundational but incomplete actions. Enacting an ordinance or gazetting a policy is necessary but insufficient. The challenge is implementation, institutional embedding, and sustained governance discipline. This chapter provides the detailed reform architecture required to convert legislative intent into operational reality, organized across eleven interconnected reform domains: regulatory architecture, market design, spectrum governance, digital regulatory modernization, investment climate, consumer protection, data governance and cybersecurity, digital inclusion, emerging technology governance, sustainability, and evidence based policymaking.

8.1 Regulatory Architecture Reform

8.1.1 The Governance Deficit

BTRC was established under the Bangladesh Telecommunication Act 2001 as an independent regulator. However, its operational autonomy has been progressively eroded through amendments, ministerial directives, and informal practices. The 2010 amendment framework transferred critical decision making authority, including licensing, tariff setting, and spectrum allocation, to the Ministry, reducing BTRC to a de facto executing agency for ministerial decisions. The Task Force documented that regulatory decisions were routinely subject to ad hoc instructions, ministerial intervention, or political influence rather than rule based governance. This dependency model stands in stark contrast to the standards articulated by the International Telecommunication Union, the OECD Principles for the Governance of Regulators, and the European Electronic Communications Code, all of which mandate clear institutional separation between policy formulation and regulatory execution.

Simultaneously, the Department of Telecommunications (DoT) lacked a well defined regulatory mandate, rendering it largely inactive despite its statutory existence. The resulting institutional vacuum created overlapping authorities with unclear accountability, where neither BTRC nor DoT possessed the mandate, capacity, or political authority to drive coherent sectoral reform.

8.1.2 The Amended Ordinance Framework

The Bangladesh Telecommunication (Amendment) Ordinance 2025 represents the most significant governance restructuring since the original 2001 Act. Its principal provisions establish the following architecture.

Restored Regulatory Autonomy. The Ordinance returns operational decision making authority to BTRC. All licensing authority other than nationally significant licences requiring independent study has been returned to the Commission. This reverses the centralization imposed by the 2010 amendments and aligns Bangladesh with the FCC, Ofcom, and TRAI models where the regulator possesses independent licensing and enforcement authority.

Prohibition of Internet Shutdowns. Section 97 permanently eliminates the scope for nationwide or localized internet shutdowns by administrative order. This directly addresses the July to August 2024 shutdown and provides statutory protection against future connectivity disruption. Bangladesh joins a small group of nations, including India following its Supreme Court ruling in *Anuradha Bhasin v. Union of India*, that have institutionalized legal protections against internet shutdowns.

Parliamentary Accountability Committee. A new accountability committee, chaired by the head of the relevant parliamentary standing committee, has been established to oversee regulatory performance. This introduces a legislative accountability layer that did not previously

exist, modelled on the relationship between the UK Parliament's Select Committee on Culture, Media and Sport and Ofcom.

Privacy Protections. The Ordinance strengthens citizens' privacy protections in alignment with international standards established by the United Nations and the ITU, narrowing the scope of punitive action to direct incitement to violence in line with the Cyber Safety Ordinance 2025.

Investment Facilitation. Licence processing timelines have been reduced and high and repetitive penalties have been scaled back, measures designed to encourage investment in the telecom sector and reduce regulatory friction.

8.1.3 DoT and BTRC Mandate Clarification

The Ordinance framework must be operationalized through a formal Cabinet or Government Order that specifies the functional boundary between DoT and BTRC. Drawing from the Task Force recommendations and international best practice, the mandate split should be structured as follows.

Department of Telecommunications (DoT): Policy formulation and advisory functions, strategic sector planning, inter ministerial coordination on digital economy matters, oversight of state owned enterprise performance, and coordination with Planning Commission on ADP projects. DoT should serve as the policy brain of the sector, responsible for long term strategic direction without involvement in day to day regulatory decisions.

BTRC: Licensing and guideline formulation, spectrum management and allocation, technical standards and enforcement, consumer protection and quality of service monitoring, market analysis and competition oversight. BTRC retains the full operational toolkit of a modern regulator.

Revenue Collection: The Task Force recommends removing revenue collection responsibility from BTRC and transferring it to the Government's appropriate financial authorities. This eliminates the inherent conflict of interest where the regulator responsible for sectoral development simultaneously extracts maximum revenue from the entities it regulates.

A joint steering committee with a public facing tracker should monitor inter agency coordination between DoT and BTRC and ensure timely implementation of decisions. The Government Order should be published within 90 days of this Strategy's adoption and the steering committee should hold its first meeting within 30 days of publication.

International benchmark: Australia's separation between the Department of Infrastructure, Transport, Regional Development, Communications and the Arts (policy) and the Australian Communications and Media Authority (regulation) provides a comparable institutional architecture where policy direction and regulatory execution operate through clearly defined mandates with formal coordination mechanisms.

8.1.4 KPI Framework: Regulatory Architecture

Indicator	Baseline (2025)	Target (2027)	Target (2030)
DoT-BTRC Government Order	Not published	Published, operational	Reviewed and updated
BTRC licensing autonomy	Ministry dependent	Full operational autonomy	Sustained
Parliamentary oversight sessions	None	Quarterly	Quarterly with published reports
Internet shutdown incidents	Multiple (2024)	Zero	Zero
Licence processing time	Variable, unpredictable	Published SLA timelines	90% within SLA



8.2 Market Design and Licensing Reform

8.2.1 The Legacy Licensing Problem

As of early 2025, BTRC had issued over 3,000 licences across approximately 23 categories, many overlapping in functionality and creating artificial silos in service provision. The fragmented regime compelled Mobile Network Operators to maintain multiple licences for converged services, resulting in increased compliance costs, duplicated infrastructure investments, and slower rollout of innovative solutions. The Task Force documented how the proliferation of licence categories, including International Gateway (IGW), International Internet Gateway (IIG), Interconnection Exchange (ICX), National Internet Exchange (NIX), and Internet Service Provider (ISP) categories, fostered rent seeking, anti competitive cartel formation, and regulatory capture. The IGW Operators Forum (IOF), formed with BTRC approval in 2015, exemplified this failure: a two tier system that held the international voice traffic market to ransom through artificial pricing structures, channelling an estimated BDT 631 crore to a connected account.

8.2.2 The Telecommunications Network and Licensing Policy 2025

The TNL Policy 2025, gazetted on 22 September 2025, represents the most ambitious licensing reform since Bangladesh's telecom liberalization. It consolidates the fragmented legacy structure into four technology neutral categories: Access Network Service Providers (ANSP), covering cellular mobile and fixed telecom operators; National Infrastructure and Connectivity Service Providers (NICSP), covering towers, fibre networks, and national transmission; International Connectivity Service Providers (ICSP), covering submarine cable, IP transit, and international bandwidth; and Non-Terrestrial Networks and Service Providers (NTNSP), covering satellite and NGSO services.

The policy abolishes the intermediary licence layers that historically inflated costs without adding meaningful value. IGW, IIG, ICX, and NIX licences will not be renewed upon expiry, with most expiring by 2027. Existing Cellular Mobile Service Providers will be migrated to the ANSP category. For the first time, the framework formally recognizes Mobile Virtual Network Operators (MVNOs), opening the market to asset light entrants that can stimulate competition and innovation without requiring massive capital expenditure on network infrastructure.

The policy introduces strict performance obligations: mobile operators must connect 50 percent of towers to fibre within 18 months and 80 percent within three years, the national fibre backbone must extend underground to every union parishad, and in disaster prone areas at least 25 percent of towers must have diesel generator backup. A National Quality of Service Dashboard will provide anonymized, near real time data on accessibility, coverage, throughput, and reliability.

8.2.3 Implementation Architecture

BTRC has begun drafting the operational guidelines necessary to translate the policy into enforceable rules. The implementation requires careful attention to several dimensions.

Managed Transition. Legacy licence holders possess legitimate expectations based on existing investments. The transition roadmap must include compensation mechanisms, migration incentives, and safeguards against stranded assets. Stage 1 begins with the Policy taking effect (September 2025). Stage 2 introduces the new licensing regime upon government approval of guidelines. Stage 3 commences in 2027, with overlap continuing until existing licences in each category expire or migrate. This graduated approach avoids the sudden disruption that has undermined telecommunications reforms in other jurisdictions.

Foreign Ownership Framework. The Policy caps foreign equity in mobile operators at 85 percent, requiring at least 15 percent local ownership. Among private operators, only Grameenphone currently meets this threshold. Robi (Axiata 61.82 percent, Bharti Airtel 28.18 percent) and Banglalink (fully owned by VEON) face significant restructuring requirements. The Strategy recommends that BTRC establish a clear timeline and mechanism for compliance, avoiding disruptive forced divestiture while ensuring meaningful local participation in the sector's value creation.

Fiscal Revenue Protection. Licence simplification reduces point of collection revenue. The government must introduce compensatory fiscal measures, including broadened base, revised fee structures, usage growth targets, or temporary levy instruments, and monitor revenue impact quarterly to preserve both regulatory sustainability and investment attractiveness.

Anti-Cartel and Competition Safeguards. The Strategy mandates telecom specific anti cartel rules. Any collective arrangement must be contractually transparent and externally audited. Market Development Expense funds or collective funds must be held in regulated accounts subject to independent audit and public disclosure. The Significant Market Power (SMP) framework should empower BTRC to mandate infrastructure sharing, regulate wholesale access pricing, and intervene to prevent market abuse.

Revenue Generation and Licensing Alignment: Following the Telecommunications Network and Licensing Policy 2025, PTD will execute a comprehensive revenue re-collection and generation plan. This includes:

- **Fiber Bank & Customer Acquisition:** Establishing a "National Fiber Bank" to map all existing dark fiber and identify potential commercial customers through partnerships with new license holders.
- **Service Evolution:** Accelerating the launch of Triple Play (Voice, Data, IPTV) and Quad Play services, including the sale of mobile devices and IoT hardware in installments supported by private sector financing.
- **MVNO Implementation:** Finalizing the technical and commercial integration for Mobile Virtual Network Operators (MVNO) to maximize the utilization of state-owned spectrum and infrastructure.

International benchmark: India's Unified License regime, introduced in 2013, consolidated over a dozen service specific licences into a single framework and catalyzed rapid mobile broadband growth. Malaysia's Unified Licence under the Communications and Multimedia Act 1998 similarly demonstrated that licence simplification, combined with technology neutrality, accelerates investment and reduces consumer prices.

8.3 Spectrum Governance Reform

8.3.1 Diagnosis

Bangladesh's spectrum allocation policy has functioned more as a revenue extraction mechanism than as an enabler of efficient utilization and universal connectivity. The Task Force documented that auctions carried rigid rules and inflated reserve prices, leaving operators overburdened and spectrum unsold. The proposed 700 MHz auction, critical for rural broadband and 5G deployment, set a base price of BDT 263 crore per MHz, far exceeding international benchmarks. If applied, spectrum costs would reach 14 to 16 percent of operator revenue compared to the 9 to 12 percent prevailing rate, actively discouraging the investment needed for network modernization. Overpriced base prices left valuable spectrum unsold, representing a pure loss of opportunity for the duration of non-utilization.

Additional structural deficiencies include fragmented band assignments, the absence of secondary spectrum trading or sharing mechanisms, limited availability of key low frequency bands, weak spectrum monitoring and enforcement capacity, and inconsistent regulatory decisions that undermined investor trust. The absence of a long term, harmonized policy framework resulted in inefficient utilization, with some frequency bands remaining underused while others faced congestion due to pricing asymmetry.

8.3.2 Reform Architecture

National Five-Year Spectrum Plan. The government should develop and publish a comprehensive Spectrum Plan that sets out frequency bands, auction schedules, coverage obligations, and technology pathways through formal public consultation. The Plan should cover IMT bands for 5G deployment (including 3.5 GHz mid-band and 700 MHz for coverage), satellite bands aligned with NGSO licensing, Industrial IoT spectrum for manufacturing and agriculture, and Fixed Wireless Access (FWA) allocations for rural broadband. The Plan should be published for public consultation within six months of this Strategy's adoption.

Technology-Neutral Spectrum Rights. Spectrum rights should be converted into technology neutral permits allowing operators flexibility to deploy 4G, 5G, or FWA services without additional administrative burdens. This aligns with ITU-R Resolution 223 (Rev.WRC-19) promoting flexible broadband spectrum use and eliminates the technology specific constraints that have historically limited operator choice.

Secondary Spectrum Markets. The Policy should enable spectrum trading, leasing, and sharing to enhance efficient utilization. A public spectrum registry should maintain transparency and accountability. Active radio network sharing among operators should be permitted under defined security and technical rules, with fee rebates for shared rural infrastructure that demonstrably reduces diesel consumption and environmental impact.

Performance-Linked Pricing. Spectrum fee structures should shift from large upfront payments to staged, performance linked payments that tie financial obligations to measurable coverage and service delivery outcomes, especially in underserved regions. Reserve prices should be benchmarked against comparable markets, with clear published justifications to prevent overpricing and unsold allocations. BTRC should maintain a public spectrum and performance dashboard, updated quarterly, to track usage, coverage, and efficiency as eligibility criteria for future spectrum awards.

International benchmark: The United Kingdom's Ofcom Spectrum Management Strategy 2021 to 2026 exemplifies a balanced approach, combining competitive auctions with coverage obligations, secondary trading markets, and technology neutral permits. India's spectrum sharing guidelines allow operators to pool spectrum for efficiency, while the European Commission's 5G Action Plan coordinated cross-border spectrum roadmaps with binding coverage targets.

8.3.3 KPI Framework: Spectrum Governance

Indicator	Baseline (2025)	Target (2027)	Target (2030)
Five-Year Spectrum Plan	None	Published, operational	Second edition
Spectrum trading framework	Not permitted	Rules published	Active trading market
Unsold spectrum (auctions)	Significant (2022)	Below 10%	Below 5%
Spectrum cost as % revenue	14-16% proposed	Below 12%	9-10% (ITU aligned)
5G mid-band assignment	Unallocated	Allocated to operators	Commercial deployment

8.4 Digital Regulatory Modernization

Governance reform is incomplete without the operational systems that make transparency, efficiency, and accountability technically achievable. The current regulatory ecosystem operates on fragmented manual processes that impede both speed and integrity.

8.4.1 Enterprise Resource Planning for BTRC

The Strategy mandates deployment of a comprehensive Enterprise Resource Planning (ERP) system to integrate all BTRC operational functions under a unified digital platform. The ERP should automate licensing and renewal processes, centralize revenue and spectrum management, monitor network quality through real time dashboards, and streamline enforcement, compliance, and consumer grievance handling. It should interlink with Customs, the National Board of Revenue, the Election Commission, BSCPLC, NTTN operators, and other agencies for cross verification of data, while supporting research, analytics, and administrative operations. Equipped with role based access, digital signatures as part of PKI, blockchain secured records, and AI assisted analytics, the ERP should enable BTRC to evolve into a data driven, responsive, and analytically empowered regulator.

BTCL has already commenced ERP procurement, with a MoU signed and ERP consultant procurement underway as of November 2025. The ERP initiative should be extended across all PTD entities as a shared platform or interoperable system, creating a unified information backbone for the entire posts and telecommunications ecosystem.

International benchmark: Singapore's Info-communications Media Development Authority (IMDA) operates a fully digital licensing and compliance platform with real time industry data analytics. The Turkish telecommunications regulator ICTA deployed an integrated regulatory management system covering licensing, spectrum, QoS monitoring, and enforcement within a single digital architecture.

8.4.2 National One-Stop Digital Permit Portal



A national one stop digital permit portal should be established to streamline site construction, licensing, and renewals through coordinated timelines, dig once scheduling, and digital tracking. The portal should cover necessary licences and imports with statutory 15 and 30 day decision clocks. A shared dig once works calendar should prevent duplicative excavation for fibre and utility deployment, reducing both cost and urban disruption. The first phase should launch within 120 days of this Strategy's adoption with an initial scope covering tower construction permits and equipment import clearances. Full scope should be achieved by 2027, covering all licence categories, renewals, spectrum applications, and interconnection approvals.

8.4.3 Digital Equipment Import Registry

A digital registry for pre-approved telecom equipment should be introduced, allowing automatic clearance of repeat shipments and reduced processing times. The registry should include a SAR testing and certification component, enabling local verification and certification of mobile handsets. This reduces reliance on foreign laboratories, prevents falsified compliance reports, and expedites clearance for both imported and locally manufactured devices. The National Equipment Identity Register (NEIR), re-launched in 2025 to curb illegal handset imports in a market where grey channel phones account for an estimated 50 to 60 percent of sales, should be integrated with the equipment registry for comprehensive device governance.



8.5 Investment Climate and Fiscal Reform

The telecommunications sector cannot fulfil its infrastructure and inclusion mandate without a stable, predictable, and investment friendly fiscal environment. The current taxation framework imposes cumulative burdens that erode profitability and deter both domestic and foreign investment.

8.5.1 Taxation Reform

Bangladesh's telecom taxation regime prioritizes short term fiscal gain over long term digital transformation. Excessive taxation erodes profitability and investment incentives, reducing the financial capacity of operators to expand networks, upgrade technologies, and enhance service quality. The Task Force documented that the taxation framework is complicated by inconsistent fiscal treatment across different operator categories, lack of harmonization between BTRC and National Board of Revenue (NBR) policies, and frequent ad hoc changes introduced through annual Finance Acts. India's GST regime, the UK's uniform VAT system, and the FCC's universal contribution framework demonstrate that rationalized tax structures can generate sustainable government revenue while simultaneously encouraging sectoral growth.

The Strategy recommends a predictable, investment oriented, and sector neutral tax structure that harmonizes fiscal obligations across all operators. This includes rationalizing VAT and supplementary duties, reducing corporate tax rates for telecom operators to regional averages, and integrating spectrum and licensing fees within a transparent long term amortization framework. Tax incentives should be introduced for infrastructure sharing, green ICT adoption, subsidized service for underserved regions, and digital innovation. Institutional coordination between BTRC and NBR should be formalized to eliminate regulatory overlap and ensure policy coherence.

8.5.2 Foreign Direct Investment Framework

The entry of Starlink in April 2025, with its 10 year operating licence, commercial launch on 20 May 2025, four local gateways installed by August 2025, and 880 Gbps of deployed capacity, demonstrates Bangladesh's capacity to attract global technology investment when regulatory processes are streamlined. This momentum must be institutionalized. The Strategy recommends clarifying foreign investment rules, maintaining stable ownership frameworks that balance local participation with international capital access, and creating a regulatory environment where licence processing timelines are predictable and transparent. The 15 percent local ownership requirement under the TNL Policy 2025 should be implemented through a phased compliance timeline that avoids forced divestiture and provides adequate transition mechanisms for affected operators.

8.6 Consumer Protection and Quality of Service

Consumer protection has been a persistent weak point in Bangladesh's telecom governance. The Task Force documented weak enforcement, absence of telecom service quality monitoring, consumer dissatisfaction, and reputational damage that deterred investment and affected sector revenue. BTRC drive tests revealed widespread service deficiencies including excessive call drop rates and poor 4G voice performance across major urban and peri-urban areas.

8.6.1 Modernized Customer Code

A modernized customer code should require clear, plain language tariff communication, explicit consent for value added services, standardized billing formats, and transparent fair usage policies. Pricing regulation should move toward a flexible yet fair structure, allowing competitive pricing while preventing predatory practices through benchmark based oversight and published reference rates.



8.6.2 National Quality of Service Dashboard

District level independent performance scorecards should be published regularly, combining drive test and crowdsourced data on quality of service indicators such as speed, latency, call drop rates, and complaint resolution. The National QoS Dashboard, mandated by the TNL Policy 2025, should provide anonymized near real time data accessible to the public. Monthly public reporting will improve user trust and create competitive pressure for operators to improve service standards.

8.6.3 Digital Complaint Management System

BTRC should establish a national digital complaint management system providing a single online and mobile platform for submitting, tracking, and escalating grievances. Complaint resolution SLAs should be published, and operator compliance should be tracked through the system. Monthly complaint data should be published to enable comparative assessment of operator performance.

Technical Monitoring: BTRC will implement advanced systems for Spectrum Occupancy, Spectrum Noise monitoring, SAR/Radiation testing, and regular network Drive Tests.

Grievance Resolution: Launch an "All-in-One" ERP/Complaint Management solution with a mandatory 7-day time-bound ticketing system for resolving customer issues.

Licensing Compliance: Strengthen institutional capacity (labs, software, and data analytics) to ensure all licensees fulfill KPI and Quality of Experience (KQoS) obligations as per the 2025 Licensing Policy.

International benchmark: TRAI (India) publishes quarterly QoS reports with operator-specific performance data, creating transparent competitive pressure. The UK's Ofcom requires operators to publish complaint statistics and resolution rates, with persistent underperformers subject to formal investigation and financial penalties.

8.7 Data Governance, Privacy, and Cybersecurity

The Task Force identified an insufficient legal framework and limited incident response capacity that leaves vital sectors such as telecommunications and energy vulnerable to cybersecurity breaches, data theft, and system intrusions, jeopardizing national security. The lack of user data privacy protection measures damages consumer trust, while surveillance and digital eavesdropping without adequate legal safeguards violates constitutional privacy rights.

8.7.1 Reform Architecture

Information Security Standards. The sector should adopt international standards for information security and privacy, specifically ISO 27001 and ISO 27701, supported by mandatory vendor attestations, independent audits, and 72 hour breach reporting requirements. All operators and major service providers should achieve ISO 27001 certification by 2028.

Data Governance Framework. Data governance reforms should ensure purpose limited access, strict retention policies, and auditable lawful interception, preserving both privacy and national security. The Draft Data Protection Ordinance should be harmonized with the Amended Telecom Ordinance to establish clear jurisdictional boundaries and consistent enforcement mechanisms. Cross border data flow provisions should align with GDPR adequacy principles to facilitate international commerce and attract technology investment.

Cybersecurity Preparedness. Cross agency readiness should be institutionalized through biannual cybersecurity and crisis management drills. A sectoral Computer Emergency Response Team (CERT) should be established or strengthened to coordinate incident response across telecommunications infrastructure. The July 2024 experience demonstrated the catastrophic consequences of infrastructure vulnerability, both from external threats and internal executive overreach.

International benchmark: The European Union's NIS2 Directive imposes mandatory cybersecurity obligations on essential service providers including telecommunications operators, with requirements for risk management, incident reporting within 24 hours, and supply chain security assessment. South Korea's KISA operates a world-class telecommunications CERT with mandatory breach reporting and cross-sector coordination protocols.

8.8 Digital Inclusion and Universal Access

Universal access reforms under this Strategy operate on two parallel tracks: restructuring the Social Obligation Fund for results based financing, and deploying complementary inclusion mechanisms that address the affordability, literacy, and accessibility dimensions of the digital divide.

8.8.1 Results-Based SOF

The SOF should be restructured to operate on a results based financing model, awarding disbursements only for verified delivery of measurable outcomes such as broadband speeds, latency, and active user adoption, confirmed by independent audits. This replaces the input based model that allowed equipment to be installed but not operationalized, as documented by the Task Force's field inspections that found routers and hotspots reported as deployed but not functional.

8.8.2 Affordability Interventions

The government should introduce affordability initiatives including data and device vouchers and zero interest handset programs for low income populations, supported by phased fiscal incentives tied to measurable affordability gains. The Strategy also endorses Teletalk's proposed SIM distribution through post offices, creating a physical touchpoint for affordable connectivity that leverages the Nagorik Sheba Bangladesh platform's 8,000 planned centres by 2030.

8.8.3 Digital Navigator Helpdesks

Locally anchored digital navigator helpdesks should be established to assist rural and marginalized users with connectivity registration, digital safety, and complaint resolution. These helpdesks should be integrated with Nagorik Sheba centres, leveraging trained local entrepreneurs as human intermediaries who bridge the digital divide for populations that cannot independently navigate online services. This addresses the critical insight from the Digital Transformation Strategy 2025 to 2030 that digital transformation cannot succeed through technology deployment alone; it requires investment in human capacity at the last mile.

International benchmark: Colombia's community digital centres provide free digital literacy training and subsidized connectivity, reaching over 10 million citizens in underserved areas. India's Common Service Centres operate a network of over 500,000 village-level outlets providing digital government services through trained operators, a model directly analogous to the Nagorik Sheba concept.

8.9 Emerging Technology Governance

8.9.1 5G Readiness

The Five Year Spectrum Plan should establish a clear 5G deployment pathway, including mid-band spectrum allocation (3.5 GHz), coverage obligations for urban and peri-urban areas, and vertical industry use cases in health, manufacturing, smart agriculture, and education. The Strategy recommends phased 5G deployment beginning with urban centres and economic zones by 2028, with progressive expansion to district headquarters and transport corridors by 2030. BTRC should operate regulatory sandboxes for emerging 5G use cases, including Intelligent Transport Systems, Industrial IoT, and Fixed Wireless Access, allowing controlled experimentation before full scale commercial deployment.

8.9.2 Satellite Connectivity

Starlink's entry in May 2025 inaugurated Bangladesh's satellite broadband era. With four gateways operational by August 2025 (two at Kaliakair Hi-Tech Park, one each in Rajshahi and Jashore) and approximately 1,800 active devices within five months, the service has demonstrated commercial viability. Around half a dozen additional global satellite providers have expressed interest. The NGSO licensing framework should be maintained as an open category, with clear fee structures, revenue sharing provisions (3 percent for years three through five, 5.5 percent from year six), and local gateway requirements that balance market access with data sovereignty and lawful interception obligations.

Starlink's application to export bandwidth from Bangladesh to neighbouring countries, including India, Nepal, Bhutan, and Myanmar, represents an opportunity for Bangladesh to position itself as a regional connectivity hub. This aligns with Bangladesh Submarine Cable Company's existing bandwidth export to northeast India and should be governed through clear IPLC guidelines that protect domestic interests while encouraging transit revenue.

8.9.3 OTT and Digital Platform Governance

The TNL Policy 2025 recognizes that ANSP licensees are permitted to offer any value added or OTT services through their networks. The Amended Telecom Ordinance extends regulatory scope to social media platforms, intermediary services, OTT services, audiovisual and streaming platforms, e-commerce, content based services, cloud providers, data centres, and AI services. While comprehensive scope is important for regulatory coherence, the Strategy cautions against regulatory overreach that could stifle innovation or centralize excessive authority. A proportionate, flexible regulatory approach should be adopted for emerging and innovative services, as the TNL Policy itself prescribes, with distinct governance tracks for telecommunications infrastructure (heavily regulated) versus digital content and platform services (lighter touch, consumer protection focused).

8.9.4 IoT and AI Governance

Governance frameworks for the Internet of Things and Artificial Intelligence should be developed through cross sectoral coordination rather than telecommunications regulation alone. The recent IoT and radio equipment import guidelines represent a useful start but must be complemented by interoperability standards, security certification requirements, and spectrum allocation for low power wide area networks. AI governance should be aligned with the National AI Policy and the Digital Transformation Strategy 2025 to 2030, with BTRC's role limited to AI applications within telecommunications networks rather than general purpose AI regulation.

8.10 Sustainability and Green ICT

Environmental sustainability should be integrated into the telecommunications policy framework through energy efficiency targets, renewable energy adoption for telecom sites and data centres, and enforceable e-waste recycling standards under producer responsibility frameworks. The TNL Policy 2025 mandates eco-friendly practices including renewable energy for towers, aligning with global trends in green telecommunications.

The Strategy recommends measurable green KPIs linked to concessional financing opportunities. Operators should be required to report annual energy consumption, renewable energy adoption rates, and e-waste management volumes. Spectrum fee rebates should be provided for shared rural infrastructure that demonstrably reduces diesel consumption. The Hathazari Green Data Centre project, structured as a PPP with ADB involvement, should serve as the pilot for Tier III certified, energy efficient data hosting that meets both domestic compliance requirements and international sustainability standards.

International benchmark: The European Green Digital Coalition commits ICT companies to science-based emissions targets. GSMA's Climate Action Taskforce tracks mobile industry progress toward net zero by 2050. Kenya's Communications Authority requires operators to report annual carbon footprint and renewable energy adoption, with penalties for non-compliance integrated into licence renewal conditions.

8.11 Evidence-Based Policymaking

8.11.1 BTRC Research Division

To embed evidence based policymaking and future foresight into telecom governance, BTRC should establish a permanent, independent research division modelled on global best practices. The division should be staffed with experts in telecommunications economics, data science, law, and engineering and should collaborate with universities, think tanks, and international institutions. It should be mandated to conduct market and spectrum economics analysis, including auction benchmarking, SMP analysis, and open access cost modelling, and to evaluate unified licensing frameworks and operate regulatory sandboxes for emerging technologies.

The Research Division should publish quarterly transparent KPIs and open datasets covering affordability (effective price per GB), rural latency, spectrum utilization, and permitting benchmarks (p50 and p90 approval times). It should possess operational autonomy to publish insights, convene stakeholder dialogues, and advise on strategic reforms without requiring prior clearance from BTRC's operational leadership.

International benchmark: Ofcom's research and evidence team publishes the annual Communications Market Report and sector-specific analysis that forms the empirical foundation for regulatory decisions. TRAI's research division conducts consultation papers on emerging issues, typically receiving hundreds of stakeholder submissions that inform final regulatory determinations. The FCC's Office of Economics and Analytics was established in 2018 to strengthen the empirical basis for US telecommunications regulation.

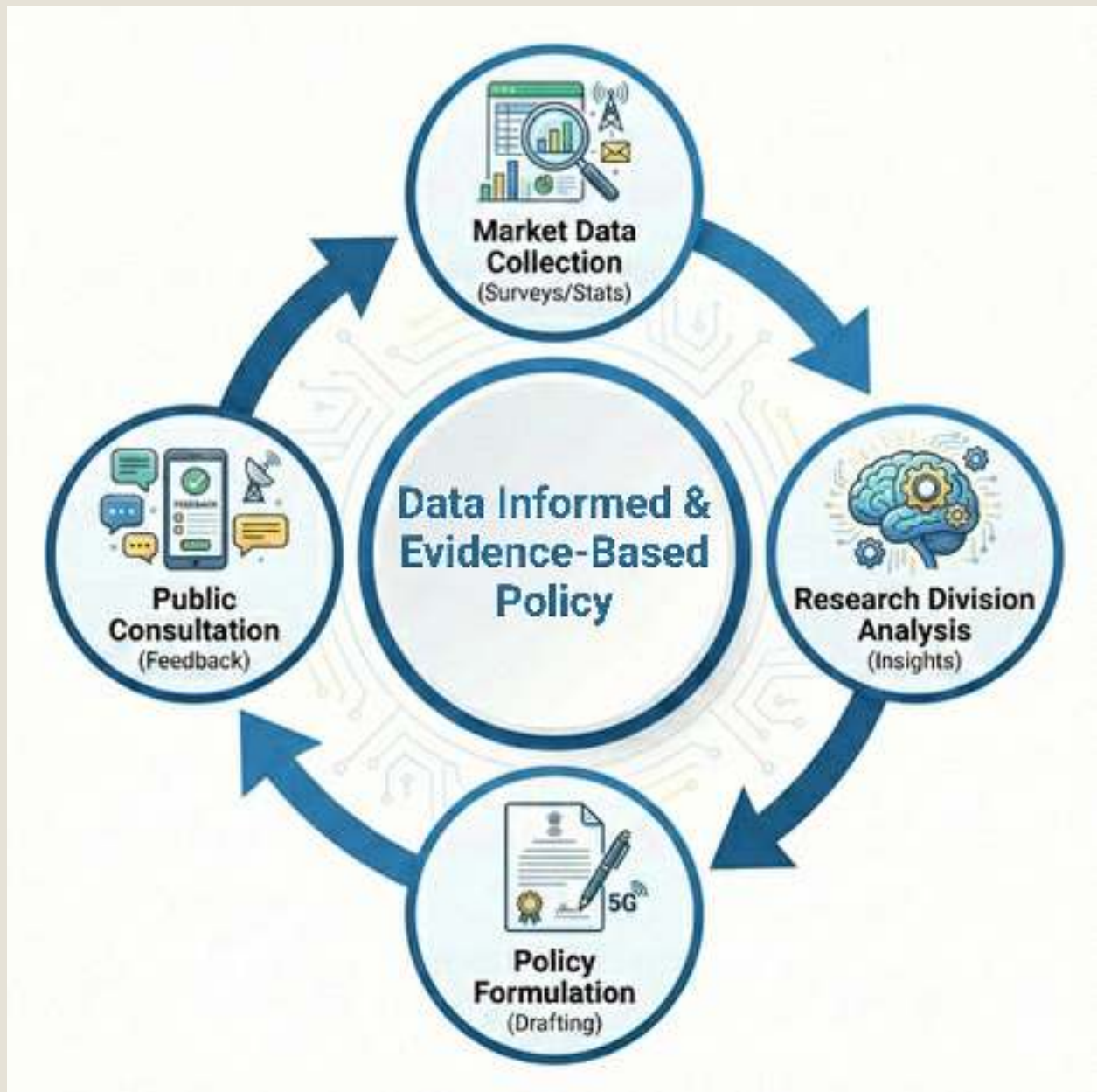
8.11.2 Centre of Excellence for Digital Public Infrastructure

An independent Centre of Excellence should be established for third party impact evaluation of SOF projects, digital public infrastructure, and long term sustainability assessment. The Centre should operate independently of BTRC and PTD, with funding from SOF allocations, to provide objective evaluation that informs both project design and policy formulation. It should be mandated to publish all evaluation findings and should develop partnerships with international research institutions for methodology development and peer review. The Centre should also serve as the institutional home for the sectoral knowledge base, maintaining comprehensive databases on project outcomes, technology assessments, market analysis, and policy evaluations that inform future decision making.

8.11.3 Telecom Reform Program Management Office

A dedicated Telecom Reform Program Management Office (PMO) should be established under PTD to oversee the implementation of reforms across all domains, address bottlenecks, coordinate cross agency actions, and publish monthly progress dashboards. The PMO should report directly to the PTD Secretary and should have the authority to convene inter-entity coordination meetings, escalate implementation failures, and recommend course corrections.

State owned entities should be required to meet clear performance contracts and Service Level Agreements (SLAs), with annual performance assessments linked to leadership renewal and resource allocation decisions.



8.12 Consolidated KPI Framework

The following table consolidates the key performance indicators across all reform domains addressed in this chapter, providing a unified monitoring framework for strategic and governance transformation.

Reform Domain	Baseline	2027 Target	2030 Target	Lead Entity
BTRC autonomy operational	Limited	Full	Sustained	PTD, BTRC
Licensing categories	23+	4	4 (refined)	BTRC
Five-Year Spectrum Plan	None	Published	2nd edition	BTRC, DoT
ERP deployment (BTRC)	None	Full deployment	Operational	BTRC
Telecom taxation reform	Ad hoc	Rationalized	Stable regime	NBR, PTD
QoS Dashboard operational	None	Operational	Real-time	BTRC
ISO 27001 adoption	Minimal	All MNOs	All licensees	BTRC
5G commercial deployment	None	Pilot (urban)	Commercial	BTRC, MNOs
BTRC Research Division	None	Established	Mature	BTRC
Reform PMO dashboards	None	Monthly published	Sustained	PTD
Internet shutdowns	Multiple	Zero	Zero	BTRC, PTD

8.13 Chapter Summary

This chapter has set out the strategic and governance reform architecture necessary to convert Bangladesh's posts and telecommunications sector from a governance constrained, rent extraction oriented system into a transparent, investment friendly, and citizen centred digital infrastructure ecosystem. The reforms are organized across eleven interdependent domains, each building on the diagnostic findings of the Task Force and the institutional reforms detailed in preceding chapters.

The regulatory architecture has been fundamentally restructured through the Bangladesh Telecommunication (Amendment) Ordinance 2025, which restores BTRC autonomy, prohibits internet shutdowns, and establishes parliamentary oversight. Operationalizing this framework requires a formal DoT-BTRC mandate split through Government Order, with DoT assuming policy and strategic oversight while BTRC retains operational regulatory authority and revenue collection transitions to the Government's financial authorities.

Market design has been reformed through the Telecommunications Network and Licensing Policy 2025, consolidating over 23 fragmented licence categories into four technology neutral categories, abolishing rent seeking intermediary layers, formally recognizing MVNOs, and introducing strict performance obligations including fibre rollout targets, QoS standards, and disaster resilience requirements. The managed transition framework protects legitimate expectations while creating a clear pathway to the simplified regime.

Spectrum governance reform addresses the fundamental misalignment between Bangladesh's revenue maximization approach and the investment enabling approach recommended by ITU guidelines. The Five Year Spectrum Plan, technology neutral permits, secondary trading markets, and performance linked pricing collectively create the conditions for efficient utilization, 5G deployment, and rural broadband expansion.

Digital regulatory modernization, through ERP deployment, the one stop permit portal, and the digital equipment registry, provides the operational infrastructure without which transparency and accountability remain aspirational. Investment climate reform, encompassing taxation rationalization and FDI framework clarification, creates the predictable fiscal environment necessary to attract and retain the capital investment that network modernization demands.

Consumer protection, data governance, and cybersecurity reforms address the trust deficit that constrains both adoption and investment. Digital inclusion reforms restructure the SOF for results based financing and deploy affordability interventions and digital navigator helpdesks through the Nagorik Sheba platform. Emerging technology governance, covering 5G, satellite connectivity (including Starlink), OTT platforms, IoT, and AI, establishes proportionate frameworks for the technologies that will define the sector's next decade.

Sustainability and green ICT reforms integrate environmental responsibility into licensing and operational requirements. Evidence based policymaking, through the BTRC Research Division, the Centre of Excellence, and the Reform PMO, ensures that the governance reforms introduced

in this chapter are monitored, evaluated, and adapted on the basis of empirical evidence rather than political expediency.

The consolidated KPI framework provides measurable accountability targets across all reform domains, enabling parliamentary oversight, civil society monitoring, and internal performance management. The Strategy's success depends not on the ambition of its architecture but on the discipline, consistency, and institutional commitment with which it is implemented across electoral cycles and leadership transitions.

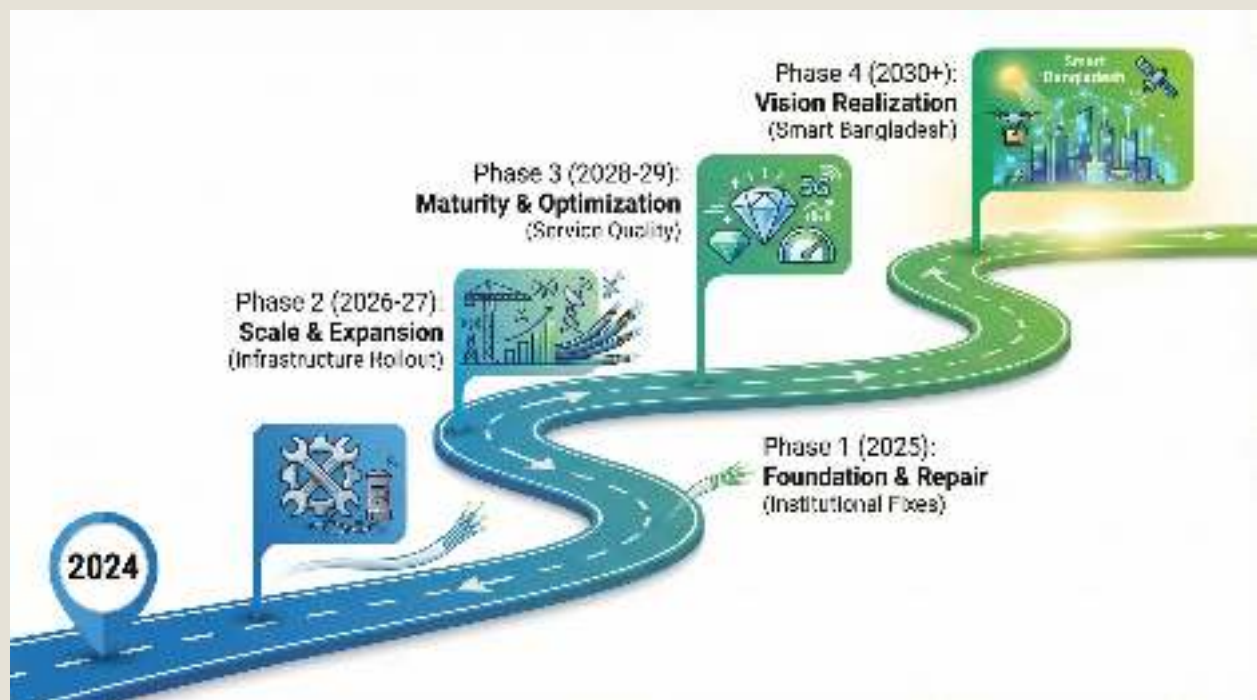
— End of Chapter 8 —

9. Implementation Roadmap

9.0 Introduction

The preceding chapters of this strategy have established the diagnostic foundations, legal architecture, institutional reforms, project corrections, citizen service ambitions, and governance frameworks that collectively define the Posts and Telecommunications Division's digital transformation agenda. This chapter consolidates those interventions into a unified, phased implementation roadmap that sequences actions across time, assigns accountability to specific entities, and establishes measurable milestones at each stage.

The roadmap is structured around three primary phases spanning 2025 to 2030, with a fourth horizon extending to 2035 for reforms and infrastructure investments whose impacts necessarily mature over longer cycles. The phasing logic follows a deliberate sequencing principle: foundational legal and institutional reforms in the early years create the enabling conditions for market restructuring, infrastructure deployment, and technology adoption in the middle years, which in turn generate the data, capacity, and trust required for advanced governance and regional positioning in the later years.



This approach reflects lessons from comparable transformation programmes internationally. The United Kingdom's Digital Strategy 2022 and South Korea's Digital New Deal both adopted three-phase models that prioritized legal clarity and institutional readiness before scaling technology deployment. India's Digital India programme similarly sequenced infrastructure

rollout after completing its Unified Licence reform in 2013, achieving measurable broadband penetration gains only after the regulatory architecture was stabilized. Bangladesh's roadmap applies the same principle to the specific conditions of the posts and telecommunications sector.

Each phase is defined not by rigid calendar deadlines alone but by the achievement of prerequisite conditions that unlock subsequent actions. Phase 2 activities, for instance, cannot proceed effectively without the Government Orders, recruitment rules, and ERP specifications that Phase 1 is designed to produce. This conditional sequencing reduces the risk of reforms stalling because institutional foundations were incomplete.

9.1 Phase 1: Foundation (2025 to 2026)

Phase 1 constitutes the most intensive reform period, requiring simultaneous action across legal, institutional, and technical domains. The central objective is to establish the legal and governance architecture upon which all subsequent phases depend.

9.1.1 Legal and Regulatory Foundations

The Bangladesh Telecommunication (Amendment) Ordinance 2025, approved in December 2025, represents the single most consequential legislative action in the sector in over a decade. Its provisions permanently prohibiting internet shutdowns, restoring BTRC operational autonomy, establishing a Parliamentary Accountability Committee, and extending regulatory scope to digital platforms must now be translated into operational reality. The immediate priorities are:

- **Government Order on BTRC Autonomy:** A formal Government Order must codify the division of responsibilities between DoT (policy advisory and SOE oversight) and BTRC (licensing, enforcement, spectrum management, and consumer protection). This order must be gazetted within the first quarter of 2026, with a joint steering committee established to manage the transition period.
- **TNL Policy 2025 Migration Framework:** The Telecommunications Network and Licensing Policy 2025, gazetted in September 2025, consolidates over twenty three legacy licence categories into four technology-neutral categories: ANSP, NICSP, ICSP, and NTNSP. A managed licence migration framework must be published by mid-2026, with existing operators given a defined twelve-month transition window supported by clear compliance schedules and dispute resolution mechanisms.
- **Postal Services Ordinance Modernization:** The Postal Services Ordinance requires amendment to reflect Bangladesh Post's evolving role as a logistics, financial services, and digital access platform. Drafting should commence in 2025, with a consultation draft circulated to stakeholders by the second quarter of 2026. The amendments should authorize competitive parcel services, digital financial products, and integration with e-commerce platforms.
- **National Network Evolution Framework:** A mandatory Five-Year Network Evolution Plan will be required for all licensees to ensure long-term technology parity. To optimize public investment, PTD will enforce a Network Development Plan Sharing protocol between Teletalk, BSCL (Satellite), and BSCPLC (Submarine Cable). This protocol will ensure equitable resource allocation, preventing duplicative infrastructure while ensuring that satellite and submarine capacity are utilized as primary backups for each other to guarantee 100% national uptime.

- **Five-Year National Spectrum Plan:** BTRC must publish a comprehensive spectrum plan by the end of 2026, addressing 5G mid-band allocation (3.3 to 3.8 GHz), secondary trading frameworks, performance-linked pricing, technology-neutral assignments, and enterprise and localized licensing provisions. The plan should include a public consultation period and incorporate ITU recommendations on spectrum efficiency.
- **MOCSLA Recruitment Rules:** The Mailing Operator and Courier Services Licensing Authority requires finalized recruitment regulations to enable permanent staffing. This legal instrument should be completed within 2025 to 2026.

9.1.2 Institutional and Human Resource Reforms

The institutional reform agenda detailed in Chapter 6 demands immediate action on several fronts during this phase:

- **SOE Board Reconstitution:** Boards of Directors across BTCL, Teletalk, BSCPLC, BSCL, BCSL, and TSS must be reconstituted to include a majority of independent members with sectoral expertise. The PTD Secretary should be separated from the Board Chair position. Target: all boards reconstituted by the end of 2026.
- **Digital HRIS Deployment:** A unified Human Resource Information System must be specified, procured, and deployed across all PTD entities by the end of Phase 1. The system should integrate personnel records, payroll, promotions, deputation tracking, and performance management into a single auditable platform.
- **Telecom Cadre Restoration:** The proposal to rebuild a dedicated Telecommunication Cadre under DoT through competitive civil service recruitment should be formalized during this phase, with the first recruitment cycle initiated by the end of 2026.
- **CAG HR Compliance Audit:** The Comptroller and Auditor General should commence the first comprehensive HR compliance audit across all PTD entities, with findings reported to Parliament. This audit cycle should be completed within 2026.
- **Performance Contracts:** The Reform Programme Management Office should establish formal performance contracts with each SOE, linking measurable KPIs to management accountability. Quarterly dashboards should be operational by mid-2026.

9.1.3 Digital Infrastructure and Systems

The digital systems that underpin regulatory modernization must be initiated during Phase 1, even if full deployment extends into Phase 2:

- **BTRC ERP Specification and Procurement:** The Enterprise Resource Planning system for BTRC, integrating licensing, revenue management, spectrum allocation, QoS

monitoring, and enforcement into a unified platform, requires detailed specification during this phase. A consultant should be engaged to develop the technical specification, with procurement initiated by the end of 2026.

- **BTCL ERP Consultant Engagement:** BTCL has already issued a tender for an ERP specification consultant. The engagement should be completed and the customized ERP specification finalized within Phase 1.
- **Fibre and IP Equipment Digital Mapping:** BTCL's ongoing digital mapping of optical fibre and IP equipment resources should be completed during this phase. The mapping has been made live on the BTCL TIP portal and must be expanded to cover all transmission assets.
- **BSCPLC SMW-6 Commissioning:** The third submarine cable, SEA-ME-WE 6, is expected to become operational in the first quarter of 2026, adding 13,200 Gbps of potential capacity. BSCPLC must complete commissioning, testing, and initial capacity sales during this phase.
- **Teletalk 4G Expansion:** Teletalk has added 1,699 new 4G sites, increasing daily data consumption from 354 to 464 terabytes (approximately 20 percent growth). The tower sharing agreement with Edotco for 2,300 sites should be operationalized during Phase 1, with site surveys and equipment installation proceeding on schedule toward the December 2026 completion target.
- **Bangladesh Post GPS Fleet Tracking:** The installation of GPS trackers on 85 mail vans has been completed. During this phase, the system should be integrated with route optimization software and the Domestic Mail Monitoring System (DMS) to enable real-time fleet management.

9.1.4 Service Delivery Foundations

- **Nagorik Sheba Centre Rollout:** The integration of postal infrastructure with the Nagorik Sheba network should commence during Phase 1, with an initial target of converting 500 post offices into Nagorik Sheba service points by the end of 2026. Each centre should offer NID, passport, tax filing, SIM registration, and basic connectivity services.
- **E-Commerce Logistics Pilot:** Bangladesh Post's reverse logistics middleware system and API integration with the Domestic Mail Monitoring System should be completed during this phase, establishing the technical foundation for two-way urban-rural e-commerce parcel delivery.
- **Cold Chain Infrastructure:** The project to install 14 chiller chambers and 84 deep freezers across postal distribution centres should be completed, enabling Bangladesh Post to handle agricultural and perishable goods logistics.

- **TSS Strategic Partnerships:** TSS has signed a Strategic Partnership Agreement with two technology firms in September 2025 and an MoU with Edotco Bangladesh in December 2025. The pilot manufacturing programmes for CCTV equipment, lithium-ion battery repair, and 4G/5G dual-band WiFi routers should produce initial outputs during Phase 1.
- **Starlink Integration:** Starlink, licensed in April 2025 and launched in May 2025, has deployed four gateways (Kaliakair, Rajshahi, Jashore) with 880 Gbps capacity and 1,800 active devices. During Phase 1, BTRC should formalize the NGSO satellite service licensing framework drawing on the Starlink precedent, with clear revenue sharing schedules (3 percent in years 3 to 5, rising to 5.5 percent thereafter) and local gateway requirements.

9.1.5 Phase 1 Consolidated Milestone Table

Reform Action	Lead Entity	Target Date	Deliverable
BTRC Autonomy Government Order	PTD, DoT	Q1 2026	Gazetted Order
TNL Policy Migration Framework	BTRC	Mid-2026	Published Framework
Postal Services Ordinance Draft	PTD, BD Post	Q2 2026	Consultation Draft
Five-Year Spectrum Plan	BTRC	End 2026	Published Plan
MOCSLA Recruitment Rules	PTD, MOCSLA	2025-2026	Gazetted Rules
SOE Board Reconstitution	PTD	End 2026	All Boards Reconstituted
Digital HRIS Deployment	PTD, All SOEs	End 2026	Operational System

BTRC ERP Specification	BTRC	End 2026	Procurement Initiated
SMW-6 Commissioning	BSCPLC	Q1 2026	Operational Cable
Teletalk Tower Sharing (2,300 sites)	Teletalk, Edotco	Dec 2026	Sites Operational
Nagorik Sheba Postal Integration (500)	BD Post	End 2026	Centres Operational
TSS Manufacturing Pilots	TSS	End 2026	Initial Production
Reform PMO Quarterly Dashboards	PTD	Mid-2026	Operational Dashboards
CAG HR Compliance Audit	CAG, PTD	End 2026	Audit Report

9.2 Phase 2: Scale and Optimize (2027 to 2028)

Phase 2 leverages the legal, institutional, and technical foundations established in Phase 1 to scale reforms across the sector. The emphasis shifts from establishing frameworks to achieving measurable outcomes in market structure, infrastructure deployment, service quality, and institutional capacity.

9.2.1 Market Structure and Regulatory Implementation

With the TNL Policy migration framework in place and the BTRC ERP system under deployment, Phase 2 should deliver tangible market structure changes:

- **Licence Migration Completion:** All existing operators should complete migration from legacy licence categories to the four TNL Policy 2025 categories by the end of 2027. The intermediary licence layer (IGW Operators Forum, IIG, ITC) should be

formally abolished, with international voice and data traffic flowing directly between ICSPs and ANSPs.

- **MVNO Launch:** BTCL and Teletalk have been progressing technical integration for Bangladesh's first MVNO service. The service should be commercially launched during Phase 2, with BTCL leveraging Teletalk's mobile network to offer quad-play services (fixed, mobile, broadband, IPTV) to its customer base.
- **BTCL IPTV Launch:** BTCL has submitted the IPTV tender documentation to the Bangladesh Public Procurement Policy Authority for review. The IPTV service should be procured, deployed, and commercially launched during Phase 2, complementing BTCL's existing fixed broadband offerings.
- **Significant Market Power Framework:** BTRC should publish and operationalize an SMP determination framework, identifying operators with significant market power in specific market segments and imposing proportionate remedies including open-access obligations, cost-oriented pricing, and reference interconnection offers.
- **Spectrum Secondary Trading:** The secondary spectrum trading framework outlined in the Five-Year Spectrum Plan should become operational during Phase 2, enabling operators to lease, trade, and refarm spectrum holdings subject to BTRC oversight.
- **National Equipment Identity Register:** The NEIR, re-launched in 2025 to address the estimated 50 to 60 percent grey market handset share, should achieve full integration with customs and mobile operator databases during Phase 2, with enforcement actions reducing illegal handset imports measurably.

9.2.2 Infrastructure Expansion

Phase 2 represents the peak infrastructure deployment period:

- **Tower Fiberization:** The TNL Policy 2025 mandates 50 percent tower fiberization within 18 months of licence issuance and 80 percent within three years. Phase 2 should see the 80 percent target achieved for major operators, supported by infrastructure sharing guidelines and open-access fibre obligations.
- **Hathazari Green Data Centre:** BTCL's PPP project to establish a Green Data Centre at Hathazari, Chattogram, with Asian Development Bank consultation services (MoU signed January 2025), should complete the feasibility study and PPP structuring during Phase 2, with construction commencing by the end of 2028.
- **BTCL Co-location Modernization:** The project to modernize BTCL's district-level telecom complexes for co-location services should be fully operational during Phase 2, converting underutilized physical assets into revenue-generating data centre and hosting facilities.

- **BCSL Optical Fibre Manufacturing Expansion:** Bangladesh Cable Shilpa Limited should complete its production line modernization to meet the increased domestic demand for optical fibre cables driven by the tower fiberization mandate and backbone expansion programmes.
- **BSCL Satellite Services Expansion:** Bangladesh Satellite Company Limited should expand its commercial satellite service portfolio across South Asia, leveraging Bangabandhu-1 capacity for DTH broadcasting, VSAT connectivity, and disaster resilience communications. Preparatory work for Bangabandhu-2 procurement should advance during this phase.
- **Fourth Submarine Cable Exploration:** BSCPLC should begin feasibility assessment for Bangladesh's fourth submarine cable to diversify international connectivity beyond the SEA-ME-WE corridor, potentially exploring connectivity through the Bay of Bengal to Southeast Asian landing points.

9.2.3 Digital Systems and Regulatory Modernization

- **BTRC ERP Full Deployment:** The integrated ERP platform should be fully operational by the end of 2028, with all licensing, revenue, spectrum, QoS, and enforcement modules live. The system should be interlinked with Bangladesh Customs, National Board of Revenue, Election Commission, and BSCPLC for automated data exchange.
- **National One-Stop Digital Permit Portal:** The single-window e-permitting system for tower construction, fibre laying, right-of-way, and equipment import approvals should be operational with statutory 15-day and 30-day SLA timelines.
- **QoS Public Dashboard:** The National Quality of Service Dashboard should be operational, publishing district-level network performance scorecards using drive-test and crowdsourced data. Monthly operator performance reports should be publicly accessible.
- **Digital Complaint Management:** A unified digital complaint platform for telecommunications consumers should be operational across web, mobile, and USSD channels, with SLA tracking and monthly operator performance publication.
- **ISO 27001/27701 Adoption:** All major telecommunications operators should achieve ISO 27001 certification during Phase 2, with the 72-hour mandatory breach reporting regime fully operational.

9.2.4 Postal Transformation

- **E-Commerce Parcel Network:** Bangladesh Post's reverse logistics middleware and API integration should enable full two-way e-commerce parcel delivery between urban and rural areas. The network should handle both last-mile delivery and returns processing.

- **National Address Management System:** The NAMS project, which has completed PEC review with conditional approval, should progress to implementation during Phase 2, establishing a standardized national addressing infrastructure that supports digital delivery and geolocation services.
- **Postal Financial Services Modernization:** EMS and EMTS services should be modernized and expanded, enabling faster merchant payment settlement and supporting e-commerce growth. Integration with digital payment gateways should be completed.
- **Revenue Generation:** Execute a revenue recovery plan based on the 2025 Licensing Policy, utilizing "Fiber Banks" and partnerships to identify new fiber customers.
- **Service Diversification:** Scale **Triple Play**, **Quad Play**, and **MVNO** services, including device sales through installment plans and private sector financing.
- **Nagorik Sheba Expansion (3,000 centres):** The postal Nagorik Sheba network should expand to 3,000 centres by the end of 2028, with each centre offering an expanded service portfolio including Teletalk SIM distribution, connectivity registration, digital safety guidance, and complaint resolution.
- **Postal Ballot Modernization:** The SOP for postal ballot services, being developed in coordination with the Election Commission, should be finalized and piloted during Phase 2, with digital tracking integration ensuring ballot chain-of-custody integrity.

9.2.5 Phase 2 Consolidated Milestone Table

Reform Action	Lead Entity	Target Date	Deliverable
Licence Migration Complete	BTRC	End 2027	All Operators Migrated
MVNO Commercial Launch	BTCL, Teletalk	2027	Service Operational
BTCL IPTV Launch	BTCL	2027	Service Operational
SMP Framework Operational	BTRC	Mid-2027	Determinations Published

Spectrum Trading Operational	BTRC	2027	Trading Framework Live
80% Tower Fiberization	Operators	End 2028	Verified Compliance
Hathazari Green Data Centre PPP	BTCL, ADB	End 2028	Construction Commenced
BTRC ERP Full Deployment	BTRC	End 2028	All Modules Live
One-Stop Permit Portal	BTRC, PTD	2027	Portal Operational
QoS Public Dashboard	BTRC	2027	Dashboard Live
ISO 27001 Major Operators	Operators, BTRC	End 2028	Certifications Achieved
NAMS Implementation	BD Post	2028	System Operational
Nagorik Sheba 3,000 Centres	BD Post	End 2028	Centres Operational
5G Urban Pilot	BTRC, Operators	2028	Pilot Launched
NEIR Full Integration	BTRC, Customs	2027	System Operational

9.3 Phase 3: Smart Governance and Maturity (2029 to 2030)

Phase 3 consolidates the reforms and investments of the preceding four years into a mature, self-sustaining governance ecosystem. The emphasis shifts from building systems to optimizing performance, embedding evidence-based decision-making, and positioning Bangladesh's posts and telecommunications sector for regional and global competitiveness.

9.3.1 Advanced Regulatory Governance

- **BTRC Research Division Maturity:** The Research Division, established during Phase 1 and 2, should be fully staffed and producing regular market analysis, spectrum economics studies, auction benchmarking assessments, and regulatory impact evaluations. Quarterly KPI publications should inform both regulatory decision-making and public accountability.
- **Centre of Excellence Operational:** The Centre of Excellence for Digital Public Infrastructure, funded through the Social Obligation Fund, should be conducting independent third-party evaluations of all SOF-funded projects and producing annual sectoral knowledge publications.
- **Evidence-Based Policymaking Culture:** By the end of Phase 3, all major regulatory decisions including spectrum assignments, licence conditions, and tariff interventions should be supported by published cost-benefit analyses, international benchmarking, and stakeholder consultation records.
- **Parliamentary Accountability Maturity:** The Parliamentary Accountability Committee established under the Amended Ordinance should be conducting regular oversight sessions, with BTRC annual reports and reform progress dashboards subject to parliamentary scrutiny.

9.3.2 Technology Deployment and Innovation

- **5G Commercial Deployment:** Following urban pilot launches in 2028, commercial 5G services should be deployed in metropolitan areas by 2029 and extended to all district headquarters by 2030. Regulatory sandboxes for 5G vertical use cases including intelligent transport systems, industrial IoT, and fixed wireless access should produce documented outcomes.
- **Satellite Connectivity Maturity:** The NGSO satellite framework, anchored by the Starlink operational precedent, should enable multiple satellite operators to serve Bangladesh, with bandwidth export to neighbouring countries (India, Nepal, Bhutan, Myanmar) through local gateways positioning Bangladesh as a regional satellite hub.
- **IoT and AI Governance:** Cross-sectoral IoT interoperability standards and security certification requirements should be operational, with LPWAN spectrum allocated and BTRC's role in telecom network AI applications clearly defined.

- **TSS Manufacturing Scale:** TSS should be producing a range of digital devices including 4G/5G WiFi routers, CCTV equipment, lithium-ion battery repair services, and computer accessories, contributing to import substitution and local value addition.

9.3.3 Service Delivery Maturity

- **Nagorik Sheba Full Network (8,000 centres):** The combined postal and community Nagorik Sheba network should reach 8,000 centres by 2030, ensuring every Union has a physical touchpoint for digital service access, connectivity registration, and complaint resolution.
- **Bangladesh Post Digital Logistics Platform:** The integrated digital logistics platform connecting e-commerce operators, courier services, payment gateways, and regulatory systems should be fully operational, with Bangladesh Post functioning as the national last-mile logistics backbone for rural commerce.
- **Postal Banking and Financial Inclusion:** Bangladesh Post's financial services portfolio should be expanded to include savings, micro-insurance, and remittance products, leveraging its network of 9,800 post offices to reach financially excluded populations.
- **Agricultural Market Connectivity:** Building on the 2025 pilot that transported 65 tons of mangoes directly to consumers, Bangladesh Post should operate a scaled agricultural produce logistics service connecting rural producers to urban and export markets.

9.3.4 Sustainability and Green ICT

- **Green Data Centre Operational:** The Hathazari Green Data Centre should be operational and Tier III certified, with renewable energy sourcing, modern containment cooling, and a carbon footprint below industry benchmarks.
- **Renewable Energy at Telecom Sites:** All operators should demonstrate measurable progress toward renewable energy adoption at tower sites, with BTRC offering spectrum fee rebates for shared rural infrastructure that reduces diesel consumption.
- **E-Waste Management:** A producer responsibility framework for electronic waste from telecom equipment should be operational, with measurable recycling rates and annual reporting requirements.

9.3.5 Phase 3 Consolidated Milestone Table

Reform Action	Lead Entity	Target Date	Deliverable
5G Metropolitan Deployment	Operators, BTRC	2029	Commercial Services Live
5G All District Headquarters	Operators, BTRC	2030	Coverage Achieved
Research Division Full Maturity	BTRC	2029	Regular Publications
Centre of Excellence Evaluations	SOF, PTD	2029	Annual Evaluations
Nagorik Sheba 8,000 Centres	BD Post, PTD	2030	Network Complete
BD Post Digital Logistics Platform	BD Post	2030	Platform Operational
Hathazari Green Data Centre	BTCL	2029-2030	Tier III Certified
ISO 27001 All Operators	BTRC	2030	100% Certified
SOF Results-Based Model Mature	BTRC	2030	Audited Outcomes
Regional Satellite Hub	BTRC, BSCL	2030	Bandwidth Export Active
Postal Financial Services Expanded	BD Post	2030	Products Operational

E-Waste Framework Operational	BTRC, DoE	2030	Annual Reporting
BTCL Co-location Revenue	BTCL	2030	Revenue Targets Met
Bangabandhu-2 Procurement Advanced	BSCL, PTD	2030	Procurement Progress

9.4 Phase 4: Long-Term Vision (2030 to 2035)

Certain investments and reforms initiated within the 2025 to 2030 framework will mature over longer horizons. This phase captures those extended-timeline commitments:

- **Bangabandhu-2 Satellite:** The procurement, construction, and launch of a next-generation communications satellite, building on the lessons and commercial base established by Bangabandhu-1, should be targeted for completion by 2033 to 2035. The satellite should incorporate higher throughput capacity, enhanced spot beam coverage, and expanded Ka-band capabilities for broadband services.
- **Fourth Submarine Cable:** If feasibility assessment during Phase 2 confirms viability, the fourth submarine cable should be commissioned by 2032 to 2034, potentially connecting Bangladesh through the Bay of Bengal to Southeast Asian landing points and diversifying international connectivity beyond the SEA-ME-WE corridor.
- **6G Research and Preparedness:** The BTRC Research Division should commence monitoring 6G standardization developments at ITU and 3GPP from 2028 onward, with Bangladesh participating in international working groups and preparing spectrum allocation frameworks for commercial deployment anticipated in the 2033 to 2035 window.
- **Universal Broadband Access:** The Social Obligation Fund's results-based financing model should drive broadband penetration toward universal access targets, with a goal of 95 percent household broadband availability by 2035, including remote and disaster-prone areas served through a combination of fibre, mobile, satellite, and community WiFi technologies.
- **Digital Sovereignty:** Bangladesh should establish comprehensive data sovereignty capabilities, with sovereign cloud infrastructure housing all critical government data, indigenous cybersecurity capacity, and a recognized international position in digital governance frameworks.
- **Regional Connectivity Hub:** Bangladesh's strategic position should be leveraged to serve as a regional digital connectivity hub, offering submarine cable landing services, satellite gateway bandwidth, and data centre hosting to neighbouring countries, generating export revenue and diplomatic leverage.

9.5 Risk Management and Mitigation

The roadmap's implementation faces several categories of risk that require proactive mitigation:

- **Political Discontinuity:** Electoral cycles and leadership transitions have historically disrupted reform continuity in Bangladesh. Mitigation measures include embedding reform commitments in gazetted legal instruments (the Amended Ordinance, TNL Policy), establishing performance contracts that survive leadership changes, and building institutional capacity that reduces dependence on individual champions.
- **Institutional Resistance:** SOE reform, board reconstitution, and HR compliance audits will encounter resistance from entrenched interests. The Parliamentary Accountability Committee and CAG audit mechanisms provide external accountability pressure, while the Reform PMO's direct reporting line to the PTD Secretary ensures escalation pathways for bottlenecks.
- **Fiscal Constraints:** The scale of infrastructure investment required may exceed available public financing. The strategy addresses this through PPP models (Hathazari Green Data Centre), revenue-sharing arrangements (Starlink), SOF results-based financing, and taxation reform to improve sectoral revenue efficiency.
- **Technology Evolution:** Rapid technology change may render specific infrastructure investments obsolete. Technology-neutral licensing, spectrum management flexibility, and the BTRC Research Division's monitoring function provide adaptive capacity to respond to emerging developments.
- **Market Concentration:** The consolidation of licence categories under the TNL Policy may inadvertently reinforce existing market power. The SMP framework, open-access obligations, and anti-cartel safeguards serve as countervailing measures.
- **Cybersecurity Threats:** As digital systems proliferate across regulatory and service delivery functions, cybersecurity vulnerabilities increase. The ISO 27001/27701 adoption mandate, 72-hour breach reporting, sectoral CERT establishment, and biannual cybersecurity drills provide layered defence.

9.6 Consolidated KPI Dashboard (2025 to 2030)

The following master KPI framework consolidates the measurable targets established across all chapters, providing a single accountability reference for the Reform Programme Management Office, the Parliamentary Accountability Committee, and PTD leadership.

Indicator	Baseline (2025)	2026 Target	2028 Target	2030 Target
BTRC Licensing Autonomy	Compromised	GO Published	Fully Operational	Institutionalized
Active Licence Categories	23+	23+ (migration)	4 (migrating)	4 (complete)
Spectrum Plan Published	None	Published	Mid-term Review	Revised Plan
Internet Shutdowns	Multiple (historical)	Zero	Zero	Zero
Tower Fiberization (%)	~35%	~45%	80%	95%+
International Bandwidth (Gbps)	~7,220	~15,000 (SMW-6)	20,000+	25,000+
Teletalk 4G Sites	~3,400	5,100+	6,500+	7,500+
Nagorik Sheba Centres (postal)	0	500	3,000	8,000
BTRC ERP Status	None	Specified	Deployed	Mature

SOE Board Independence	Politicized	Reconstituted	Independent Majority	Institutionalized
ISO 27001 Operator Coverage	~0%	Initiated	Major Operators	100%
5G Commercial Service	None	Plan Published	Urban Pilot	District HQs
QoS Public Dashboard	None	Specified	Operational	Mature
Research Division Publications	None	Established	Quarterly Reports	Full Programme
SOF Results-Based Model	Input-Based	Framework Designed	Pilots Running	Full Deployment
BD Post E-Commerce Integration	None	API Integration	Full Network	Platform Mature



9.7 Chapter Summary

This roadmap translates the strategic analysis, legal frameworks, institutional reforms, project corrections, and governance architectures developed across the preceding eight chapters into a sequenced, accountable implementation plan. The four-phase structure, spanning 2025 to 2035, reflects the principle that sustainable transformation requires patient institution-building before it can yield measurable technology and service outcomes.

Phase 1 (2025 to 2026) establishes the legal and institutional foundations: the Amended Ordinance and TNL Policy translated into operational Government Orders, SOE boards reconstituted, digital systems specified and procured, and the Reform PMO's accountability dashboards operational. Phase 2 (2027 to 2028) scales these foundations into market structure changes, infrastructure deployment, and regulatory system maturity, with the BTRC ERP, QoS Dashboard, licence migration, MVNO services, and tower fiberization targets all achieving measurable completion. Phase 3 (2029 to 2030) consolidates gains into evidence-based governance, 5G commercial deployment, postal service transformation, sustainability targets, and regional positioning. Phase 4 (2030 to 2035) captures longer-horizon investments including Bangabandhu-2, a potential fourth submarine cable, 6G preparedness, and universal broadband access.

The risk management framework acknowledges the real constraints of political discontinuity, institutional resistance, fiscal limitations, and technology evolution, while identifying the specific legal, institutional, and financial instruments designed to mitigate each risk category.

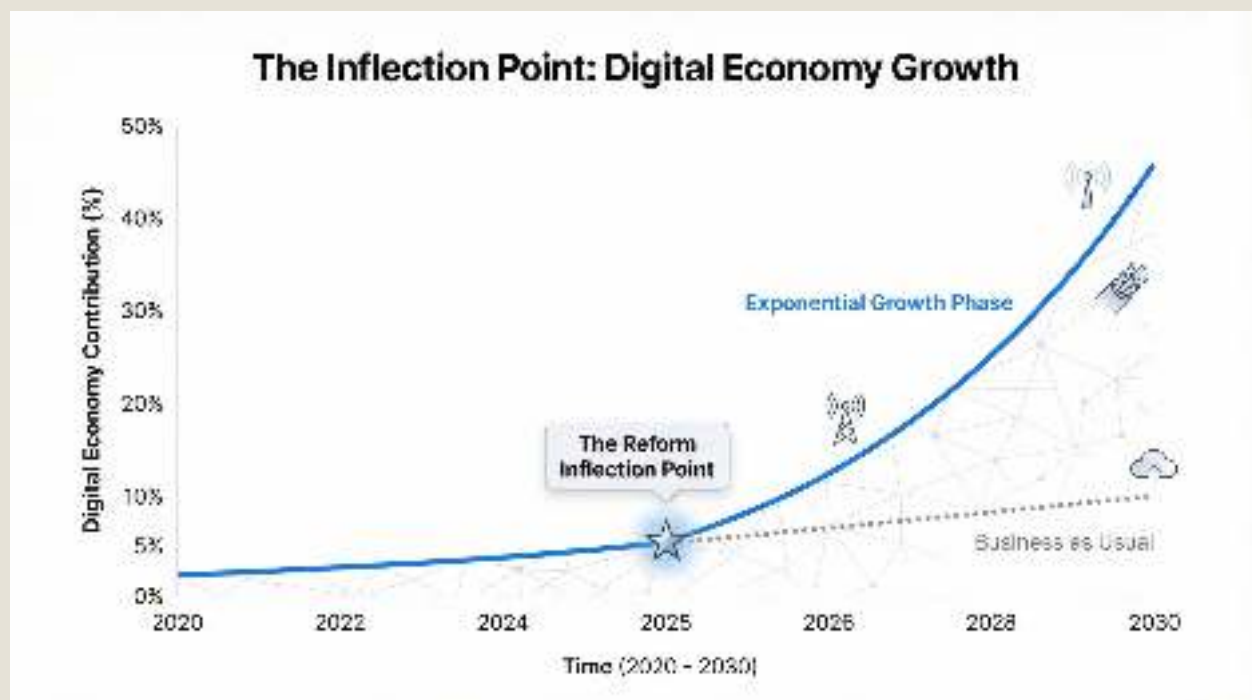
The consolidated KPI dashboard provides a single reference point for monitoring progress across all reform domains, ensuring that the strategy's ambitions are anchored in measurable, time-bound commitments. Implementation discipline, sustained political commitment, and institutional professionalism will ultimately determine whether this roadmap delivers the connected, inclusive, and competitive posts and telecommunications sector that Bangladesh requires.

— End of Chapter 9 —

Conclusion

This strategy document has undertaken a comprehensive examination of the Posts and Telecommunications Division's institutional ecosystem, regulatory architecture, infrastructure assets, service delivery networks, and governance frameworks. The analysis has been grounded in evidence gathered through structured institutional visits, document review, media analysis, public consultations, and crowdsourced citizen input. The findings have been cross-referenced with international benchmarks, regional comparisons, and the reform trajectories of comparable jurisdictions to ensure that the recommendations are both ambitious and implementable.

The diagnosis presented across these chapters reveals a sector of immense strategic importance that has been undermined by cumulative governance failures. The erosion of BTRC's operational autonomy through the 2010 amendments, the proliferation of over twenty three fragmented licence categories enabling intermediary rent extraction, the systematic mispricing of spectrum to maximize short-term revenue at the expense of deployment efficiency, the politicization of SOE leadership appointments, and the conversion of the Social Obligation Fund from a digital inclusion instrument into a discretionary allocation mechanism have collectively constrained the sector's capacity to serve Bangladesh's development needs.



The strategy does not merely catalogue these deficits. It proposes a coherent reform architecture that addresses root causes rather than symptoms. The Bangladesh Telecommunication (Amendment) Ordinance 2025, approved in December 2025, permanently prohibits internet shutdowns, restores BTRC autonomy, establishes Parliamentary accountability, and extends

regulatory scope to the digital economy. The Telecommunications Network and Licensing Policy 2025, gazetted in September 2025, replaces the fragmented legacy licensing regime with four technology-neutral categories that eliminate intermediary layers and link licence conditions to measurable performance obligations. These legislative instruments provide the legal foundation for everything that follows.

The institutional reform agenda addresses the human and organizational dimensions that no legislative change can resolve by itself. SOE board reconstitution with independent expertise, digital HRIS deployment for transparent personnel management, restoration of the Telecommunication Cadre, CAG compliance audits, performance contracts, and whistleblower protections together constitute a comprehensive institutional integrity framework. These are not abstract governance prescriptions. They respond to specific, documented patterns of irregular appointments, unauthorized deputation extensions, and governance manipulation that the Task Force identified across PTD entities.

The project reform chapter demonstrates that accountability is possible even for legacy investments burdened by procurement irregularities, scope deviation, and implementation delays. The disciplined application of third-party verification, cost re-baselining, component cancellation where feasibility has failed, and results-based financing for ongoing commitments provides a replicable methodology for restoring fiscal discipline in public sector telecommunications investment.

The strategy's engagement with infrastructure and service delivery extends well beyond the traditional boundaries of telecommunications regulation. The recognition that Bangladesh Post's 9,800 post offices represent the country's most extensive physical service network, and that their integration with the Nagorik Sheba programme, e-commerce logistics, cold chain infrastructure, agricultural market connectivity, postal ballot modernization, and financial inclusion services can transform an institution conventionally associated with letter delivery into a cornerstone of citizen-centric service design, represents a significant strategic reorientation. Similarly, the positioning of TSS as a domestic manufacturing platform for digital devices, the expansion of BSCPLC's submarine cable capacity with SMW-6, the formalization of NGSO satellite licensing through the Starlink precedent, and the structuring of BTCL's Hathazari Green Data Centre as a PPP demonstrate that the strategy treats each PTD entity as a contributor to a unified sectoral vision rather than an isolated organizational improvement project.

The governance reform architecture consolidates these entity-level interventions into sector-wide frameworks. Regulatory architecture reform through the DoT-BTRC mandate clarification, market design modernization through the TNL Policy, spectrum governance through the Five-Year Plan, digital regulatory systems through the BTRC ERP and one-stop permit portal, consumer protection through the QoS Dashboard and complaint management system, data governance through ISO 27001/27701 adoption and breach reporting, digital inclusion through

results-based SOF financing, emerging technology governance through 5G readiness and satellite frameworks, sustainability through green ICT targets, and evidence-based policymaking through the Research Division, Centre of Excellence, and Reform PMO together constitute an integrated governance system where each component reinforces the others.

The implementation roadmap sequences these reforms across four phases spanning 2025 to 2035, with a deliberate logic that prioritizes legal and institutional foundations before scaling infrastructure deployment and technology adoption. This sequencing reflects international evidence that reform programmes which attempt to deploy advanced technologies before stabilizing the regulatory and institutional environment consistently underperform.

The strategy is candid about risks. Political discontinuity, institutional resistance, fiscal constraints, technology evolution, market concentration, and cybersecurity threats each pose genuine challenges to implementation. The strategy addresses these through embedded legal instruments that survive leadership changes, external accountability mechanisms including Parliamentary oversight and CAG audits, diversified financing through PPP, revenue sharing, and SOF models, technology-neutral regulatory design, competition safeguards, and layered cybersecurity obligations. These are not guarantees of success. They are the conditions most likely to sustain reform momentum across the inevitable disruptions of a six-year implementation period.

Bangladesh's posts and telecommunications sector stands at an inflection point. The legislative, policy, and institutional reforms initiated in 2025 represent the most comprehensive restructuring of the sector since the corporatization of the Bangladesh Telegraph and Telephone Board. The Amended Ordinance's prohibition of internet shutdowns, the TNL Policy's elimination of intermediary licence layers, the Starlink licensing that brought NGSO satellite services to Bangladesh within weeks, the tower sharing agreements that are expanding mobile coverage through commercial rather than public investment, and the postal reforms that are repositioning the country's oldest public service network for digital-age relevance all demonstrate that meaningful reform is not only possible but already underway.

The challenge is not whether reform can begin. It has begun. The challenge is whether the institutions, legal frameworks, and accountability mechanisms established through this strategy can sustain reform across electoral cycles, leadership transitions, fiscal pressures, and the inevitable resistance of entrenched interests. This document is designed to provide the analytical foundation, policy architecture, and implementation discipline required to meet that challenge. The consolidated KPI framework, the Reform PMO, the Parliamentary Accountability Committee, and the Research Division's monitoring function together ensure that progress, or the absence of progress, will be visible, measurable, and accountable.

The vision is straightforward: a posts and telecommunications sector that delivers affordable, reliable, and inclusive connectivity to every citizen, operates under transparent and predictable

governance, attracts private investment through regulatory certainty, contributes to Bangladesh's economic competitiveness through world-class infrastructure, and positions the country as a credible participant in global digital governance. This strategy provides the path from aspiration to implementation. The responsibility for walking that path belongs to the institutions, leaders, and public servants who will translate these words into measurable outcomes for the people of Bangladesh.

Annex

Note to Successor

আমার আন্তরিক সালাম এবং মোবারকবাদ গ্রহণ করুন।

০৫ মার্চ ২০২৫ থেকে ১৪ই ফেব্রুয়ারি ২০২৬ সময়ে ডাক, টেলিযোগাযোগ ও আইসিটি মন্ত্রণালয়ের দায়িত্ব পালন করি। এই সময়ে ডাক ও টেলিযোগাযোগ বিভাগ থেকে গৃহীত গুরুত্বপূর্ণ উদ্যোগ, সংশ্লিষ্ট প্রতিষ্ঠানসমূহের সংস্কার ও আধুনিকায়ন কার্যক্রম, এবং চলমান ও পাইপলাইন কাজের ধারাবাহিকতা সম্পর্কে আপনাকে অবহিত করার অভিপ্রায় থেকে লিখছি। এই নোটের উদ্দেশ্য হলো সিদ্ধান্তের ধারাবাহিকতা বজায় রাখা, বিগত শাসনামলের কার্যমোগত ব্যর্থতা থেকে শিক্ষা নেওয়া, এবং চলমান সংস্কারগুলোকে প্রাতিষ্ঠানিকভাবে স্থায়ী করা।

১। ডাক ও টেলিযোগাযোগ বিভাগের সামগ্রিক দৃষ্টিভঙ্গি ছিল রাষ্ট্রায়ত্ত্ব ও নিয়ন্ত্রক প্রতিষ্ঠানগুলোকে আধুনিক, প্রযুক্তি নির্ভর এবং বাণিজ্যিকভাবে টেকসই ভিত্তির উপর দাঁড় করানো। একইসাথে, টেলিযোগাযোগ খাতে নীতিগত স্বচ্ছতা, বিনিয়োগবান্ধব পরিবেশ, আন্তর্জাতিক সংযোগ ক্ষমতা, জাতীয় অবকাঠামোর রেজিলিয়েন্স, এবং গ্রাহক পর্যায়ে সেবার মান নিশ্চিত করা।

বিগত শাসনামলে খাতটি দীর্ঘ সময় ধরে একটি পরিকল্পনাহীনতা, প্রাতিষ্ঠানিক দুর্বলতা এবং রেগুলেটরি ক্যাপচারের মধ্যে চলেছে। বড় বড় প্রকল্প নেওয়া হয়েছে কিন্তু বাস্তবসম্মত মাস্টার প্ল্যান, ডিম্যান্ড অ্যাসেসমেন্ট, এবং টেকসই রেভিনিউ মডেল ছাড়া। ফলে অনেক অবকাঠামো সম্পদ আন্ডারইউটাইজড থেকেছে, সেবার মান উন্নত হয়নি, এবং বিনিয়োগকারীদের আস্থা কমেছে। আমাদের সময়ে লক্ষ্য ছিল এই ধারা ভাঙা এবং প্রতিটি প্রতিষ্ঠানে ফলাফলভিত্তিক কর্মপরিকল্পনা, নির্দিষ্ট টাইমলাইন, এবং নিয়মিত মনিটরিংয়ের সংস্কৃতি প্রতিষ্ঠা করা। ভবিষ্যতেও এই ধারাবাহিকতা বজায় থাকলে খাতের দীর্ঘদিনের কার্যমোগত সমস্যা কমবে এবং সেবার মান ও আর্থিক শৃঙ্খলা উল্লেখযোগ্যভাবে উন্নত হবে।

২। বিটিআরসি-র সংস্কার কার্যক্রম, নীতিগত আধুনিকায়ন এবং ভবিষ্যৎ টেলিযোগাযোগ রূপরেখা বাস্তবায়ন বিষয়ে আপনার বিশেষ মনোযোগ আকর্ষণ করছি। দেশের টেলিযোগাযোগ খাতকে বৈশ্বিক মানে উন্নীত করতে এবং একটি বিনিয়োগবান্ধব স্মার্ট ইকোসিস্টেম গড়ে তুলতে গত এক বছরে বিটিআরসিতে যে সাহসী ও পদ্ধতিগত সংস্কার শুরু করেছি, তা স্থায়ী করতে হলে এটিকে প্রাতিষ্ঠানিক নীতি, গাইডলাইন ও ডিজিটাল সিস্টেমে রূপ দিতে হবে।

বিগত শাসনামলে নিয়ন্ত্রকের স্বাধীনতা ক্ষয় হয়েছে, মন্ত্রণালয়কেন্দ্রিক সিদ্ধান্ত, অনিয়মিত নিয়োগ ও প্রভাব বিস্তার, এবং স্বচ্ছতা ঘাটতির কারণে রেগুলেটরি বিশ্বাসযোগ্যতা কমেছে। লাইসেন্সিং কার্যমোের অতিরিক্ত খণ্ডিত অবস্থা এবং মধ্যস্বত্বভোগী স্তরগুলো বাজারে অপ্রয়োজনীয় খরচ ও রেন্ট এক্সট্রাকশন তৈরি করেছে। সেবার মান নিয়ে অভিযোগ ব্যবস্থাও ছিল দুর্বল, ফলে গ্রাহক স্বার্থ সুরক্ষা যথাযথভাবে হয়নি। এখনও প্রায় ১২ হাজারের উপর অভিযোগ অনিশ্পন্ন রয়েছে।

(ক) স্পেকট্রাম ম্যানেজমেন্ট ও স্পেকট্রাম মূল্যের যৌক্তিকীকরণ কার্যক্রমের আওতায় আন্তর্জাতিক উত্তম চর্চার সাথে সামঞ্জস্য রেখে ৭০০ মেগাহার্টজ ব্যান্ডের তরঙ্গ নিলাম সম্পন্ন করা হয়েছে। ঘনবসতিপূর্ণ দেশে ডিপ ইনডোর সেবা বিস্তৃতির জন্য ৭০০ মেগাহার্টজ অপরিহার্য। ৮০০/৯০০ মেগাহার্টজ তরঙ্গে ভয়েস কলের আধিক্য থাকায় বিদ্যমান ১৮০০/২১০০ মেগাহার্টজে ৪জি/৫জি সেবা দেয়া হচ্ছে, যার কাভারেজ অত্যন্ত সীমিত। ASMS ও TMS ব্যবস্থার মাধ্যমে সারাদেশে স্পেকট্রামের ব্যবহার রিয়েল টাইমে পর্যবেক্ষণ, মনিটরিং ও নিয়ন্ত্রণ কার্যক্রম অব্যাহত রয়েছে। ভবিষ্যৎ রূপরেখা অনুযায়ী অব্যবহৃত ও কম ব্যবহৃত তরঙ্গ পুনর্গঠন করে পুনরায় নিলামের উদ্যোগ গ্রহণের পরিকল্পনা রয়েছে এবং স্পেকট্রাম বরাদ্দকে পারফরম্যান্স লিঙ্কড করার দিকে যেতে হবে।

(খ) পর্বতসম দূরত্ব, অটল সংকল্প, অবিচল নিষ্ঠা এবং সর্বোচ্চ অঙ্গীকারের সমন্বয়ে অন্তর্বর্তীকালীন সরকার বাংলাদেশে টেলিযোগাযোগ আইনে এক মৌলিক সংস্কার সম্পন্ন করেছে। এই সংশোধন কেবল একটি আইনি পরিবর্তন নয়- এটি রাষ্ট্র ও নাগরিকের সম্পর্কে নতুন ভিত্তিতে দাঁড় করানোর এক সাহসী পদক্ষেপ।

এই আইনের উল্লেখযোগ্য বৈশিষ্ট্যসমূহ হচ্ছে-

মতপ্রকাশের স্বাধীনতার সুরক্ষা:

পূর্ববর্তী আইনে হেইট স্পিচকে সরাসরি অপরাধ হিসেবে গণ্য করা হয়েছিল। সংশোধিত আইনে সেটি পুনর্বিবেচনা করা হয়েছে। এখন কেবল তখনই অপরাধ হিসেবে গণ্য হবে, যখন বক্তব্যের সাথে সহিংসতা উসকে দেওয়ার সরাসরি সম্পর্ক থাকবে। নাগরিকের বাকস্বাধীনতা ও মতপ্রকাশের অধিকার সম্মুখত রাখাই এই পরিবর্তনের মূল লক্ষ্য।

নিয়ন্ত্রকের ক্ষমতা প্রায় শতভাগ পুনর্বহাল:

মন্ত্রণালয়ের পূর্বানুমোদনের বাধ্যবাধকতা উল্লেখযোগ্যভাবে কমানো হয়েছে। ঘনঘন প্রদত্ত লাইসেন্স, পারমিট, এনলিস্টমেন্ট, ট্যারিফ পরিবর্তন, মনিটরিং ও এনফোর্সমেন্টসহ অধিকাংশ কার্যকরী ক্ষমতা পুনরায় বিটিআরসি-র নিকট ন্যস্ত করা হয়েছে।

শুধুমাত্র জাতীয় অর্থনীতির কৌশলগতভাবে গুরুত্বপূর্ণ কিছু ‘once-in-a-time’ লাইসেন্সের ক্ষেত্রে—স্বাধীন ইন্ডাস্ট্রি-একাডেমিয়া যৌথ গবেষণার ভিত্তিতে মন্ত্রণালয়ের পর্যালোচনার শর্ত রাখা হয়েছে।

বিনিয়োগবান্ধব কাঠামো:

দেশি-বিদেশি বিনিয়োগনির্ভর এই খাতে আস্থা ফিরিয়ে আনতে সকল জরিমানা এক-তৃতীয়াংশে নামিয়ে আনা হয়েছে। ফলে বাংলাদেশের টেলিযোগাযোগ খাত বিনিয়োগকারীদের ডিউ-ডিলিজেন্স ও ফিজিবিলিটি মূল্যায়নে আর নেতিবাচক সংকেত দেবে না।

স্বচ্ছতা ও জবাবদিহি নিশ্চিতকরণ:

মন্ত্রণালয় ও বিটিআরসি’র কার্যক্রম পর্যালোচনার জন্য একটি উচ্চক্ষমতাসম্পন্ন কোয়ালিটি-জুডিশিয়াল স্বচ্ছতা ও জবাবদিহি কমিটি গঠন করা হয়েছে, যা গৃহীত সিদ্ধান্তসমূহ পোস্ট-ফ্যাক্টো রিভিউ করবে।

সংসদীয় জবাবদিহি:

নির্বাচিত জনপ্রতিনিধিদের কাছে জবাবদিহি নিশ্চিত করতে সংসদীয় কমিটির নিকট নিয়মিত প্রতিবেদন দাখিলের বাধ্যবাধকতা যুক্ত করা হয়েছে।

ইন্টারনেট বন্ধ নিষিদ্ধ:

এই আইনের সবচেয়ে ঐতিহাসিক দিক- বাংলাদেশে ইন্টারনেট বন্ধ করার ক্ষমতা আইনি কাঠামো থেকে প্রত্যাহার করা হয়েছে। নাগরিকদের অধিকার ও সংযোগ, অবাধ তথ্যপ্রবাহ ও ডিজিটাল অর্থনীতির ধারাবাহিকতা এখন আইনি সুরক্ষা পেয়েছে।

আন্তর্জাতিক মানের আইনসম্মত নজরদারি ব্যবস্থা:

পূর্ববর্তী অস্বচ্ছ ও নিকৃষ্ট নজরদারি কাঠামোর পরিবর্তে (Article 97 সংশোধন) আন্তর্জাতিক মানদণ্ডসম্মত lawful interception ব্যবস্থা প্রবর্তন করা হয়েছে।

নজরদারিকে ‘জরুরি’ ও ‘অ-জরুরি’- এই দুই ভাগে বিভক্ত করা হয়েছে।

প্রি-অ্যাপ্রভাল ও পোস্ট-ফ্যাক্টো রিভিউ, কেস-বাই-কেস স্ট্যান্ডার্ড অপারেটিং প্রসিডিউর, সময়সীমা নির্ধারণ, ইভেন্ট লগিং, অ্যাক্সেস কন্ট্রোল- এই সবকিছু স্পষ্টভাবে কাঠামোবদ্ধ করা হয়েছে।

স্বরাষ্ট্র মন্ত্রণালয়কে বিস্তারিত বিধিমালা প্রণয়নের নির্দেশ প্রদান করা হয়েছে।

যে দেশে একসময় টেলিকম আইনের নিবর্তনমূলক ধারার অপব্যবহার করে বেআইনি নজরদারি চালানো হয়েছে, ইন্টারনেট বন্ধ করে নাগরিকদের কণ্ঠরোধ করা হয়েছে, এবং যার পরিণতিতে অসংখ্য গুম ও প্রাণহানির অভিযোগ উঠেছে-সেই দেশে টেলিযোগাযোগ আইনের এমন মানবিক, দায়িত্বশীল ও ন্যায়ভিত্তিক সংশোধন নিঃসন্দেহে এক ঐতিহাসিক ঘটনা।

আমি দৃঢ়ভাবে বিশ্বাস করি, এই মৌলিক সংস্কার অন্তর্বর্তীকালীন সরকারের অন্যতম শ্রেষ্ঠ অর্জন হিসেবে ইতিহাসে স্থান করে নেবে। এটি শুধু একটি আইনি পরিবর্তন নয়- এটি রাষ্ট্র ও নাগরিকের সম্পর্ক পুনর্গঠনের একটি সাহসী পদক্ষেপ। এই মৌলিক সংস্কার অন্তর্বর্তীকালীন সরকারের অন্যতম বড় অর্জন হিসেবে ইতিহাসে স্থান পাবে।

(গ) নতুন টেলিকম লাইসেন্সিং পলিসি ও গাইডলাইন ২০২৫ এর আওতায় লাইসেন্সিং কাঠামো সহজীকরণ করা হয়েছে এবং উদ্যোক্তা ও অপারেটরদের জন্য ওয়ান স্টপ সার্ভিস চালু হয়েছে। লাইসেন্স অবলিগেশন বাস্তবায়নে প্রাতিষ্ঠানিক সক্ষমতা বৃদ্ধি, যেমন ল্যাব ফ্যাসিলিটি, সফটওয়্যার টুল, ডেটা অ্যানালাইসিস সক্ষমতা, এবং ডেটাভিত্তিক KPI ও পারফরম্যান্স অ্যানালাইসিস চালু করা দরকার। KQoS, ড্রাইভ টেস্ট, স্পেকট্রাম অকুপেশন, নয়েজ, SAR ও রেডিয়েশন টেস্টিং ব্যবস্থা শক্তিশালী করতে হবে।

(ঘ) এসওএফ তহবিলের ব্যবস্থাপনায় অতীতে দুর্বলতা ছিল। অনেক ক্ষেত্রে লক্ষ্যভিত্তিক ও ফলাফলভিত্তিক ব্যয় নিশ্চিত হয়নি। বর্তমানে এসওএফকে ফলাফলভিত্তিক ব্যবস্থায় নিয়ে আসার দিকনির্দেশনা রয়েছে, যাতে বাস্তব আউটকাম যাচাই ছাড়া অর্থ ছাড় না হয়।

(ঙ) গ্রাহক অভিযোগ নিষ্পত্তির জন্য সময়সীমাভিত্তিক টিকেটিং সফটওয়্যার ও একীভূত অভিযোগ ব্যবস্থার কাজটি দ্রুত এগিয়ে নিতে হবে, যাতে প্রতিটি অভিযোগ নির্দিষ্ট সময়ের মধ্যে নিষ্পত্তি নিশ্চিত করা যায় এবং মাসিকভাবে জেলা ও অপারেটরভিত্তিক অভিযোগ ও QoS রিপোর্ট প্রকাশ করা সম্ভব হয়।

(চ) বিটিআরসির নেতৃত্বে এবং আইন শৃংখলা বাহিনী, মোবাইল এবং নেটওয়ার্ক অপারেটরদের কোওর্ডিনেশনের মাধ্যমে সাইবার স্পেসে জুয়া, জালিয়াতিসহ বিভিন্ন অপরাধ দমনের উদ্যোগ আমি নিয়েছি। এই কার্যক্রম চালু রাখা অত্যন্ত জরুরি সাধারণ নাগরিকদের নিরাপত্তার স্বার্থে।

(ছ) বাংলাদেশের একটা বড় সমস্যা হয়ে দাঁড়াতে পারে ই-ওয়েস্ট। আমি বিটিআরসির কাছে চিঠি দিয়েছি যাতে ইউরোপিয়ান ইউনিয়নের মত ইউনিভার্সাল চার্জিং ক্যাবলের স্ট্যান্ডার্ড তৈরী করে এনফোর্স করা হয়, যাতে এই খাতে ই-ওয়েস্টের পরিমাণ কমে আসে, বাজে কোয়ালিটির চার্জিং প্রোডাক্ট দেশে আসা বন্ধ হয় এবং দেশেই এই ধরনের ডিভাইস/চার্জার তৈরীর উদ্যোগ নেয়া হয়। এই লক্ষ্যে বিটিআরসি, বাংলাদেশ কম্পিউটার সমিতি, মোবাইল ফোন প্রস্তুতকারী এবং বাজারজাতকারী কোম্পানি এবং বাণিজ্য মন্ত্রণালয় (মোবাইল বা ল্যাপটপ বাদেও অন্য ছোটখাট কমার্শিয়াল ইলেকট্রনিক প্রোডাক্টে এই ফ্রেমওয়ার্ক এনফোর্স করতে) একসাথে কাজ করতে পারে।

এছাড়া, এনইআইআর (National Equipment Identity Register) বাস্তবায়নের মাধ্যমে এমএফএস স্ক্যামিং, জুয়া, ক্রিপ্টো, মোবাইল চুরি/ছিনতাইসহ অবৈধ ও ক্লোন হ্যান্ডসেট নিয়ন্ত্রণ কার্যক্রমকে ধারাবাহিক রাখতে হবে। ভবিষ্যৎ চ্যালেঞ্জ মোকাবিলায় ডিজি রিসার্চ সেন্টার, এনজিএসও নীতিমালা চূড়ান্তকরণ, এবং সাইবার ও ডেটা সুরক্ষা মান বাস্তবায়নে আপনার ধারাবাহিক মনোযোগ প্রয়োজন।

৩। বাংলাদেশ সাবমেরিন ক্যাবল পিএলসি দেশের আন্তর্জাতিক ব্যাল্ডউইথ সক্ষমতা এবং নিরবচ্ছিন্ন ইন্টারনেট সংযোগ নিশ্চিত করার ক্ষেত্রে কৌশলগত প্রতিষ্ঠান। বিগত শাসনামলে কোম্পানির সক্ষমতা থাকা সত্ত্বেও বিপুল পরিমাণ ক্যাপাসিটি অব্যবহৃত ছিল এবং হোলসেল ব্যাল্ডউইথ প্রাইসিং জনসাধারণের ইন্টারনেট খরচ কমাতে সহায়ক ছিল না। আমরা পলিসি ইন্টারভেনশনের মাধ্যমে টিটিসি:সাবমেরিন ব্যাল্ডউইথ ৫০:৫০ করে ১.৯ টেরাবাইট ক্যাপাসিটিকে ৪.৫ টেরাবাইট ক্যাপাসিটিতে উন্নীত করা হয়েছে। পাশাপাশি সেবার মান উন্নয়ন, ট্যারিফ সমন্বয় এবং বকেয়া আদায়ের মাধ্যমে আর্থিক ভিত্তি শক্তিশালীকরণে কোম্পানিটি যে পদক্ষেপ নিয়েছে তা ইতিবাচক। ডিমাল্ড নোট পরিশোধের ২৪ ঘণ্টার মধ্যে আপগ্রেড এবং ৭ দিনের মধ্যে ওয়েট সেগমেন্ট চালুর মতো ফাস্ট সার্ভিসের প্রতিশ্রুতি গ্রাহক সন্তুষ্টি বৃদ্ধিতে সহায়ক হবে।

অন্যদিকে, টেলিকম খাতে কর, শুল্ক, সম্পূরক শুল্ক ও ভ্যাটের কারণে সেবা ব্যয় বৃদ্ধি এবং বাজারে অস্থিতিশীল মূল্য পরিস্থিতির সৃষ্টি হচ্ছে, যা ভোক্তা পর্যায়ে ইন্টারনেটের মূল্য সহজলভ্য করার জাতীয় লক্ষ্যকে বাধাগ্রস্ত করছে। এ বিষয়ে অর্থ মন্ত্রণালয় ও জাতীয় রাজস্ব বোর্ডের সঙ্গে সমন্বিত আলোচনা করে ভারসাম্যপূর্ণ কর কাঠামো নির্ধারণ জরুরি।

ভবিষ্যৎ পরিকল্পনার অংশ হিসেবে আন্তর্জাতিক ব্যাল্ডউইথ সক্ষমতা বৃদ্ধি ও নেটওয়ার্ক রেজিলিয়েন্স নিশ্চিত করতে নতুন সাবমেরিন কেবল সংযোগ স্থাপন এবং একাধিক রুটে রিডান্ডেন্সি নিশ্চিত করা জরুরি। একইসঙ্গে কুয়াকাটা ও কক্সবাজারকে স্থলভিত্তিক ফাইবার নেটওয়ার্কের মাধ্যমে ল্যান্ড কানেক্টিভিটি প্রদানের উদ্যোগ নিলে বিকল্প রুট তৈরি হবে এবং ল্যান্ডিং পয়েন্টগুলোর মধ্যে নেটওয়ার্ক নিরবচ্ছিন্নতা ও নিরাপত্তা বাড়বে।

৪। টেলিটক বাংলাদেশ লিমিটেড রাষ্ট্রায়ত্ত মোবাইল অপারেটর হিসেবে সরকারের ডিজিটাল রূপান্তর, টেলিযোগাযোগ অবকাঠামো শক্তিশালীকরণ এবং জনবান্ধব সেবা সম্প্রসারণে কৌশলগত দায়িত্ব পালন করে যাচ্ছে। অতীতে টেলিটকের বাজার অংশীদারিত্ব কম ছিল, সেবার মান ও নেটওয়ার্ক সম্প্রসারণে স্ববিরতা ছিল, এবং বাণিজ্যিক সিদ্ধান্তে জবাবদিহিতা ও কার্যকর মার্কেটিং ঘাটতি ছিল। আমাদের সময়ে নেটওয়ার্ক উন্নয়ন, নতুন প্রযুক্তি সংযোজন এবং বিক্রয় ও বিপণন কার্যক্রমে গতি আনার চেষ্টা করা হয়েছে। আমার স্বল্প সময়ে টেলিটকের ফোরজি টাওয়ার ১১০০ বেড়েছে। আরও শাস্রয়ী রেন্টাল মডেলে ১৩০০ টাওয়ার বাড়ানোর কাজ চলছে, ছয় মাসের মধ্যে যা সম্পন্ন হবে। পুরানো ১০০০ সাইটে ব্যাটারি ইন্সটল করা হয়েছে, যাতে বিদ্যুৎ চলে গেলে নেটওয়ার্ক চলে না যায়।

অত্যন্ত গুরুত্বপূর্ণ বিষয়, টেলিটককে আমরা ৭০০ মেগাহার্টজ ব্যান্ডে ১০ মেগাহার্টজ তরঙ্গ দেয়া হয়েছে। তবে, এর ফি বাবদ কোম্পানিকে কিস্তিতে অর্থ পরিশোধ করতে হবে। দুর্গম এলাকা, পাবলিক বিশ্ববিদ্যালয় ও মহাসড়কে ৪জি কভারেজ সম্প্রসারণের লক্ষ্যে নতুন প্রকল্প সবুজ পাতায় অন্তর্ভুক্ত করে ডিপিপি প্রণয়ন করা হয়েছে, যা নির্ধারিত সময়সীমার মধ্যে অনুমোদন ও বাস্তবায়ন জরুরি। টাওয়ার শেয়ারিং, ব্যাকহল ফাইবারাইজেশন, এবং QoS ভিত্তিক KPI চালু রেখে টেলিটকের অপারেশনাল দক্ষতা বৃদ্ধি করা দরকার। ক্লাউড ফোন, এমএফএস-এর মাধ্যমে রিচার্জ সুবিধা এবং পোস্ট অফিসের মাধ্যমে অনলাইনে সিম সরবরাহ কার্যক্রমের ফলে টেলিটকের রাজস্ব ও গ্রাহক সন্তুষ্টিতে ইতিবাচক প্রভাব পড়বে বলে প্রত্যাশা করা যায়।

একইসাথে, গৃহীত সিদ্ধান্ত, অনুমোদিত কর্মপরিকল্পনা এবং নির্ধারিত টাইমলাইন অনুসারে অবশিষ্ট কার্যক্রমসমূহ যথাসময়ে সম্পন্ন করা অত্যন্ত গুরুত্বপূর্ণ। বিশেষ করে যেসব কার্যক্রম সরাসরি গ্রাহক অভিজ্ঞতা, নেটওয়ার্ক অ্যাভেইলিবিটি, রাজস্ব প্রবৃদ্ধি এবং প্রতিযোগিতামূলক অবস্থানের সঙ্গে সংশ্লিষ্ট, সেগুলোর বাস্তবায়নে ধারাবাহিকতা বজায় রাখতে হবে।

৫। বাংলাদেশ টেলিকমিউনিকেশনস কোম্পানি লিমিটেড (বিটিসিএল) বিষয়ে চলমান সংস্কার ও প্রযুক্তিগত আধুনিকায়ন কর্মসূচি অব্যাহত রাখা জরুরি। বিগত শাসনামলে বিটিসিএল-এর বহু সম্পদ একেজো থেকেছে, এমভিএনও/ট্রিপল/কোয়ার্ড প্লেন-র মতো ব্যবসায়িক মডেল আপডেট হয়নি, এবং প্রকল্প ব্যবস্থাপনা ও সেবার মানে কাঙ্ক্ষিত উন্নতি আসেনি। বিটিসিএল-কে কেবল ল্যান্ডফোন কোম্পানি হিসেবে দেখলে অলাভজনক হওয়ার ঝুঁকি থাকে; কিন্তু টাওয়ার শেয়ারিং এবং ফাইবার অপটিক নেটওয়ার্কের ডিজিটাল ম্যাপিংয়ের মাধ্যমে এটি একটি ডিজিটাল অবকাঠামো হাব হিসেবে আবির্ভূত হতে পারে। আইপিটিভি এবং এমভিএনও-এর মতো আধুনিক সেবাগুলো বিটিসিএল-কে আয়ের নতুন দিগন্ত উন্মোচনে সহায়তা করবে। তবে, বিটিসিএল এর ১০টি লাইসেন্স থেকে রেভিনিউ আনার পরিকল্পনা তৈরি করতে হবে এবং পাবলিক-প্রাইভেট পার্টনারশিপ মডেল চালু করতে হবে।

বিশেষ করে, তেজগাঁও ও মহাখালীতে টেলিকম এবং সফটওয়্যার টেকনোলজি কমপ্লেক্স তৈরির পরিকল্পনাগুলো বিটিসিএল-কে একটি প্রযুক্তিভিত্তিক শক্তিশালী প্রতিষ্ঠানে পরিণত করার সম্ভাবনা তৈরি করেছে। ইআরপি সফটওয়্যার বাস্তবায়ন হলে প্রতিষ্ঠানের অভ্যন্তরীণ প্রশাসনে অনিয়ম রোধ এবং দক্ষতা বৃদ্ধি সম্ভব হবে। এসব কাজ নির্ধারিত সময়ে সম্পন্ন করতে ধারাবাহিক তদারকি প্রয়োজন এবং প্রকল্প পরিবর্তন বা ক্রয় সিদ্ধান্তে স্বচ্ছতা ও ভ্যালু ফর মানি নিশ্চিত করতে হবে।

৬। বাংলাদেশ স্যাটেলাইট কোম্পানি লিমিটেড (বিএসসিএল) দেশের মহাকাশ ও টেলিযোগাযোগ সক্ষমতার অনন্য স্মারক হলেও বিগত শাসনামলে এর বাণিজ্যিক সম্ভাবনা ও কৌশলগত ব্যবহারে উল্লেখযোগ্য পরিকল্পনাহীনতা ছিল। বিশেষ করে অন্যান্য রাষ্ট্রীয় টেলিকম প্রতিষ্ঠানের সাথে সমন্বিত পরিকল্পনার অভাবে স্যাটেলাইটের বিশাল সক্ষমতা দীর্ঘকাল অব্যবহৃত ছিল। আমাদের সময়ে এই ধারা বদলে আমরা স্যাটেলাইট সক্ষমতার বাণিজ্যিক ব্যবহার এবং দুর্যোগকালীন নেটওয়ার্ক রেজিলিয়েন্স বৃদ্ধিতে বিএসসিএল-কে সক্রিয় করেছি। বর্তমানে এনজিএসও (NGSO) নীতিমালা চূড়ান্তকরণ এবং স্যাটেলাইট ইন্টারনেটের প্রেক্ষাপটে রেগুলেটরি ক্রিয়ারেন্স ও বাজার বিন্যাস সম্পন্ন করার কাজ অত্যন্ত গুরুত্বের সাথে এগিয়ে চলেছে।

ভবিষ্যৎ চ্যালেঞ্জ মোকাবিলায় আপনার মেয়াদে বিএসসিএল-এর বাণিজ্যিক সক্ষমতা বৃদ্ধিতে এবং আধুনিক প্রযুক্তির সাথে খাপ খাইয়ে নিতে ধারাবাহিক মনোযোগ প্রয়োজন। বিশেষ করে স্যাটেলাইট ইন্টারনেটের প্রবর্তন এবং দুর্যোগ ব্যবস্থাপনায় বিএসসিএল-এর সক্ষমতাকে প্রাতিষ্ঠানিক রূপ দেওয়ার পাশাপাশি সংশ্লিষ্ট প্রতিষ্ঠানগুলোর সাথে রিসোর্স অপটিমাইজেশন ও টেকসই নেটওয়ার্ক উন্নয়ন নিশ্চিত করতে হবে। বিএসসিএল-কে কেবল রাষ্ট্রীয় গৌরবের স্মারক নয়, বরং একটি দক্ষ রাজস্ব উৎপাদনকারী ও কৌশলগত সম্পদে পরিণত করার এই প্রক্রিয়া অব্যাহত রাখা জরুরি। সমুদ্রের ইকোনোমিক জোন ও বর্ধিত ইকোনোমিক জোনে ডিপ সী ফিসিং নৌযান ও জেলে নৌকা/ভেসেল সমূহের জন্য কমিউনিকেশনে সহায়তা করতে এস ও এফ ফান্ডের মাধ্যমে বিএসসিএল-কে অর্থায়ন করা হয়েছে, যেখানে স্যাটেলাইট ও প্ল্যাটফর্ম সেবার পাশাপাশি স্বল্প মূল্যে ডিভাইস দেয়া হবে।

স্যাটেলাইট ইমেজ প্রকিউরমেন্টে খরচ কমাতে ড্রোন পলিসি ইমপ্লিমেন্ট করার মাধ্যমে স্যাটেলাইট এবং ড্রোন ফুটেজ রিপোজিটরি নিয়ে কাজ করা শুরু করা যাতে পারে যাতে এই রিসোর্সগুলো বিভিন্ন সেক্টরে (রাজউকের মনিটরিং, ভূমির ম্যাপিং, দুর্যোগ ব্যবস্থাপনা, বন সংরক্ষণ ইত্যাদি) রিইউজ হতে পারে।

৭। বাংলাদেশ ডাক বিভাগের আধুনিকায়ন, লজিস্টিকস সেবার ডিজিটাল রূপান্তর এবং জনবল ব্যবস্থাপনায় গৃহীত কার্যক্রমগুলোর ধারাবাহিকতা বজায় রাখা জরুরি। বিগত শাসনামলে ডাক বিভাগ দীর্ঘদিন ধরে স্থবিরতা, শূন্যপদ জট, কাগজ নির্ভর প্রক্রিয়া, এবং সমসাময়িক লজিস্টিকস বাজারের সাথে তাল মেলাতে না পারার সমস্যায় ছিল। ফলে ই-কমার্স ও পার্সেল ডেলিভারির বাজারে পোস্ট অফিসের অংশীদারিত্ব কমেছে এবং বড় নেটওয়ার্কটি পর্যাপ্তভাবে ব্যবহার হয়নি। ইডি কর্মচারীদের জন্য নতুন নীতিমালা প্রণয়ন ও ভাতা বৃদ্ধির প্রস্তাব মন্ত্রণালয়ে অনুমোদনের অপেক্ষায় রয়েছে। জনবল ব্যবস্থাপনায় স্বচ্ছতা আনতে ডাটা অ্যানালিটিক্স ভিত্তিক পারফরম্যান্স অ্যানালাইসিস পদ্ধতি ব্যবহারের পরিকল্পনা নেওয়া হয়েছে।

জাতীয় লজিস্টিক্স ট্র্যাকিং প্ল্যাটফর্ম (CLTP) অনবোর্ড কার্যক্রম চলমান। ডোমেস্টিক মেইল ট্র্যাকিং ব্যবস্থা আরও শক্তিশালী করা এবং মেইলিং ও কুরিয়ার সার্ভিসের দক্ষতা বৃদ্ধিতে বিশেষ টিম গঠনের উদ্যোগ নেওয়া হয়েছে।

নাগরিক সেবা কেন্দ্র স্থাপন ও আধুনিকায়নের অংশ হিসেবে গুলশান ও উত্তরাসহ একাধিক কেন্দ্রে উন্নত সেবা চালু হয়েছে এবং পোস্টাল সেন্টারগুলোকে আধুনিক সেন্টারে রূপান্তরের কাজ এগিয়ে চলেছে। কাস্টমার কমপ্লেইন ম্যানেজমেন্ট এবং ডিজিটাল অ্যাড্রেস ম্যানেজমেন্টের জন্য নিজস্ব সোর্স কোড রিপোজিটরি তৈরির কাজ চলছে।

ডাক সেবা অধ্যাদেশ ও প্রশাসনিক সংস্কার কার্যক্রমের অংশ হিসেবে ফি রিফর্ম বা পুনর্নির্ধারণ, এবং সামগ্রিক দক্ষতাকে KPI ভিত্তিক মূল্যায়নের দিকনির্দেশনা রয়েছে। একইসাথে, পোস্টাল ব্যালট সেবাকে অধিকতর নির্ভরযোগ্য ও যুগোপযোগী করতে নির্বাচন কমিশনের সাথে সমন্বয় করে SOP প্রণয়ন ও বাস্তবায়ন চলমান। ডাক বিভাগের ফিলাটেলি সার্ভিস এবং পোস্টাল ব্যালট সেবা পুনরিস্তীবিভ করা হয়েছে। জাতীয় নির্বাচনে পোস্টাল ব্যালট প্রসেসিং এর কাজে ডাক বিভাগ আন্তরিকতার সাথে কাজ করেছে। ভবিষ্যতে পোস্টাল ব্যালটের সার্ভিস আরো উন্নত, ইন্ক্লুসিভ এবং ইউজার ফ্রেন্ডলি করার উদ্যোগ নেয়া যেতে পারে।

উল্লেখ্য, বাংলাদেশ ডাক আইন হালনাগাদ ও সংশোধনের কাজ সম্পন্ন হয়েছে এবং আইনটিকে সম্পূর্ণ নতুন রূপ দিয়ে ‘ডাকসেবা অধ্যাদেশ, ২০২৬’ নামে Post Office Act, 1898 প্রতিস্থাপিত হয়েছে। এতে লাইসেন্সিং কর্তৃপক্ষকে আইনিভাবে ক্ষমতায়িত করে বাণিজ্যিক ডাক ও কুরিয়ার অপারেটরদের লাইসেন্সিং ও শৃঙ্খলা সংক্রান্ত নিয়ন্ত্রণ কাঠামো শক্তিশালী করার লক্ষ্য নির্ধারণ করা হয়েছে। এই আইন অনুযায়ী সংস্কার কার্যক্রম সময়মতো সম্পন্ন করা প্রয়োজন, কারণ অতীতে নিয়ন্ত্রক দুর্বলতার কারণে বাজারে অনিয়ম ও সেবার মানে বৈষম্য তৈরি হয়েছিল।

৮। শিল্প খাতে বাংলাদেশ ক্যাবল শিল্প (বাকেশি) এবং বাংলাদেশ টেলিফোন শিল্প সংস্থা (টেশিস)-কে দেশীয় উৎপাদন ক্ষমতা ও প্রযুক্তি স্বনির্ভরতার কৌশলগত প্ল্যাটফর্ম হিসেবে প্রতিষ্ঠা করা আমাদের অন্যতম অগ্রাধিকার ছিল। বিগত শাসনামলে এই প্রতিষ্ঠানগুলোতে ধারাবাহিক বিনিয়োগ এবং কার্যকর কৌশলগত অংশীদারিত্বের চরম ঘাটতি ছিল; ফলে আমদানিনির্ভরতা বৃদ্ধি পেয়েছে এবং স্থানীয় শিল্প সম্ভাবনাগুলো যথাযথভাবে কাজে লাগেনি। এই স্ববিরতা কাটাতে আমরা বাকেশি ও টেশিস উভয় প্রতিষ্ঠানেই আধুনিকায়ন ও বহুমুখীকরণ কার্যক্রম শুরু করেছি। বাংলাদেশ ক্যাবল শিল্প (বাকেশি) এবং বাংলাদেশ টেলিফোন শিল্প সংস্থা (টেশিস) কে দেশীয় উৎপাদন ক্ষমতা ও প্রযুক্তি স্বনির্ভরতার কৌশলগত প্ল্যাটফর্ম হিসেবে ধরে রাখা হয়েছে। বিগত শাসনামলে দেশীয় উৎপাদন ক্ষমতা ও প্রযুক্তি উন্নয়নে ধারাবাহিক বিনিয়োগ এবং কৌশলগত পার্টনারশিপে ঘাটতি ছিল। ফলে আমদানিনির্ভরতা বেড়েছে এবং স্থানীয় শিল্প সম্ভাবনা কাজে লাগেনি।

বাকেশিতে সুপার এনামেলড কপার ওয়্যার প্ল্যান্টের বাণিজ্যিক উৎপাদন শুরু হয়েছে। পাওয়ার ক্যাবল উৎপাদন ক্ষমতা বৃদ্ধি, বাজার সম্প্রসারণ, CAT 6 ক্যাবল উৎপাদনের পরিকল্পনা এবং ফাইবার ক্যাবল আধুনিকায়নকে মূল লক্ষ্য হিসেবে নির্ধারণ করা হয়েছে। কপার ও অ্যালুমিনিয়াম রড প্ল্যান্টের সম্ভাব্যতা যাচাই চলছে। ইলেকট্রনিক্স এক্সেসরিজ উৎপাদনে টেশিস ও বাকেশি যৌথভাবে কাজ করার সিদ্ধান্ত গৃহীত হয়েছে। সবুজ সনদ প্রাপ্তি এবং সোলার প্যানেল স্থাপনের কার্যক্রমও শুরু হয়েছে।

টেশিসে সফটওয়্যার ইনকিউবেশন সেন্টার স্থাপন, ব্যাটারি রিফারিশমেন্ট ও রিপেয়ার সেন্টার, আধুনিক ডিভাইস উৎপাদনে কৌশলগত পার্টনারশিপ, CCTV ও সার্ভেলেন্স সরঞ্জাম, 4G বা 5G ডুয়াল ব্যান্ড রাউটার সংযোজন, স্মার্ট ইলেকট্রিক মিটার ও IoT ডিভাইস উৎপাদন, হাই টেক পার্ক ভিত্তিক শিল্প কার্যক্রম এবং ই-বাইক উৎপাদনের মতো উদ্যোগসমূহ এগোচ্ছে। এ উদ্যোগগুলোর টাইমলাইন, গুণগত মান, সাপ্লাই চেইন এবং বাজারে কার্যকর অনুপ্রবেশ নিশ্চিত করতে ধারাবাহিক তদারকি প্রয়োজন এবং প্রাকিউরমেন্টে ভেন্ডর নিউট্রালিটি ও মান নিয়ন্ত্রণকে অগ্রাধিকার দিতে হবে।

৯। ইতোমধ্যে বিটিসিএল পরীক্ষামূলকভাবে এমভিএনও (MVNO, Mobile Virtual Network Operator) নেটওয়ার্ক চালুর অনুমোদন পেয়েছে। BTCL টিম MVNO চালুর জন্য কাজ করছে, দ্রুতই এটার পাইলট শুরু হবে ইনশাআল্লাহ। দায়িত্ব নেয়ার পরে সব ধরনের টেকনোলজি আনলকের উদ্যোগ নেয়া হয়েছে।

এই ধারাবাহিকতায় বাংলাদেশে প্রথমবারের মতো চালু হয়েছে ফাইভ-জি। VoLTE রোল আউট শুরু হয়েছে পুরোদমে। ভয়েস ওভার ওয়াইফাই (VoWiFi) লঞ্চ হয়েছে। শুরু হচ্ছে প্রাইভেট ফাইভ-জি। আমি আশা করি, দ্রুতই লার্জ পাবলিক ইনডোর (LPI), স্মল সেল, নেটওয়ার্ক স্লাইস এবং হট স্পট-এর মতো টেলিকম সার্ভিস গুলোও শুরু হবে। নতুন পলিসিতে এসব প্রযুক্তিকে উন্মুক্ত করে দেওয়া হয়েছে।

কানেক্টিভিটিকে কেন্দ্র করে টেলিকম খাতকে জিম্মি করে রেখেছে প্রায় সব লেয়ারের দুর্বল সত্ত্বানে ও অঙ্গতানে। ফ্যাসিস্ট গোষ্ঠী টেলিকম কানেক্টিভিটিকে ব্যবহার করেছে টোল প্লাজা হিসেবে। আমরা চাই কানেক্টিভিটি হাইওয়েতে কোনো টোল প্লাজা থাকবে না। অতি সামান্য বিনিয়োগ করে কেউ বিশাল ইকোসিস্টেম এবং ইনভেস্টমেন্ট সার্কেলে রেন্ট সিকিং করতে পারবে না। এজন্য আমরা ফাইবার উন্মুক্ত করে দিয়েছি ও দিচ্ছি, যাতে এক্সেস টু ফাইবার এবং বান্ধ ডেটা ক্লোকে কোনো অবকাঠামোগত বাধা না আসে।

আমি আশা করি, দেশের কানেক্টিভিটি নির্ভর টেলিকম ধীরে ধীরে ডিজিটাল সার্ভিস বেজড ইনডাস্ট্রিতে রূপান্তরিত হবে। যেখানে এড টেক, অ্যাগ্রো-টেক, গভ-টেক, ফিন-টেক, হেলথ-টেক, স্টার্টাপ, ওটিটি বেজড ডিজিটাল সার্ভিসের অমিত সম্ভাবনা তৈরি হবে। টেলিকম ও ইন্টারনেট-কানেক্টিভিটিতে পড়ে থাকায় দেশের ডিজিটাল সার্ভিস ও ডিজিটাল ট্রান্সফর্মেশনের বিকাশ বাধাগ্রস্ত হয়েছে। এখানে মোবাইল ইন্টারনেট ও ব্রডব্যান্ডের বাজার এখনো মেগাবাইট/গিগাবাইট কেন্দ্রিক। অথচ দরকার ছিল সেবা কেন্দ্রিক যেখানে বেসিক প্যাকেজের উপর শিক্ষার্থী, পেশাজীবী, সেবা গ্রহীতার জন্য আলাদা আলাদা প্যাকেজ, থাকবে স্পোর্টস ও এন্টারটেইনমেন্ট প্যাকেজ, থাকবে সিকিউর ইন্টারনেট প্যাকেজ, আইওটি, মিশন ক্রিটিক্যাল কমিউনিকেশন, ইন্ডাস্ট্রি ক্রিটিক্যাল কমিউনিকেশন, কিংবা গেমিং কিংবা সফটওয়্যার ডেভেলপমেন্ট কিংবা ক্লাউড ও কম্পিউটিং সার্ভিস। তারা আমাদের বামন করে রেখেছিল।

নতুন 'Telecommunications Network and Licensing Policy 2025' কানেক্টিভিটি থেকে সার্ভিসে রূপান্তরের ফিলোসফিতেই গড়ে তোলা হয়েছে। ধীরে ধীরে ইন্ডাস্ট্রির সব প্লেয়ারকে সার্ভিসে মাইগ্রেট করা ই গ্র্যান্ড ডিজাইন। টেলিকমকে কানেক্টিভিটি ব্যবসায় সীমাবদ্ধ করে আওয়ামীলীগ এই ইন্ডাস্ট্রিকে একটা ইনোভেশনহীন, স্তনশূন্য এবং বর্তুলাকারে সার্ভিস সংকীর্ণ করে রেখেছিল। এই বামন দশা থেকে মুক্তির যাত্রা শুরু হয়েছে। এর চূড়ান্ত সাফল্য হবে যখন আমরা ডিজিটাল পাবলিক ইনফ্রাস্ট্রাকচার (DPI) এর ইন্টারকানেক্টিভিটি হাব এবং ইলেকট্রনিক আইডেন্টিটি অথেন্টিকেশন লেয়ার গড়ে তোলার ডিজিটাল ট্রান্সফর্মেশন প্রজেক্ট, ন্যাশনাল এপিআই এক্সচেঞ্জ এবং সেক্টরাল ইন্টার-অপারেবিলিটি এক্সচেঞ্জ প্রজেক্টগুলো শেষ করতে পারবো।

আবার ডেটা ওশান এবং ডিজিটাল সার্ভিস ইকোসিস্টেমে ব্যক্তিগত উপাত্ত ও গোপনীয়তা ব্যবস্থাপনায় শৃঙ্খলা আনতে আমরা করেছি বাংলাদেশের প্রথম 'পার্সোনাল ডেটা প্রটেকশন অর্ডিন্যান্স ২০২৫' এবং 'ন্যাশনাল ডেটা গভর্নেন্স অর্ডিন্যান্স ২০২৫'। অর্থাৎ লিগ্যাল ফ্রেইম ওয়ার্ক ফাউন্ডেশন, কানেক্টিভিটি টু সার্ভিস ট্রান্সফর্মেশন, ডিজিটাল পাবলিক ইনফ্রাস্ট্রাকচার, ইলেক্ট্রনিক আইডি ম্যানেজমেন্ট এবং নাগরিক সেবার মতো ইন্টারকানেক্টিভিটি হাব বেজড সার্ভিস ইনোভেশন। সবমিলে এ এক মহাযাত্রা, যার দ্বার আমরা উন্মোচন করে যাচ্ছি।

আমরা দুর্বৃত্তায়ন ও অচলায়তন ভেঙে সার্ভিস বেজড ইকোসিস্টেম গড়ছি, দীর্ঘমেয়াদে এর সুফল পাবে দেশের মানুষ এবং সার্বিক অর্থনীতি। আপনার সদয় বিবেচনার জন্য সবিনয় অনুরোধ, উপযুক্ত সংস্কার ও আধুনিকায়ন কার্যক্রমগুলোকে একই ধারাবাহিকতায় এগিয়ে নেবেন এবং সংশ্লিষ্ট প্রতিষ্ঠানগুলোকে নিবিড় ও সময়াবদ্ধ পর্যবেক্ষণের মধ্যে রাখবেন।

আপনার উত্তরোত্তর সাফল্য কামনা করছি।